

REGULARIZED IMAGE RECONSTRUCTION IN PARALLEL MRI WITH MATLAB®

Joseph Suresh Paul

Raji Susan Mathew



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Joseph Suresh Paul and Raji Susan Mathew

Medical Image Computing and Signal Processing Laboratory
Indian Institute of Information Technology and Management-Kerala
Trivandrum, India



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Preface

Image reconstruction in parallel magnetic resonance imaging (pMRI) is a mathematical process that generates combined images using a limited number of data samples acquired simultaneously from an array of receiver coils. Based on the nature of under-sampling and trajectories employed for acquisition, several variants of reconstruction methods have evolved since the advent of pMRI. Two major categories of reconstruction methods are those requiring prior or self-calibration, and the iterative type methods which do not require calibration data. Regularization methods form an integral part in both variants of reconstruction approaches as they involve inversion of ill-posed matrices of large sizes.

This book provides a detailed discussion on the key issues and challenges relating to the choice and usage of regularization methods applied to pMRI reconstruction algorithms. The main motivation behind the writing of this book is to bring forward some of the unique identifiable features of several types of regularization techniques applied to pMRI reconstruction. The book summarizes key aspects required for judiciously choosing regularization parameters required for artefact- and noise-free reconstruction, with relevance to each of the specific reconstruction variants. A broad spectrum of pMRI reconstruction algorithms are discussed along with case studies and MATLAB® codes. Case studies are presented with experimentally acquired data from both Siemens and GE scanners using different types of MR sequences.

This book targets two groups of readers. The first group includes researchers and students involved in optimization and development of pMRI reconstruction methods. The latter group of readers includes the practitioners of medical imaging and MR technologists working in MRI centers and industries who require knowledge of image reconstruction for their projects. Readers are not presumed to have an advanced background in mathematics, but they are expected to have a basic understanding of linear algebra and probability theory. Readers of this book are also expected to have basic knowledge of signal and image analysis.

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Authors

Joseph Suresh Paul is currently a professor at the Indian Institute of Information Technology and Management–Kerala (IIITM-K), India. He obtained his PhD in Electrical Engineering from the Indian Institute of Technology, Madras, India, in 2000. His research is focused on magnetic resonance (MR) imaging from the perspective of accelerating image acquisition, with the goal of enhancing clinically relevant features using filters integrated into the reconstruction process. His other interests include mathematical applications to problems in MR image reconstruction, compressed sensing, and super-resolution techniques for MRI. He has published a number of articles in peer-reviewed international journals of high repute.

Raji Susan Mathew is currently pursuing her PhD in the area of magnetic resonance (MR) image reconstruction. She received a bachelor degree in Electronics and Communication Engineering from Mahatma Gandhi University, Kottayam, and a master's degree in Signal Processing from the Cochin University of Science and Technology, Kochi, in 2011 and 2013, respectively. She is a recipient of the Maulana Azad National Fellowship (MANF) from the University Grants Commission (UGC), India. Her research interests include regularization techniques for MR image reconstruction and compressed sensing.



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Parallel MR Image Reconstruction

1.1 Basics of MRI

Magnetic resonance imaging (MRI) is a non-invasive medical imaging technique that uses the magnetic property of materials for visualization of soft tissues. As opposed to other imaging modalities, such as computed tomography (CT) or ultrasound, MRI utilizes the electromagnetic properties of hydrogen nuclei present in fat and water molecules. Once hydrogen atoms are placed in a powerful external magnetic field B_0 , the nuclear spins are aligned with the direction of the magnetic field. As the number of protons aligning with the field is slightly larger, there exists a net magnetization in the direction of the main magnetic field [1]. The precession frequency of the spins is proportional to the main magnetic field, that is, $\omega_0 = \gamma B_0$, where γ is a gyromagnetic ratio constant. This precession frequency which is in the MHz range is called Larmor or resonance frequency.

1.1.1 Basic Elements of an MR System

The MR system consists of four essential components, including:

1. A uniform, steady magnetic field B_0 . In a typical clinical MR system, this magnetic field strength is either 1.5 or 3.0 T. When an object to be imaged is placed in this field, a macroscopic or bulk magnetization M_0 is established.
2. A pulsed, high power, radio-frequency (RF) magnetic field B_1 . The magnetization in some specific region or all of the volume is excited using this pulse, giving rise to a detectable signal.
3. A switched, linear, magnetic field gradient G . This field is superimposed on the uniform field B_0 to provide the spatial differentiation required to form an image. There are three orthogonal gradients (G_x, G_y, G_z) , which can combine to yield any gradient direction in three-dimensional (3D) space.
4. RF receiver coil used to detect the MR signal from the sample. The received signal is processed to produce the image.

1.1.2 Static Magnetic Field B_0

During MR data acquisition, the patient lies in a strong static magnetic field. Typically, the methods to generate this field can be broadly classified into three: fixed magnets, resistive magnets in which current is passed through a traditional coil of wire, and superconducting

magnets. However, as fixed and resistive magnets are generally restricted to field strengths below 0.4 Tesla (T), they cannot generate the higher field strengths necessary for high-resolution imaging. Therefore, superconducting magnets are used in most high-resolution imaging systems. The superconducting magnets are made from coils of superconducting wire, and they require the coils to be soaked in liquid helium to reduce their temperature to a value close to absolute zero.

1.1.3 RF Magnetic Field B_1

The RF pulses required to resonate the hydrogen nuclei are generated using an MRI transmitter coil. The range of frequencies in the transmit excitation pulse together with the magnitude of the slice-selection gradient field decide the width of the imaging slice. Typically, a transmit pulse produces an output signal with a relatively narrow bandwidth of ± 1 kHz. The time-domain waveform that is required to produce this narrow frequency band resembles a sinc function. This waveform is generated digitally at baseband and then up-converted using a mixer to the appropriate centre frequency. Very-high-speed, high-resolution digital-to-analog converters (DACs) can be used for direct RF generation of transmit pulses up to 300 MHz. Therefore, waveform generation and up-conversion over a broad band of frequencies can be accomplished in the digital domain.

1.1.4 RF Receiver

Due to the capability of RF receiver to process signals from multiple receiver coils, modern MRI systems use more than two receivers to process the signals. The bandwidth of the received signal is dependent on the magnitude of the gradient field and is typically less than 20 kHz. Modern receiver systems use high-input-bandwidth, high-resolution, 12–16-bit analog-to-digital converters (ADCs) with sample rates up to 100 MHz, which eliminates the need for analog mixers in the receive chain by directly sampling the signals.

1.1.5 Gradient Fields

The gradient field used in MRI is typically a small magnetic field applied along the longitudinal axis linearly varying in space. In the presence of gradient fields, the total magnetic field at a given location is given by:

$$\mathbf{B}(\mathbf{r}) = (B_0 + G_x x + G_y y + G_z z) \mathbf{k} = (B_0 + \mathbf{G} \cdot \mathbf{r}) \mathbf{k}, \quad (1.1)$$

Here, $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, denotes the spatial location of the spin, and the vector $\mathbf{G} = [G_x \ G_y \ G_z]$ the field gradient, with G_x , G_y and G_z representing the gradient strengths along x , y , and z directions, respectively. As the overall main magnetic field varies in space, the spins inside the volume have spatially dependent resonance frequencies, expressed as:

$$\omega(\mathbf{r}) = \gamma (B_0 + \mathbf{G} \cdot \mathbf{r}) \quad (1.2)$$

The addition of a gradient field to the static B_0 field enables the resultant total magnetic field to vary linearly in space. The Larmor frequencies also vary with spatial location as they are proportional to the field.

1.1.6 Slice Selection

Slice selection is a technique of isolating a single plane in the object being imaged, by exciting the spins in that plane only. This is achieved by applying an RF pulse, which affects only a limited part of the nuclear magnetic resonance (NMR) spectrum, in the presence of a linear field gradient along the direction in which the slice is to be selected. This can be explained in the following three steps involving application of:

1. Gradient pulse: A gradient magnetic field is applied in the z-axis superimposed on the background magnetic field. Since the frequency of precession depends on the magnetic field, the nuclei will have different frequencies throughout the z-axis.
2. Slice-select pulse: The magnetization of the nuclei is flipped to the transverse plane by the application of an RF pulse. The RF pulse frequency should be the same as the Larmor frequency of the nuclei in order to flip the precession of the nuclei. As the Larmor frequency of nuclei is different along the z-axis, a slice of interest can be selected by altering the frequency of the RF pulse.
3. Reset pulse: Due to the variation of frequencies along the gradient, the nuclei begin to precess out of phase. Therefore, the nuclei are to be reset before selecting the next slice. This is achieved by temporarily reversing the gradient or applying 180° RF pulse to reverse the precessional frequencies.

These steps for slice selection are illustrated in Figure 1.1. There are certain factors which affect the slice properties. These include RF pulse bandwidth, RF pulse frequency and gradient strength. The RF pulse bandwidth is the range of frequencies within the pulse. Large bandwidth results in the selection of a larger slice, and vice versa. Varying the RF pulse frequency moves the selected slice up or down the z-axis. The size of the slice is determined by altering the gradient strength.

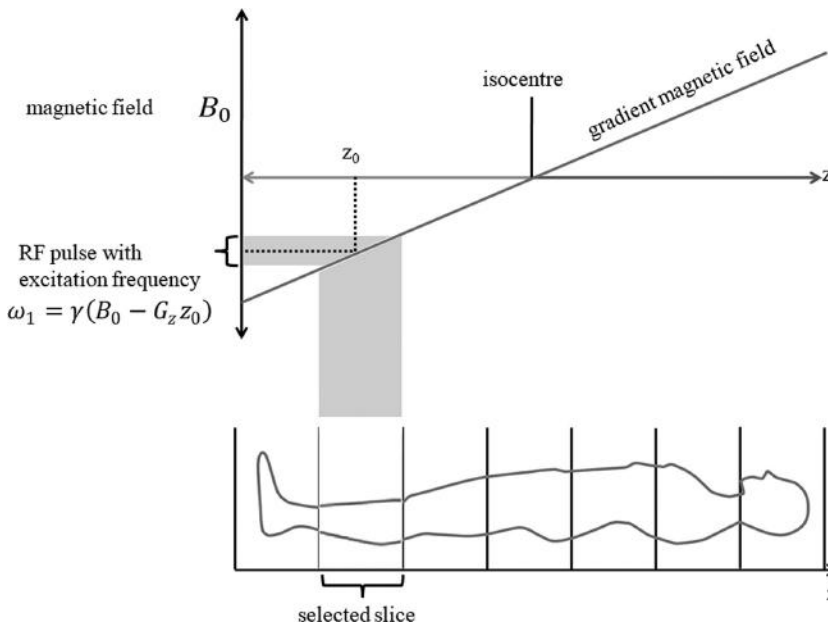


FIGURE 1.1

Slice selection and selective excitation.

1.1.7 Generation of FID

Upon application of an RF pulse at the resonance frequency, the atoms absorb energy and tilt the net magnetization onto the xy plane. Also, the spins associated with the individual protons begin to rotate about the z-axis at the same phase. When the RF transmitter is switched off, the protons return back to the equilibrium state, that is, from traverse plane (xy) to longitudinal plane (z). This relaxation is characterized by the time constant called spin-lattice relaxation time, denoted by T_1 . With M_{z0} and $M_z(t)$ denoting the equilibrium magnetization and the longitudinal magnetization at time t , respectively, the recovery of longitudinal magnetization is modeled as a function that grows exponentially with time constant T_1 . This is expressed mathematically as:

$$M_z(t) = M_{z0} (1 - e^{-t/T_1}). \quad (1.3)$$

Due to the spin-spin interactions and variations in B_0 , the spins eventually lose their transverse magnetization. The time that characterizes their return to the equilibrium state is called spin-spin relaxation time, denoted by T_2 [1]. With M_{xy0} denoting the transverse magnetization at equilibrium, the transverse magnetization M_{xy} is obtained as follows:

$$M_{xy}(t) = M_{xy0} e^{-t/T_2}. \quad (1.4)$$

The changes in magnetization induce an alternating current in the receiver coils, and this signal is referred to as free induction decay (FID). FID is a short-lived sinusoidal electromagnetic signal generated immediately following the 90° RF pulse. This signal is induced in the receiver coil by the rotating component of the magnetization vector in the xy plane, which crosses the coil loops perpendicularly. The FID is one among the four basic types of NMR signals generated in different ways. The other types include the gradient echo (GRE), induced using one RF pulse and gradient reversal; the spin echo (SE), using two RF pulses; and the stimulated echo generated using the application of three or more RF pulses.

1.1.8 Imaging

In one-dimensional (1D) imaging, a frequency-encoding (FE) gradient pulse is applied immediately after application of the 90° -RF pulse, as shown in Figure 1.2. The resulting FID signal is simultaneously digitized along with application of a readout pulse. The readout is enabled synchronously with the FE gradient pulse. Application of this gradient along the x-direction, modulates the precession frequency of spins as a linear function of their positions along the x-axis. This causes the spin frequency to vary spatially:

$$\omega(x) = \gamma(xG_x + B_0). \quad (1.5)$$

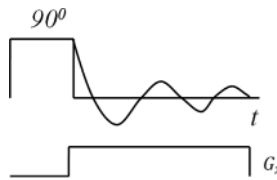


FIGURE 1.2

Application of gradient in 1D imaging. Following the excitation with a 90° RF pulse, the FID signal is read off in the presence of the FE gradient.

The net signal generated from an ensemble of spins can be modeled as the summation of their contributions from each location along the x-direction. In the case where the spin density is modeled as a continuous function of space, the resultant signal can be expressed as:

$$s(t) = kM_0 e^{-j\gamma B_0 t} \int_{-\infty}^{\infty} \rho(x) e^{-t/T_2(x)} e^{-j\gamma x G_x t} dx \quad (1.6)$$

In the signal equation, kM_0 can be removed by using appropriate normalization/scaling. Also, with the assumption that $T_2 \gg t$, or by quadrature detection, the factor $e^{-j\gamma B_0 t}$ can be removed. Thus, the new signal equation takes the following form:

$$s(t) = \int_{-\infty}^{\infty} \rho(x) e^{-j\gamma x G_x t} dx \quad (1.7)$$

Denoting $k_x = \gamma G_x t / 2\pi$, and rewriting Equation (1.7) as a function of k_x yields:

$$R(k_x) = \int_{-\infty}^{\infty} \rho(x) e^{-j2\pi k_x x} dx \quad (1.8)$$

The integral in Equation (1.7) represents the Fourier transform of the spin density function. The acquired MR data can be thought of as being defined in the spatial frequency domain (Fourier space). It follows that the spin density can be obtained by taking the inverse Fourier transform of the signal as:

$$\rho(x) = F^{-1}\{R(k_x)\}. \quad (1.9)$$

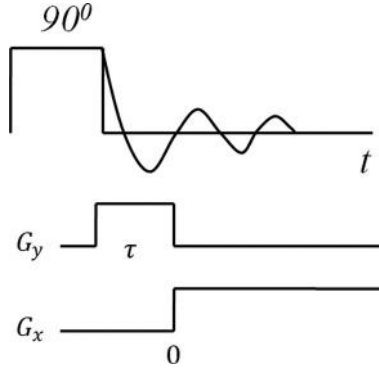
For two-dimensional (2D) image acquisition, the pulse sequence must involve two gradients. Suppose a 90° pulse excites spins within a transverse slice at a particular value of z . In order to acquire a spatially differentiated signal from the slice, gradients in the x and y directions are to be applied as shown in [Figure 1.3](#). In contrast to 1D imaging, a G_y gradient is additionally applied for a fixed time τ , before the start of the acquisition. Then the spins along y are dephased by

$$\theta = \gamma y G_y \tau. \quad (1.10)$$

After switching off the gradient, the acquisition begins and the frequency differentiation along the y axis is also turned off. Alternatively, the differentiation in the phase is acquired over a short duration of the G_y pulse. Therefore, the G_y gradient is referred to as a phase-encoding (PE) gradient.

The FID is sampled in the presence of G_x after turning off the y gradient. The spatial frequencies denoted by k_x and k_y can be defined as follows:

$$\begin{aligned} k_x &= \frac{\gamma G_x t}{2\pi} \\ k_y &= \frac{\gamma G_y \tau}{2\pi}. \end{aligned} \quad (1.11)$$

**FIGURE 1.3**

Application of gradients in 2D imaging. Following excitation with a 90° pulse, a phase-encoding gradient G_y is applied in the y direction for a time τ . At the end of that time, the phase acquired by parts of the sample along the y -axis will vary in proportion to the value of y . Then, the FID signal is read off in the presence of the frequency-encoding gradient G_x .

For one FID signal, there is only one value of the vertical spatial frequency k_y . The sequence must be repeated several times to obtain other values of k_y , with the values of G_y stepped up or down with each onset of the FID signal generation. Once the required set of data is obtained, the data can be arranged to form a 2D array. The signal equation in the 2D case can then be expressed as:

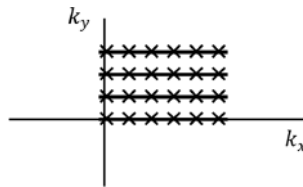
$$R(k_x, k_y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \rho(x, y) e^{-j2\pi(k_x x + k_y y)} dx dy, \quad (1.12)$$

where ρ varies in the xy -plane. The inverse 2D Fourier transform of the array in (1.12) reconstructs the spin density in the image space as follows:

$$\rho(x, y) = F^{-1} \{ R(k_x, k_y) \}. \quad (1.13)$$

The sampled signal is arranged along horizontal lines in the $(k_x k_y)$ -plane, as shown in [Figure 1.4](#). The horizontal line with $k_y = 0$ represents the FID signal sampled with $G_y = 0$. For a gradient step ΔG_y , the separation between the sampled horizontal lines in the $(k_x k_y)$ -space is given by

$$\Delta k_y = \frac{\gamma \Delta G_y \tau}{2\pi}. \quad (1.14)$$

**FIGURE 1.4**

The trajectory in the $k_x k_y$ plane resulting from the application of the Fourier technique of [Figure 1.3](#). Each line corresponds to an increment of G_y . The crosses denote the sample points.

Until now, the discussion was limited to the acquisition in the first quadrant of the (k_x, k_y) -plane. Similarly, one could sample the fourth quadrant and complete the positive half-plane. This is achieved by allowing the G_y gradient to take on negative values. The negative half-plane is acquired by allowing G_x to take negative values. Because each step change in the gradient requires a new FID signal and the associated waiting time TR , sampling all the quadrants becomes a lengthy process. In practice, when the gradient pulses are rectangular, the spatial frequencies are obtained by integration over the gradient pulse duration. For time varying gradient pulses,

$$k_x = \frac{\gamma}{2\pi} \int_0^t G_x(t') dt', \quad (1.15)$$

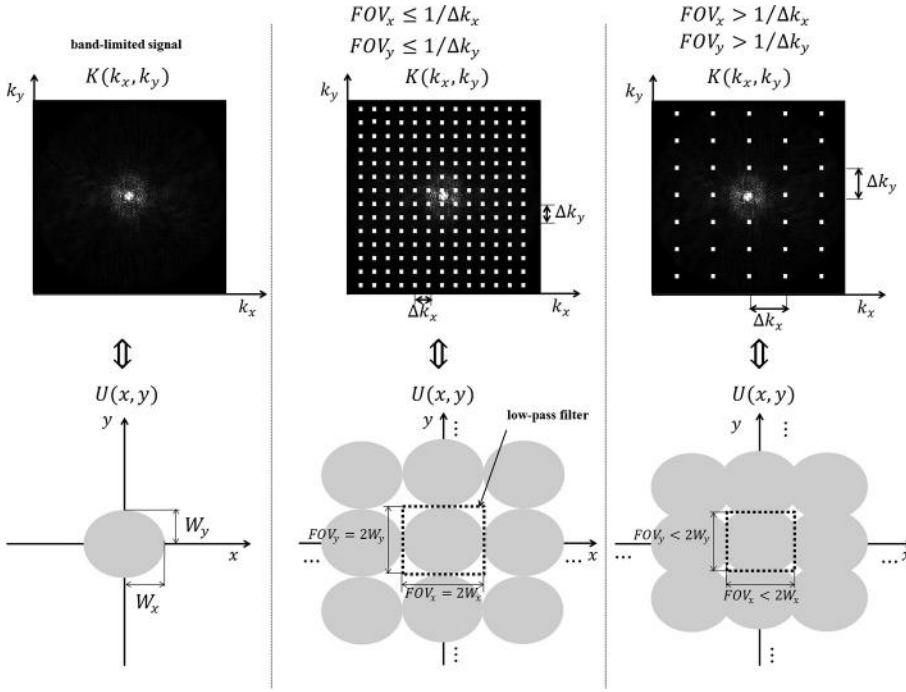
and

$$k_y = \frac{\gamma}{2\pi} \int_0^\tau G_y(\tau') d\tau'.$$

1.2 Nyquist Limit and Cartesian Sampling

For better interpretation of Nyquist limits for k -space sampling, let us first recall the Nyquist sampling theorem for a 1D time domain signal. Suppose the highest frequency component for a 1D analog signal is $f_{\max}$. According to the Nyquist theorem, the sampling frequency must be at least $2f_{\max}$, in order to perfectly recover the underlying signal. The multiplication of the time domain signal with the sampling train is equivalent to a convolution in the frequency domain, wherein the frequency spectrum of the 1D time domain signal is convoluted with the Fourier transform of the impulse train. However, as the Fourier transform of the impulse train is a scaled impulse train with twice the separation in time domain, the convoluted signal in the Fourier domain will have the frequency spectrum of the 1D signal replicated at each of the impulse. Therefore, if the separation between samples in the time domain is chosen as $t_s = 1/2f_{\max}$, then the 1D signal can be recovered back from its corresponding frequency spectrum by applying a low-pass filter to the frequency spectrum. In case of oversampling also, the signal can be recovered as the separation between the impulses is larger. However, if the sampling frequency f_s is less than that of the Nyquist rate, then the frequency spectra overlap and the signal cannot be recovered by using low-pass filtering.

Extension of this idea to MR acquisition needs a slightly different interpretation. This is because sampling in MRI is performed in the k -space, which itself is a frequency domain. Therefore, the bandwidth is interpreted as a quantity pertaining to the maximum spatial information contained in the imaging object. In other words, this is simply the extent of spatial support of the object in the imaging plane. Consider an elliptical object with dimensions $2W_x$ and $2W_y$ along the x - and y -axes, respectively. The Nyquist sampling steps in k -space would then be $\Delta k_{xN} = 2\pi/2W_x$ and $\Delta k_{yN} = 2\pi/2W_y$, respectively. If the sampling steps are reduced further below the Nyquist limits Δk_{xN} and Δk_{yN} , the repetition intervals of the imaging area, referred to as the field of view (FOV), would increase in relation to the object dimensions. This is the case with oversampling when $\text{FOV}_x/2W_x > 1$, and $\text{FOV}_y/2W_y > 1$. [Figure 1.5](#) illustrates the repetitions of the imaging plane when the k -space is sampled at the Nyquist rate and below. The leftmost panel shows the k -space, which

**FIGURE 1.5**

k -space sampling and Nyquist criterion. The k -space sampling interval Δk_x and maximum extent $k_{x\max}$ are related to the FOV_x and voxel size Δx , respectively, of the reconstructed image in x -direction. Similarly, Δk_y and $k_{y\max}$ are related to the FOV_y and Δy .

is band limited, and the corresponding full object in the image space. The band limits in the x and y dimensions are FOV_x and FOV_y , respectively. The central panel shows the sampling grid on the k -space, and the corresponding object space with $\text{FOV}_x = 2W_x$ and $\text{FOV}_y = 2W_y$. As in the 1D case, the object space is repeated periodically with a repetition interval equal to the FOV. Here, the object images do not overlap, and hence the object of interest can be recovered by application of a low-pass filter. The rightmost panel illustrates the case in which $\text{FOV}_x < 2W_x$ and $\text{FOV}_y < 2W_y$. In this case, the object images overlap, and a perfect recovery of the object is not possible.

FOV depends on the strengths of magnetic gradients, their durations and the acquisition time. By acquiring data at uniform intervals Δt_s in time, the smallest FE step in the k -space will be $\Delta k_x = \gamma G_x \Delta t_s / 2\pi$. This in turn determines the size of FOV. FOV should be set appropriately to cover the whole imaged object. Since the encoding function is periodic, parts of the imaged object outside the FOV are misinterpreted by aliasing as being inside the FOV. While designing an MRI acquisition, appropriate choices of the FOV and the spatial resolution (i.e., voxel size) are to be determined. As the magnitude of the gradient fields cannot be increased beyond a limit, the total acquisition time is proportional to the number and length of the scan lines acquired, in the case of Cartesian imaging. In this case, a larger FOV (or smaller voxels) in the PE direction requires more scan lines. However, acquisition time is unaffected by the spacing between samples within a scan line, and arbitrarily large FOV in the FE direction can be achieved for free. Therefore, the longest dimension of the volume can be chosen as FE direction so that one can oversample the k -space in that direction to avoid any possibility of aliasing.

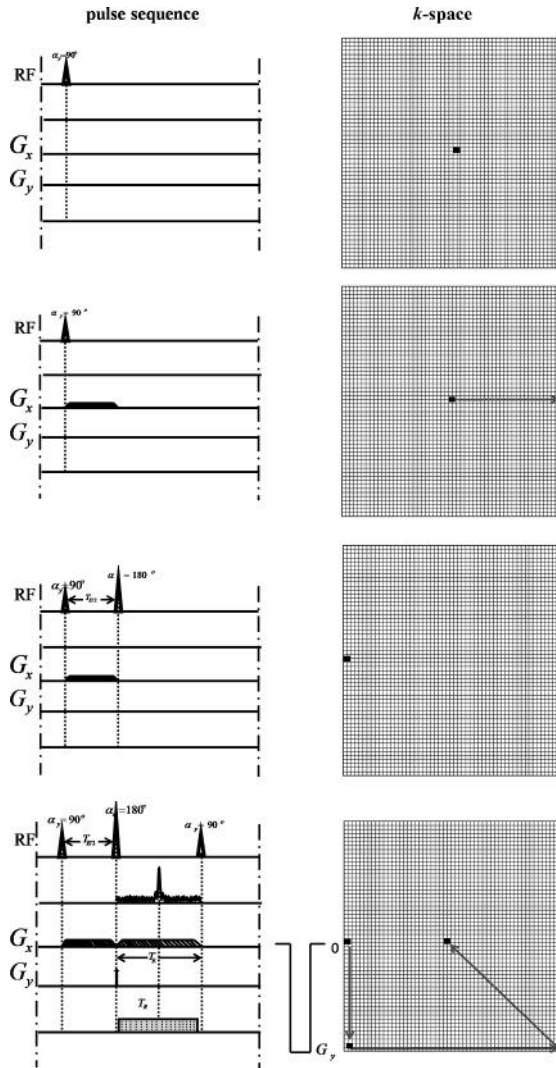
1.3 Pulse Sequencing and k -Space Filling

A pulse sequence refers to the repeated application of a set of RFs and gradient pulses for which the time duration between the successive repetitions together with the gradient pulse amplitude and shape will influence the resultant contrast and signal intensities of the MR image. There are computer programs that control all the hardware aspects of MRI acquisition process. Generally, the pulse sequences are defined using the repetition time (TR), the echo time (TE), the inversion time (TI) (if using inversion recovery) in milliseconds, and flip angle in case of a gradient echo sequence. TR is the time between two consecutive RF excitation pulses. A long TR allows the magnetization in tissues to relax and return to their original values. Alternatively, a short TR results in the protons from some tissues not having fully relaxed to realign with the main magnetic field before the next measurement. TE represents the time from the centre of the RF pulse to the centre of the echo. Larger extent of dephasing resulting from longer TE can lead to reduced signal intensities. In contrast with a shorter TE , the spins will remain in phase for a longer time, thereby reducing the extent of dephasing. In the inversion recovery sequence, a conventional SE sequence is preceded by a 180° inverting pulse. Here the inversion time TI refers to the time between the 180° inverting pulse and the 90° pulse. Flip angle is the angle by which the net magnetization vector is rotated by an RF pulse into the transverse plane. Flip angle depends on the particular pulse sequence utilized, and it is critical in determining the signal intensity as well as the image contrast.

SE is the most common pulse sequence used in MR imaging based on the RF refocusing of spins. This sequence uses 90° RF pulses to excite the magnetization and 180° RF pulses to refocus the spins and generate signal echoes. A schematic representation of pulse sequencing using SE and the corresponding k -space filling is shown in [Figure 1.6](#). The top panel shows application of a 90° RF excitation pulse. At this time point, there are no gradient fields, and the k -space location is at the zero frequency point, as shown in the right panel. In the next step, an FE gradient pulse with amplitude G_x is applied along the x -direction. With this, the k -space point traverses toward the $+k_x$ direction. Panels in the third row show the 180° RF pulse applied along the longitudinal axis that causes rotation of spins in the (x, y) -plane about the x -axis. Correspondingly, the k -space location shifts instantly from the $+k_{x,\max}$ position to the $-k_{x,\max}$ position on the extreme left side of the k -space. To traverse the lower portion of k -space, a PE pulse with maximum negative swing is applied, as shown in the bottom left panel. Together with this, an FE pulse of amplitude G_x is applied for a duration TE . The refocusing RF pulse, together with the FE gradient pulse, fills the echo samples along the lowest PE line. It is seen that the echo amplitude is maximum as the k -space trajectory passes through the centre of the line (i.e., the k_y -axis). Before the next repetition interval, the magnetizations relax and the k -space location again starts from the centre. This process is repeated with application of G_y gradients of different pulse amplitudes, sampled discretely at intervals Δk_y as dictated by the Nyquist criterion.

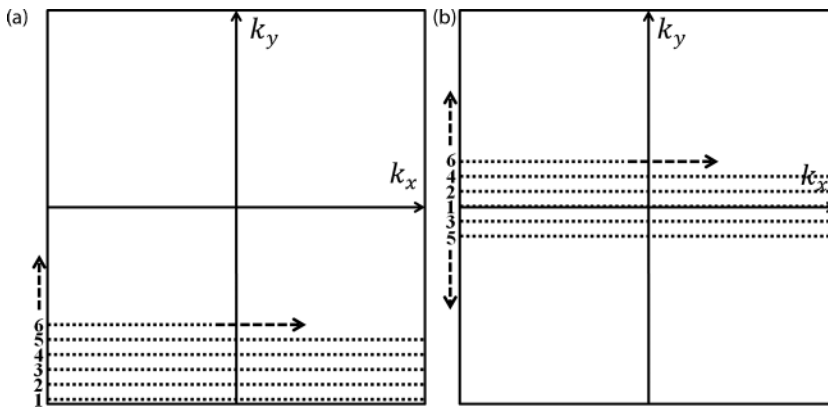
1.3.1 Cartesian Imaging

In conventional MR data acquisition, MR imaging is performed by acquiring k -space data along a Cartesian or rectilinear trajectory. Cartesian MRI refers to the acquisition of a Cartesian grid of samples of the spatial Fourier transform of the bulk magnetization.

**FIGURE 1.6**

The steps in a typical SE pulse sequence. Corresponding to each step, the k -space trajectories and location are shown in the right side.

Here data is sampled in a line-by-line fashion on a rectangular grid. The main benefit of this type of sampling trajectory is that images can be directly reconstructed using fast Fourier transform (FFT). The spacing and extent of the Cartesian grid have a direct relation to the acquisition time, FOV and voxel size of the resulting image. In Cartesian acquisition, the number of PE steps can be reduced by an acceleration factor or reduction factor R by increasing the distance between the k -space lines. If only one line is left between two k -space lines, the MR data acquisition is said to be accelerated by a factor of $R=2$. The reduction of sampling along the PE direction of the k -space results in a reduced FOV in that direction with associated fold-over artifacts.

**FIGURE 1.7**

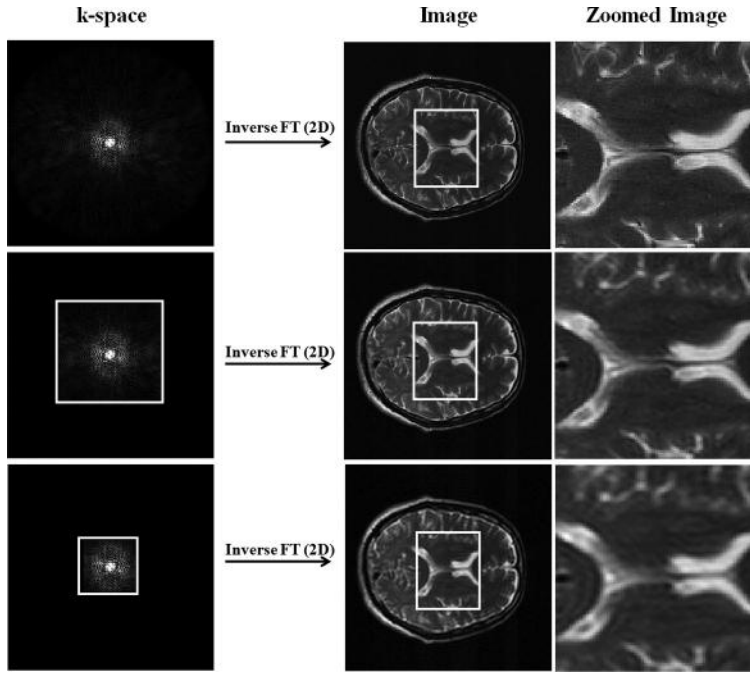
Cartesian k -space acquisition. (a): Linear phase-encoding order or standard MR acquisition, (b): low-high or centric k -space order.

The position of the line being filled in the k_y direction is determined by the PE gradient. The amplitude of the PE gradient is incremented in steps such that the next adjacent line in the k -space is filled successively starting from one edge of the k -space until the opposite edge is reached. This is illustrated in Figure 1.7 and is referred to as a linear phase-encoding order [2]. In some of the dynamic applications, including contrast enhanced angiography, a different ordering of the PE steps is adopted to retain the contrast information at the onset of acquisition. In that case, the PE gradient is incremented from zero, but with an alternating sign, starting from the centre of the k -space in steps to acquire lines in extreme edges. This type of k -space sampling in the PE direction is referred to as the centric phase encode order or low-high k -space order.

1.3.2 k -Space Features

The central k -space region carries low spatial frequency information, and peripheral data carries high-frequency information required to improve the spatial resolution. The nominal spatial resolution of the image can be improved by extending the data collection farther from the k -space origin. Since a large chunk of the image information is contained in the low spatial frequencies, the addition of high spatial frequency information can only sharpen the image without affecting the contrast or the basic shape features [3]. This is illustrated in Figure 1.8 using k -space truncation with two square windows of different sizes. With a smaller square window used for truncation, the reconstructed image is blurred due to loss of high-frequency signals near the periphery of the k -space. However, with inclusion of higher frequencies using a larger window, the resolution of the reconstructed image is improved.

A k -space line in 2D Fourier imaging refers to the samples of MR signal received at a particular PE level. Since the k -space position is encoded by the gradient amplitude, sampling through the Fourier space is accomplished by changing the gradient shape and amplitude over time. Every point in the raw data matrix carries partial information for the complete image. The outer rows of the raw data matrix corresponding to the high spatial frequencies provide information regarding the borders and contours of the image, whereas the inner rows of the matrix with low spatial frequencies provide general contrast information of the image. In real MR experiments, the k -space signal is acquired in the discretized form. Consequently, the discrete Fourier transform (DFT) is used for image reconstruction from k -space.

**FIGURE 1.8**

k -space and image are Fourier transform (FT) pairs. Panels in the top row show a completely filled k -space and the resulting image. The middle and bottom rows illustrate the k -space and image obtained after truncating the k -spaces with large and small rectangular windows. The effects of truncation are clearly seen in the zoomed versions of the images shown in the rightmost panels.

As inferred from [Figure 1.8](#), with a greater net strength of the phase-encoding gradient, the data is mapped to a farther position from the k -space centre. Based on the polarity of the gradient pulse, the mapping can be either toward the upper or lower directions. Thus, the duration of the PE gradient governs the location of data on the vertical axis of the k -space. Alternatively, with a greater net strength of the FE gradient, the data samples tend to be located farther from the k -space centre. The data is mapped to the right side of the k -space if the gradient is positive, or in the left direction if the gradient is negative. Thus, the strength of the FE gradient determines the location of data on the horizontal axis of k -space. After the acquisition of the entire k -space, inverse Fourier transform can be used to reconstruct the image. With uniform Cartesian acquisition, the k -space data is filled during the scan so that one phase-encoding line is collected per TR . An extensive coverage of the effect of discrete k -space sampling are provided in [\[1,4\]](#).

Scanning time increases with an increase in the number of PE lines. In the case of 3D imaging, second and third dimensions are acquired using phase-encoding. Compared to frequency-encoding, phase-encoding is a much slower process. As an example, with a TR of 500 ms, collection of 256 PE lines require 128 seconds in comparison to 8 ms for acquisition of 256 samples in one PE line [\[5\]](#). This indicates that by reducing the number of acquired PE lines, the data acquisition time can be significantly reduced. The image reconstructed from full k -space can achieve the best quality as all signals have been sampled to fill the entire k -space. Considering the scanning time required, k -space reconstruction is carried out from under-sampled k -space, in which only a limited number of k -space samples are acquired.

1.3.3 Non-Cartesian Imaging

1.3.3.1 Data Acquisition and Pulse Sequencing

In non-Cartesian data acquisition, the trajectory on which the k -space samples are acquired is not a Cartesian grid. The trajectory is manipulated by proper modification of the FE and PE gradients. The commonly employed non-Cartesian trajectories include radial and spiral patterns. This section discusses the pulse sequencing and gradient patterns employed in such non-Cartesian trajectories. For acquisition of data along radial trajectories, the encoding gradients in the n 'th excitation cycle are given by:

$$\begin{aligned} G_{n,x} &= G \cos(\phi_n) \\ G_{n,y} &= G \sin(\phi_n), \end{aligned} \quad (1.16)$$

where $\phi_n = \arctan\left(\frac{G_{n,y}}{G_{n,x}}\right)$. Accordingly, the sequence begins by moving from the origin of the k -space to the edge of a disk in k -space (e.g., point 'a' in Figure 1.9a) before starting data acquisition. The corresponding k -space locations are given by:

$$\begin{aligned} k_x &= \gamma G(t - t_0) \cos(\phi_n); \quad t_0 < t < \frac{T_{acq}}{2} + t_0 \\ k_y &= \gamma G(t - t_0) \sin(\phi_n), \end{aligned} \quad (1.17)$$

where T_{acq} is the acquisition time. Figure 1.9a shows the k -space trajectory and the corresponding readout gradients.

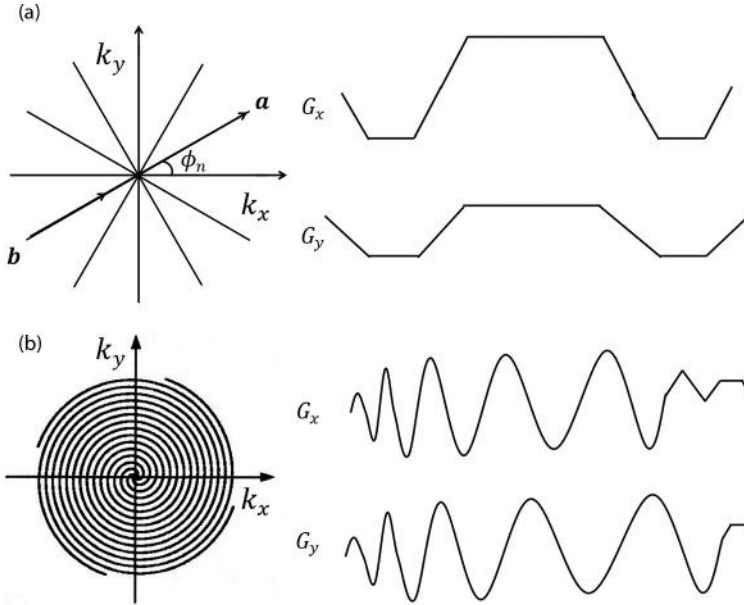


FIGURE 1.9

Non-Cartesian data acquisition trajectories and pulse sequences for (a) radial data acquisition trajectory and (b) spiral data acquisition trajectory.

The k -space location at point ' $\mathbf{a}'$ is given by:

$$\mathbf{k}_a = \gamma G \frac{T_{acq}}{2} (\cos(\phi_n), \sin(\phi_n)). \quad (1.18)$$

Then data is collected by scanning across to the other edge of the disk along a line passing through the origin of the k -space. The 180° pulse moves the trajectory to the point ' $\mathbf{b}'$ so that

$$\mathbf{k}_b = -\mathbf{k}_a. \quad (1.19)$$

The k -space locations during acquisition are given by:

$$\begin{aligned} k_x &= \gamma G \cos(\phi_n)(t - TE); |t - TE| < T_{acq} / 2 \\ k_y &= \gamma G \sin(\phi_n)(t - TE). \end{aligned} \quad (1.20)$$

This defines a radial line starting at $\mathbf{k}_b$ and ending at $\mathbf{k}_a$, passing through the k -space origin. The gradients can be easily rotated on subsequent acquisitions to acquire lines at different angles through the origin until the entire k -space is filled. This sequence is also referred to as projection reconstruction (PR) imaging.

Another example of a non-Cartesian imaging sequence is the spiral acquisition trajectory in which the entire k -space data can be acquired using either a single trajectory or multiple trajectories. As shown in [Figure 1.9b](#), this method uses a pair of increasing sinusoidal gradients to traverse through the k -space in a spiral trajectory. The main advantage of using this type of acquisition is that it completely eliminates the rapid gradient switching associated with other pulse sequences discussed in the previous sections. A spiral trajectory in the (k_x, k_y) -plane is defined by:

$$\mathbf{k}(t) = A\omega(t)e^{i\omega(t)}, \quad (1.21)$$

where $\mathbf{k} \triangleq k_x + ik_y$, and $\omega(t)$ is some function of time. In the case of multiple spiral trajectories comprising of N_i spirals, the trajectory of the n 'th spiral is given by $\mathbf{k}(t) = A\omega(t)e^{i[\omega(t) + \varphi_0]}$, where $\varphi_0 = 2\pi(n-1)/N_i$. The gradients are optimized to reach the maximum radius of the k -space in a given amount of time. Then the trajectory is rewound to the origin in order to produce a predictable residual magnetization.

1.3.3.2 Transformation from Non-Cartesian to Cartesian Data

Non-Cartesian acquisition involves sampling performed on a non-Cartesian grid. For the non-Cartesian acquisition, a direct inverse Fourier transform cannot yield the desired image. Therefore, the data points are first interpolated to a Cartesian grid before the application of the inverse Fourier transform by using the regridding principle [6–9].

The non-Cartesian sampled data has to be convolved with a regridding kernel for interpolating the data points on a rectangular grid. The resultant samples are evaluated at $N \times N$ grid points with a spacing Δk . For interpolation in k -space, an optimal kernel window is an infinite sinc function as the corresponding multiplication of image with a rectangular function does not distort the desired reconstructed image. However, for practical purposes, convolution with an infinite kernel window is not feasible. Therefore, a kernel with compact support is used instead. Based on the systematic assessment of the performance of different kernel windows [9–11], the Kaiser-Bessel window is used as a standard kernel for regridding in MR imaging.

A brief description of the gridding algorithm in the 1D case is provided here. This can be extended to higher dimensions in a straightforward manner. With the original non-Cartesian data denoted by $\hat{K}_{nc}(k_l)$, and the convolution kernel denoted by $\Gamma(k_l)$, the regridded Cartesian k -space is given by:

$$\hat{K}_c(k) = \sum_j \hat{K}_{nc}(k_j) \cdot \Gamma(k_l - k_j) = \hat{K}_{nc}(k_j) * \Gamma(k_l), \quad (1.22)$$

where k_l denotes the k -space location on the Cartesian grid for 2D case, k_j denotes an arbitrary k -space position which is not on a Cartesian grid and $*$ denotes the discrete convolution. As most of the non-Cartesian trajectories use higher sampling densities at lower frequencies, some form of density compensation is first applied prior to regridding. Therefore, each sample data point has to be weighted with a density compensation function $DCF(k_j)$. The density compensation function (DCF) is a measure of the inverse of the k -space sampling density. The regridded k -space is then given by:

$$\hat{K}_{c-dc}(k_l) = (DCF(k_j) \cdot \hat{K}_{nc}(k_j)) * \Gamma(k_l), \quad (1.23)$$

where $\hat{K}_{c-dc}(k_l)$ denotes the regridded Cartesian k -space data obtained using the density compensated non-Cartesian data. The inverse discrete Fourier transform of the density compensated, regridded data is then computed to obtain the reconstructed image signal $\hat{U}_{dc}(x)$ as:

$$\hat{U}_{dc}(x) = F^{-1} \left((DCF(k_j) \cdot \hat{K}_{nc}(k_j)) * \Gamma(k_l) \right), \quad (1.24)$$

The most widely used approach for estimating the sampling density in MR image reconstruction is the Voronoi diagram [7,12]. For radial sampling, analytical representation of the sampling density is also possible with the Ram-Lak filter [13]. After interpolation using the convolution operation, the regridded data is transformed to the image space using the inverse discrete Fourier transform. Deapodization or roll-off correction is then performed to compensate for the undesired intensity modulations resulting from the convolution with a finite sized kernel. In this correction step, the reconstructed image is divided by the inverse Fourier transform of the regridding kernel (denoted by $\gamma(x)$). The dampening of the outer parts of the reconstructed image due to intensity modulation is corrected with the deapodization procedure. The roll-off corrected image is given by:

$$\hat{U}_{dc-de}(x) = \hat{U}_{dc}(x) \cdot \frac{1}{\gamma(x)}. \quad (1.25)$$

1.4 Parallel MRI

In classical MRI, a single receiver coil with an approximately homogeneous spatial sensitivity is used. Parallel pMRI is a technique to increase the speed of MRI acquisition, by using the spatial information from an array of coils simultaneously. In pMRI, the signal is acquired using an array of n_c receiver coils with distinct spatial sensitivities (coil array) [14]. The sensitivity of the coil depends on the coil size and shape. Although large RF coils are employed to detect the signal from large areas, these coils provide low signal-to-noise

ratio (SNR). While small coils are more sensitive, their sensitivity decreases with distance from the coil. Surface coils, which are very commonly used in pMRI, are directly placed onto the surface of the object to be imaged. They are effective for imaging the regions close to the surface, and hence the name. Phased array coils are formed from different arrangements of multiple surface coils. They are capable of imaging larger regions. Although the use of phased array coils for MRI dates back to the late 1980s, the design motive was largely intended toward building arrays with reduced coil-to-coil coupling [15,16]. A major benefit of the phased array coils is their ability to generate high SNR signals with reduced motion artifacts. Roemer et al. [14] implemented the first phased array MRI system in the form of two inductively coupled resonant rings, electrically isolated from each other with a decoupling element connected between the two rings. In their approach, the signal localization is mainly achieved by differentially weighting the signals received from the two coils, thereby reducing the need for elaborate and complex gradient-encoding steps. SNR enhancement was achieved by means of the decoupling circuitry. In the absence of mutual coupling for n_C independent coils, the SNR is increased by a factor of $\sqrt{n_C}$. A detailed description of phased array MRI is provided in the review article by Wright and Wald [17].

In multi-channel acquisition, all coils work in parallel and the acquisition time remains the same as that of a single coil. Additional information about the spatial position of the received signal is obtained using the array of coils. Neglecting the inhomogeneities of sensitivity in the body coil and noise in the receivers, the image U_c retrieved by the c 'th receiver from the coil array is related to the body-coil image U as:

$$U_c(x, y) = U(x, y)C_c(x, y) \quad (1.26)$$

where $C_c(x, y)$ is the sensitivity of the c 'th coil and $U(x, y)$ is the image retrieved by the homogeneous-sensitivity coil. The coil sensitivities are smoothly varying functions which, when multiplied with the original spin density, give the individual coil images. With s and r representing k -space and spatial coordinates in 3D, the received k -space signal for the c 'th coil is given by:

$$Y_c(s) = \iint \hat{C}_c(r) \rho(r) e^{-j2\pi sr} dr, \quad (1.27)$$

where $\hat{C}_c(r)$ is the estimated coil sensitivity of the c 'th coil and $\rho(r)$ is the unknown spin density function.

1.4.1 Coil Combination

MR acquisition using phased array coils requires an algorithm to combine the individual complex channel images into a single composite image with full FOV. The optimal method to combine the channel images for maximum SNR is the spatial matched filter [14]. Then the reconstruction process involves the application of a filter vector m to coherently sum the n_C individual complex coil images $\{U_c(x, y), c=1, \dots, n_C\}$ into the combined composite image using:

$$U(x, y) = \sum_{c=1}^{n_C} m_c^* U_c(x, y), \quad (1.28)$$

where complex vector $\mathbf{m}_c$ may vary as a function of the spatial coordinates. For a complex vector $\mathbf{b}$ representing the magnitude and phase of the transverse magnetic field created by each coil and the array noise correlation matrix $\mathbf{R}_n$ at the given pixel location, the matched filter is obtained as:

$$\mathbf{m} = \mathbf{R}_n^{-1} \mathbf{b}. \quad (1.29)$$

The array noise correlation matrix can be directly measured by performing a pre- or post-scan noise calibration. Also, at each location within the FOV, the accuracy of the signal-dependent term $\mathbf{b}$ is highly dependent on the precise knowledge of the complex near-field sensitivity pattern. As this information is difficult to obtain in practice, the use of suboptimal reconstruction techniques that do not require detailed coil field map information is used [14,18,19].

Due to this difficulty associated with a practical matched filter implementation, the ‘sum-of-squares’ (SoS) technique [14] has become a preferred choice for combining multi-channel MR images. Generally, the noise correlations are ignored and the SoS reconstruction can be obtained as the square root of the sum of the squared individual coil magnitude images. With σ_c denoting the normalized root mean squares (RMS) noise level (i.e., $\sum 1/\sigma_c^2 = 1$) for the c ’th coil image, the array noise power statistics can be incorporated in the SoS reconstruction as [20]:

$$U(x, y) = \sqrt{\sum_{c=1}^{n_c} \left(\frac{U_c(x, y)}{\sigma_c} \right)^2}. \quad (1.30)$$

1.5 MR Acceleration

Although, MRI has many benefits not associated with other imaging modalities, imaging speed is still one of the challenges for MRI applications. Due to the slow scanning speed in MRI, any improvement to accelerate MR data acquisition can greatly improve the quality of patient care. A further limitation in terms of imaging speed is the number of PE steps required to form an image. This is especially true in cases where high-resolution images are required and the repetition times are long because of the specific image contrast needed. Besides specialized pulse sequences intended for rapid acquisition, other schemes for this purpose include pMRI and compressed sensing (CS).

1.5.1 Acceleration Using Pulse Sequences

With the advancements in MR hardware, pulse sequence physicists have developed many fast MRI sequences in parallel that adds to the reduction in scanning time. For example, in echo planar imaging (EPI), a complete encoding of the FOV is achieved by a single RF excitation (or pair) [21]. The PE steps are acquired using the rephasing gradients instead of repeated 180° RF pulses that follows the $90^\circ/180^\circ$ in a SE sequence. This is achieved by rapidly reversing the FE gradient. The reversal can be also done in a sinusoidal way. In a single-shot EPI, with all the PE steps are acquired in one TR.

Like EPI, the rapid imaging with refocused echoes (RARE), also known as fast spin-echo (FSE) or turbo spin-echo (TSE) technique, can also acquire the entire 2D image from a single excitation. As the magnetization is continually refocused during formation of spin echoes by application of a 180° refocusing pulse between each line of k -space, the pulse sequence is more robust to the local field inhomogeneities. In the SE sequence, the signal decays as a function of T_2 and not T_2^* . RARE is slightly longer than EPI due to the time required for the 180° refocusing pulse used between lines of k -space.

Another strategy called steady-state free precession (SSFP) MRI is a type of gradient echo MRI pulse sequence in which a train of excitation pulses that are separated by a constant time interval (TR) is used. This sequence is noted for its application in the dynamic/cine assessment of cardiac function. A low tip angle allows short TR values, resulting in reduced imaging time.

1.5.2 Acceleration Using Sampling Schemes

The scanning time can be reduced by an appropriate selection of the data acquisition mode and k -space sampling scheme. The k -space sampling trajectory refers to the path used for navigating the sampling locations for data acquisition. This path illustrates the acquisition strategy and has direct influence on the artifacts induced. Therefore, the image reconstruction algorithm should be employed accordingly. Although, the most popular k -space trajectory is Cartesian, other trajectories that include radial, spiral, or rosette type patterns are also employed for data acquisition. The k -space coverage also varies depending on the sampling pattern. If multiple receiver coils and parallel imaging reconstruction techniques are used, the aliasing artifacts could be reduced effectively or even removed, thereby improving the actual scanning time. The imaging speed can be accelerated by acquiring only a portion of the k -space data. Using parallel imaging, k -space data acquisition can be accelerated by means of under-sampling. However, as the Nyquist sampling theorem is violated, aliasing artifacts are introduced in the images. The structure of the artifacts depends on the sampling trajectory used.

The under-sampled acquisition using pMRI requires multiple coils to cover the entire FOV. Each coil records signals from the imaged object in parallel. The basic idea of parallel MR imaging can be summarized as follows. Reduced FOV acquisition covering a portion of the entire image is obtained from each coil, and the images from all the coils are finally combined to produce the full FOV image. Each element of the phased coil array possesses a separate independent receiver system. This enables the reconstruction of each channel image separately. In order to optimally reconstruct array coil images with high SNR, explicit knowledge of the coil sensitivities at all pixel locations is required [14]. Theoretically, imaging time can be reduced by a factor equal to the number of array coil elements [22]. In actual reconstruction methods, the decreased SNR and increased image non-uniformities limit the acceleration factor well below the theoretical maximum acceleration factor R .

If a dataset with a one-coil system is sufficient to reconstruct an image, then a dataset acquired with a multi-coil system using an identical sampling trajectory in k -space will contain redundant information. This is justified using the framework of a generalized linear inverse problem in which the number of equations is increased, while the number of unknowns remains the same. In parallel imaging, image acceleration is achieved by exploiting this redundancy in spatial encoding. Furthermore, the requirement to traverse only a subset of the original k -space locations results in a proportionally faster acquisition.

1.5.3 Under-Sampled Acquisition and Sampling Trajectories

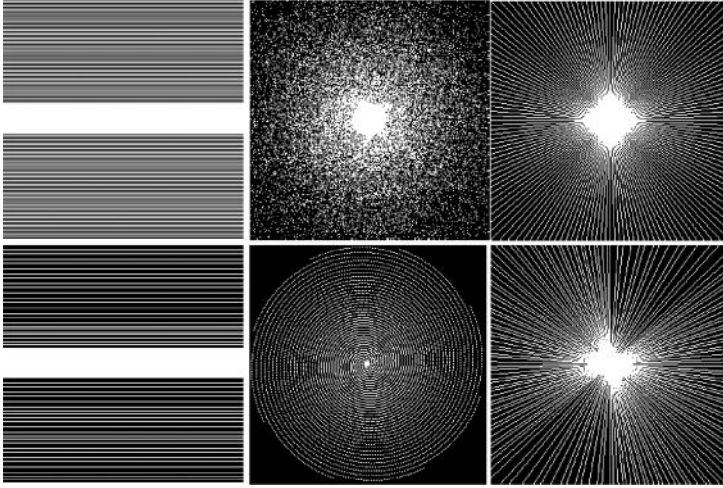
Under-sampled acquisition refers to the collection of only a subset of data samples in the k -space, instead of a full acquisition. The acceleration factor in parallel imaging (PI) is limited by the number of coils. If the k -space is under-sampled in a regular manner by missing out every alternate line of k -space in a PE direction, the number of repetition times required (and hence the total acquisition time) can be halved. This type of under-sampling is called uniform under-sampling. Most of the pulse sequences currently used for clinical imaging purposes are Cartesian. Reconstruction procedures for such acquisitions are relatively simple compared to non-Cartesian. In addition, reconstructions from Cartesian sampling are robust to many sources of system imperfections. In addition to the uniform Cartesian sampling, several other Cartesian patterns are also used in practice such as pseudo-random under-sampling suitable for application to CS-based methods. Poisson-disk sampling (PDS) provides an even but random distribution of samples [23,24].

In radial acquisition, the central k -space region is oversampled and continuously updated due to the overlapping “spokes” that repeatedly pass through this region. This redundant information can be exploited to detect and correct for any movement of the subject during the acquisition if the signal from the centre of the k -space changes between views. Unlike uniform acquisition in which the central region contributes more, all radial spokes contribute equally to the image in radial acquisition. Consequently, motion on just one or a few radial views is not likely to severely degrade the image quality [25].

A spiral sampling scheme makes efficient use of the gradient system hardware, and it is used for rapid real-time imaging applications that include cardiovascular, renal, breast, spectroscopy, and functional neuro-imaging [26,27]. The efficiency in dynamic applications is attributed to its intrinsic moment-nulling motion compensation [26,28] and efficient use of gradient power [29]. In spite of these advantages, spiral methods are not commonly used as a routine on clinical scanners due to the complexity involved in application of gradient waveforms required to generate the spiral trajectory. Furthermore, reconstruction from such non-Cartesian trajectories is more complicated, and this requires sophisticated algorithms such as filtered back-projection [30] or k -space interpolation schemes [10].

To improve the fidelity of the image, more samples are acquired at the lower frequencies compared to the outer k -space using variable-density sampling [31,32]. The variable-density sampling patterns are achieved based on either the distance measure [32] or the golden ratio [33,34]. The distance-based scheme chooses random samples with sampling density scaling according to a power of distance from the origin. In Cartesian acquisition, the variable-density sampling is achieved with the k -space under-sampled only along the PE direction. The random PE lines are chosen successively based on the golden ratio. In each step, the region between two successive acquired lines having the largest separation is identified. If more than one such region exists, the one nearest to the k -space centre is chosen. Then a new PE line is selected so that it divides this region according to the golden ratio.

For radial acquisition also, the sampling patterns can be either uniform or random angular sampling schemes. The randomness can be achieved by choosing the spokes at random angular intervals based on the golden ratio [35]. For spiral acquisition, the sampling pattern consists of an under-sampled interleaved variable-density spiral trajectory [36]. The sampling schemes typically used in MR acquisition are shown in Figure 1.10.

**FIGURE 1.10**

Sampling masks used for MR acceleration. Top row from left to right show uniform undersampling with acceleration factor $R = 2$ and ACS lines=32, random sampling with 30% acquired samples and radial sampling with 80 spokes equally spaced. Bottom row from left to right show 1D-linear variable density sampling mask based on golden ratio with 80 PE lines of which 32 are from the central k-space region, spiral Sampling mask, and radial Sampling using 80 spokes and angular separation determined by the golden ratio.

1.5.4 Artifacts Associated with Different Sampling Trajectories

In uniform k -space acquisition, the distance between the PE lines is increased by an acceleration factor R . The number of acquired PE lines is reduced without affecting the maximum gradient intensity. This reduces the FOV and causes aliasing artifacts. The nature of the artifacts that occurred as a result of reduced acquisition is illustrated using Cartesian uniform under-sampling (sub-sampling) in Figure 1.11. In the example shown in the figure, acceleration factor is chosen to be $R = 2$, $R = 3$ and $R = 4$ (all in row dimension).

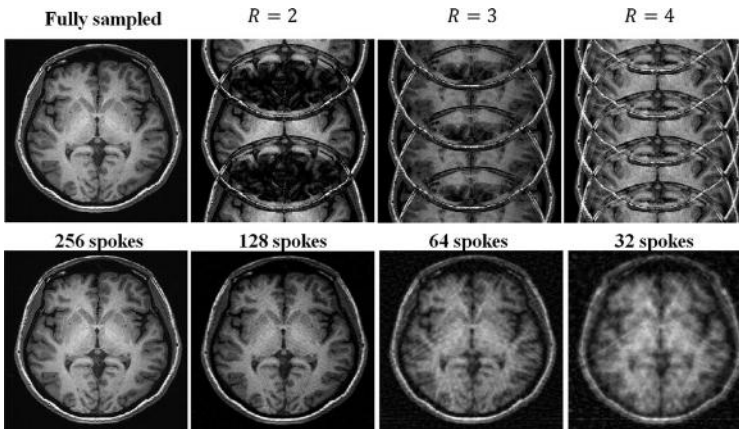
**FIGURE 1.11**

Illustration of artifacts in uniform and radial subsampling. The top row shows images obtained with Cartesian under-sampling, fully sampled, $R = 2$, 3 and 4, respectively. The bottom row shows images obtained from radial sampling with 256, 128, 64 and 32 spokes. Reducing the number of spokes results in streaking artifacts.

For radial sampling, the reconstructed images are shown for 256, 128, 64 and 32 spokes for a 256×256 image. In order to satisfy the Nyquist-Shannon sampling theorem in the case of radial sampling, $N\pi/2$ spokes are required [4]. The spokes described above result in acceleration factors of $R \approx 1.57$, $R \approx 3.14$, $R \approx 6.28$ and $R \approx 12.57$, respectively. The sub-sampled Cartesian imaging leads to backfolding artifacts, whereas sub-sampling of radial trajectories yields images with streaking artifacts.

1.6 Parallel Imaging Reconstruction Algorithms

Improving the imaging speed using multiple receiver coils dates back to the late 1980s and early 1990s [37–40]. In the first approach to pMRI by Hutchinson and Raff [41], acquisition of N k -space lines employed N surface coils placed around the object. In such a large multi-element array, the coil size must be small and the coupling between adjacent coils should be negligible. A novel sub-encoding strategy involving a multiple coil receiver system was proposed by Ra and Rim [42] to reduce the number of PE steps to be acquired by a factor L . The resultant image was an aliased one which was unfolded by using the distance-dependent sensitivity differences among n_c receiver coils where $n_c \geq L$. The variable n_c is chosen as a small number so that each channel has a higher sensitivity with less decoupling. The rate of reduction in the time required for data acquisition in this type of approach is proportional to the number of receiver coils. In this approach, the number of coils is independent of the required image resolution.

Another simple approach involving two receive coils was put forward by Carlson and Minemura [43], in which half the phase-encoding steps were acquired as in a standard MRI sequence. To mitigate the coil coupling problems partially, receivers with no intrinsic coupling were employed. The first channel was called a primary coil with a common head or body coil, and the second one was called a secondary coil with a gradient response. The sensitivity of the secondary channel coil was kept zero at the centre of PE direction, and varied linearly in either direction.

A notable development following this period was the introduction of the simultaneous acquisition of spatial harmonics (SMASH) method by Sodickson and Manning [44]. This method, which uses linear combinations of coil sensitivity profiles to generate low-order spatial harmonics of the desired FOV, was the first experimentally successful pMRI technique. Sodickson et al. then presented a successful in-vivo implementation of pMRI using the SMASH technique [45], setting the course for further developments in the field of parallel imaging. One year later, Pruessmann et al. introduced the concept of sensitivity encoding (SENSE) [46], which is strongly related to the early proposals by Kelton [47], and Ra and Rim [42]. SENSE uses an SNR optimized matrix inversion along with an initial estimate of the coil sensitivity mapping. Since the development of SENSE, a cohort of parallel imaging reconstruction methods have been implemented with the goal of improving the accuracy of reconstruction [18,44,46,48–50].

Parallel imaging techniques generally fall into two broad categories: (1) methods in which reconstruction takes place in the image domain that requires an unfolding or inversion procedure such as SENSE [46,51], and (2) methods in which reconstruction takes place in k -space, such as generalized auto-calibrating partially parallel acquisitions (GRAPPA) [48], iterative self-consistent parallel imaging reconstruction (SPIRiT) [50], and parallel reconstruction using null operations (PRUNO) [52]. Hybrid techniques in which a combination of both are used in parallel imaging using eigenvector maps (ESPIRiT) [53]. Detailed reviews on parallel imaging methods can be found in the literature [54–56].

The main objective of all pMRI strategies is to reconstruct high-quality diagnostic images through acquisition of reduced number of k -space samples. The correlation between the signals acquired from different channels is exploited in PI to reduce the number of samples below the Nyquist limit. Image reconstruction in pMRI is a mathematical process that generates combined images using a limited number of data samples acquired simultaneously from an array of receiver coils. The methods employed for reconstruction determine the reconstructed image quality based on measures signifying the resultant SNR and remnant artifacts. With constraints on permissible time limits for acquisition and the number of data samples to be collected, it is desirable to reconstruct images having the lowest possible noise without sacrificing accuracy and spatial resolution.

1.6.1 Image-Based Reconstruction Methods

In image-based reconstruction methods, the unaliasing process is done after taking the Fourier transform. However, the image-based reconstruction techniques require a prior knowledge of the coil sensitivities. The estimates of coil sensitivities can be obtained from a separate calibration scan or utilizing the calibration data, which is contained in the acquisition. The commonly used image-based reconstruction methods include SENSE [46] and partially parallel imaging with localized sensitivity (PILS) [57]. When the k -space data is under-sampled by a factor R , the FOV of the corresponding image is reduced by $1/R$. Hence the entire information is contained in a smaller area. Due to the fold-over-type aliasing artifacts, the image-based reconstruction methods involve the unaliasing procedure to obtain the full FOV image.

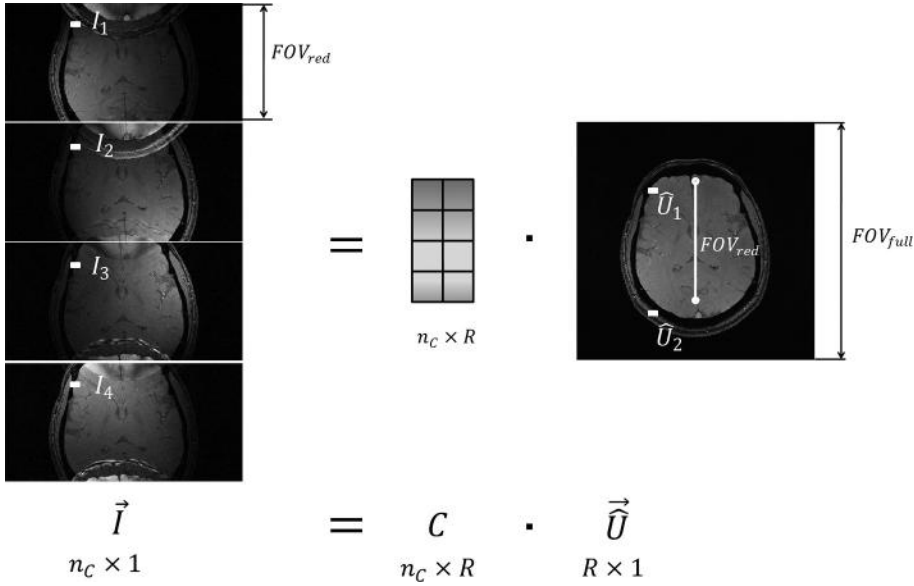
1.6.1.1 SENSE

SENSE is based on the principle that the image reconstructed from a k -space under-sampled in the PE direction can be modeled as a superposition of periodic repetitions of the original full FOV image $\hat{U}$. As inferred from Figure 1.5, if the FOV is chosen to be small with respect to the imaged object, the sampling theorem is violated, and aliasing occurs. Therefore, if the spacing Δk_x of the sampling function is increased by a factor of 2, as in the case of an acceleration factor $R=2$, the FOV is halved. Each pixel I^c in the image obtained from under-sampled k -space data acquired with coil c consists of a superposition of R pixels from $\hat{U}$, which are weighted according to the values of the coil sensitivities C^c at the corresponding pixel positions l :

$$I^c = \sum_{l=1}^R C_l^c \cdot \hat{U}_l \quad (1.31)$$

The index l corresponds to the number of pixels that are super-imposed, and c counts from 1 to the number of coils n_c . For $n_c=4$ and $R=2$, the pixels from all n_c coils can be stacked in a single complex signal vector $\vec{I}$ with dimension $n_c \times 1$, as shown in Figure 1.12. This can also be obtained with super-imposed pixels from the full FOV image $\hat{U}$, which form an $R \times 1$ vector $\vec{\hat{U}}$. With C representing an $n_c \times R$ matrix consisting of the coil sensitivities of all n_c coils at the R super-imposed positions, we can obtain a system of linear equations. In matrix form, the system of equations can be written as:

$$\vec{I} = C \cdot \vec{\hat{U}}. \quad (1.32)$$

**FIGURE 1.12**

SENSE reconstruction for $R=2$ and $n_c = 4$. Pixels from all channels are stacked in a single complex signal vector $\vec{I}$ of size $n_c \times 1$. Similarly, all superimposed pixels from the full FOV image $\hat{U}$ are stacked to form an $R \times 1$ vector $\vec{\hat{U}}$. With C representing the $n_c \times R$ coil sensitivity matrix the system of linear equations can be written in matrix form.

With the knowledge of C , the values of the original full FOV image $\vec{\hat{U}}$ can be unfolded using:

$$\vec{\hat{U}} = \text{pinv}(C) \cdot \vec{I}, \quad (1.33)$$

where $\text{pinv}(C)$ denotes the pseudoinverse computed as $(C^H C)^{-1} C^H$. This step is repeated for all pixels in the reduced FOV image. Here SENSE reconstruction with only 1D sub-sampling is discussed. However, it is also possible to use the SENSE method with 2D under-sampling when a second PE direction is used, as in 3D imaging [58].

As evident from the reconstruction steps, the SENSE method requires knowledge of the coil sensitivity matrix C . The simplest method to estimate the coil sensitivity information is to perform a reference scan prior to the accelerated acquisition. The reference scan is a fully sampled 3D acquisition with low spatial resolution. This pre-scan can take approximately 15 seconds. As the coil sensitivities are smooth functions, they can be captured with such a low-resolution scan with the additional advantage of high SNR. This can be performed with either the body coil or the phased array coil. The images obtained from each coil of the receiver array represent a modulation of the image $\hat{U}$ with the corresponding coil sensitivity C^c . In case of the underlying image $\hat{U}$ acquired using the body coil, the coil sensitivity estimate C^c can be directly obtained by normalizing the coil array images with the body coil image.

The reference scan can lead to inaccuracies in the estimated coil sensitivities due to the patient motion in the accelerated scan compared to the pre-scan. This led to the development of auto-calibrated methods which do not require the explicit knowledge of coil sensitivities. Even with the precise knowledge of the sensitivity matrix, SENSE reconstruction can be performed only if the matrix inversion in Equation (1.33) is possible. The number

of superimposed pixels must be always less than or equal to the number of receive coils to avoid the formation of an underdetermined system of equations. This implies that if the coil sensitivities were not sufficiently non-uniform and distinct, the matrix C would be rank-deficient, and the resultant SENSE reconstruction would not be able to rectify the aliasing present in the images using these sensitivities alone.

Although the SENSE method can be used with arbitrary coil configurations, the phased array coil should deliver independent information along the dimension corresponding to the under-sampling direction. With significant overlap between different coils, the matrix inversion in Equation (1.33) becomes error prone as the rows of the sensitivity matrix C become linearly dependent. This linear dependency between the rows of C can lead to noise amplification in the reconstructed image. The corresponding noise amplification is quantified by means of the so-called geometry factor (g -factor) [59]. Analytically, the g -factor g_f is defined as:

$$g_f = \frac{SNR_{\text{full}}}{\sqrt{R} \cdot SNR_{\text{red}}}, \quad (1.34)$$

where SNR_{full} and SNR_{red} correspond to the SNR of the fully sampled and the reduced FOV images, respectively. The noise amplification is expected to vary spatially, due to the spatial inhomogeneity induced by the coil sensitivities. Therefore, g -factors are usually computed for each voxel in the full FOV. The noise amplification and aliasing artifacts in SENSE can be reduced using Tikhonov regularization [60,61], a sparsity-promoting l_1 -norm, or a low-rank matrix prior using the nuclear norm [62].

As the sensitivities are not known in practice, they need to be extracted from the acquired data. As an alternative to the coil sensitivities obtained from a body coil image [63–65], the auto-calibration signal (ACS) data collected by dense sampling of the low-frequency portion of the k -space can be used to obtain an approximate estimation of the coil sensitivity images. An SoS combination of low-resolution images, collected from the ACS data of all channels, is used to estimate the low-resolution coil sensitivity maps [66–68] by the pixel-wise division of the individual coil images by the SoS image. Some recent approaches based on sub-space methods, such as ESPIRiT [53], LOTUS [69] and the eigen vector-based approach by Wang et al. [70], obtain an estimate of the relative coil sensitivities relying on the eigen value decomposition of the ACS data.

The errors present in the coil sensitivity estimation cause SENSE to yield a combined image with visible aliasing artifacts. In order to minimize the effect of coil sensitivity errors on the SENSE reconstruction, a variant of SENSE called JSENSE [71] was introduced. In JSENSE, both sensitivities and the image are jointly estimated and refined using a low-degree polynomial basis representation of the sensitivity function. In SENSE reconstruction, pronounced non-uniform noise enhancement can be observed with increasing acceleration. Due to the highest amount of overlap of the coil sensitivities, the central parts of the image exhibit the highest noise amplification.

1.6.2 k -Space Based Reconstruction Methods

In the k -space-based reconstruction, the missing k -space values are estimated using interpolation. The commonly used k -space-based reconstruction methods include SMASH [44], GRAPPA [48], AUTO-SMASH [18], and the variable-density (VD) AUTO-SMASH [49]. The auto-calibrating pMRI techniques which require a self-calibration try to fit a model using the calibration data. The model then estimates a set of filter coefficients which are used for estimation of the unknown k -space values in the outer k -space regions.

1.6.2.1 SMASH

SMASH is an auto-calibrating method that attempts to interpolate the missing frequencies in the under-sampled k -space by approximating the k -space frequency shifts with linear combinations of acquired data across coils. In its basic form, SMASH uses a linear array of surface coils to generate multiple composite sensitivity variations. The linear array is required because the net sensitivity of the array is designed to have a sinusoidal shape. There are no strict response requirements for the individual coil elements in the array. Using a coil array comprised of multiple elements, multiple harmonics may be generated from a single dataset. With acquisition of a reduced number of PE lines, the remaining lines of k -space may be reconstructed using linear combinations of component coil signals. For a total of N spatial harmonics, N different lines of k -space can be reconstructed with different strengths of the PE gradient. Therefore, the total signal matrix can be acquired in a fraction $1/N$ of the usual acquisition time.

The procedure for SMASH estimation is explained using a 1D example in which a signal $M[m]$ of length N and k -space $K(n)$; ($n=0, \dots, N-1$) is under-sampled by a factor R . Therefore, the PE index n for the under-sampled k -space $K_c(n)$ assume values $n=0, \dots, N/R-1$. The spacing in the fully sampled k -space is denoted by Δk and that in the under-sampled k -space by $R\Delta k$. From the basic Fourier transform property, a shift in the frequency domain is equivalent to multiplication of the signal by a complex exponential in the image domain, that is, $e^{-j2\frac{\pi}{N}rm} M[m] \leftrightarrow K(n+r)$ for $r=0, \dots, R-1$. The SMASH method consists of finding weights $w_{c,r}$ such that:

$$\sum_{c=1}^{n_c} w_{c,r} C_c(m) \approx e^{-\frac{j2\pi rm}{N}}, \quad (1.35)$$

where C_c denotes the coil sensitivity of c 'th coil. If the sensitivities are assumed to be known, the weights $w_{c,r}$ can be determined by solving the system of linear equations in Equation (1.35). In SMASH, an expression to generate shifted k -space lines $K(Rn+r)$ is obtained from the weighted combinations of $K_c(n)$. The weighted combination of under-sampled channel k -spaces can be represented as:

$$\sum_{c=1}^{n_c} w_{c,r} K_c(n) = \sum_{m=0}^{N-1} \sum_{c=1}^{n_c} w_{c,r} M(m) C_c(m) e^{-\frac{j2\pi Rnm}{N}}. \quad (1.36)$$

From Equation (1.36), the fully sampled k -space is determined as:

$$K(Rn+r) = \sum_{m=0}^{N-1} M(m) e^{-\frac{j2\pi(r+Rn)m}{N}} = \sum_{c=1}^{n_c} w_{c,r} K_c(n). \quad (1.37)$$

The estimate of the combined image $M(m)$ is the inverse DFT of the full interpolated k -space. SMASH is robust to errors in the coil sensitivity estimates and computationally efficient as it only requires R linear system inversions compared to SENSE, which requires N/R linear system inversions. The limitation of SMASH is the quality of the approximation in Equation (1.35). As the acceleration rate increases (large R), the slowly varying coil sensitivities can only be a poor match for the high-frequency complex exponentials.

GRAPPA is a more robust extension to the SMASH method, incorporating several fundamental differences. In GRAPPA, a small set of calibration data with frequency spacing

corresponding to the full FOV called ACS lines is used, instead of relying on the coil sensitivity estimates explicitly. Uniform under-sampling is performed in the regions outside the calibration region. This form of VD-sampling scheme increases the complexity of the SNR analysis of the reconstructed images. This is because the central lines of k -space provide most of the contrast information in an image and possess most of the image power that primarily affects SNR. Higher frequencies that carry the edge information are under-sampled and the noise distribution will also be non-uniform across the image. As the under-sampling is not uniform, the central k -space lines are weighted more heavily than the higher spatial frequencies.

Similar to AUTO-SMASH [18], which is a variant of SMASH, the lines in the fully sampled region are used to fit the interpolation weights. The ACS lines are usually chosen at the centre of k -space, which is known to have high SNR. GRAPPA utilizes multiple neighboring acquired k -space data (source) points to interpolate the missing (target) points, yielding an improved reconstruction.

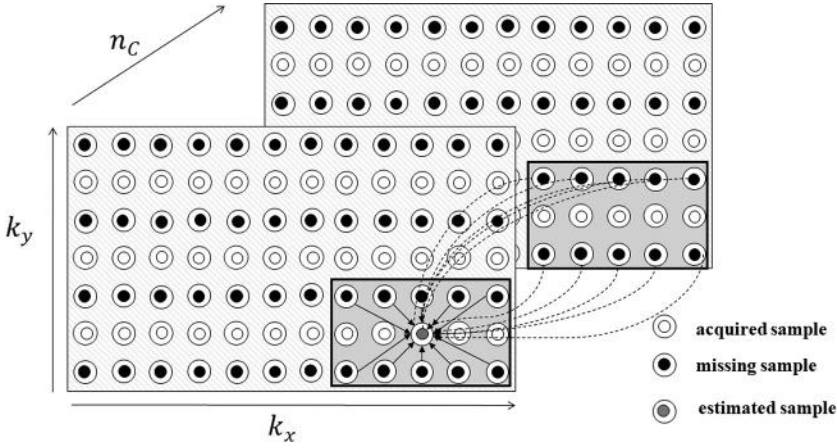
1.6.2.2 GRAPPA

GRAPPA is a coil-by-coil parallel image reconstruction that estimates missing data in k -space using the additionally acquired ACS data. In GRAPPA reconstruction, the unacquired k -space samples are obtained by fitting the reduced dataset to the ACS data. This fitting process involves solving the GRAPPA coefficients using an overdetermined linear system of equations (LSE) in the least-squares (LS) sense. GRAPPA is the most widely employed k -space-based pMRI technique in MR scanners. Later, a generalized iterative version of GRAPPA called SPIRiT was introduced; it works efficiently regardless of the sampling trajectories employed.

For the ease of description of both GRAPPA and SPIRiT theory, it is assumed that the acquired k -space K^l is uniformly under-sampled in the PE direction with an acceleration factor R , and fully sampled in the central portion with n_{ACS} number of ACS lines. Estimation of the unacquired k -space samples in l -th channel $\hat{K}^l(k_x, k_y + r\Delta k_y)$ for $r = 1, \dots, R-1$, are carried out using a linear combination of the fully sampled k -space points $K^c(k_x, k_y)$ from all the n_c channels, as given by:

$$\hat{K}^l(k_x, k_y + r\Delta k_y) = \sum_{c=1}^{n_c} \sum_{b=-N_{b1}}^{N_{b2}} \sum_{a=-N_{a1}}^{N_{a2}} \mathbf{z}_r^l K^c(k_x + b\Delta k_x, k_y + a\Delta k_y), \quad (1.38)$$

where $\mathbf{z}_r^l$ represents a vector of l -th coil coefficients of size $(N_{b2} + N_{b1} + 1) \times (N_{a2} + N_{a1} + 1) \times n_c \times 1$, obtained by fitting a kernel of size of $(N_{b2} + N_{b1} + 1)$ (along phase-encode) $\times (N_{a2} + N_{a1} + 1)$ (along readout) with samples from all n_c coils. Using the GRAPPA equations for all acquired k -space points from ACS, Equation (1.38) can be equivalently represented in a matrix form. The filter weights can then be computed from the matrix representation using standard estimation methods such as the LS. If the under-sampling pattern were extended into the region of ACS lines, a resultant unacquired location with offset $r\Delta k_y$ from the nearest acquired location with respect to the zero-phase encoding line could be obtained. A kernel centered at this location may now be constructed as in Equation (1.38). A schematic representation of the reconstruction processes is shown in Figure 1.13. The acquired k -space data in each coil (black circles) are weighted by the respective GRAPPA coefficients fit to estimate the unacquired target data (grey circle). Each coil k -space is under-sampled with $R-1$ unacquired lines and the fully acquired ACS data.

**FIGURE 1.13**

2D GRAPPA: Missing k -space data are synthesized from neighboring acquired data. The synthesizing kernel depends on the specific sampling pattern in the neighborhood of the missing point. The reconstruction of a point is independent of the reconstruction of other missing points.

Each row of the calibration matrix Π is formed by arranging the k -space samples at all the acquired locations within the kernel in some pre-determined order. Figure 1.14 illustrates the calibration matrix formation, with the under-sampling pattern extended into the region of ACS lines. Here N_{acq} denotes the number of acquired k -space points in the ACS. A sample kernel of size $2 \times 3 \times n_c$ centered at a missing k -space location $(k_x, k_y + r\Delta k_y)$ with $r = 1$ is also shown. Similarly, the observation vector $\mathbf{k}_{ur}^l$ is formed by arranging the k -space samples in the ACS of a given channel- l at all unacquired locations, with offsets indexed by r . Then the fitting equation can be expressed as:

$$\mathbf{k}_{ur}^l = \Pi \mathbf{z}_r^l. \quad (1.39)$$

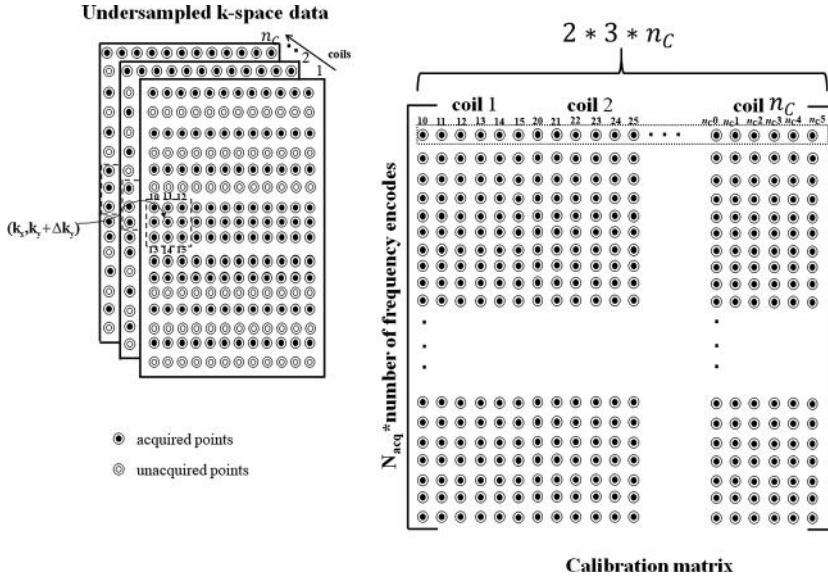
Using Equation (1.39), the calibration process consists of computing the LS solution as follows:

$$\mathbf{z}_r^l = (\Pi^H \Pi)^{-1} \Pi^H \mathbf{k}_{ur}^l, \quad (1.40)$$

where H denotes the Hermitian transpose. As GRAPPA incorporates more information into the estimation, the size of the kernel can be increased to provide an improved fit. During the reconstruction process, a similar kernel centered at each missing k -space location in the accelerated region is first constructed. The unacquired k -space samples can now be estimated using:

$$\hat{K}^l(k_x, k_y + r\Delta k_y) = \mathbf{k}^T(k_x, k_y) \mathbf{z}_r^l \quad (1.41)$$

where $\mathbf{k}(k_x, k_y)$ represents a query vector whose elements are the acquired (source) k -space samples within the kernel. As shown in Figure 1.14, the elements are arranged following the same sequence as that used for calibration. After estimation of missing lines using GRAPPA, each channel k -space is Fourier transformed to generate the channel images and then combined using any magnitude reconstruction procedure such as the SoS.

**FIGURE 1.14**

Calibration matrix formation. A sample calibration kernel at k -space location $(k_x, k_y + \Delta k_y)$ for $R=2$ is also shown.

GRAPPA can be extended to 2D or 3D under-sampling in different ways [72]. In 3D acquisition, sub-sampling is performed in only two PE dimensions. Therefore, the inverse Fourier transform of the data along the frequency-encoded direction can be computed and the resulting 2D slices can be processed individually.

Due to the larger number of source points in two dimensions, calibration becomes even more important, and thus an increase in the number of ACS lines can result in increased reconstruction quality. In conventional GRAPPA, the missing k -space samples are estimated from known samples on a Cartesian grid. However, GRAPPA can be extended to non-uniform or non-Cartesian data. For estimating the missing k -space samples from non-uniformly under-sampled data, the patterns of source and target points vary from block to block, and therefore different interpolation kernels are computed for calibration depending on the pattern encountered. GRAPPA can also be extended to non-Cartesian k -space trajectories using the re-gridding procedure, followed by conventional GRAPPA.

As with other pMRI reconstruction methods, the noise amplification can be significant. The GRAPPA g -factors can be analytically computed by considering interpolation in the k -space as multiplication in the image domain [73]. Since an analytical expression for the g -factors does not exist, the multiple replica method is generally adopted for the g -factor computation. This method involves several full and reduced FOV acquisitions and averaging the observed noise in the reconstructions to estimate the g -factors. However, such an approach may be also impractical for human subjects as tens or hundreds of repetitions may be required to yield statistically significant g -factors. Alternatively, the g -factors are estimated using the pseudo multiple replica method. This method uses only one set of reduced-FOV data acquisition and performs Monte Carlo simulations with synthetic additive noise to approximate the multiple replica estimate [74].

With higher acceleration factors, GRAPPA reconstruction suffers from aliasing artifacts and noise amplification. Several attempts have been made to improve the quality of

GRAPPA reconstruction by simultaneous reduction of noise amplification and aliasing. These include reconstruction based on calibration matrix modification [75–81] and calibration based on different regularization approaches [82–84]. A more detailed review of these GRAPPA variants can be found in [85].

1.6.2.3 SPIRiT

SPIRiT is a generalized iterative version of GRAPPA that works efficiently regardless of the sampling trajectories employed for data acquisition. The main difference compared to GRAPPA is that both fitting and synthesis SPIRiT kernels consist of acquired as well as unacquired samples [50]. Another major difference in comparison to GRAPPA is that the source k -space consist of samples estimated during reconstruction that were not actually acquired. Due to the shift-invariant property of the SPIRiT filter, the same filter weights are applied to the entire k -space. Additional sparsity-promoting constraints can be easily incorporated in SPIRiT. Figure 1.15 illustrates the Cartesian SPIRiT reconstruction. The central sample is computed as a weighted sum of all k -space samples in the neighborhood, regardless if they are acquired or unacquired. For unacquired samples, estimates from previous iterations are used during SPIRiT reconstruction. In SPIRiT, the coil coefficients are estimated from ACS using the following:

$$\hat{K}^l(k_x, k_y) = \sum_{c=1}^{n_C} \sum_{b=-N_{b1}}^{N_{b2}} \sum_{a=-N_{a1}}^{N_{a2}} \mathbf{z}^l K^c(k_x - b, k_y - a). \quad (1.42)$$

Similar to Equation (1.39), a matrix representation of this model can be obtained, but with the difference that the number of rows of Π and $\mathbf{k}_u^l$ is now equal to the total number of samples within ACS. In GRAPPA, this is determined based on the number of acquired samples only.

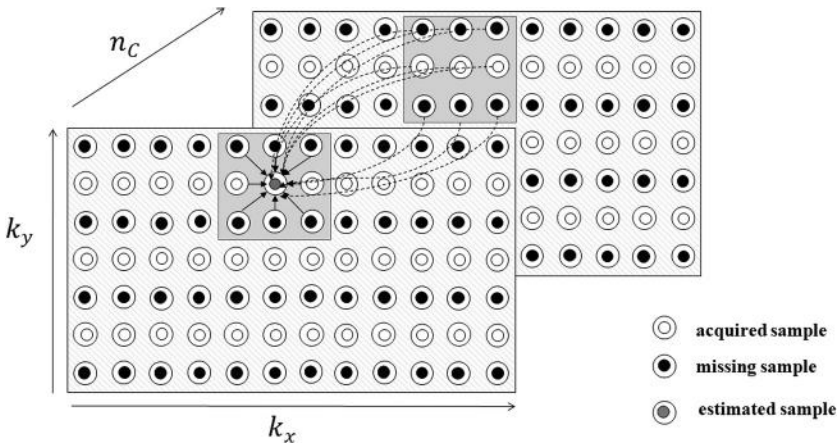


FIGURE 1.15

Cartesian SPIRiT reconstruction. The reconstruction of each point on the grid depends on its entire neighborhood. Reconstruction includes estimation of both acquired and missing samples.

Given the under-sampled k -space K^l with ACS lines and the coil coefficient vectors, the unacquired k -space samples in the accelerated region (outside ACS) can be estimated iteratively using the following steps:

1. Initialize $\hat{K}^{l(0)}$ with all unacquired k -space values of K^l as zeros and setting $t = 0$.
2. Using Equation (1.42), reconstruct all k -space points $K^{l(t)}$ in the accelerated region of $\hat{K}^{l(t)}$.
3. Synthesize an error k -space $\mathcal{E}^{l(t)}$ with non-zero consistency errors only at acquired locations (k_x, k_y) and zero elsewhere. Consistency errors at acquired locations are computed using $\mathcal{E}^{l(t)}(k_x, k_y) = K^l(k_x, k_y) - K^{l(t)}(k_x, k_y)$.
4. Compute $U^{l(t)} = \mathcal{F}^{-1}(K^{l(t)}) + \gamma \mathcal{F}^{-1}(\mathcal{E}^{l(t)})$, where $\mathcal{F}^{-1}$ is the inverse Fourier operator and $0 < \gamma \leq 1$ is a data consistency relaxation parameter.
5. Compute $\hat{K}^{l(t)} = \mathcal{F}(U^{l(t)})$ and go to Step 2 until $\|U^{l(t)} - U^{l(t-1)}\|_2 < \text{tol}$.

1.6.2.4 Regularization in Auto-calibrating Methods

In auto-calibrating methods, regularization is generally incorporated in the calibration phase of the reconstruction procedure. A set of self-consistent simultaneous equations is formed out of the acquired k -space data in the ACS. These simultaneous equations are written in matrix form, and the interpolation weights are computed via a matrix inversion operation. If the simultaneous equations involved in the matrix inversion are similar, then the resultant solutions become extremely sensitive to small changes in the observation vector. For example, if the two equations are quite similar, and the difference between them is of the same order as that of the noise in the system, the correct solution cannot be found because all the solutions will then be dominated by the influence of noise.

Matrix inversion is deeply related to the condition number of the system of equations. Condition number is the ratio of the maximum to the minimum singular values of the calibration matrix, and it controls the error propagation in linear computations. If the encoding matrix is well conditioned, then the inversion can be done with acceptable amplification of noise. Alternatively, an ill-conditioned system of equations with a disproportionately large condition number results in abrupt changes in the solution, even with small perturbations in the input. Hence, small noise perturbations in the acquired k -space data can produce large variations in the reconstructed image. In addition, non-ideal conditions resulting from higher acceleration factors, reduced number of ACS, large coil numbers, increased acceleration or high input SNR can result in over-fitting of the training data, with consequent increase in the mean squared error (MSE) during the estimation phase. A more appropriate interpretation is that when the number of parameters (filter coefficients) increases in relation to the available training samples, the MSE increase is mostly contributed by the bias introduced during calibration. The filter size is mainly determined based on the number of coils, kernel size and acceleration factor R .

All aforementioned problems relating to degradation of image quality can be addressed by the introduction of regularization constraints and additional information about the underlying object to obtain robust reconstruction. Effectiveness of the regularized solution is highly dependent on a suitable choice of the regularization parameter λ . While a smaller value of λ (under-regularization) leads to residual noise or streaking artifacts in the image, a larger λ (over-regularization) removes the image features with an increase in loss of resolution.

Regularization is considered to be effective only in situations where substantial noise reduction can be achieved without worsening the aliasing present in LS reconstruction. Any technique for choosing regularization parameter in the absence of information about error level and its statistics can pose difficulties in arriving at an optimal reconstruction. Consequently, inferences obtained using different selection strategies can lead to suboptimal reconstructions. Although reconstruction based on auto-calibration can provide satisfactory image quality with dense sampling, constraints to reduce the scan duration results in k -space with low sampling density and reduced SNR. In such situations, reconstruction using auto-calibration methods often results in images with artifacts. A better choice in this case is the incorporation of prior information that the underlying object is sparse in some transform domain [86].

1.6.3 CS MRI

CS MRI deals with the application of CS techniques to reconstruct the missing k -space points on a sampling grid and satisfies certain predetermined properties which are elaborated upon in subsequent sections. This utilizes the underlying principle in CS, requiring the acquired signal to be sparse or compressible [87,88]. In particular, the theory of CS in MRI relies on the principle that an image with a sparse representation in some known transform domain can be recovered from highly incomplete random measurements. A signal is said to be sparse if only a small fraction of coefficients is significant.

As imaging speed is one of the challenges in MRI due to the slow data acquisition process, application of CS to MRI possesses the potential for significant scan time reductions. MR signals are shown to satisfy two key factors required for successful application of CS [31,32]. These include (1) medical images are compressible by sparse coding in an appropriate transform domain (e.g., wavelet transform, finite difference domain), and (2) MRI scanners directly acquire frequency samples of the underlying image rather than direct pixel samples in the image domain.

From the general CS theory, the measurements $\mathbf{y} \in \mathbb{C}^M$ obtained by sampling a signal $\mathbf{s} \in \mathbb{C}^N$ using a sensing matrix $\phi_{M \times N}$ are given by:

$$\mathbf{y} = \phi \mathbf{s}. \quad (1.43)$$

For $M < N$, it becomes difficult to recover $\mathbf{s}$ by solving the underdetermined system of equations. With the assumption that $\mathbf{s}$ can be sparsely represented in transform domain Ψ , then $\mathbf{s}$ can be represented using the transform domain coefficients ω as follows:

$$\mathbf{s} = \Psi \omega. \quad (1.44)$$

In order for the compressed sensing algorithms to perform well, the sensing matrix should satisfy the incoherence properties, restricted isometry properties (RIPs) or null space properties (NSPs). The sensing matrix ϕ with normalized column vectors should be a low-coherence matrix, with a coherence measure $\mu(\phi)$ defined as [89] follows:

$$\mu(\phi) = \max_{1 \leq i \neq j \leq N} |\langle \phi_i, \phi_j \rangle|, \quad \mu(\phi) \in \left[\sqrt{\frac{N-M}{M(N-1)}}, 1 \right]. \quad (1.45)$$

As an alternative, for an S_k -sparse vector $\mathbf{s}$, the sensing matrix ϕ should satisfy the RIP [90]. The RIP stipulates:

$$(1 - \delta_{S_k}^\phi) \|\mathbf{s}\|_2^2 \leq \|\phi \mathbf{s}\|_2^2 \leq (1 + \delta_{S_k}^\phi) \|\mathbf{s}\|_2^2, \quad (1.46)$$

where $\delta_{S_k}^\phi$ denotes the isometry constant $0 \leq \delta_{S_k}^\phi < 1$. Using the discrete form of signal representation encompassing the continuous spatial information about spin density in MRI [1,91], the signal $\mathbf{s}$ can be represented as follows:

$$s(\mathbf{r}) = \sum_{n=1}^N s_n \varphi(\mathbf{r} - \mathbf{r}_n), \quad (1.47)$$

where $\varphi(\mathbf{r})$ denotes the voxel basis function and $\mathbf{r}_n$ the voxel grid. For N discrete locations (voxels), the signal vector $s(\mathbf{r})$ can be written in vector form as $\mathbf{s} = [s_1, s_2, \dots, s_n, \dots, s_N]^T$. Spatial information is then mapped onto the k -space measurements $y(\boldsymbol{\kappa})$ with a finite Fourier integral as follows:

$$y(\boldsymbol{\kappa}_m) = \int_{\mathbb{R}^2} s(\mathbf{r}) e^{-i2\pi \mathbf{r} \cdot \boldsymbol{\kappa}_m} d\mathbf{r} = \int_{\mathbb{R}^2} \sum_{n=1}^N s_n \varphi(\mathbf{r} - \mathbf{r}_n) e^{-i2\pi \mathbf{r} \cdot \boldsymbol{\kappa}_m} d\mathbf{r}; \quad m = 1, 2, \dots, M \quad (1.48)$$

where $\mathbb{R}^2$ is the 2D MR image over which integration is performed and $y(\boldsymbol{\kappa}_m)$ is the value at the m 'th k -space location $\boldsymbol{\kappa}_m$. For acquisition at M number of k -space locations, the measurement vector becomes:

$$\mathbf{y} = [y(\boldsymbol{\kappa}_1), y(\boldsymbol{\kappa}_2), \dots, y(\boldsymbol{\kappa}_m), \dots, y(\boldsymbol{\kappa}_M)]^T. \quad (1.49)$$

Comparing Equation (1.48) with Equation (1.43), the encoding matrix ϕ becomes:

$$\phi = \int \varphi(\mathbf{r} - \mathbf{r}_n) e^{-i2\pi \mathbf{r} \cdot \boldsymbol{\kappa}_m} d\mathbf{r}. \quad (1.50)$$

In CS MRI, the encoding matrix ϕ is denoted as the under-sampled Fourier operator F_u obtained by setting the measurements at unacquired k -space locations to zero.

1.6.3.1 CS-Based MR Image Reconstruction Model

Recent developments in CS have introduced sparsity regularization techniques, which have gained significant attention in MR image reconstruction from highly under-sampled k -spaces. The CS implementations are based on the following assumptions: (1) the underlying object has a sparse representation in a known transform domain (such as wavelet transform, discrete cosine transform, finite difference etc.), (2) the artifacts resulting from k -space under-sampling are incoherent/noise-like and (3) the use of a nonlinear reconstruction can enforce both the sparsity of image representation and consistency with the acquired data. According to the theory of compressed sensing, the solution corresponds to the one that minimizes the l_0 -norm (number of non-zero elements) of the sparse representation of the signal. However, this is an NP-hard (non-deterministic polynomial time hard) problem. Alternatively, this problem can be replaced by the convex relaxation of the l_0 -norm.

The locations of missing samples are represented using a binary sampling mask $S(k_x, k_y)$. The sampling mask is chosen to obtain incoherent aliasing interference in the sparse transform domain [32]. The resultant under-sampled k -space with unacquired locations filled with zeros can be expressed as follows:

$$K_u(k_x, k_y) = \begin{cases} K(k_x, k_y), & \text{if } S(k_x, k_y) = 1 \\ 0, & \text{if } S(k_x, k_y) = 0 \end{cases} \quad (1.51)$$

where K is the fully sampled k -space. Using the under-sampled Fourier operator (F_u) defined as

$$F_u U = S \odot F U, \quad (1.52)$$

where $\odot$ denotes the Hadamard product and F the Fourier transform operator, the under-sampled k -space of an image U can be represented as:

$$K_u = F_u U. \quad (1.53)$$

The ill-posed nature of the inversion model in Equation (1.53) calls for inclusion of a penalty term in the minimization of $\|K_u - F_u U\|_2$, thereby enabling the use of regularization strategies. In this case, a priori information that the underlying object can be sparsely represented in some transform domain can be used to regularize the problem [92–95].

1.6.3.2 Sparsity-Promoting Regularization

The solutions of sparsity-promoting regularization techniques are obtained using an under-determined system of linear equations as opposed to the over-determined system in the auto-calibrating methods. A feasible solution is obtained by constraining the solution to be sparse and enforcing the consistency condition imposed by the under-sampled Fourier operator. In the constrained form, the minimization problem becomes:

$$\min_U \|U\|_0 \quad \text{s.t.} \quad K_u = F_u U. \quad (1.54)$$

However, the presence of l_0 -norm makes this minimization problem computationally intractable. A more tractable problem can be obtained using the technique known as convex relaxation [96], which replaces the l_0 -norm with the l_1 -norm. This is because the l_1 -norm is the closest convex function to the l_0 -norm. With convex relaxation, an approximate solution to Equation (1.54) can be obtained using:

$$\min_U \|U\|_1 \quad \text{s.t.} \quad K_u = F_u U. \quad (1.55)$$

Considering the presence of noise in the model, Equation (1.55) takes the following form:

$$\min_U \|U\|_1 \quad \text{s.t.} \quad \|K_u - F_u U\|_2 \leq \varepsilon, \quad (1.56)$$

where ε is a small numerical value, typically set to the noise level. The unconstrained Lagrangian form of Equation (1.56) is:

$$\min_U \|K_u - F_u U\|_2^2 + \alpha \|U\|_1, \quad (1.57)$$

where α denotes the regularization parameter. Since U is reconstructed, the unconstrained formulation in Equation (1.57) is referred to as the analysis approach. Alternatively, in the synthesis approach the transform domain coefficients are reconstructed directly. Therefore, CS optimization problems exist in different forms depending on the constrained or unconstrained (Lagrangian) cases, and on the analysis or synthesis forms. The minimization problem with incorporation of regularization can be expressed using the synthesis-based reconstruction model in wavelet domain as [97,98] follows:

$$\min_{\hat{v}} \frac{1}{2} \|K_u^c - F_u \Psi' \hat{v}\|_2^2 + \alpha \|\hat{v}\|_1, \quad (1.58)$$

where $\hat{v}$ denotes the transform domain coefficients, α represents the trade-off between sparsity and data fidelity, and Ψ and Ψ' denote the transform operator and its adjoint operator, respectively. For an optimization problem defined in Equation (1.58), a non-linear thresholding operator [32,92–94] is used to recover the significant coefficients of the sparse transform, thereby enabling the image reconstruction. A detailed review of the sparsity-promoting regularization approaches are provided in Chapter 2.

1.6.4 CS Recovery Using Low-Rank Priors

The theory of CS has been generalized to the matrix case, with the rank replacing sparsity to recover an unknown signal from a limited number of measurements [99]. For a measurement vector $\mathbf{b} \in \mathbb{C}^p$, the low-rank matrix CS problem addresses the reconstruction an unknown low-rank matrix $X \in \mathbb{C}^{M \times N}$. The measurements are obtained via a linear operator $A: \mathbb{C}^{M \times N} \rightarrow \mathbb{C}^p$. The underlying inverse problem is ill-posed as the number of measurements is much smaller than the number of unknowns. Similar to the case with sparse recovery of vectors, the number of measurements sufficient for unique and stable recovery is approximately the same order as the number of degrees of freedom in the unknown low-rank matrix. Under certain conditions, the recovery can be accomplished by polynomial-time algorithms [100].

The low-rank matrix reconstruction problem arises in a wide range of applications. One major application is the low-order system identification [101,102]. In this, the order of a linear time invariant system determines the rank of a Hankel matrix which is formed from the system output and the linear measurement is obtained from the sampled response of the system generated using some random known input. The low-rank matrix recovery has been applied to the dynamic MRI reconstruction [103–106]. Using a partially separable model, the sequence of vectorized image frames are modeled as a low-rank matrix [107].

1.6.4.1 Low-Rank CS-Based MR Image Reconstruction Model

The recovery of a low-rank matrix from the knowledge of a small number of its linear measurements has significant implications for MR acceleration because the underlying low-rank matrix X can be perfectly recovered from its measurements $\mathbf{b} = AX$ by solving the constrained optimization problem [100,108]:

$$X^* = \min_X \frac{1}{2} \|\mathbf{b} - AX\|_2^2 \text{ such that } \text{rank}(X) \leq r. \quad (1.59)$$

Since the rank constraint significantly reduces the number of degrees of freedom, it is an effective means of regularizing the underlying inverse problem. In the context of CS, it is shown that the low-rank matrix recovery using rank minimization requires a random measurement scheme with the number of measurements equal to two to four times higher than the number of degrees of freedom [100]. Also, rank minimization for the matrix case is shown to be similar to that of l_0 -norm minimization for the vector case. The l_1 -norm as a convex approximation to the l_0 -norm is analogous to the nuclear norm as a convex approximation to the rank function. The nuclear norm of an r -rank matrix $X = USV^T$ denoted by $\|X\|_*$, is the sum of the singular values of X , that is, $\|X\|_* = \sum_{ii} S_{ii}$. Using convex relaxation, the matrix recovery problem is formulated as follows:

$$X^* = \min_X \frac{1}{2} \|b - AX\|_2^2 + \lambda \|X\|_*. \quad (1.60)$$

In the case of MRI, the matrix X can be recovered from the under-sampled k -space data as the solution to a spectrally regularized minimization problem of the following form:

$$X^* = \min_X \frac{1}{2} \|b - AX\|_2^2 + \lambda \Psi(X), \quad (1.61)$$

where $\Psi(X)$ denotes an appropriate spectral penalty. Typically, the general class of Schatten p -functionals can be used as specified by:

$$\Psi(X) = (\|X_p\|)^p = \sum_{i=1}^{\min(m,n)} \sigma_i^p. \quad (1.62)$$

The spectral penalty in Equation (1.62) simplifies to the nuclear norm for $p = 1$. If $p \leq 1$, this penalty ceases to be a norm and is non-convex. For more information on the use of similar non-convex semi-norms in the context of vector recovery, the readers may refer to articles [109–112].

The convex relaxations to solve NP-hard problems offer guaranteed performance under the conditions specified by the sparsity-restricted isometry property [86] and the rank-restricted isometry property (R-RIP) [99,100], respectively. Although the l_1 -norm minimization is formulated as a linear program (or a quadratically constrained linear program for the noisy case), the nuclear norm minimization corresponds to a convex semi-definite program (SDP). Although several methods have been proposed for solving large-scale SDP using rank minimization, the dimension of the problem is restricted and some of these methods do not guarantee convergence to the global minimum.

Several experimental studies have shown that the nonconvex optimization-based approaches often lead to more accurate reconstruction than their convex counterparts, although at the cost of higher computational complexity. Thus, more efficient solutions can be exploited by combining the benefits of both high-order dependencies among sparse coefficients, and non-convex optimization at the cost of increased computational complexity.

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2

Regularization Techniques for MR Image Reconstruction

2.1 Regularization of Inverse Problems

Regularization methods play a key role in finding the solution of inverse problems. In inverse problems, the observed output of a model is used to infer values of the parameters that characterize the model. Connected with the increasing interest in inverse problems in various engineering applications for the past 50 years, the development and analysis of regularization methods have become a necessity due to their ill-posed nature. In particular, absence of continuous dependence between the measured data and the model parameters in the presence of measurement errors always results in an approximate solution to the inverse problem. Regularization methods constitute procedures applied for analysis of such modified problems, resulting in the introduction of some prior knowledge weighted by a regularization parameter that encodes the degree of approximation.

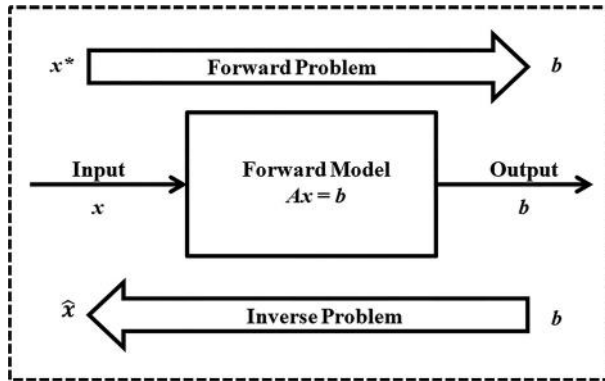
A canonical example of an ill-posed inverse problem at the abstract level is the linear operator equation:

$$Ax = b, \tag{2.1}$$

where b denotes the measurement vector; x denotes the model parameters; and A is a linear operator, whose generalized inverse $A^\dagger$ is unbounded. A typical example is an image restoration problem in which a blurred image captured by a camera system is to be restored. The direct or forward problem includes taking a sharp image. With measurement error due to noise, the observed blurred image is represented by $b = \bar{b} + n$. Given the blurred image b and the blurring operator A , the process of finding an approximation $\hat{x}$ of the unblurred sharp image x is an inverse problem, often referred to as the image deconvolution problem [1]. A schematic representation of the forward problem, forward model and corresponding inverse problem is shown in [Figure 2.1](#).

Mathematically, a regularization method can be considered as a form of parametric approximation R_λ of $A^\dagger$, with improved stability properties. For linear regularization methods, R_λ constitutes a family of bounded linear operators converging point wise to $A^\dagger$ as $\lambda \rightarrow 0$. For noisy data, the regularization parameter λ is dependent on the noise level δ , defined as a bound for the noise in the deterministic case and some form of variance in a stochastic framework. One of the main questions that arise for noisy data is the proper choice of λ required for convergence.

Although a complete description of linear regularization methods can be provided based on spectral decompositions of the operators, recent developments related to

**FIGURE 2.1**

Schematic representation of the forward problem, forward model and corresponding inverse problem. The forward problem takes an unobservable input x^* and returns an observable output b . The forward problem is modeled by the forward model. The inverse problem tries to find an estimate of $\hat{x}$ of x^* .

variational methods such as total variation techniques [2–4] or sparsity and compressed sensing [5–7] have led to growing interest in nonlinear regularization methods, with a nonlinear approximation of the mapping R_λ . With magnetic resonance (MR) image reconstruction presented as an inverse problem, images reconstructed using a certain regularization method and unknown noise level pose many challenges with regard to quantitatively validating the image quality. This is especially true when the ground truth image is not available.

2.2 MR Image Reconstruction as an Inverse Problem

In parallel imaging, only a limited number of k -space samples/Fourier measurements is acquired by a magnetic resonance imaging (MRI) scanner. In this case, the scanner records the Fourier coefficients (K_u) of the true image (U) using the forward model. Generally, the approximation ($\hat{U}$) of the true image U can be obtained by inversion of the Fourier transform. However, a useful image cannot be obtained by mere inversion of the Fourier transform with only limited Fourier measurements. This is the image reconstruction inverse problem in MRI.

In this case, a solution for the corresponding inverse problem is sought using various complex image reconstruction strategies with specifically defined mathematical procedures. The procedures are carried out to generate the combined image of the region/slice of interest from a limited number of k -space data samples acquired simultaneously from an array of receiver coils. Depending on the adopted strategy of reconstruction, the underlying mathematical procedures vary widely. For example, in auto-calibrating parallel MRI, the inverse problem arises in the calibration step. The resultant system of equations is over-determined. The direct inversion may not be preferred due to the numerical instability of the solution. In the case of very high under-sampling rates, non-linear reconstruction strategies relying on compressed sensing (CS) methods are usually preferred. Such sparsity-promoting formulations result in an under-determined problem and require use of iterative schemes for obtaining useful solutions. As both types of strategies come under the broad class of ill-posed inverse problems, the introduction of prior information about the underlying signal is required to obtain feasible and stable solutions.

2.3 Well-Posed and Ill-Posed Problems

The linear model in Equation (2.1) is well-posed [8] if its solution satisfies the following three conditions [9]:

1. Solution exists
2. Solution is unique
3. Solution depends continuously on the data

If any of the above-mentioned conditions are not satisfied, then the problem is ill-posed. The first two conditions depend on the feasibility of the problem, but the third one is dependent on the stability of the solution. The first two can be addressed by using a Moore-Penrose pseudo-inverse [10]. However, the third case is more difficult to deal with, and the solution exhibits huge variations with small variations on the observed data leading to strong instabilities of the solution. The condition number, rank of a matrix and the Picard criterion [11] can be used as analytical tools to determine whether a given problem is ill-posed.

2.3.1 Moore-Penrose Pseudo-Inverse

The Moore-Penrose pseudo-inverse is a generalization of the matrix inverse when the matrix is not invertible. For an invertible matrix, the Moore-Penrose pseudo-inverse is same as the matrix inverse. The Moore-Penrose pseudo-inverse (denoted as A^+) of an $M \times N$ matrix is defined by the unique $N \times M$ matrix satisfying the following four criteria:

1. $AA^+A = A$
2. $A^+AA^+ = A^+$
3. $(AA^+)^T = AA^+$
4. $(A^+A)^T = A^+A$

For an $M \times N$ matrix A of full rank (N) with $M > N$, the pseudo-inverse is computed as

$$A^+ = (A^T A)^{-1} A^T. \quad (2.2)$$

Using Equation (2.2), the solution that is closest in the least squares sense is given by:

$$\mathbf{x} = A^+ \mathbf{b}. \quad (2.3)$$

2.3.2 Condition Number

With reference to the linear model in Equation (2.1), the condition number $\kappa(A)$ acts as a magnifying factor for perturbations in either A or $\mathbf{b}$. The condition number is obtained as

the ratio of the largest to smallest singular value of A . The singular value decomposition (SVD) of A gives:

$$A = USV^H, \quad (2.4)$$

where U and V are unitary matrices, and S is a diagonal matrix with diagonal entries $\sigma_1 \geq \sigma_2 \geq \sigma_3 \cdots \geq \sigma_N$. The extent of ill-conditioning is determined based on how large is the value of the condition number. Based on the nature of decay of the singular values, the ill-posed problems can be classified into discrete ill-posed problems and rank deficient problems [12]. In discrete ill-posed problems, the numerical rank of A is not well-defined and the singular values decay gradually to zero. In rank deficient problems, the rank of A is well-defined, with a large difference between the smallest and largest singular values. As the condition number is a proxy measure of amplification of errors in A or $\mathbf{b}$, a low value of the condition number indicates that the errors are less amplified. In practical situations, however, this definition of ill-posedness is not sufficient to obtain a useful solution. In particular, two additional problems arise for real-world applications: (1) the forward linear model is only an approximation of the actual process, and so the coefficients in $\hat{\mathbf{x}}$ may not be a good approximation of $\bar{\mathbf{x}}$, and (2) the presence of random noise in the measured data $\mathbf{b}$. These two problems lead to numerical instabilities in the solution, which in turn result in ill-posedness. Such problems often arise in noise suppression problems where one tries to suppress unwanted disturbances in the given data.

2.3.3 Picard's Condition

In most of the ill-posed problems, the singular values decrease rapidly to zero, with the number of singular values having small values being generally more in number. Also, with increasing index i of the singular value σ_i , there will be an increasing number of sign changes in the singular vectors $\mathbf{u}$ and $\mathbf{v}$. This means that, in the inverse problem, one can expect a high amplification of the high-frequency components represented by the small singular values. More specifically, this corresponds to the Picard's condition, which states that $\mathbf{u}_i^T \mathbf{b}$ should be less than σ_i or $\frac{\mathbf{u}_i^T \mathbf{b}}{\sigma_i} < 1$. If this condition is violated, the inverse problem can become numerically ill-posed.

The inverse problems are usually unstable since small changes in $\mathbf{b}$ result in large changes in the solution $\mathbf{x}$, which is undesirable. This means that a small difference in $\mathbf{b}$ can cause a large difference between $\bar{\mathbf{x}}$ and $\hat{\mathbf{x}}$. The third condition of well-posedness requiring that the determination of $\mathbf{x}$ from $\mathbf{b}$ is continuous holds true only if the operator A is compact and bounded. However, in practice, the inverse operator is generally not compact and bounded. As a result, the inverse problem becomes unstable, indicating that the solution does not continuously depend on the data. In the inversion process, the measurement errors at high frequencies are highly amplified due to inversion of small singular values. Therefore, in the presence of measurement errors, the generalized inverse cannot be directly determined using Equation (2.2) without introduction of some form of prior information to obtain a reliable solution [13].

The prior information may be any valuable assumption about the underlying signal. Typical prior information used in regularized reconstruction can be image smoothness, gradient information or sparse representation in some transform domain. As discussed earlier, the contribution of this prior information in finding the solution is controlled by adjusting the value of the regularization parameter. With effective incorporation of the prior information, regularization can restore the stability of the reconstruction algorithm.

2.4 Types of Regularization Approaches

Regularization methods can be broadly classified into two different groups. The first group includes methods which obtain the desired solution by reducing the search space, and the second group includes those which minimize some penalized criterion that measures the closeness of the solution to the underlying data [14]. The basis for the first category of regularization approaches is rooted in the fact that most inversion algorithms involve a complex forward model with a large number of parameters needed for accomplishing the desired accuracy of the solution. However, this may not be the best choice for inversion since the model parameters can be correlated due to the physics of the underlying forward problem. Consequently, there is a need to reduce the number of parameters used to solve the inverse problem. In the linear case, this is related to the dimension of the null space of the linear operator.

In the second category of regularization approaches, additional prior information about the solution is modeled as a penalization term added to the criterion measuring the closeness of the estimate to the underlying data. Broadly, the penalization is imposed either by quadratic or non-quadratic constraints that represent prior information [15–18]. While Tikhonov-like quadratic regularization [19–22] leads to a closed-form solution that can be numerically implemented efficiently, non-quadratic regularization schemes are generally non-smooth; that is, they may not be differentiable everywhere. Non-quadratic methods come under the purview of non-linear optimization problems that use iterative algorithms for determining the solution.

2.4.1 Regularization by Reducing the Search Space

This approach restricts the solution dimensions by first decomposing the solution space into sub-spaces of reduced dimensions and then reconstructs by eliminating those sub-spaces dominated by noise. The regularization methods can be based on some kind of “canonical decomposition” such as the QR factorization; in which the matrix is decomposed into a product, where is an orthogonal matrix and is an upper triangular matrix or the SVD [12, 23, 24]. Truncated SVD (TSVD) is a popular regularization technique in which SVD components contaminated by noise are chopped off. If A is assumed to be of rank r , this is accomplished by forcing to zero the singular values of A that are less than σ_r . Here the regularization parameter k is the number of unity rank components in the reduced rank approximation of A , which is equal to r if the rank is known a priori. With S_k denoting the diagonal singular value matrix obtained by setting the last $(N - k)$ singular values to zero, the reduced rank approximation $A_k = US_kV^H$. Using TSVD, the approximate solution is then obtained as follows:

$$\hat{\mathbf{x}}_{TSVD} = A_k^+ \mathbf{b} = \sum_{j=1}^k \frac{\mathbf{u}_j^T \mathbf{b}}{\sigma_j} \mathbf{v}_j. \quad (2.5)$$

2.4.2 Regularization by Penalization

A penalized regularization of the inverse problem is achieved by minimizing the optimization problem:

$$\hat{\mathbf{x}} = \min_{\mathbf{x}} [D(A\mathbf{x}, \mathbf{b}) + \lambda R(\mathbf{x})], \quad (2.6)$$

where $D(\cdot)$ denotes the data fidelity term and $R(\cdot)$ the penalty term. The parameter $\lambda > 0$ is the regularization parameter that balances the solution between the data fidelity term $D(Ax, b)$ and the penalization term $R(x)$. With $\lambda \rightarrow 0$, the regularized solution tends to the least-squares (LS) estimator, and as $\lambda \rightarrow \infty$, the solution tends to the one obtained by minimizing the penalization term alone. Over-fitting is avoided by limiting the absolute value of the parameters in the model. This can be done by adding a term to the cost function that imposes a penalty based on the magnitude of the model parameters. The nature of the solution will differ depending on the type of norm used to measure the magnitude.

2.5 Regularization Approaches Using l_2 Priors

In l_2 regularization, we shrink the weights by computing the Euclidean norm of the weight coefficients. By adding l_2 regularization terms to the cost function, large model coefficients (weights) are penalized, thereby reducing the complexity of the model. Since the penalty is proportional to the weight squared, l_2 regularization penalizes the larger weights more by choosing a higher value of the regularization parameter. Consequently, l_2 regularization serves to protect the solution from outliers in the data and also shrinks the model to fewer coefficients without over-fitting.

For functions with large residuals, the sensitivity of any penalty-based regularization method to outliers is dependent on the (relative) value of the penalty function. With convex penalty functions, the ones that are least sensitive are those for which the value of the regularization function grows linearly for large magnitudes of the coefficients. Penalty functions with this property are sometimes called robust, since the associated penalty function approximation methods are much less sensitive to outliers or large errors than, for example, a simple least-squares.

2.5.1 Tikhonov Regularization

In the absence of noise, A is invertible and the Moore-Penrose pseudo-inverse is identical with the inverse matrix if the matrix is of full rank. In this case, the solution can be found in a straightforward way using the pseudo-inverse and SVD. For problems in which A is not of full rank, the solution can be obtained only in an LS sense. The information that the noise distribution is a multivariate Gaussian type with known covariance matrix $\Sigma = \text{cov}(n)$ can be used to find a solution of the linear system. In order to solve rank-deficient problems, the LS approach is commonly used. This is achieved by minimizing the residual function $R(x)$, which is the weighted sum of squares of the components of n . As $n = b - Ax$, the residual function $R(x)$ is computed using:

$$R(x) = (b - Ax)^T \Sigma^{-1} (b - Ax). \quad (2.7)$$

The corresponding solution is obtained by minimizing $R(x)$. Since $R(x)$ is the exponent of the likelihood function, minimizing $R(x)$ has an equivalent effect of maximizing the likelihood probability density function (PDF). In this case, the analytic solution or the maximum likelihood (ML)-solution can be computed by equating the gradient vector of R with respect to x to a zero vector:

$$\nabla_x R = -(\mathbf{b}^T \Sigma^{-1} - \mathbf{x}^T A^T \Sigma^{-1}) A = 0. \quad (2.8)$$

Using the complex conjugate of the gradient, we get the back-projected problem:

$$A^T \Sigma^{-1} A \mathbf{x} = A^T \Sigma^{-1} \mathbf{b} \quad (2.9)$$

In order to obtain an explicit solution for Equation (2.9), $A^T \Sigma^{-1} A$ should be invertible. The solution of the weighted complex LS problem can be obtained by multiplying Equation (2.9) with $(A^T \Sigma^{-1} A)^{-1}$ from the left. It can be shown that $(A^T \Sigma^{-1} A)^{-1}$ can be replaced by the pseudo-inverse of $\Sigma^{-\frac{1}{2}} A$ (as shown in the appendix to this chapter) [25]. The latter can be computed efficiently with the help of the SVD if the size of the matrix is not too big. The weighted LS solution not only minimizes the sum-of-squares functional $R(\mathbf{x})$ but also solves the original linear system. However, if $\Sigma^{-\frac{1}{2}} A$ has linear dependent columns due to the presence of redundant information, $\Sigma^{-\frac{1}{2}} A (\Sigma^{-\frac{1}{2}} A)^+$ is not an identity matrix any longer. The solutions $\mathbf{x}$ of the linear system, even if it exists, are not unique. Although, the minimizer $\hat{\mathbf{x}}$ of the weighted LS functional can be computed using the pseudo-inverse, it does not solve the original linear system, as in the full-rank case. A rank-deficient problem can be transformed into a full-rank problem by the addition of a regularization term.

Tikhonov–Phillips regularization puts an additional constraint in the form of a matrix B on the solution [26–28]. The regularization matrix can be either an identity matrix or a covariance matrix of the parameters $\mathbf{x}$. Since the regularization parameter λ weights the penalizing function, the effectiveness of the regularized solution depends on a suitable choice for the value of λ . Tikhonov–Phillips regularization was developed independently in the early 1960s by Tikhonov and Phillips. It is called Tikhonov regularization (TR) and introduces a regularity requirement into the formulation of the optimization problem. The solution with this form of regularization is obtained as follows:

$$\hat{\mathbf{x}}_{TR} = \min_x \left\{ (\mathbf{b} - A\mathbf{x})^H C^{-1} (\mathbf{b} - A\mathbf{x}) + \lambda \|B\mathbf{x}\|_2^2 \right\}. \quad (2.10)$$

The first term $\|\mathbf{b} - A\mathbf{x}\|_2^2$ defines the precision of the solution, that is, how well the computed solution $\mathbf{x}$ fits the noisy data $\mathbf{b}$. If this term is too big, then $\mathbf{x}$ does not solve the problem. On the other hand, if this norm is too small, it means that $\mathbf{x}$ solves the initial problem by including noise in the fit. This should be avoided as well, because we do not want to fit noise in the data. The second term $\|B\mathbf{x}\|_2^2$ defines the regularity of the solution. The purpose of minimizing the regularity term is to suppress the high-frequency components by controlling the norm $\|B\mathbf{x}\|_2^2$. The balance between the terms is controlled by the factor λ . Larger values of λ give more weight to the regularity term. In similar lines with Equation (2.9), the full-rank regularized back-projected problem can be formulated as:

$$(A^H C^{-1} A + \lambda B) \mathbf{x} = A^H C^{-1} \mathbf{b} \quad (2.11)$$

From this, the solution of the Tikhonov regularized weighted LS solution is obtained as:

$$\hat{\mathbf{x}}_{TR} = (A^H C^{-1} A + \lambda B)^{-1} A^H C^{-1} \mathbf{b} \quad (2.12)$$

The inverse matrix of Equation (2.12) can be computed either with SVD or directly. In TR, a damping is added to each SVD component of the solution, thus effectively filtering out the components corresponding to the small singular values. The choice of the regularization parameter is crucial to yield a good solution.

The computation of the pseudo-inverse matrix in Equation (2.12) is exhausting for high-dimensional problems because direct inversion as well as SVD requires $O(N^3)$ operations for an $N \times N$ matrix, which requires either too much time or memory. Also since Equation (2.12) is a convex optimization problem, any iterative algorithm for solving this class of problems can be used for obtaining the solution. Such iterative methods can avoid decompositions associated with increased number of equations, thereby significantly reducing the computational cost associated with the decomposition. Methods like conjugate gradient LS (CGLS), generalized minimum residual (GMRES) method, quasi-minimal residual (QMR) method and other Krylov sub-space methods come under this category.

2.5.2 Conjugate Gradient Method

The conjugate gradient (CG) method for solving large sets of linear equations with symmetric positive definite coefficients was proposed by Hestenes and Stiefel in 1952 [29]. Application of the CG method does not require computation of the eigenvalues of the system, and can be used as a more direct method for finding the solution of a linear equation [30,31]. Furthermore, the implementation of the CG algorithm is simple and requires little storage space.

A matrix is symmetric if $A^T = A$, and positive definite if $x^T A x > 0$. If the matrix is symmetric, the eigenvalues of A are real and eigenvectors associated with distinct eigenvalues are orthogonal. The matrix A is positive definite (or positive semi-definite) if and only if all eigenvalues of A are positive (or non-negative). The operator arising from the quadratic functional $Q(x) = \frac{1}{2} x^T A x - b^T x$ is always symmetric. This symmetric matrix may be positive definite (PD), negative definite, singular point (positive-indefinite) and saddle point (indefinite). A symmetric positive or negative definite matrix operator of a quadratic function guarantees a strictly convex function, which in the first place shows that it is an increasing or decreasing function and in turn gives rise to a unique minimum or maximum. The CG algorithm does not converge directly in the case of negative definite matrices. This is the reason that the CG requires a symmetric positive definite (SPD) matrix.

In the CG method, A -conjugacy means that a set of nonzero vectors $\{p_0, p_1, p_2, \dots, p_{n-1}\}$ are conjugate with respect to the SPD matrix A , that is, $p_i^T A p_j \forall i \neq j$. The reason why such A conjugate sets are important is that we can minimize the function in N steps by successively minimizing it along each of the directions. One of the main benefits of CG is that it reaches the solution in at most N steps if the system we are trying to solve has N linear equations with N unknowns. If the underlying system is ill-posed, then A has a cluster of small singular values with values decaying to zero. As inferred from Equation (2.6), the SVD components corresponding to the small singular values amplify the noise, and the solution becomes unstable.

The CG iteration has an intrinsic regularizing effect. The CG method picks up the SVD components corresponding to the highest singular values with the iteration number as the regularization parameter. The spectral components corresponding to the largest eigenvalues converge faster. Consequently, as the number of iteration increases, the degree of regularization decreases with inclusion of more and more singular values. Thus, the CG algorithm exhibits an intrinsic regularization when stopped well before the convergence to the LS solution. This is a semi-convergence behavior because the iterate approaches the desired exact solution during the early phase of iterations and converges to some undesired vector in the later phase.

The CG algorithm starts with an initial guess of the solution $\mathbf{x}_0$, with an initial residual $\mathbf{r}_0$, and with an initial search direction that is equal to the initial residual: $\mathbf{p}_0 = \mathbf{r}_0$. The initial search direction vector ($\mathbf{p}_0$) is chosen to be same as the residual ($\mathbf{r}_0$) computed as:

$$\mathbf{p}_0 = \mathbf{r}_0 = \mathbf{b} - A\mathbf{x}. \quad (2.13)$$

The residual is computed successively at each step as $\mathbf{r}_k = \mathbf{b} - A\mathbf{x}_k$. The solution in each iteration is computed as the sum of the previous solution and a multiple of the search direction. The multiple of the search direction, called the step size, is also updated in each iteration. The search direction is updated using the residuals computed in that iteration and the previous direction. Since the search direction is linearly independent of previous search directions, the main benefit of CG iterations is that there is no need to store all previous search directions [32,33]. The algorithmic steps to be followed for CG are summarized below.

Algorithm 2.1 CG

Iterate for $i = 1, \dots, \text{MaxIter}$:

- 1) Step length $\alpha_i = \frac{|\mathbf{r}_i|^2}{\mathbf{p}_i^T A \mathbf{p}_i}$
- 2) Approximate solution $\mathbf{x}_{i+1} = \mathbf{x}_i + \alpha_i \mathbf{p}_i$
- 3) Residual $\mathbf{r}_{i+1} = \mathbf{r}_i - \alpha_i A \mathbf{p}_i$
- 4) Improvement at step i $\beta_i = \frac{|\mathbf{r}_{i+1}|^2}{|\mathbf{r}_i|^2}$
- 5) Search direction $\mathbf{p}_{i+1} = \mathbf{r}_{i+1} + \beta_i \mathbf{p}_i$
- 6) Repeat until $\mathbf{r}_{i+1} < \text{tol}$

end

Here, the residual vector computes the error in $A\mathbf{x} = \mathbf{b}$ and not the error in $\mathbf{x}$ itself. In an ordinary iteration, $\mathbf{x}_i$ is corrected as:

$$\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{r}_i \quad (2.14)$$

For the system of linear equations, $A\mathbf{x} = \mathbf{b}$, consider the first two iterates of the iteration $\mathbf{x}_{i+1} = (I - A)\mathbf{x}_i + \mathbf{b}$, with $\mathbf{x}_1 = \mathbf{b}$.

$$\begin{aligned} \mathbf{x}_2 &= (I - A)\mathbf{b} + \mathbf{b} = 2\mathbf{b} - A\mathbf{b} \\ \mathbf{x}_3 &= (I - A)\mathbf{x}_2 + \mathbf{b} = 3\mathbf{b} - 3A\mathbf{b} + A^2\mathbf{b}. \end{aligned} \quad (2.15)$$

Here, Equation (2.15) for any $\mathbf{x}_j$ is obtained as a combination of $\mathbf{b}, A\mathbf{b}, \dots, A^{j-1}\mathbf{b}$, which can be computed quickly by multiplying the previous term each time by A . This involves only one matrix multiplication per iteration.

All algorithms that work on such combinations are referred to as Krylov sub-space (KS) methods. Krylov suggested that there might be other, better combinations for $\mathbf{x}_j$ than the

particular one in Equation (2.15). It is possible to express these combinations using the basis matrix $K_j = [\mathbf{b} \ A \mathbf{b} \ A^2 \mathbf{b} \dots A^{j-1} \mathbf{b}]$ and the column space $\mathcal{K}_j$ of the basis vectors as;

1. The residual vector is orthogonal to $\mathcal{K}_j$ (conjugate gradient)
2. The residual vector has minimum norm for $\mathbf{x}_j$ in $\mathcal{K}_j$ (GMRES and minimum residual (MINRES))
3. Residual vector is orthogonal to a different space $\mathcal{K}_j(A^T)$ (biconjugate gradients)
4. The error vector has minimum norm (symmetric LQ method that uses the LQ factorization, where L is lower triangular matrix and Q is an orthogonal (or unitary) matrix (SYMMLQ).)

2.5.3 Other Krylov Sub-space Methods

Krylov sub-space methods are iterative methods in which the m th step provides an approximation to the solution $\mathbf{x}_m \in \mathbf{x}_0 + \mathcal{K}_m$. This approximation is found by requiring $\mathbf{x}_m$ to be the minimizer of some functional as described in one of the four strategies given above [34]. With an initial vector $\mathbf{x}_0$ and residual vector $\mathbf{r}_0 = \mathbf{b} - A\mathbf{x}_0$, a solution in the m th step can be obtained as $\mathbf{x}_m \in \mathbf{x}_0 + \mathcal{K}_m(A, \mathbf{r}_0)$, that satisfies a projection or minimizing condition. A unique solution can be obtained by imposing any one of the following conditions:

1. Ritz-Galerkin approach: $\mathbf{r}_n \perp \mathcal{K}_n(A, \mathbf{r}_0)$
2. Petrov-Galerkin approach: $\mathbf{r}_n \perp \mathcal{K}_n(A^H, \mathbf{r}_0^*)$
3. Minimal residual approach: $\mathbf{r}_n = \min_{\mathbf{x} \in \mathbf{x}_0 + \mathcal{K}_n(A, \mathbf{r}_0)} \|\mathbf{b} - A\mathbf{x}\|$

Here, $\mathbf{r}_0^*$ is an arbitrary vector such that $(\mathbf{r}_0^*, \mathbf{r}_0) \neq 0$. For a Hermitian coefficient matrix, the Lanczos process involving a three-term recurrence relationship can be used to ensure that one among the above-mentioned conditions is satisfied for generating the (bi-)orthogonal basis vectors of $\mathcal{K}_n(A, \mathbf{r}_0)$. Since the Lanczos process cannot be used for the non-Hermitian case, the Arnoldi process or the Bi-Lanczos process is used, even neither one satisfies the recurrence relation or condition for orthogonality. The Arnoldi process can generate the orthogonal basis with low computational costs and memory. The Krylov sub-space methods that use Lanczos processes for Hermitian linear systems are CG (Ritz-Galerkin approach), conjugate residual (CR) and minimum residual (MINRES) approach. For non-Hermitian systems, GMRES and Bi-CG (Petrov-Galerkin approach) use the Arnoldi process and Bi-Lanczos, respectively.

2.5.3.1 Arnoldi Process

The Arnoldi process is an orthogonal projection method onto $\mathcal{K}_m$ applied to general non-Hermitian matrices. This method was first introduced as a procedure for reducing a dense matrix into the Hessenberg form that involves a unitary transformation. As pointed out by Arnoldi [35], the eigenvalues of the Hessenberg matrix, obtained with fewer steps than N , can provide accurate approximations to the eigenvalues of the original matrix. An advantage of this approach is that it leads to an efficient method for approximating the eigenvalues of large sparse matrices and hence can be used for obtaining the solution of large sparse linear systems of equations. Arnoldi's procedure uses the stabilized Gram-Schmidt process to generate a sequence of orthonormal vectors, $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$, called the Arnoldi vectors, such that for every N , the vectors $\mathbf{v}_1, \dots, \mathbf{v}_N$ span the Krylov sub-space $\mathcal{K}_m$.

Algorithm 2.2 Arnoldi Algorithm

Initialize $\mathbf{v}_1$ as arbitrary vector such that $\|\mathbf{v}_1\|_2 = 1$.

Repeat for $j = 1, 2, 3, \dots$

$h_{i,j} = (A\mathbf{v}_j, \mathbf{v}_i), i = 1, 2, \dots, j,$

$$\hat{\mathbf{v}}_{j+1} = A\mathbf{v}_j - \sum_{i=1}^j h_{i,j} \mathbf{v}_i$$

$$h_{j+1,j} = \|\hat{\mathbf{v}}_{j+1}\|_2$$

$$\mathbf{v}_{j+1} = \frac{\hat{\mathbf{v}}_{j+1}}{h_{j+1,j}}$$

If V_k is the $N \times k$ matrix whose columns are the l_2 -orthonormal basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$, then $H_k = V_k^T A V_k$ is the upper $k \times k$ Hessenberg matrix whose entries are the scalars $h_{i,j}$ generated by the Arnoldi algorithm. If P_k is used to denote the orthogonal projector onto K_k , and A_k the section of A in K_k , then it can be inferred that H_k is same as the matrix representation of A_k in the basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$. Thus Arnoldi method is a form of Galerkin type that approximates the eigenvalues of A by those of H_k . The Galerkin approach for solving the linear system $A\mathbf{x} = \mathbf{b}$, uses the l_2 -orthogonal basis V_k to obtain an approximate solution of the form $\mathbf{x}_k = \mathbf{x}_0 + \mathbf{z}_k$, where $\mathbf{x}_0$ is an initial guess of the solution, and $\mathbf{z}_k$ is a member of the Krylov sub-space. For k steps of Algorithm 2.2, the Galerkin condition yields:

$$\mathbf{z}_k = V_k \mathbf{y}_k, \quad \text{where } \mathbf{y}_k = H_k^{-1} \|\mathbf{r}_0\|_2 \mathbf{e}_1, \quad (2.16)$$

where $\mathbf{e}_1$ denotes the Euclidean basis vector $\mathbf{e}_1 = (1, 0, 0, \dots, 0)^T$ [36].

2.5.3.2 Generalized Minimum Residual (GMRES) Method

The GMRES method, first introduced by Saad and Schultz in 1986 [37], can be used for arbitrary (non-singular) square matrices, as opposed to the classical iterative solvers that are applicable only to either diagonally dominant or positive definite matrices. Since this is a projection method in which K_m is the m th Krylov sub-space with $\mathbf{v}_1 = \frac{\mathbf{r}_0}{\|\mathbf{r}_0\|_2}$, it can minimize the residual norm over all vectors in $\mathbf{x}_0 + K_m$. Implementation of k steps of Arnoldi's method generates an l_2 -orthonormal system V_{k+1} and a $(k+1) \times k$ matrix $\tilde{H}_k$ whose only non-zero entries are the elements $h_{i,j}$. Hence $\tilde{H}_k$ is the same as H_k except for an additional row whose only nonzero element is $h_{k+1,k}$ in the $(k+1, k)$ 'th position. The vectors $\mathbf{v}_i$ and the matrix $\tilde{H}_k$ satisfy the relation:

$$A V_k = V_{k+1} \tilde{H}_k. \quad (2.17)$$

To solve the LS problem using GMRES, an approximate solution $\mathbf{x}_k$ is computed which minimizes the residual norm:

$$\min_{\mathbf{z} \in K_n} \|\mathbf{r}_0 - A\mathbf{z}\|_2. \quad (2.18)$$

With $\beta = \|\mathbf{r}_0\|_2$, and using Equation (2.16), the above equation can be represented as a function of $\mathbf{y}$. The cost function expressed in terms of $\mathbf{y}$ takes the form:

$$J(\mathbf{y}) = \|\beta \mathbf{v}_1 - AV_k \mathbf{y}\|_2 \quad (2.19)$$

Further, using Equation (2.17) in Equation (2.19):

$$J(\mathbf{y}) = \|V_{k+1}(\beta \mathbf{e}_1 - \tilde{H}_k \mathbf{y})\|_2. \quad (2.20)$$

Using the fact that V_{k+1} is l_2 -orthonormal:

$$J(\mathbf{y}) = \|(\beta \mathbf{e}_1 - \tilde{H}_k \mathbf{y})\|_2. \quad (2.21)$$

Since the approximate solution is of the form $\mathbf{x}_k = \mathbf{x}_0 + V_k \mathbf{y}$, and $\mathbf{y}_k$ denotes the $\mathbf{y}$ that minimizes $J(\mathbf{y})$, the solution of the LS problem in Equation (2.18) in the k 'th step is given by:

$$\mathbf{x}_k = \mathbf{x}_0 + V_k \mathbf{y}_k. \quad (2.22)$$

The algorithmic steps for GMRES are summarized in Algorithm 2.3. The convergence rate of GMRES is dependent on the distribution of the eigenvalues of A in the complex plane. Fast convergence can be achieved if the eigenvalues are clustered away from the origin. Besides the condition number of A , the eigenvalue distribution is therefore a more significant determinant of the convergence of the CG method.

Algorithm 2.3 GMRES

Initialize $\mathbf{r}_0 = \mathbf{b} - A\mathbf{x}$, $\beta = \|\mathbf{r}_0\|_2$, $\mathbf{v}_1 = \mathbf{r}_0 / \beta$

Iterate for $j = 1, \dots, \text{MaxIter}$

$$h_{i,j} = (A\mathbf{v}_j, \mathbf{v}_i); i = 1, 2, \dots, j$$

$$\hat{\mathbf{v}}_{j+1} = A\mathbf{v}_j - \sum_{i=1}^j h_{i,j} \mathbf{v}_i$$

$$h_{j+1,j} = \|\hat{\mathbf{v}}_{j+1}\|_2$$

$$\mathbf{v}_{j+1} = \frac{\hat{\mathbf{v}}_{j+1}}{h_{j+1,j}}$$

Get the approximate solution

$$\mathbf{x}_k = \mathbf{x}_0 + V_k \mathbf{y}_k, \text{ where } \mathbf{y} = H_k^{-1} \|\mathbf{r}_0\|_2 \mathbf{e}_1.$$

2.5.3.3 Conjugate Residual (CR) Algorithm

The CR algorithm is derived from GMRES for the particular case where A is Hermitian [38]. In this case, the residual vectors should be orthogonal. The vectors Ap_i 's ; $i = 0, 1, \dots$, are also orthogonal. It has the same structure as CG; that is, the residual vectors are now conjugate to each other, hence, the name of the algorithm. By choosing the initial search direction vector to be same as the residual, as in Equation (2.13), the steps followed in the CR algorithm can be summarized as shown in Algorithm 2.4.

Algorithm 2.4 CR

Initialize $r_0 = b - Ax$.

Iterate for $i = 1, \dots, \text{MaxIter}$

$$\text{step length } \alpha_i = \frac{r_i^T A r_i}{(A p_i)^T A p_i}$$

Approximate Solution $x_{i+1} = x_i + \alpha_i p_i$

Residual $r_{i+1} = r_i - \alpha_i A p_i$

$$\text{Improvement at step } i \beta_i = \frac{r_{i+1}^T A r_{i+1}}{r_i^T A r_i}$$

Search direction $p_{i+1} = r_{i+1} + \beta_i p_i$

Repeat until $r_{i+1} < \text{tol}$

end

Although CR and CG algorithms typically exhibit similar convergence behavior, the CG method is often preferred for applications involving large data matrices because the CR algorithm requires one more vector update, that is, $2N$ more operations and one more vector of storage than the CG method. Although the linear CG method has a faster convergence [39], it suffers from computational difficulties in each iteration to find the step size and the descent direction [40]. In this regard, the simplicity of the Landweber method makes it a potential candidate for real-time implementation despite its inherent disadvantage of slow convergence.

2.5.4 Landweber Method

The Landweber method is an example of the so-called gradient methods in which a new approximation to the solution at each step is obtained by modifying the old one in the

direction of the gradient of the discrepancy functional. Here, the solution for the forward model is obtained by minimizing the cost function:

$$e(\hat{\mathbf{x}}) = \frac{1}{2} \|\mathbf{b} - A\hat{\mathbf{x}}\|^2, \quad (2.23)$$

where $\hat{\mathbf{x}}$ is an estimate of $\mathbf{x}$. With numerical optimization, the line search method used to minimize the cost function is given by:

$$\hat{\mathbf{x}}_{i+1} = \hat{\mathbf{x}}_i + \alpha_i \mathbf{s}_i, \quad (2.24)$$

where α_i is the step size and $\mathbf{s}_i$ is the descent direction. The main difference between various line search methods is based on the way to choose this descent direction. The Landweber method is a special case of the line search methods that uses the steepest descent method in each iteration to find the required descent direction [41–45]. The steepest descent is obtained by taking the partial derivative of the cost function with respect to $\hat{\mathbf{x}}$. Accordingly, the Landweber method finds the solution using the update equation:

$$\mathbf{x}_{LW}^{(k+1)} = \mathbf{x}_{LW}^{(k)} + \lambda A^H (\mathbf{b} - A\mathbf{x}_{LW}^{(k)}), \quad (2.25)$$

where LW stands for Landweber method and λ is a predetermined parameter such that:

$$0 < \lambda < \frac{2}{\sigma_1^2}, \quad (2.26)$$

where σ_1 is the largest singular value of the matrix A [46]. Whereas a larger value of λ yields faster convergence with increased risk of instability, a smaller value leads to slower convergence. Despite the slow convergence rates of the Landweber method compared to other methods like the CG algorithm, the simplicity of the method makes it a more suitable choice in some cases. The iterative method needs to be stopped at an appropriate iteration number as it semi-converges in the case of ill-posed problems. This is because the iterates approach a regularized solution during the first few iterations and then become unstable.

2.6 Regularization Approaches Using l_1 Priors

When the system of equations is underdetermined, it has no unique solution. In this case, the prior information that the signal has a sparse representation can be used to optimize the problem. A matrix or a vector is said to be s -sparse when it has only s non-zero coefficients. This means that the actual information content in that signal is less in the sparse representation and requires only a few samples to uniquely represent it in an ideal condition. This is the main idea behind the CS where the number of samples required to represent a signal is viewed as a function of the actual information content rather than the highest frequency assumption in the Nyquist sampling criterion. For an under-determined system, a unique solution cannot be obtained directly. However, consider that $\mathbf{x}$ is s -sparse

and we know the locations of the non-zero entries (Ω) of x . Now, if $s < M$, the problem becomes over-determined and the solution can be obtained using:

$$x_{\Omega} = \left(A_{\Omega}^T A_{\Omega} \right)^{-1} A_{\Omega}^T b, \quad (2.27)$$

where A_{Ω} represents the operator matrix with only those columns corresponding to Ω . But in practical scenarios, we do not know Ω . In such cases, the s elements can be present anywhere along the locations of x . Therefore, a brute force method is needed to solve the under-determined system. In this case, the solution is obtained by minimizing the NP-hard problem

$$\min_x \|x\|_0 \quad \text{such that} \quad Ax = b. \quad (2.28)$$

Since, the l_1 -norm is the closest convex approximation to the l_0 -norm [5,47], one can find an approximate solution by minimizing the l_1 -norm of the signal using its sparse representation.

Other information which we can use for sparse recovery includes group-sparsity and row-sparsity. A vector is said to be group-sparse if its elements can be divided into a set of groups and only a few groups among them are non-zero. Similarly, a matrix is said to be row-sparse if only a few rows of that matrix are non-zeros. A row-sparse matrix could be equivalently represented in the group-sparse form by vectorizing it. The row- and group-sparse information could be incorporated to find the best solution to the under-determined system by minimizing a mixed $l_{2,1}$ -norm. The l_2 -norm ensures that the groups having non-zero coefficients are not sparsified while l_1 -norm signifies that only a few groups are non-zeros. In all cases, the images may not be sparse in the image domain requiring the application of sparsifying transforms to make the image sparse. The choice of sparsifying transform plays an important role in determining the number of samples to be acquired for the CS reconstruction to perform well. The two measures to be considered while selecting the sparsifying transform include the sparsity and the coherence measure. The sparsification in the transform domain is defined as the rate at which its transform coefficients decay. A faster decay corresponds to the ability of the transform operator to approximate an image using a few significant coefficients. Coherence is a measure for concentration in the measurement operator. That is, if the elements in the measurement operator are well distributed and the measurement operator is dense, the coherence is small. This is important because if both the image to be recovered and measurement matrix are sparse, there is a high probability that no information about the signal will be recorded [48]. For an MRI signal, the sparsifying operator that we select should be such that it is maximally incoherent with the measurement operator (Fourier) and should make the image as sparse as possible in the sparse domain. Based on these two measures, one of the best sparsifying transforms that one could use is the wavelet transform. However, even though the Fourier transform is incoherent with the wavelet operator at finer scales, it is coherent at courser scales. Because of this, when we use the wavelet as the sparsifying transform, the sampling should be denser at the low-frequency region compared to the high-frequency region.

Another popular choice of transform sparse domain is finite differencing. It represents an image by its gradients. The sparsifying transform could be either using a tight frame or orthogonal basis. A distinguishing feature is that for tight frames, the transform operators

do not satisfy the condition $\Psi^T \Psi = \Psi \Psi^T$. An example of orthogonal transform is the discrete wavelet transform (DWT) and tight frame transform is the finite differencing. With DWT, the data in the wavelet domain can be transformed back to image domain by applying the inverse transform. But it is not possible with the tight frame transformations like finite differencing. Depending on whether the transform is orthogonal or tight frame, we need to use a synthesis or analysis model. The analysis-based model is where one seeks the signal $\mathbf{x}$ whose transformed coefficients are sparse. In the synthesis-based model, we directly seek the sparse coefficients that solve the under-determined linear model. The signal is then obtained by the inverse transform operation. Using transform domain sparsity, the l_1 -regularization may be solved using the synthesis model by minimizing the l_1 -norm of the image in the sparse domain, as follows:

$$\min_w \|\mathbf{w}\|_1 \quad (2.29)$$

$$\text{subject to } \|\mathbf{b} - A\Psi^T \mathbf{w}\|_2^2 \leq \varepsilon,$$

which is same as the following unconstrained Lagrangian form:

$$\min_w \|\mathbf{b} - A\Psi^T \mathbf{w}\|_2^2 + \lambda \|\mathbf{w}\|_1. \quad (2.30)$$

The analysis model of Equation (2.29) is:

$$\min_x \|\mathbf{b} - A\mathbf{x}\|_2^2 + \lambda \|\Psi \mathbf{x}\|_1 \quad (2.31)$$

For an optimization problem defined in Equation (2.31), a non-linear thresholding operator [16,49–51] can recover the significant coefficients of the sparse transform, thereby recovering the lost image information.

2.6.1 Solution to l_1 -Regularized Problems

Consider the l_1 -regularization problem that minimizes a twice-differentiable loss function having the general form:

$$\min_x f(\mathbf{x}) \triangleq L(\mathbf{x}) + \lambda \|\mathbf{x}\|_1, \quad (2.32)$$

where $L(\mathbf{x})$ is a twice-differentiable loss function with a lower bound. Since the combined objective function $f(\mathbf{x})$ is non-differentiable due to the presence of zero values, standard unconstrained optimization strategies cannot be used for obtaining a solution. Therefore, a wide variety of algorithms have been proposed to solve such problems. Based on the approach used for handling the non-differentiability of the objective function, the methods can be broadly divided into three categories: (1) sub-gradient methods, (2) unconstrained approximations and (3) constrained optimization methods. In the sub-gradient methods, a direct optimization strategy is employed, and sub-gradients are used at the points of non-differentiability. In the unconstrained strategy, instead of directly optimizing the non-differentiable function, a smooth surrogate objective function is optimized. Alternatively,

the unconstrained non-differentiable problem is transformed to an equivalent differentiable problem with constraints in the constrained optimization method. The following section briefly discusses the outline of the three different approaches.

2.6.1.1 Sub-gradient Methods

In this strategy, the objective function is treated as a non-smooth optimization problem, with the use of sub-gradients of the objective function at the non-differentiable points. For a parameter vector $\mathbf{x}$ to be a local minima, this means that the zero vector must be an element of the set containing all sub-gradients at $\mathbf{x}$ denoted by $\partial f(\mathbf{x})$ [52]. Since the signum function is the sub-differential of the absolute value function $|x_i|$ and assumes the sign of x_i for a non-zero x_i , the optimality conditions can be expressed as follows:

$$\begin{cases} \nabla_i L(\mathbf{x}) + \lambda \text{sign}(x_i) = 0; & |x_i| > 0 \\ |\nabla_i L(\mathbf{x})| \leq \lambda; & x_i = 0. \end{cases} \quad (2.33)$$

The sub-gradient $\nabla_i^s f(x_i)$ for each x_i is selected as follows:

$$\nabla_i^s f(x_i) \triangleq \begin{cases} \nabla_i L(\mathbf{x}) + \lambda \text{sign}(x_i); & |x_i| > 0 \\ \nabla_i L(\mathbf{x}) + \lambda; & x_i = 0, \nabla_i L(\mathbf{x}) < -\lambda \\ \nabla_i L(\mathbf{x}) - \lambda; & x_i = 0, \nabla_i L(\mathbf{x}) < \lambda \\ 0; & -\lambda \leq \nabla_i L(\mathbf{x}) \leq \lambda. \end{cases} \quad (2.34)$$

This sub-gradient yields a descent direction on the objective function for a sub-optimal $\mathbf{x}$. Therefore, several methods have been developed that use this sub-gradient as a surrogate for the gradient. Some of the methods that comes under this class includes the coordinate descent, active set methods and orthant-wise descent methods. These methods are developed with the assumption that $\nabla^2 f(\mathbf{x}) = \nabla^2 L(\mathbf{x})$, although it is not strictly true if x_i is 0.

2.6.1.1.1 Coordinate Descent

The general implementation of this method includes choosing some coordinate i and finding the optimal value of x_i , given the remaining parameters. This procedure is repeated until some termination criterion is met. A coordinate descent algorithm for $L(\mathbf{x}) \triangleq \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2$ called the shooting algorithm was proposed [53]. In this algorithm, the coordinate is chosen by repeatedly passing through the variables in order, and the optimal value is found using a simple analytic solution, obtained by setting x_i to 0, and $x_i \leftarrow x_i - \nabla_i^s f(x_i) / \nabla_{ii}^2 L(\mathbf{x})$. Although the coordinate descent algorithms are simple to implement, they are more effective if the variables have largely independent effects on the objective. The performance of the algorithm is degraded as the variables become more dependent. These methods may require so many coordinate updates that implementation becomes impractical.

2.6.1.1.2 Active Set Methods

From the definition of the sub-gradient vector, it is evident that $-\nabla^s f(\mathbf{x})$ is equivalent to the steepest descent direction in the case of smooth optimization. Also since $\nabla^2 f(\mathbf{x}) = \nabla^2 L(\mathbf{x})$,

any element of the sub-differential can be taken for the first derivative. Thus, the convergence can be accelerated by using Newton-like iteration of the form:

$$x \leftarrow x - \beta T^{-1} \nabla^s f(x), \quad (2.35)$$

where $T = \nabla^2 L(x)$ and step size $\beta > 0$ is selected so that it satisfies some decrease rule. However, there is no guarantee that the direction $-T^{-1} \nabla^s f(x)$ is a descent direction at the non-differentiable point. This means that it is impossible to find a $\beta > 0$ that makes this update improve on the objective function. However, this method of update can be used for the non-zero variables in the context of an active set method that uses the following two steps:

1. Update the non-zero variables.
2. Given the active set of zero-valued variables, optimize the non-zero variables.

Using $\mathcal{W}_s$ to denote the working set, Step 2 is implemented using the Newton-like method:

$$x_{\mathcal{W}_s} \leftarrow x_{\mathcal{W}_s} - \beta T_{\mathcal{W}_s}^{-1} \nabla_{\mathcal{W}_s}^s f(x), \quad (2.36)$$

that provides a descent direction so that all elements of $x_{\mathcal{W}_s}$ are non-zero. Step 1 can be implemented using different methods such as the grafting procedure [54]. Active set methods have been applied to specific loss functions such as least squares [55] and logistic regression [56]. Active set methods work well when there is a good guess of the initial active set. In the absence of a good initial guess of the active set, a large number of iterations are required.

2.6.1.1.3 Orthant-wise Descent Quasi-Newton Method

Unlike the active set method, the orthant-wise descent method restricts the updated parameter to be within certain orthants to keep differentiability of the l_1 -regularized problem [57]. The main difference is the use of a projection operator that projects the Newton-like direction so that it is guaranteed to be a descent direction. Using the projection operator, the update equation takes the following form:

$$x \leftarrow \mathcal{P}_o \left[x - \beta \mathcal{P}_s \left[T^{-1} \nabla^s f(x) \right] \right], \quad (2.37)$$

where $\mathcal{P}_s$ is the operator that projects the Newton-like direction and $\mathcal{P}_o$ projects the updated value onto the orthant containing x . The zero-valued variables are assigned to the orthant that the steepest descent direction points into. In the absence of a good guess for the active set, the orthant-wise descent method is a better alternative to the active set methods because the orthant-wise descent method can make more changes to the set of non-zero variables. However, after the identification of the set of non-zero variables, the orthant-wise descent does not apply Newton's method directly on this subset of variables due to the presence of the operator $\mathcal{P}_s$.

2.6.1.2 Constrained Log-Barrier Method

In this method, the original non-differentiable problem is reformulated as a differentiable problem with constraints which have the same minimizer. In the constrained form, each variable x_i is expressed as the sum:

$$x_i = x_i^+ - x_i^-. \quad (2.38)$$

Also, $x_i^+ \geq 0$ and $x_i^- \geq 0$. Since $|x_i| = x_i^+ - x_i^-$, the optimization problem can be recast into the following form:

$$\begin{aligned} \min_{x^+, x^-} L(x^+ - x^-) + \lambda \sum_i [x_i^+ + x_i^-], \\ \text{s.t. } \forall_i x_i^+ \geq 0, x_i^- \geq 0. \end{aligned} \quad (2.39)$$

A major disadvantage of this approach is the increase in the number of variables in the optimization problem [58,59]. Instead of using a smooth approximation of the objective function, a barrier function is added in order to keep x_i^+ and x_i^- sufficiently positive. The unconstrained approximations are parameterized in terms of a scalar μ , and take the following form:

$$g(x^+, x^-, \mu) \triangleq L(x^+ - x^-) + \lambda \sum_i [x_i^+ + x_i^-] - \mu \sum_i \log x_i^+ - \mu \sum_i \log x_i^-. \quad (2.40)$$

Note that the log-barrier functions tend to infinity if any coordinate of the vectors x^+ and x^- approach zero. Therefore, the constraints in Equation (2.39) should be $x_i^+ > 0, x_i^- > 0$ in order for the iterations to stay inside the constraint set. When μ approaches zero, the minimizer of $g(x^+, x^-, \mu)$ converges to the minimizer of $f(x)$.

2.6.1.3 Unconstrained Approximations

Instead of dealing directly with $f(x)$ that tries to solve a non-smooth optimization problem, unconstrained approximation methods replace $f(x)$ with a twice-differentiable surrogate objective function denoted by $g(x)$. Note that the minimizer of $g(x)$ is sufficiently close to the minimizer of $f(x)$. Thus, an unconstrained optimization method can be used to minimize $g(x)$ directly. Some methods under this class include smoothing, and bound optimization.

2.6.1.3.1 Smoothing

The non-differentiable optimization can be transformed into a differentiable optimization by replacing the non-differentiable term with a differentiable approximation. In the epsL1 approximation by Lee et al. [60], the l_1 -norm is replaced by the sum of multi-quadratic functions as follows:

$$g(x) \triangleq L(x) + \lambda \sum_i \sqrt{x_i^2 + \epsilon}. \quad (2.41)$$

Since $\lim_{\epsilon \rightarrow 0} \sqrt{x_i^2 + \epsilon} = |x_i|$, the twice differentiable function $g(x)$ is chosen such that $\lim_{\epsilon \rightarrow 0^+} g(x) = f(x)$. Therefore, for a sufficiently small value of ϵ , the minimizer of $g(x)$ will numerically correspond to the minimizer of $f(x)$. With this approximation, Newton iterations can be directly applied to minimize $g(x)$ as follows:

$$x \leftarrow x - \beta T^{-1} \nabla^s f(x), \quad (2.42)$$

where $T \triangleq \nabla^2 g(x)$. However, using $r(x) \triangleq \Sigma_i \sqrt{x_i^2 + \epsilon}$, the diagonal terms of $\nabla^2 r(x) = \epsilon / (\sqrt{x_i^2 + \epsilon})$ and off-diagonal elements as zeros. Hence for a non-zero x_i such that $x_i \gg \epsilon$, the diagonal elements can be approximated by $\nabla^2 r(x) \approx \epsilon / |x|^3$, implying that a positive number becomes very small as $|x|$ increases. If $x_i = 0$, then, $\nabla^2 r(x) \approx \sqrt{\epsilon}^{-1}$, yielding a relatively larger value. The convergence of Newton iterations is affected when the condition number of $\nabla^2 r(x)$ becomes very large and when the optimal solution contains some $x_i \gg \epsilon$ and some $x_i = 0$.

2.6.1.3.2 Bound Optimization

In the bound optimization approach, the regularizer is replaced with a convex upper bounding function which is exact at the current value of x instead of replacing the regularizer with a fixed smooth function. Consequently, the method alternates between two simple steps until some termination criterion is met. The alternating steps consist of computing the convex upper bounding function $g(x)$, followed by minimizing $g(x)$. A typical example is the bound on the square root function given by [61]:

$$\sqrt{u} \leq \frac{\sqrt{u}}{2\sqrt{v}} + \frac{\sqrt{v}}{2}. \quad (2.43)$$

Denoting the value of x_i from the previous iteration as z_i and $|x_i| = \sqrt{x_i^2}$, the bounding function becomes:

$$\sqrt{u} \leq \frac{1}{2} \frac{x_i^2}{|z_i|} + \frac{1}{2} |z_i|, \quad (2.44)$$

where the equality becomes exact when $z_i = x_i$. The bound optimization algorithm can be implemented by alternating between a step consisting of setting $z \leftarrow x$, followed by a Newton step on the twice differentiable cost function:

$$g(x) \triangleq L(x) + \lambda \sum_i \frac{1}{2} \frac{x_i^2}{|z_i|} + \frac{1}{2} |z_i|. \quad (2.45)$$

Since $g(x)$ has the form of l_2 -regularization, the update has a closed form for a quadratic $L(x)$. In the probabilistic context, this algorithm arises as an expectation maximization (EM) algorithm [62], with the z_i update corresponding to the E-step and optimizing $g(x)$ given z corresponding to the M-step. The method is implemented using an iteratively reweighted least squares (IRLS) update. The EM algorithm can be treated as a special case of the majorization-minimization algorithm.

2.6.1.3.3 Majorization-Minimization (MM)

MM involves replacing a difficult minimization problem with a sequence of simpler minimization problems that generates a sequence of vectors $\mathbf{x}^{(k)}$; $k=0,1,2,\dots$ and converges to a desired solution [63,64]. For the minimization of $J(\mathbf{x}) = \|\mathbf{b} - A\mathbf{x}\|_2^2 + \lambda \|\mathbf{x}\|_1$ using the MM approach, one starts with some initial guess $\mathbf{x}^{(k)}$ as the minimum of $J(\mathbf{x})$ and tries to find a new vector $\mathbf{x}^{(k+1)}$, such that $J(\mathbf{x}^{(k+1)}) < J(\mathbf{x}^{(k)})$. The MM approach chooses a new function that majorizes $J(\mathbf{x})$, and then minimizes the new function to get $\mathbf{x}^{(k+1)}$.

For ease of explanation, the new function is denoted as $G(x)$. The new function is such that the $G(x) > J(x)$; that is, $G(x)$ majorizes $J(x)$. $G(x)$ also becomes equal to $J(x)$ at $x = x^{(k)}$. Now $x^{(k+1)}$ is found by minimizing $G(x)$. For convenience of notation, the majorizing function is denoted as $G^k(x)$, as it changes with iteration. The algorithmic procedure is summarized in Algorithm 2.5.

Algorithm 2.5 MM

- 1) Initialize $k, x^{(0)}$
 - 2) Choose $G^k(x)$
 - (i) $G^k(x) > J(x)$ for all x
 - (ii) $G^k(x^{(k)}) = J(x^{(k)})$
 - 3) Set $x^{(k+1)}$ as minimizer of $G^k(x)$
 - 4) $k = k + 1$ and go to step 2
-

2.6.1.3.3.1 Landweber Iteration The MM approach can be used to minimize $J(x)$ instead of solving the system of linear equations. For this purpose, a $G^k(x)$ satisfying $G^k(x^{(k)}) = J(x^{(k)})$ is used so that the non-negative function to be added to $J(x)$ is zero at $x = x^{(k)}$. For this, consider the minimization of the simple objective function:

$$J(x) = \|b - Ax\|_2^2 \quad (2.46)$$

With further simplification, Equation (2.46) can also be expressed as:

$$J(x) = b^T b - 2b^T Ax + x^T A^T A x \quad (2.47)$$

As Equation (2.47) is differentiable and convex, the minimizer can be obtained by differentiating $J(x)$ with respect to x and equating to zero. This yields the solution:

$$x = (A^T A)^{-1} A^T b. \quad (2.48)$$

Here, the minimizer is obtained by solving a linear system of equations. However, solving these equations may not be easy due to the larger matrix size and ill-posed nature of the matrix involved. Alternately, the MM approach can be used to minimize $J(x)$ instead of solving the system of linear equations. Accordingly, the $G^k(x)$ that satisfies the MM conditions is given by:

$$G^k(x) = \|b - Ax\|_2^2 + (x - x^{(k)})^T (\alpha I - A^T A) (x - x^{(k)}), \quad (2.49)$$

where α is chosen to be equal to or greater than the maximum eigenvalue of $A^T A$. The minimizer in the next iteration $\mathbf{x}^{(k+1)}$ is then obtained by minimizing $G^k(\mathbf{x})$. Minimizing $G^k(\mathbf{x})$ yields:

$$\frac{\partial}{\partial \mathbf{x}} G^k(\mathbf{x}) = 2A^T \mathbf{b} - 2(\alpha I - A^T A) \mathbf{x}^{(k)} + 2\alpha \mathbf{x} = 0. \quad (2.50)$$

The solution is then obtained by updating $\mathbf{x}^{(k)}$ using:

$$\mathbf{x} = \mathbf{x}^{(k)} + \frac{1}{\alpha} A^T (\mathbf{b} - A \mathbf{x}^{(k)}). \quad (2.51)$$

This update equation obtained by using the MM procedure is known as the Landweber iteration. The Landweber iteration can be used only if the objective function contains differentiable convex functions. However, if the objective function contains terms for which differentiation cannot be obtained directly, then a proximal operator becomes necessary to obtain the solution.

2.6.1.3.3.2 Soft Thresholding Soft thresholding is just the proximal mapping of the l_1 -norm. Assuming $A = I$ in the minimization function $J(\mathbf{x})$, and denoting $f(\mathbf{x}) = \lambda \|\mathbf{x}\|_1$, the proximal mapping of f is defined as follows:

$$\text{prox}_f(\mathbf{x}) = \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{b} - \mathbf{x}\|_2^2 + \lambda \|\mathbf{x}\|_1. \quad (2.52)$$

The optimality condition for Equation (2.52) is:

$$0 \in \nabla \left(\frac{1}{2} \|\mathbf{b} - \mathbf{x}\|_2^2 \right) + \partial \lambda \|\mathbf{x}\|_1 \Leftrightarrow 0 \in \mathbf{b} - \mathbf{x} + \lambda \partial \|\mathbf{x}\|_1. \quad (2.53)$$

As the l_1 -norm is separable, the cases for $x_i \neq 0$ and $x_i = 0$ can be considered separately. First, consider the case when $x_i \neq 0$. Then $\partial \|x_i\|_1 = \text{sign}(x_i)$ and the optimum x_i^* is obtained as:

$$0 = x_i - b_i + \lambda \text{sign}(x_i) \Leftrightarrow x_i^* = b_i - \lambda \text{sign}(x_i^*). \quad (2.54)$$

Also if $x_i^* < 0$, then $b_i < -\lambda$ and if $x_i^* > 0$, then $b_i > \lambda$. Thus $|b_i| > \lambda$ and $\text{sign}(x_i^*) = \text{sign}(b_i)$. Therefore:

$$x_i^* = b_i - \lambda \text{sign}(b_i). \quad (2.55)$$

In the second case when $x_i = 0$, the sub-differential of the l_1 -norm is the interval $[-1, 1]$ and the optimality condition is:

$$0 \in -b_i + \lambda [-1, 1] \Leftrightarrow b_i \in [-\lambda, \lambda] \Leftrightarrow |b_i| \leq \lambda. \quad (2.56)$$

Combining both cases:

$$[\text{prox}_f(\mathbf{x})]_i = x_i^* = \begin{cases} 0; & |b_i| \leq \lambda \\ b_i - \lambda \text{sign}(b_i); & |b_i| > \lambda. \end{cases} \quad (2.57)$$

This implies that the minimizer of $f(\mathbf{x})$ is obtained by applying the soft-threshold rule to $\mathbf{b}$ with threshold λ . The solution vector is then obtained by applying the soft-thresholding operator defined as:

$$\mathbf{x} = \text{soft}(\mathbf{b}, \lambda) = \begin{cases} \mathbf{b} + \lambda, & \text{if } \mathbf{b} \leq -\lambda \\ 0, & \text{if } |\mathbf{b}| < \lambda \\ \mathbf{b} - \lambda, & \text{if } \mathbf{b} \geq \lambda. \end{cases} \quad (2.58)$$

This can be more compactly expressed as:

$$\text{soft}(\mathbf{b}, T) = -\text{sign}(\mathbf{b}) \max(0, |\mathbf{b}| - \lambda). \quad (2.59)$$

The solution for $J(\mathbf{x}) = \|\mathbf{b} - A\mathbf{x}\|_2^2 + \lambda \|\mathbf{x}\|_1$ can be obtained by using the soft thresholding operator iteratively using the iterated soft-thresholding algorithm (ISTA).

2.6.1.3.3.3 ISTA With the MM approach, a majorizer $G^k(\mathbf{x})$ can be found that coincides with $J(\mathbf{x})$ at $\mathbf{x} = \mathbf{x}^{(k)}$ and easily minimized. For the cost function with l_1 -norm penalty, the $G^k(\mathbf{x})$ satisfying the MM conditions is given by:

$$G^k(\mathbf{x}) = \|\mathbf{b} - A\mathbf{x}\|_2^2 + (\mathbf{x} - \mathbf{x}^{(k)})^T (\alpha I - A^T A) (\mathbf{x} - \mathbf{x}^{(k)}) + \lambda \|\mathbf{x}\|_1. \quad (2.60)$$

Separating the terms containing $\mathbf{x}$:

$$G^k(\mathbf{x}) = \alpha \left\| \mathbf{x}^{(k)} + \frac{1}{\alpha} A^T (\mathbf{b} - A\mathbf{x}^{(k)}) - \mathbf{x} \right\|_2^2 + \lambda \|\mathbf{x}\|_1 + C, \quad (2.61)$$

where C is a constant with respect to $\mathbf{x}$. Using the soft-thresholding operator, the solution to Equation (2.61) is obtained as:

$$\mathbf{x}^{(k+1)} = \text{soft} \left(\mathbf{x}^{(k)} + \frac{1}{\alpha} A^T (\mathbf{b} - A\mathbf{x}^{(k)}), \frac{\lambda}{2\alpha} \right), \quad (2.62)$$

where α is greater than the maximum eigenvalue of $A^T A$. This algorithm based on the MM approach is referred to as the thresholded Landweber (TL) or ISTA [65,66]. The steps used in the algorithm are summarized in Algorithm 2.6.

Algorithm 2.6 ISTA

- 1) Initialize $\mathbf{x}^{(0)}$, $k = 1$.
 - 2) Find Land weber update $\mathbf{x} = \mathbf{x}^{(k)} + \frac{1}{a} A^T (\mathbf{b} - A\mathbf{x}^{(k)})$
 - 3) Perform the soft thresholding operation $\mathbf{x}^{(k+1)} = \text{soft}\left(\mathbf{x}, \frac{\lambda}{2a}\right)$
 - 4) Stop if converged. Otherwise goto step 2
-

Although different algorithms have been put forward for solving general linear inverse problems subject to l_1 -regularization, ISTA is the simplest nonparametric method that is mathematically proven to converge [48,67,68].

2.6.1.3.3.4 Sub-band Adaptive ISTA (SISTA) The sub-band adaptive iterative shrinkage/thresholding algorithm (SISTA) [69] is a generalized form of ISTA used to increase the convergence rate particularly for signal recovery in which sparsity is enforced using the wavelet transform. Unlike ISTA, where all sub-bands are treated uniformly, SISTA uses different thresholds and update steps for different sub-bands of the wavelet transformed image. With Ψ denoting the synthesis operator for the wavelet frame acting on the frame coefficients $\mathbf{w}$, an estimate of $\mathbf{x}$ using wavelet regularization can be obtained by constraining the estimate to have a sparse representation, such that $\hat{\mathbf{x}} = \Psi\mathbf{w}$. SISTA introduces a vector $\mathbf{a} = (a_1, a_2, \dots, a_J)$ of step sizes, and a diagonal operator $\Lambda_a = \text{diag}(\mathbf{a})$ such that it multiplies the j th sub-band of the wavelet coefficients by a_j ; that is, $(\Lambda_a \mathbf{w})_j = a_j w_j$. SISTA modifies the ISTA algorithm as outlined in Algorithm 2.7.

Algorithm 2.7 SISTA

- 1) Initialize $\mathbf{x}^{(0)}$, $\mathbf{w}^{(0)}$, $k = 1$.
 - 2) Find Landweber update $\mathbf{w} = \mathbf{w}^{(k)} + \Lambda_a^{-1} \left(\Psi^* A^* (\mathbf{b} - \Psi \mathbf{w}^{(k)}) \right)$
 - 3) Perform the soft thresholding operation $\mathbf{w}^{(k+1)} = \text{soft}\left(\mathbf{w}, \frac{\lambda}{2a_j}\right)$
 - 4) Stop if converged. Otherwise go to step 2
-

Since the only change in SISTA compared to ISTA is the use of a weighting matrix Λ_a , SISTA will be the same as ISTA when all elements of $\mathbf{a}$ are equal.

2.6.1.3.3.5 Fast Iterated Soft-Thresholding Algorithm (FISTA) The inherent slow convergence of ISTA prompted researchers to find ways to accelerate ISTA in large-scale convex optimization problems. These efforts have led to the use of a general class of multistep methods that utilize the result of past iterations to speed up convergence, as in two-step iterative

shrinkage/thresholding (TwIST) [70], and FISTA [71]. FISTA increases the speed of ISTA by introducing a proximity measure in the ISTA algorithm. The proximity measure used here is a specific linear combination of the solutions estimated in the previous two iterations. For a constant step size a , the fast proximal gradient method of Beck and Teboulle [71], known as FISTA, includes the following:

$$\begin{aligned} \mathbf{x}^{(k+1)} &= \text{prox}\left(\mathbf{w}^{(k)} - a\nabla J\left(\mathbf{w}^{(k)}\right)\right) \\ t^{k+1} &\leftarrow \frac{1}{2}\left(1 + \sqrt{1 + 4\left(t^k\right)^2}\right) \\ \mathbf{w}^{(k+1)} &\leftarrow \mathbf{x}^{(k+1)} + \frac{t^k - 1}{t^{k+1}}\left(\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\right). \end{aligned} \tag{2.63}$$

Here, the proximal step is not just performed on the previous point $\mathbf{x}^{(k)}$ but at a point $\mathbf{w}^k$ that is a specific linear combination of the two previous points $\{\mathbf{x}^{k+1}, \mathbf{x}^k\}$. The algorithmic steps in FISTA are summarized in Algorithm 2.8.

Algorithm 2.8 FISTA

- 1) Initialize $\mathbf{x}^{(0)}, \mathbf{w}^{(k)}, k = 1, t_1 = 1$
 - 2) Find Landweber update $\mathbf{x} = \mathbf{w}^{(k)} + \frac{1}{a}A^T\left(\mathbf{b} - A\mathbf{w}^{(k)}\right)$
 - 3) Perform the soft thresholding operation $\mathbf{x}^{(k+1)} = \text{soft}\left(\mathbf{x}, \frac{\lambda}{2a}\right)$
 - 4) Perform controlled over relaxation $t_{k+1} = \frac{1 + \sqrt{1 + 4\left(t^k\right)^2}}{2}$
 $\mathbf{w}^{(k+1)} = \mathbf{x}^{(k+1)} + \frac{t^k - 1}{t^{k+1}}\left(\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\right)$
 - 5) Stop if converged. Otherwise go to step 2
-

2.7 Linear Estimation in pMRI

In general, all pMRI algorithms make use of the model in which a vector of observations are expressed in terms of a linear operator A , an unknown vector of nonrandom parameters $\mathbf{x}$, and a noise vector $\mathbf{n}$, as discussed in the beginning of this chapter. Unless otherwise mentioned, the noise vector is assumed to be characterized by a zero-mean Gaussian

distribution with covariance matrix Σ_{cov} . The measurement vector $\mathbf{b}$ based on this linear model can be expressed as:

$$\mathbf{b} = A\mathbf{x} + \mathbf{n}. \quad (2.64)$$

For the model in Equation (2.64), the Cramer-Rao bound (CRB) gives the lower bound on the variance of the estimator $\hat{\mathbf{x}}(\mathbf{b})$ for $\mathbf{x}$. Traditionally, the application of CRB has been confined to the class of unbiased estimators for which the expected value of any estimator of a parameter equals the true parameter value. An example is the familiar LS (Moore-Penrose) pseudo-inverse solution discussed earlier. The CRB denoted by $\Lambda_{\text{unbiased}}$ for the covariance matrix of any unbiased estimator has been theoretically shown to satisfy:

$$\Lambda_{\text{unbiased}} \geq (A^T \Sigma_{\text{cov}}^{-1} A)^{-1}, \quad (2.65)$$

where $I_b(\mathbf{x}) = A^T \Sigma_{\text{cov}}^{-1} A$ is the Fisher information matrix for the linear model with added Gaussian noise.

However, recent work has extended this bound to include the class of biased estimators such as Generalized autocalibrating partially parallel acquisitions (GRAPPA). In the context of parallel imaging, bias is introduced with the purpose of achieving better conditioning of matrix inversion, with the result that an inexact reconstruction is obtained. To characterize the inexactness of the reconstruction, a bias gradient matrix B is defined as:

$$B = A_{\text{biased}}^{-1} A - I_n. \quad (2.66)$$

The biased estimator given by $\hat{\mathbf{x}}_{\text{biased}}(\mathbf{b}) = A_{\text{biased}}^{-1} \mathbf{b}$, is then characterized by the bias vector:

$$\begin{aligned} \mathbb{b}(\mathbf{b}) &= \mathbb{E}(\hat{\mathbf{x}}_{\text{biased}}(\mathbf{b})) - \mathbf{x} \\ &= \mathbb{E}(A_{\text{biased}}^{-1} A \mathbf{x}) + \mathbb{E}(A_{\text{biased}}^{-1} \mathbf{n}) - \mathbf{x} \\ &= A_{\text{biased}}^{-1} A \mathbf{x} - \mathbf{x} = B \mathbf{x} \end{aligned} \quad (2.67)$$

By definition, $A_{\text{biased}}^{-1} = (I_n + B)A^{-1}$, so the unbiased CRB in Equation (2.65) can be modified to compute the biased CRB as:

$$\Lambda_{\text{unbiased}} \geq (I_n + B) (A^T \Sigma_{\text{cov}}^{-1} A)^{-1} (I_n + B)^T \quad (2.68)$$

To understand the effect of bias in GRAPPA, for example, it is helpful to describe the regression and prediction process in terms of latent variables. Specifically, the observed variables A and $\mathbf{b}$ derived from the auto-calibration signal (ACS) data with measurement noise can be related to the unobserved latent variables $\bar{A}$ and $\bar{\mathbf{b}}$ representing the true, noise-free counterparts. In terms of the latent variables, the observed variables can be expressed as:

$$\begin{cases} A = \bar{A} + \delta A \\ \mathbf{b} = \bar{\mathbf{b}} + \delta \mathbf{b} \\ f : \bar{\mathbf{b}} = \bar{A} \mathbf{x} \end{cases} \quad (2.69)$$

where δA and $\delta \mathbf{b}$ represent measurement noise present in the ACS data and are assumed to be independent of the true value $\bar{A}$ and $\bar{\mathbf{b}}$, and $\mathbf{x}$ denotes the underlying true coefficients that satisfy the linear relationship in the absence of noise. With $\hat{\mathbf{x}}$ denoting the coefficients estimated from the LS fitting:

$$\hat{\mathbf{x}} = \left[(\bar{A} + \delta A)^T (\bar{A} + \delta A) \right]^{-1} (\bar{A} + \delta A)^T (\bar{\mathbf{b}} + \delta \mathbf{b}), \quad (2.70)$$

the bias in the coefficients is given by $\delta \mathbf{x} = \hat{\mathbf{x}} - \mathbf{x}$. Even if there are an infinite number of measurements, there is still a bias in the LS estimator. This is because the estimated coefficients converge to:

$$\hat{\mathbf{x}} = \frac{\mathbf{x}}{\left(1 + \frac{\sigma_b^2}{\sigma_A^2} \right)}, \quad (2.71)$$

where the noise in A and $\mathbf{b}$ is assumed to have zero mean and variance of σ_A^2 and σ_b^2 , respectively. The effects of bias on the estimated coefficients $\hat{\mathbf{x}}$ is dependent on the noise present in both A and $\mathbf{b}$. The bias in GRAPPA coefficients can be large at high acceleration factors due to ill-conditioned A . This is true since bias is not a linear function of noise in the ACS, as inferred from the errors-in-variable problem in regression.

2.7.1 Regularization in GRAPPA-Based pMRI

Poor conditioning of the calibration matrix will result in low-quality reconstruction due to the biased nature of the estimator. Ill-conditioning can result from reduced number of ACS, large coil numbers, or increased acceleration. These non-ideal conditions can result in over-fitting of training data, with a consequent increase in the mean squared error (MSE) during GRAPPA estimation. The number of filter coefficients as determined based on the number of coils, kernel size and acceleration factor R also is a contributing factor. A more appropriate interpretation is that, when the number of parameters (filter coefficients) increases in relation to the available training samples, the MSE increase is mostly contributed by the bias introduced during calibration. Typically, the training data size as determined from the number of ACS is much larger than the filter size. In this situation, maximum likelihood estimation (MLE) performs well due to its asymptotic optimality as determined from the inverse of the Fisher information matrix. However, under non-ideal imaging conditions, MLE performs poorly as the training data size reduces in relation to that of the filter. Therefore, to get an overall lower MSE, the variance of the estimator can be reduced by bringing the l_2 -norm of the filter coefficients close to zero. This calls for introduction of a penalty term in the calibration process, thereby enabling the use of a general class of regularization strategies to improve the overall MSE in a deterministic sense. Some examples of regularization approaches applied to GRAPPA are discussed below.

2.7.1.1 Tailored GRAPPA

In contrast to normal GRAPPA, tailored GRAPPA [72] assumes unequal contribution of the acquired neighbors for estimation of the unacquired k -space samples. The individual

contributions are determined based on coil array configuration and field-of-view (FOV) orientation. Consequently, the calibration matrix in normal GRAPPA is assumed to consist of redundant information, leading to the contribution of fewer signals and more noise into the reconstruction. By eliminating the contribution from a pre-determined number of low singular values as in Equation (2.5), the most linearly independent columns in the calibration matrix are made to contribute more to the fitting process.

The number of independent columns, however, determines the goodness of fit. Because the sensitivities overlap only in the phase-encoding (PE) direction in acquisitions employing planar arrays, the performance of this method becomes equivalent to that of normal GRAPPA. Also, as the signal redundancies increase mostly with increasing number of coils, this method is particularly effective computationally due to sufficient goodness of fit even with the reduced number of columns in the calibration matrix.

2.7.1.2 Discrepancy-Based Adaptive Regularization

Instead of directly truncating the singular values as in tailored GRAPPA, application of Tikhonov regularization mitigates the noise amplification effect in the GRAPPA coefficients by filtering the singular values using smooth roll-off filter factors. Application of Tikhonov regularization to Equation (1.39), modifies the GRAPPA fitting equation as follows:

$$\mathbf{k}_{u_r}^l = (\Pi + \lambda \mathbf{I}) \mathbf{z}_r^l, \quad (2.72)$$

where $\lambda > 0$ is the regularization parameter, and $\mathbf{I}$ is the identity matrix. With $\Pi = \mathbf{U}\mathbf{S}\mathbf{V}^H$, where $\mathbf{U}$ and $\mathbf{V}$ are unitary matrices and $\mathbf{S}$ a diagonal matrix with diagonal entries $\sigma_1 \geq \sigma_2 \geq \sigma_3 \dots \geq \sigma_N$, the Tikhonov solution is obtained as:

$$\mathbf{z}_r^l = \sum_{j=1}^r \frac{\mathbf{u}_j^T \mathbf{k}_{u_r}^l}{\sigma_j^2 + \lambda^2 / \sigma_j} \mathbf{v}_j, \quad (2.73)$$

where the smooth roll-off filter factors are given by $f_j = \sigma_j^2 + \lambda^2 / \sigma_j^2$. In the discrepancy-based adaptive regularization, the estimate of the errors in the observation vector $\mathbf{k}_{u_r}^l$ is used to find the optimal parameter that minimizes the fitting error, as in the discrepancy principle. This is typically achieved using a search procedure that estimates the optimal value corresponding to the knee-point of an L-curve that plots l_2 -norm of the solution versus fitting error for each value of the search parameter. The L-curve method is usually unacceptable for GRAPPA, mainly due to the difficulty in observing L-shapes. As a result, bluntly locating the corner using the maximal curvature criterion becomes difficult.

According to the discrepancy principle [73], the fitting error corresponding to each value of the regularization parameter is considered to be the sum of a model error δ_0 and noise-related error δ_e . While the model error is determined based on the error due to the LS fit, the noise-related error is dependent on the number of ACS (n_{ACS}) and the noise variance estimated from a noise scan without radio frequency (RF) transmission. The optimal solution is the one corresponding to the L-curve corner at which the fitting error becomes equal to the sum of the model error and the noise-related error. As the optimization is achieved using the noise-related error calculation, the fitting accuracy of this method is strongly dependent on the n_{ACS} .

2.7.1.3 Penalized Coefficient Regularization

The penalized coefficient (PC) regularization is a comparatively recent regularization technique introduced to GRAPPA calibration. In this approach, the fitting coefficients are weighted with different penalty factors, depending on the relative displacement from source data to target data in the k -space [74]. Instead of using an identity matrix as the regularization matrix, a diagonal matrix that consists of entries contributing to varying penalty weighting is used for the coefficient fitting. With the weighting factor p defined as the k -space displacement from acquired data toward the fitting ACS data, the regularization matrix takes the following form:

$$B = \lambda \text{diag}(PF(p)), \quad (2.74)$$

where $\text{diag}(PF(p))$ denotes the diagonal matrix in which the diagonal entries are defined by a penalty function $PF(p)$. A particular form of the penalty function is:

$$PF(p) = \|p\|^2. \quad (2.75)$$

With $\|p\| = 1$ for the nearest k -space points, the distant k -space coefficients are represented by $\|p\| > 1$.

The extent of noise suppression is determined by the structure of the regularization matrix used. For example, when an identity matrix is used, all the source data is penalized equally. A source point nearer to the target point and farthest from the target point is given the same penalty. But actually, the source point nearest to the target point contributes more toward the lower spatial harmonics of coil sensitivity. This is utilized in PC regularization by penalizing the calibration weights based on the displacement from the source data to the target data. Thus, the noise error is suppressed by modulating the higher order coefficients. The superior performance of PC compared to other GRAPPA methods becomes more conspicuous when the kernel size increases in the PE direction.

2.7.1.4 Regularization in GRAPPA Using Virtual Coils

This method is preferred when the background phase disturbs the conjugate symmetry of individual coil k -spaces. The background phase arises due to the phase profile of a receiver coil, or modifications in shim settings. This effect becomes more pronounced when the coil sensitivity profiles are non-homogenous, typically when the FOV is large. For a single homogenous coil with $R = 2$, the g -factor is analytically described as a function of the phase difference between superimposed pixels of cyclically shifted images. For a phase difference of $\pi/2$, the superimposed pixels are perfectly separable with a g -factor corresponding to 1. In the non-ideal situations where this condition is not satisfied, the virtual coil concept can be useful to achieve GRAPPA reconstruction without additional noise enhancement due to presence of the background phase. In the virtual coil approach, the complex conjugate of the original signal is also included in the GRAPPA kernel [75–77]. The complex conjugate included in the reconstruction is considered to have been generated from a set of virtual coils. This is effectively equivalent to an increase in the number of coils to $2n_c$. The main advantage is the effective reduction in the g -factor, when the Fourier symmetry is absent in each individual coil k -space. However, inclusion of virtual coils slows the reconstruction process for acquisition with a large number of coil arrays.

The complex conjugate included in the calibration will add additional phase information to the reconstruction process. The signal from the l th coil is obtained by weighting the spin density distribution $\rho(x)$ with its complex coil sensitivity $C^l(x)$. This leads to an additional phase in the coil image. This phase is dependent on the local field, pulse sequence and shape of the RF pulse. The k -space resulting from virtual coil can be represented as follows:

$$K^{l*}(k) = FT[\rho(x) \cdot C^l(x)] \quad (2.76)$$

where FT denote the Fourier transform.

With inclusion of the virtual coils, the k -space signal in the j th coil becomes:

$$K^j(k_x, k_y + \eta \Delta k_y) = \sum_{l=1}^{n_c} \sum_{b=-l_1}^{l_1} K^l(k_x, k_y + Rb \Delta k_y) z_\eta^l + \sum_{l=1}^{n_c} \sum_{b=-l_1}^{l_1} K^{l*}(k_x, k_y + Rb \Delta k_y) z_\eta^l \quad (2.77)$$

From Equation (2.77), the actual coil and virtual coil signals can then be related using:

$$K^{j+n_c}(k) = K^j(k) \quad (2.78)$$

In the regularization approach, different penalizing factors are applied to the real and virtual channels. Based on the proportion in which the real and virtual channels are penalized, the penalty function is chosen as:

$$PF(p) = \begin{cases} \lambda; & \text{if } p \text{ is from real channel} \\ \kappa; & \text{if } p \text{ is from virtual channel} \end{cases} \quad (2.79)$$

The virtual channels can be strongly penalized when $\kappa > \lambda$. When the $\kappa \gg \lambda$, then the virtual coils are fully suppressed, and the reconstructed image is almost the same as that from standard LS reconstruction.

2.7.1.5 Sparsity-Promoting Calibration

In the sparsity-promoting calibration for GRAPPA, the kernel is first convolved with the zero-filled k -space to obtain the full FOV reconstructed multi-channel k -space [78]. Then the norm is evaluated on the sparse transform of the image obtained from the fully reconstructed k -space. The sparsity-promoting transforms used are total variation and the four-level bi-orthogonal '9-7' DWT. The optimization problem for sparsity promoting calibration is formulated as:

$$\hat{Z} = \min_Z \frac{1}{2} \|K_s^{ACS} Z - K_t^{ACS}\|_F^2 + \lambda \|\Psi F^{-1} \text{GRAPPA}(K_u, Z)\|_{1,2}, \quad (2.80)$$

where K_s^{ACS} and K_t^{ACS} denote the source and target points in the ACS, Z is the calibration coefficients, Ψ is the sparsifying transform operator and $\text{GRAPPA}(K, Z)$ is the GRAPPA reconstructed k -space. Using the half-quadratic minimization scheme [79,80], an iterative form of the solution is obtained:

$$\hat{Z}^{(i+1)} = \min_Z \frac{1}{2} \|K_s^{ACS} Z - K_t^{ACS}\|_F^2 + \frac{\lambda}{2} \left\| \left(\Delta^{(i)} \right)^{1/2} \Psi F^{-1} \text{GRAPPA}(K_u, Z) \right\|_F^2, \quad (2.81)$$

where $\Delta^{(i)}$ is the diagonal weight matrix $\Delta_{n,n}^{(i)} = \| [W_{n,1}^{(i)} \dots W_{n,P}^{(i)}] \epsilon \|_2^{-1}$, $W^{(i)}$ is the sparse transform obtained using $W^{(i)} = \Psi F^{-1} \text{GRAPPA}(K_u, \hat{Z}^{(i)})$, and ϵ is a small positive value. Since a typical slice of an MRI volume consists of as many as 256×256 voxels, the filled k -space consists of approximately 2^{16} rows. For a 32-channel coil, the full k -space possesses 2^{21} entries. Furthermore, with higher acceleration factors, Z has almost as many entries as the k -space, resulting in a large-scale optimization problem with over 1 million variables. Therefore, iterative methods are used to solve this problem. Although several iterative methods exist, the iterative algorithm for sparse least-squares problems (LSMR) method [81] is used because it guarantees a monotonic decrease in both the normal residual, like the CG method, and the least-squares residual (for example, LSQR: an algorithm for sparse linear equations and sparse least squares). The LSMR method is a descent method for minimizing $\|Ax - b\|_2^2$ which requires performing matrix-vector multiplication involving A and A^H . Since the GRAPPA reconstruction step consists of convolution of the data with the different kernels, the adjoint GRAPPA reconstruction operation $\text{GRAPPA}^*(K_u, Y)$ involves convolution of GRAPPA reconstructed k -space $Y = \text{GRAPPA}(K_u, Z)$ with the conjugated and reversed acquired k -space data. Efficient implementation of the above operations can be performed using the fast Fourier transform (FFT) [78]. The algorithm is summarized in Algorithm 2.9.

Algorithm 2.9 Sparsity-Promoting GRAPPA Calibration

- 1) Initialize $K_u, K_s^{ACS}, K_t^{ACS}, \hat{Z}^{(0)}, \lambda, \epsilon, k, tol, maxIter$
 - 2) Set $f^{(0)}$ to (2.80) evaluated for $Z = \hat{Z}^{(0)}$
 - 3) for $k = 1 : maxIter$
 - 4) $W^{(i-1)} \leftarrow \Psi F^{-1} \text{GRAPPA}(K_u, \hat{Z}^{(i-1)})$
 - 5) $\Delta_{n,n}^{(i-1)} \leftarrow \left\| [W_{n,1}^{(i-1)} \dots W_{n,P}^{(i-1)}] \epsilon \right\|_2^{-1}$ for $n = 0, 1, \dots, N-1$
 - 6) Perform LSMR to solve (2.81) for $\hat{Z}^{(i)}$ using $\Delta^{(i-1)}$
 - 7) Compute $f^{(i)}$ using $\hat{Z}^{(i)}$ in (2.80)
 - 8) if $f^{(i-1)} - f^{(i)} \leq tol$, then

break

 endif
 - endfor
- $\hat{Y} \leftarrow \text{GRAPPA}(K_u, \hat{Z}^{(i)})$
-

2.7.1.6 KS-Based Calibration

In KS-based methods, regularization is achieved through projection onto the Krylov sub-space. This is applicable to large-scale regularization problems in which the regularization matrix B in Equation (2.74) is chosen using a form of orthogonal projection [82]. The solution is achieved in less than k iterations, where k stands for the product of kernel size and number of coils (number of columns of the calibration matrix Π) [83]. The calibration equation is converted to normal equations by pre-multiplying with Π^T . This yields a modified form of the calibration equation [84]:

$$G\mathbf{z}_r^l = \mathbf{g}_{ur}^l, \quad (2.82)$$

where $G = \Pi^T \Pi$ and $\mathbf{g}_{ur}^l = \Pi^T \mathbf{k}_{ur}^l$. The solution is then computed iteratively as $\mathbf{z}_{rk}^l = \mathbf{z}_{r0}^l + V_k \mathbf{y}_k$, where $\mathbf{z}_{r0}^l$ is an initial guess of the GRAPPA coefficients, V_k is the l_2 -orthogonal basis for B_k , and $\mathbf{y}_k$ is any vector of a k -dimensional sub-space as defined in Equation (2.16). When expressed as a function of $\mathbf{y}_k$, the calibration problem can be formulated as:

$$\min_{\mathbf{y}_k} \|\beta_k \mathbf{e}_k - B_k \mathbf{y}_k\|_2, \quad (2.83)$$

where β_k is the k' th scalar projection of $\mathbf{g}_{ur}^l$ onto V , and $\mathbf{e}_k$ is k' th the Euclidean basis vector. Based on the structure of B , the method is classified as CG (B is symmetric tridiagonal) or GMRES (B is upper Hessenberg).

2.8 Regularization in Iterative Self-Consistent Parallel Imaging Reconstruction (SPIRiT)

Compared to GRAPPA, which uses only consistency with calibration, SPIRiT relies on both calibration consistency as well as data consistency. Calibration consistency is achieved by the application of consistency between every point on the entire neighborhood under consideration as well as its entire neighborhood across all coils. With k denoting the entire k -space data for all coils, and Z a matrix containing the calibration kernels in the appropriate locations, the consistency equation can be expressed as:

$$\mathbf{k} = Z\mathbf{k}. \quad (2.84)$$

With the assumption that the reconstructed data is always Cartesian, the additional consistency with data acquisition can be represented as:

$$\mathbf{k}_{NC} = D\mathbf{k} \quad (2.85)$$

where D is an operator that transforms the reconstructed Cartesian data $\mathbf{k}$ to the k -space points $\mathbf{k}_{NC}$ along the accelerated trajectory. For Cartesian acquisitions, the operator D acts as a sampling matrix that chooses only the acquired k -space locations, whereas for non-Cartesian sampling, it represents an interpolation matrix. To combine the concepts of both parallel imaging and compressed sensing for enabling faster high-resolution MR imaging,

the sparsity constraints can also be included in the cost function. This results in the following modified cost function:

$$\min \|k_{NC} - Dk\|^2 + \gamma \|(Z - I)k\|^2 + \sum_i \alpha_i R_i(k), \quad (2.86)$$

where the parameter γ establishes a trade-off between the two consistency constraints, with larger values enforcing higher calibration consistency and smaller values imposing higher data consistency. $R(x)$ is a penalty function that incorporates the prior knowledge about the k -space or the image [85]. Denoting W as a data-weighting function, $\nabla(\cdot)$ as a finite-difference operator, and $\Psi(\cdot)$ as a wavelet operator, the following penalties can be applied to the SPIRiT formulation:

$$R(k) = \|k\|_2 \quad (2.87a)$$

$$R(k) = W\|k\|_2 \quad (2.87b)$$

$$R(k) = \|\Delta(\text{IFFT}(k))\|_1 \quad (2.87c)$$

$$R(k) = \|\Psi(\text{IFFT}(k))\|_1 \quad (2.87d)$$

Here Equation (2.87a–d) corresponds to the Tikhonov, weighted Tikhonov, TV and wavelet-based regularization penalties, respectively. The first two are based on l_2 -norm regularization, whereas the last two come under the l_1 -norm regularization approach. Accordingly, the latter variants are referred to as the L1-SPIRiT (l_1 -norm based Iterative Self-consistent Parallel Imaging Reconstruction from Arbitrary k-Space) [86,87].

2.9 Regularization for Compressed Sensing MRI (CSMRI)

The application of the CS principle in MR image reconstruction uses the sparsity in wavelet and finite differencing domain to improve the accuracy of reconstruction from randomly under-sampled k -space data. Following the formulation in [Section 1.6.3.2](#), the optimal solution is the minimizer of the cost function:

$$\min_{\hat{w}} \frac{1}{2} \|K_u - F_u \Psi' \hat{w}\|_2^2 + \lambda \|\hat{w}\|_1, \quad (2.88)$$

where w denotes the wavelet coefficients. Since the cost function in Equation (2.88) is similar to that in Equation (2.62), the nonlinear shrinkage operation ISTA can be used to solve the minimization problem. With application of ISTA, the solution takes the form of a Landweber update followed by soft thresholding. With the use of sparsity in the wavelet domain for example, the updated wavelet coefficients in the Landweber step is given by:

$$\hat{w}^{(k)} = w^{(k-1)} + \Psi \left(F_u' \left(K_u - F_u \left(\Psi' w^{(k-1)} \right) \right) \right), \quad (2.89)$$

where $\hat{w}$ and w denote the wavelet coefficients before and after soft thresholding. Following the Landweber update, application of the soft-thresholding function gives:

$$w^{(k)} = \text{soft}\left(\hat{w}^{(k)}, \lambda\right). \quad (2.90)$$

Iterative application of the steps in Equations (2.89) and (2.90) yields an optimal solution. Although ISTA is the simplest strategy for solving problems involving non-differentiable convex penalties, it suffers from the problem of a slow convergence rate. Acceleration is achieved by methods that either use information from past iterations, such as the TwIST [70] and FISTA [71], or methods that optimize the wavelet parameters, such as the multi-level TL (MLTL) [88,89] and SISTA [69]. Fast-weighted ISTA (FWISTA) uses a combination of the two types [65]. The FWISTA algorithm is a sub-band adaptive version of FISTA. In FWISTA, the FISTA algorithm is generalized using a parametric weighted norm, where the difference with respect to FISTA resides in using the SISTA step [65]. Also, FWISTA can be simply adapted in order to impose a monotonic decrease of the objective function value. Since there are no additional matrix-vector multiplications per iteration in FWISTA when compared to ISTA, SISTA and FISTA, the computational cost of an iteration is expected to be equivalent. Due to the aforementioned advantages, FWISTA remains the preferred choice for CSMRI.

The previous discussions deal with the implementation of CSMRI using orthogonal wavelets. Although the aforementioned CS algorithms using orthogonal wavelet basis [49,90,91] are useful in both the algorithm design and theoretical analysis due to the minimal complexity offered by such realizations, redundant representation offered by wavelet frames have been found to be more effective in terms of edge recovery and noise removal [92,93]. Approaches using redundant representations include undecimated or shift-invariant wavelet frames [94–96], patch-based methods [89,92,97,98], and over-complete dictionaries [99]. The redundancy offered by such representations leads to robust image representations with introduction of additional benefits. In the case of wavelets, redundancy enables shift-invariant property, which enables the suppression of artifacts induced by Gibbs phenomena near the image discontinuities [94,100]. In patch-based methods, redundancy comes from the overlapping of image patches. This results in better noise removal as well as artifact suppression [92,97,98,101]. The use of over-complete dictionaries is also redundant, resulting in better representation of image features, while making the representations relatively sparser than the orthogonal dictionaries. Most of the redundant representation systems fall under the broad category of tight frame systems.

Although there are various mathematical formulations for sparsity using redundant representations, the redundancy often brings in more variables than the dimension of the MR image. This makes the implementation of numerical algorithms more complex, particularly in cases where the redundancy is high and the dataset is large. Examples include three-dimensional (3D) imaging, dynamic MRI and diffusion tensor imaging. In the case of the synthesis-based sparsity formulation with redundant transform domain representations, the challenge mainly arises from the computational complexity. Contrary to this, analysis-based formulation requires storage of both the redundant coefficients and MR images in each iteration.

To overcome these disadvantages, a projected iterative soft-thresholding algorithm (pISTA) was introduced in the tight frame domain that approximates a solution of the analysis-based sparsity formulation using ISTA. The implementation of pISTA also avoids

the storage of a full set of redundant transform coefficients thereby enabling its application to large-scale MR image reconstruction. The pISTA algorithm is further accelerated by incorporation of FISTA steps resulting in projected fast ISTA (pFISTA) [102]. For the sake of clarity, we first introduce some basic concepts of tight frames prior to describing the pFISTA algorithm for CSMRI.

A set of vectors $\{\mathbf{d}_j\}_{j=1,2,\dots,J}$ is called a frame of $\mathbb{C}^N$ if there exist positive real numbers A, B such that:

$$A\|\mathbf{x}\|_2^2 \leq \sum_j |\langle \mathbf{x}, \mathbf{d}_j \rangle|^2 \leq B\|\mathbf{x}\|_2^2 \quad (2.91)$$

where $\langle \mathbf{x}, \mathbf{d}_j \rangle$ is the inner product between $\mathbf{x}$ and $\mathbf{d}_j$. Since a frame $\{\mathbf{f}_k\}_{k \in \mathbb{N}}$ for the Hilbert space $\mathbb{C}^N$ can span the whole space, every signal $f \in \mathbb{C}^N$ can be represented as a linear combination of $\{\mathbf{f}_k\}_{k \in \mathbb{N}}$. If $A=B$, then such a frame is called a tight frame. In this description, Ψ and $\Psi^* = [\mathbf{d}_1, \mathbf{d}_2, \dots, \mathbf{d}_J]$ denotes the analysis and synthesis operators associated with a frame. With reference to a given frame Ψ^* , a frame Φ is called the dual frame if:

$$\Phi\Psi = \mathbf{I} \quad (2.92)$$

In cases where Φ is not unique, a canonical dual frame is defined as:

$$\Phi = (\Psi\Psi^*)^{-1} \Psi^*. \quad (2.93)$$

Using the dual frame concept, the analysis and synthesis models for CSMRI can be represented as:

$$\min_U \frac{1}{2} \|K_u - F_u U\|_2^2 + \lambda \|\Psi U\|_1 \quad (2.94a)$$

$$\min_\alpha \frac{1}{2} \|K_u - F_u \Phi \alpha\|_2^2 + \lambda \|\alpha\|_1, \quad (2.94b)$$

where α contains the coefficients of an image under the representation of a tight frame Φ . In the pFISTA algorithm, the analysis model in Equation (2.94a) is first expressed in terms of an equivalent synthesis-like model as:

$$\min_{\alpha \in \text{Range}(\Psi)} \frac{1}{2} \|K_u - F_u \Phi \alpha\|_2^2 + \lambda \|\alpha\|_1. \quad (2.95)$$

The constraint in the synthesis-like analysis model in Equation (2.95) is handled using an indicator function defined as follows:

$$d(\alpha) = \begin{cases} 0; & \alpha \in \text{Range}(\Psi) \\ \infty; & \alpha \notin \text{Range}(\Psi) \end{cases} \quad (2.96)$$

Using the indicator function, the unconstrained model of Equation (2.95) can be expressed as:

$$\min_{\alpha} \frac{1}{2} \|K_u - F_u \Phi \alpha\|_2^2 + \lambda \|\alpha\|_1 + d(\alpha). \quad (2.97)$$

By incorporating the proximal mapping operator, we have:

$$\begin{aligned} \alpha_{k+1} &= \text{prox}_{\gamma g}(\alpha_k - \gamma \nabla f(\alpha_k)), \\ &= \min_{\alpha} \lambda \gamma \|\alpha\|_1 + \frac{1}{2} \|\alpha - \alpha_k - \gamma \nabla f(\alpha_k)\|_2^2 + d(\alpha), \\ &= \min_{\alpha \in \text{Range}(\Psi)} \lambda \gamma \|\alpha\|_1 + \frac{1}{2} \|\alpha - \alpha_k - \gamma \nabla f(\alpha_k)\|_2^2, \end{aligned} \quad (2.98)$$

where $f(\alpha) = \frac{1}{2} \|K_u - F_u \Phi \alpha\|_2^2$, $g(\alpha) = \lambda \|\alpha\|_1 + d(\alpha)$, and γ is the step size. However, it is difficult to compute an analytical solution to the approximate functional in Equation (2.98) due to the presence of the constraint $\alpha \in \text{Range}(\Psi)$. Without this constraint, we have:

$$\tilde{\alpha}_{k+1} = \text{soft}(\alpha_k - \gamma \nabla f(\alpha_k), \lambda \gamma). \quad (2.99)$$

With inclusion of the constraint, the solution is obtained using the orthogonal projection operator $P_{\text{Range}(\Psi)}$ onto $\alpha \in \text{Range}(\Psi)$ as follows:

$$\alpha_{k+1} = P_{\text{Range}(\Psi)}(\tilde{\alpha}_{k+1}). \quad (2.100)$$

Substituting the coefficients $\tilde{\alpha}_k$ and $\tilde{\alpha}_{k+1}$ with images $U_k = \Phi \tilde{\alpha}_k$ and $U_{k+1} = \Phi \tilde{\alpha}_{k+1}$, we have:

$$U_{k+1} = \Psi^* \text{soft}(\Psi(U_k + \gamma F_u'(K_u - F_u U_k)), \gamma \lambda). \quad (2.101)$$

The steps used in the algorithm are summarized in Algorithm 2.10.

Algorithm 2.10 pFISTA

1) Initialize $U_0, \hat{U}_0$ $k = 1, t_1 = 1$.

2) Compute $U_{k+1} = \Psi^* \text{soft}(\Psi(U_k + \gamma F_u'(K_u - F_u U_k)), \gamma \lambda)$

3) Perform controlled over relaxation $t_{k+1} = \frac{1 + \sqrt{1 + 4(t^k)^2}}{2}$

4) Compute $\hat{U}_{k+1} = U^{(k+1)} + \frac{t^k - 1}{t^{k+1}}(U^{(k+1)} - U^{(k)})$

5) Stop if converged. Otherwise go to step 2

The implementation of pFISTA is simple as it introduces only one tunable parameter, that is, the step size γ . However, it has been shown empirically that this parameter will not affect the image quality. For attaining lower reconstruction errors and faster convergence in tight frames-based MR image reconstruction, γ is typically set to 1. Generally, the algorithm that converges to an approximate solution instead of the exact solution provided by the analysis model is called the balanced sparse model [103–105]. The use of balanced sparse models in CSMRI yields reconstructed images with image quality comparable to those of analysis models [95].

Appendix

This appendix provides the mathematical proof for replacement of $(A^T \Sigma^{-1} A)^{-1}$ with the pseudo-inverse of $\Sigma^{-\frac{1}{2}} A$. For obtaining an analytical solution to Equation (2.9), $(A^T \Sigma^{-1} A)$ should be invertible. This is true when $\Sigma^{-\frac{1}{2}} A$ has full column rank. Then $\Sigma^{-\frac{1}{2}} A$ is also invertible and has full rank. An explicit solution for Equation (2.9) can then be obtained by multiplying Equation (2.9) with $(A^T \Sigma^{-1} A)^{-1}$ from the left:

$$\begin{aligned} \hat{\mathbf{x}} &= (A^T \Sigma^{-1} A)^{-1} A^T \Sigma^{-1} \mathbf{b} \\ &= \left(A^T \Sigma^{-\frac{1}{2}} \Sigma^{-\frac{1}{2}} A \right)^{-1} A^T \Sigma^{-\frac{1}{2}} \Sigma^{-\frac{1}{2}} \mathbf{b} \\ &= \left(\Sigma^{-\frac{1}{2}} A \right)^{-1} \left(A^T \Sigma^{-\frac{1}{2}} \right)^{-1} A^T \Sigma^{-\frac{1}{2}} \Sigma^{-\frac{1}{2}} \mathbf{b} \end{aligned} \quad (2.102)$$

Since $(A^T \Sigma^{-\frac{1}{2}})^+ A^T \Sigma^{-\frac{1}{2}} = 1$, the solution is obtained as:

$$\hat{\mathbf{x}} = \left(\Sigma^{-\frac{1}{2}} A \right)^+ \Sigma^{-\frac{1}{2}} \mathbf{b} \quad (2.103)$$

The pseudo-inverse of $\Sigma^{-\frac{1}{2}} A$ is then obtained as:

$$\left(\Sigma^{-\frac{1}{2}} A \right)^+ = (A^T \Sigma^{-1} A)^{-1} A^T \Sigma^{-\frac{1}{2}} \quad (2.104)$$

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Regularization Parameter Selection Methods in Parallel MR Image Reconstruction

3.1 Regularization Parameter Selection

The accuracy of the regularized output depends on the selection of the appropriate regularization parameter value. The parameter selection methods can be broadly classified into three basic types based on the criterion used for choosing the value of the regularization parameter [1,2]: a priori methods, a posteriori methods, and data-driven methods. Since a priori methods need information about the solution which is not known generally, they are generally not suitable for practical applications. A major difference between the other two methods is that, while a posteriori methods use an estimate of the noise present in the data, the data-driven methods require the noisy observed data as an input. In general, all parameter selection methods rely on computing an associated function $P(n)$ that defines the parameter either as the point at which P falls below some threshold, or the minimum point of the function [3]. Most methods of the first type have their origins and analysis in a deterministic setting and employ a sensitive tuning mechanism. In contrast, the second category of methods originate either from a stochastic framework, or they are based simply on heuristic ideas.

For the linear model $Ax = b$, with data $b = \bar{b} + n$, the unknown solution x and the solution computed for the parameter λ denoted by x_λ^n , the total error incurred in computing a solution using regularization is bounded by:

$$\|x - x_\lambda^n\|_2 \leq \|x - x_\lambda^0\|_2 + \|x_\lambda^0 - x_\lambda^n\|_2. \quad (3.1)$$

The first term on the right-hand side (RHS) of Equation (3.1) is the error that is generated due to a regularization operator acting on the noise-free observed data $\bar{b}$, and it is referred to as the regularization error. Since the regularization operator can be viewed as a perturbation of the generalized inverse of A , the perturbation decreases as λ increases. For regularization methods such as spectral cut-off and Tikhonov regularization, the regularization error is bounded by

$$\|x - x_\lambda^0\|_2 \leq \varphi(\lambda), \quad (3.2)$$

where $\varphi(\lambda)$ is a decreasing function which depends on the smoothness assumptions on the solution and the regularization method. Likewise, the second term, referred to as the propagated noise error, is also bounded by:

$$\|x_\lambda^0 - x_\lambda''\|_2 \leq n\varrho(\lambda), \quad (3.3)$$

where ϱ is a known increasing function of λ . Figure 3.1 shows the plot of the total error $\|x - x_\lambda''\|_2$ computed as a function of λ . Since the bound in Equation (3.1) is not directly computable, the parameter selection methods in turn aim at obtaining an estimate that minimizes the total error. Thus, the complexity of a parameter selection method depends on the dimension of the underlying problem, the condition number of A , the nature of ill-posedness and the noise model. Any technique for choosing the regularization parameter in the absence of information about the noise level and its statistics can pose difficulties in arriving at an optimal solution [4].

In the context of parallel magnetic resonance imaging (MRI), the criterion for optimum parameter selection is determined based on a compromise between the signal-to-noise ratio (SNR) and artifacts. While under-regularization leads to residual noise or streaking artifacts in the image, over-regularization removes the image features and enhances aliasing.

Broadly, the parameter selection strategies can be classified into those based on the discrepancy principle [5], generalized cross validation (GCV) [6], Lagrange multiplier method [7] and prediction error-based estimators [8,9]. In the discrepancy principle-based methods, the parameter selection is achieved by matching the norm of the residual to an upper bound that depends on the noise level. This dependence is in the form of a linear relationship in which the proportionality factor is fixed heuristically. The GCV method was developed to overcome this problem [10]. Due to the bilevel nature of the problem for non-quadratic functions, the GCV-based parameter selection is presented as the primal dual sub-problems in which the primal problem is posed as a quadratic regularization problem. However, the inversion calculation of huge matrices involved in each step of the parameter estimation algorithm makes the implementation computationally complex. The minimum number of measurements to ensure optimal performance needs to be empirically determined by the user. If the available number of input measurements is less than the required number, then noise may prevail in the regularized solution. Due to reports of failure in some model parameter selection problems, this algorithm does not have a wide range of

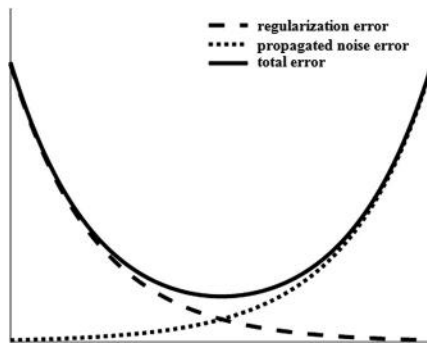


FIGURE 3.1

Total error $\|x - x_\lambda''\|_2$, regularization error $\|x - x_\lambda^0\|_2$ and propagated noise error $\|x_\lambda^0 - x_\lambda''\|_2$ as a function of the regularization parameter λ .

applications [11]. The other disadvantages of this approach are that, as part of a search procedure, the GCV function is to be evaluated several times in each step and requires averaging the regularization parameters obtained in each primal iteration.

Another duality-based regularization approach was proposed [7] in which the solution to the original minimization problem is obtained by solving the Lagrangian dual problem. In this approach, the parameter is updated in each iteration using the bisection method for a given initial value of the regularization parameter. The value of the initial parameter in turn is computed using Tikhonov approximate solution in which an unknown step size parameter is involved [7]. A further disadvantage of this approach is that the estimated value of the regularization function is to be determined heuristically.

In addition, several splitting schemes have been proposed to solve the non-quadratic regularization. An alternating direction method of multipliers (ADMM) was proposed by Ng et al. in [12] in which the current estimate is projected onto a feasible set by solving a constrained least-squares (LS) problem. Newton's iterative scheme is employed to compute the parameter in each iteration and requires another set of inner iterations to update the parameter. However, the efficiency of this adaptation strategy relies completely on prior knowledge of the noise variance similar to any other discrepancy principle-based parameter selection strategy [5].

Finally, in the prediction error-based methods, a proxy measure is computed as a function of the regularization parameter to predict the difference between the computed solution and the ground truth solution called the prediction error (risk). Then the optimal parameter is estimated as the minimizer of this proxy measure. The design and operation of these algorithms also incorporate some form of heuristics. Since the selection methods in general depend on the type of penalizing function used, the parameter selection methods for each type of regularization is discussed separately in this chapter.

3.2 Parameter Selection Strategies for Tikhonov Regularization

In autocalibrating pMRI, the Tikhonov regularization is used predominantly for estimation of filter weights in Generalized autocalibrating partially parallel acquisitions (GRAPPA) and iterative self-consistent parallel imaging reconstruction (SPIRiT) calibration, as described in Sections 2.5 and 2.7. In this form of regularization applied to the calibration problem, the cost functional takes the form $\min_x \{\|b - Ax\|_2^2 + \lambda \|Bx\|_2^2\}$. The resultant solution depends not only on the training data but also on the regularization parameter λ and regularization matrix B . Although the matrix B is commonly chosen to be an identity matrix, it is meaningful to select B to be a scaled finite difference approximation or a scaled orthogonal projection if the desired solution has some specific known properties [2,13–15]. If the regularization matrix used in the Tikhonov regularization is without an exploitable structure, then the solution is computed using generalized singular value decomposition (GSVD).

Typically, the regularization matrix B can be rectangular and is usually chosen as an approximation of the first- or second-order derivative operators to impose a certain degree of smoothness on the solution. For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{p \times n}$ with $m \geq n \geq p$, the decomposition using GSVD gives:

$$A = USX^{-1} \text{ and } B = VMX^{-1}, \quad (3.4)$$

where $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{p \times p}$ have orthonormal columns such that $U^T U = I_m$ and $V^T V = I_p$, $X \in \mathbb{R}^{n \times n}$ is nonsingular and S and M are of the form:

$$S = \begin{bmatrix} S_p & 0 \\ 0 & I_{n-p} \end{bmatrix}, \text{ and } M = \begin{bmatrix} M_p & 0 \end{bmatrix}. \quad (3.5)$$

The matrices $S_p = \text{diag}(\sigma_i)$ and $M_p = \text{diag}(\mu_i)$ are both $p \times p$ diagonal matrices whose diagonal elements satisfy $\sigma_i^2 + \mu_i^2 = 1$ and ordered such that $0 \leq \sigma_1 \leq \sigma_2 \leq \dots \leq \sigma_p$, $1 \geq \mu_1 \geq \mu_2 \geq \dots \geq 0$. The generalized singular values γ_i of the matrix pair (A, B) are defined as:

$$\gamma_i = \frac{\sigma_i}{\mu_i}, \quad \text{for } i = 1, 2, \dots, p. \quad (3.6)$$

In particular, if $B = I_n$, the generalized singular values are the ordinary singular values of A . Using the generalized singular values, the Tikhonov regularized solution can be expressed as:

$$\mathbf{x}_\lambda = \sum_{i=1}^p \frac{\gamma_i^2}{\gamma_i^2 + \lambda} \frac{\mathbf{u}_i^T \mathbf{b}}{\sigma_i} \mathbf{x}_i + \sum_{i=p+1}^n (\mathbf{u}_i^T \mathbf{b}) \mathbf{x}_i \quad (3.7)$$

In the above GSVD representation, the filter factor $\gamma_i^2 / (\gamma_i^2 + \lambda)$ dampens, or filters out the contributions to $\mathbf{x}_\lambda$ by the generalized singular values γ_i . Since $\sigma_i = \gamma_i (1 - \sigma_i^2) \approx \gamma_i$ for all $\sigma_i \ll 1$, and since the largest perturbations of the LS solution are associated with the smallest σ_i , the regularized solution $\mathbf{x}_\lambda$ will be less sensitive to perturbations than the LS solution. In order for the Tikhonov regularization to have a solution that is as accurate as possible, it is necessary to determine the value of the optimal regularization parameter. There are many methods to determine such a parameter value, including the discrepancy principle [16], unbiased predictive risk estimator (UPRE) method [17], Stein's unbiased risk estimation [18], quasi-optimality criterion [19], the GCV [20] and the L-curve method [21]. For the general-form Tikhonov, the GSVD of (A, B) can be used to express the solution to the Tikhonov cost function as a spectral filtered solution [22]. In what follows, this fact is utilized to develop efficient implementation schemes for each selection method to compute an optimal regularization parameter.

3.2.1 Discrepancy Principle

If the ill-posed problem is consistent, and only the observation vector is perturbed, selection of the regularization parameter is achieved using the discrepancy principle. According to the discrepancy principle, the residual norm is always equal to the upper bound d_e prescribed by the noise level present in data. Based on this bound, the regularized solution is obtained such that [23]:

$$\|A\mathbf{x}_\lambda - \mathbf{b}\|_2 = d_e, \quad \text{s.t. } \|\mathbf{e}\| \leq d_e, \quad (3.8)$$

where e is the error in the measurement vector $\mathbf{b}$. In cases where the noise levels are not well defined, it is not straightforward to determine the upper bound. With the assumptions

that (1) the coefficients $|u_i^T \bar{\mathbf{b}}|$ decay to zero faster than γ_i ; (2) for $\mathbf{b} \in \mathbb{R}^m$, the perturbation vector $\mathbf{e}$ has zero mean and covariance matrix $\sigma_0^2 I_m$; and (3) the norm of $\mathbf{e}$ satisfies $\|\mathbf{e}\| < \|\bar{\mathbf{b}}\|$, Equation (3.8) has an optimal solution point just right of the corner point of the L-curve at $(\sigma_0 \sqrt{m-n+p}, \|L\mathbf{x}_\lambda\|)$ [21]. Since the discrepancy principle requires a priori knowledge about the noise level in the observed data, an estimate of the noise level σ_0 is obtained by monitoring the following function [10]:

$$\mathcal{G}(\lambda) = \frac{\|A\mathbf{x}_\lambda - \mathbf{b}\|_2^2}{T(\lambda)}, \quad (3.9)$$

where $T(\lambda)$ can be considered as the degrees of freedom (DOF), defined as:

$$T(\lambda) = \text{trace} \left(I_m - A(A^T A + \lambda^2 B^T B)^{-1} A^T \right) = m - n + \sum_{i=1}^p \frac{\lambda^2}{\gamma_i^2 + \lambda^2}. \quad (3.10)$$

An interesting fact about the function in Equation (3.9) is that, when $\mathcal{G}$ is plotted as function of λ^{-1} , the value of $\mathcal{G}$ first decreases, and then levels off at a plateau, which gives an estimate of σ_0^2 . Hence the $\|\mathbf{e}\|^2$ can be estimated as m times of the value at the plateau of $\mathcal{G}$. In practice, the variance of the residuals of the LS solution can also be used as a quick estimate of the noise level. As the error in the data goes to zero, the parameter λ selected using the discrepancy principle also goes to zero, and hence the corresponding regularized solution $\mathbf{x}_\lambda$ converges to the exact solution $\mathbf{x}^*$ [2,23]. However, note that the discrepancy principle is very sensitive to an under-estimation of the noise level. Therefore, the application of the discrepancy principle is limited to cases in which the noise level can be estimated with high fidelity [15].

3.2.2 Generalized Discrepancy Principle (GDP)

GDP takes into account the errors in the matrix A as well as the incompatibility measure d_0 [23] that represents the norm of that component of $\mathbf{b}$ which is outside the range of A . Also, the system becomes consistent with $d_0 = 0$ since:

$$d_0 \equiv \|(I_m - UU^T)\mathbf{b}\| \quad (3.11)$$

Using GDP, the regularization parameter λ is chosen so that:

$$\|A\mathbf{x}_\lambda - \mathbf{b}\|_2 = d_0 + d_e + d_E \|\mathbf{x}_{LS}\|_2, \quad (3.12)$$

where $\mathbf{x}_{LS}$ is the LS solution, E is the errors in the matrix A , and d_E is the upper bound of $\|E\|_2$. Estimates of d_E can be computed using the statistical information about the errors in E as discussed in Edelman [24] and Hansen [25]. Equation (3.12) can also be expressed in terms of the upper bound $\Delta_{E,B}$ for $\max_{B\mathbf{x} \neq 0} \{\|E\mathbf{x}\| / \|B\mathbf{x}\|\}$ as:

$$\|A\mathbf{x}_\lambda - \mathbf{b}\|_2 = d_0 + d_e + \Delta_{E,B} \|B\mathbf{x}_\lambda\|_2. \quad (3.13)$$

In general, $\Delta_{E,B}$ corresponds to the largest singular value of the pair (E, B) and in particular, if $B = I_n$, then $\Delta_{E,B} = d_E$. In terms of the L-curve, the sharpness of the corner becomes more pronounced if $d_E \neq 0$. This also confirms the claim that the discrepancy principle over-smooths the real solution [10,26].

3.2.3 Unbiased Predictive Risk Estimator (UPRE)

The UPRE approach, also known as the C_L method, was first applied for the selection of the optimal parameter λ in Tikhonov regularization problems [17]. Since the optimal parameter should minimize the error between the Tikhonov regularized solution $\mathbf{x}_\lambda$ and the exact solution $\mathbf{x}^*$, UPRE is based on minimization of the predictive error:

$$\frac{1}{m} \|P_\lambda\|^2 = \frac{1}{m} \|A\mathbf{x}_\lambda - A\mathbf{x}^*\|^2, \quad (3.14)$$

where m is the number of elements in $\mathbf{b}$ and λ_{opt} is computed so that:

$$\lambda_{opt} = \min_{\lambda} \left\{ \frac{1}{m} \|P_\lambda\|^2 \right\}. \quad (3.15)$$

However, as the predictive error is difficult to estimate in practical situations without the knowledge of the true solution $\mathbf{x}$, an unbiased estimator is necessary to estimate the value of λ_{opt} in Equation (3.15). With unbiased estimation, the parameter can be computed by using the information of the measurable residual $\mathbf{r}_\lambda = A\mathbf{x}_\lambda - \mathbf{b}$ and the statistical estimator of the mean squared norm of the error σ_0^2 as [17]:

$$UPRE(\lambda) = \frac{1}{m} \|\mathbf{r}_\lambda\|^2 + \frac{2\sigma_0^2}{m} \text{trace}(A_\lambda) - \sigma_0^2, \quad (3.16)$$

where $A_\lambda = A(A^T A + \lambda I)^{-1} A^T$ and $\mathbb{E}(UPRE(\lambda)) = \mathbb{E}\left(\frac{1}{m} \|P_\lambda\|^2\right)$. Under this assumption, Equation (3.15) becomes:

$$\lambda_{UPRE} = \min_{\lambda} \{UPRE(\lambda)\}. \quad (3.17)$$

Equation (3.16) can be directly computed if the singular value decomposition (SVD) decomposition of A is available. However, in many large-scale problems, it is expensive to compute the SVD of A . In such cases, Lanczos procedure can be used to approximate the eigenvalues of the large-scale system matrix by a small matrix as proposed by Kilmer and D. P. O'Leary [27].

3.2.4 Stein's Unbiased Risk Estimation (SURE)

Although it has been shown that, for certain cases, UPRE is an unbiased estimate of mean squared error (MSE) [28], it is restricted to the case of a linear estimator [29,30] and is largely empirical [31]. An alternative predictive risk-based parameter selection criterion is SURE, which provides an unbiased estimate of the MSE, leading to an automatic selection of the parameter [18]. As the unbiased nature of SURE is mathematically

established, it is nonempirical compared to UPRE. The closeness of SURE to the true MSE is further supported by the law of large numbers for large data size, as in the case of images.

Although the method was initially developed for model selection in linear regression, it was later adapted for finding the solution of inverse problems. SURE aims to minimize the predictive risk, similar to Equation (3.14). With $\bar{\mathbf{b}} = \mathbf{A}\mathbf{x}$, the observed vector $\mathbf{b}$ can be expressed as:

$$\mathbf{b} = \bar{\mathbf{b}} + \mathbf{n}. \quad (3.18)$$

If $\hat{\mathbf{b}}$ is used to denote an estimate of $\bar{\mathbf{b}}$ such that $\hat{\mathbf{b}} = \mathbf{b} + \mathbf{e}_{\bar{\mathbf{b}}}(\mathbf{b})$, where $\mathbf{e}_{\bar{\mathbf{b}}}(\mathbf{b})$ is a weakly differentiable function, the mismatch between $\hat{\mathbf{b}}$ and $\bar{\mathbf{b}}$ can be computed using the mean squared risk of $\hat{\mathbf{b}}$, as follows:

$$R_{\lambda} = \|\hat{\mathbf{b}} - \bar{\mathbf{b}}\|_2^2. \quad (3.19)$$

In SURE, an unbiased estimate of the risk for estimation using $\hat{\mathbf{b}}$ is given by:

$$\hat{R}_{\lambda} = -m\sigma_0^2 + \|\mathbf{e}_{\bar{\mathbf{b}}}(\mathbf{b})\|_2^2 + 2\sigma_0^2 \nabla_{\mathbf{e}_{\bar{\mathbf{b}}}}(\mathbf{b}), \quad (3.20)$$

where $\mathbf{e}_{\bar{\mathbf{b}}}(\mathbf{b})$ is a measure for the fitness of the estimate $\hat{\mathbf{b}}$ to the observation $\mathbf{b}$, and $\nabla_{\mathbf{e}_{\bar{\mathbf{b}}}}(\mathbf{b}) = \sum_i \partial \mathbf{e}_{\bar{\mathbf{b}}}(\mathbf{b}) / \partial b_i$. As σ_0^2 is known or accurately estimated, the SURE parameter is estimated as:

$$\lambda_{\text{SURE}} = \min_{\lambda} \{\hat{R}_{\lambda}\}. \quad (3.21)$$

Since $\mathbf{e}_{\bar{\mathbf{b}}}(\mathbf{b})$ is linearly dependent on the data, the computation of the gradient in Equation (3.20) is straightforward for the case of standard Tikhonov regularization.

3.2.5 Bayesian Approach

The Bayesian approach provides a statistical interpretation for regularization that avoids the need for a separate regularization parameter selection procedure. In this approach, the unknown signal $\mathbf{x}$ and noise component $\mathbf{n}$ are both considered to be random fields. Therefore, the cost function in Equation (2.6) can be considered as consisting of a data-dependent term called the log-likelihood function and a prior model dependent only on the signal component $\mathbf{x}$. Using the Bayes rule and monotonicity properties, the solution can be obtained by seeking the maximum a posteriori (MAP) estimate of $\mathbf{x}$ that maximizes the posterior density $p(\mathbf{x}|\mathbf{b})$:

$$\hat{\mathbf{x}}_{\text{MAP}} = \max_{\mathbf{x}} p(\mathbf{x}|\mathbf{b}) = \max_{\mathbf{x}} \ln p(\mathbf{b}|\mathbf{x}) + \ln p(\mathbf{x}) \quad (3.22)$$

With the assumption that $\mathbf{n}$ and $\mathbf{x}$ are Gaussian random variables, $\mathbf{x} \sim N(0, \mathbf{C}_x)$ and $\mathbf{n} \sim N(0, \mathbf{C}_n)$, $\ln p(\mathbf{b}|\mathbf{x}) \propto -\frac{1}{2} \|\mathbf{b} - \mathbf{A}\mathbf{x}\|_{\mathbf{C}_n^{-1}}^2$ and $\ln p(\mathbf{x}) \propto -\frac{1}{2} \|\mathbf{x}\|_{\mathbf{C}_x^{-1}}^2$. This leads to the following alternative formulation:

$$\hat{\mathbf{x}}_{\text{MAP}} = \min_x \|\mathbf{b} - A\mathbf{x}\|_{C_n^{-1}}^2 + \|\mathbf{x}\|_{C_x^{-1}}^2. \quad (3.23)$$

The solution to Equation (3.23) can be obtained by solving the normal equation:

$$(A^T C_n^{-1} A + C_x^{-1}) \hat{\mathbf{x}}_{\text{MAP}} = A^T C_n^{-1} \mathbf{b}. \quad (3.24)$$

With the noise and prior models represented in terms of uncorrelated random variables so that $C_n = c_n I$ and $C_x = c_x I$, Equation (3.23) becomes:

$$\hat{\mathbf{x}}_{\text{MAP}} = \min_x \|\mathbf{b} - A\mathbf{x}\|_2^2 + \frac{c_n}{c_x} \|\mathbf{x}\|_2^2. \quad (3.25)$$

This representation corresponds to the Tikhonov regularization, with $B = I$ and $\lambda = c_n / c_x$. In this formulation, the problem of regularization parameter estimation is exchanged for a problem of statistical modeling through the use of probability densities $p(\mathbf{b}|\mathbf{x})$ and $p(\mathbf{x})$. The trade-off between the data-dependent term and the prior model is achieved by modeling the relative uncertainties in the processes $\mathbf{b}$ and $\mathbf{x}$ [32]. In many cases, the densities $p(\mathbf{b}|\mathbf{x})$ and $p(\mathbf{x})$ are obtained using the physical observation or direct experimental investigation. However, the specification of these densities is still a difficult task for some problems.

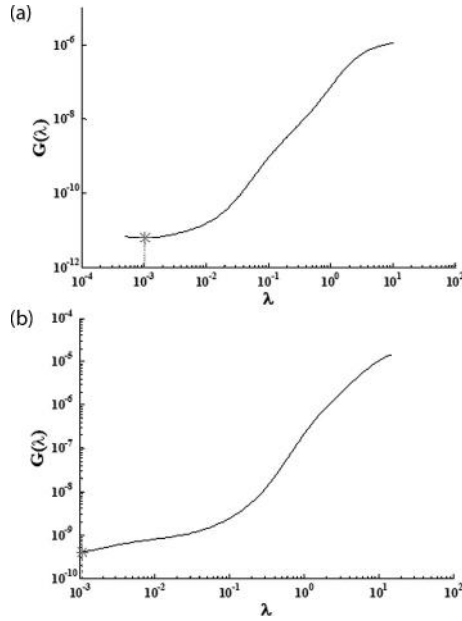
3.2.6 GCV

The method of GCV was introduced by Wahba [10] as a means to overcome the inconvenience of noise estimation. This method is a rotation-invariant version of ordinary leave-one-out cross validation. In this approach, the parameter is chosen by minimizing the GCV functional with an estimated linear model [20]:

$$G(\lambda) = \frac{\|A\mathbf{x}_\lambda - \mathbf{b}\|_2^2}{(T(\lambda))^2}, \quad (3.26)$$

where the numerator is the squared residual norm and the denominator is a squared effective number of DOF computed as in Equation (3.10). However, since GCV is derived for the case of additive white Gaussian noise, it is expected that GCV can yield under-regularized solutions if the noise model deviates from the normality assumption. As the GCV function may exhibit very flat minima in some cases, it becomes difficult to determine a unique value for the regularization parameter.

In general, the GCV function yields minimum near zero regularization, so that generalized cross-validation may yield under-smoothed estimates [33]. This can be checked by constructing bounds for the regularization parameters, such as forming a priori guesses of the magnitude of the residuals [15]. Figure 3.2a shows the GCV values plotted as a function of λ used for calibration of GRAPPA coefficients. The optimum regularization parameter is obtained at the point where $G(\lambda)$ is minimum. Optimum parameter is indicated using an asterisk (*). Figure 3.2b shows the GCV values used for calibration of SPIRiT coil coefficients. Due to the presence of correlated errors in the measurement vector, the optimum parameter in Figure 3.2b corresponds to the very first point in the curve.

**FIGURE 3.2**

GCV function plotted as a function of λ for (a) GRAPPA and (b) SPIRiT calibration.

3.2.7 Quasi-optimality Criterion

The quasi-optimality criterion is a heuristic parameter choice strategy which is generally used in cases where a good knowledge of noise in the model is available. Using this criterion, the parameter is selected as the minimizer of the quasi-optimality function [23]:

$$Q(\lambda) \equiv \left\| \lambda^2 \frac{d\mathbf{x}_\lambda}{d\lambda^2} \right\| = \frac{1}{2} \left\| \lambda \frac{d\mathbf{x}_\lambda}{d\lambda} \right\|. \quad (3.27)$$

Although the quasi-optimality criterion is one of the simplest and most efficient selection strategies, the convergence rate of estimation based on the quasi-optimality criterion is slower compared to those based on the discrepancy principle. For practical implementations, the partial derivative of the regularized solution based on the current value of the regularization parameter λ_j is approximated using [34]:

$$\left. \frac{\partial \mathbf{x}_\lambda}{\partial \lambda} \right|_{\lambda=\lambda_i} \approx \sum_j a_{ij} \mathbf{x}_{\lambda_j}, \quad (3.28)$$

where the coefficients a_{ij} are computed using the backwards difference formula [35]:

$$a_{ii} = c_i (\lambda_i - \lambda_{i-1})^{-1}$$

and

$$a_{i,i-1} = -c_{i-1} (\lambda_i - \lambda_{i-1})^{-1}, \quad (3.29)$$

where $c_i = \langle \mathbf{x}, \mathbf{x}_{\lambda,i}^n \rangle / \|\mathbf{x}_{\lambda,i}^n\|_2^2$ is the minimizer of $\|\mathbf{x} - c_i \mathbf{x}_{\lambda,i}^n\|_2$. Note that $Q(\lambda)$ seeks to find a good compromise between the regularization error and perturbation error [21]. The parameter chosen using this approach is exactly the same as that corresponding to the L-curve method.

3.2.8 L-Curve

Another approach of regularization parameter selection that uses a noise-free estimation is the L-curve method [36]. This method is based on the log-log plot of the solution norm versus residual norm, with λ as the free parameter. With increasing λ values, $\|B\mathbf{x}_\lambda\|_2$ decreases, and the residual norm $\|A\mathbf{x}_\lambda - \mathbf{b}\|_2$ increases. The flat and the steep parts of the L-curve correspond to solutions dominated by regularization errors and perturbation errors, respectively. The vertical portion corresponds to solutions where the solution norm is sensitive to variations in the regularization parameter. The horizontal part of the L-curve corresponds to solutions where the residual norm is very sensitive to changes in λ . This is because $\mathbf{x}_\lambda$ is dominated by the regularization error as long as $\mathbf{b}$ satisfies the discrete Picard condition. In cases where the log-log curve has a characteristic L-shape, the regularization parameter is chosen at the point corresponding to the maximum curvature.

Although the practical use of such plots was first suggested by Lawson and Hanson [37], similar plots also appear in other studies [36,38–41]. In practice, there are several situations in which the L-curve method fails to converge, without explicitly exhibiting an ‘elbow’ [42,43]. With Π denoting the calibration matrix, $\mathbf{k}_{u_r}^l$ denoting the observation vector and $\mathbf{z}_r^l$ denoting the coil coefficients, the L-curve as in Tikhonov regularization for GRAPPA is shown in Figure 3.3a. Along similar lines, Figure 3.3b

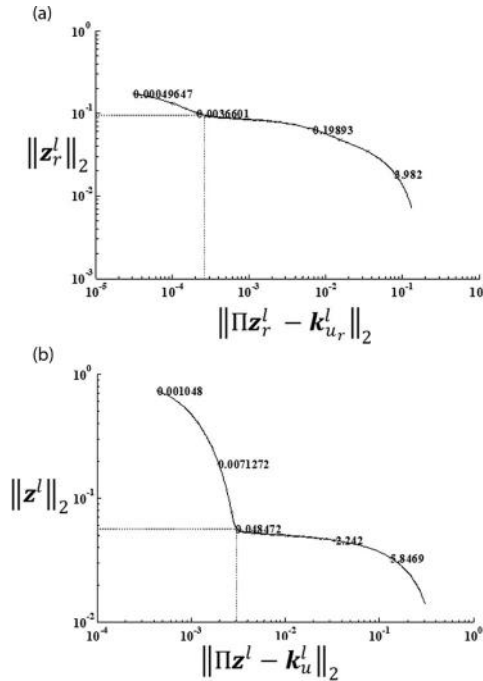


FIGURE 3.3

L-curves for (a) GRAPPA and (b) SPIRiT calibration. The SPIRiT calibration reveals a sharper L-curve compared to GRAPPA.

shows the L-curve obtained for SPIRiT calibration with $\mathbf{k}_u^l$ and $\mathbf{z}^l$ denoting the observation vector and coil coefficients, respectively.

3.3 Parameter Selection Strategies for Truncated SVD (TSVD)

As regularization using TSVD is achieved by neglecting those components of the solution that correspond to the smallest singular values, the number of singular values (k) used in rank approximation serves as the regularization parameter. The optimal parameter k can be computed using Picard's criterion, discrepancy principle, GCV or the L-curve. The Picard's criterion is described in [Section 2.3.3](#), and the truncation parameter k is chosen as the first singular value index for which the Picard condition is satisfied. Based on applying the discrepancy principle, the regularization parameter k is chosen so that the solution $\mathbf{x}_k$ obtained using the TSVD method satisfies the following:

$$\|A\mathbf{x}_k - \mathbf{b}\|_2 = d_e. \quad (3.30)$$

Likewise, the GCV-based regularization parameter k is chosen to minimize the GCV function:

$$G(k) = \frac{\|A\mathbf{x}_k - \mathbf{b}\|_2^2}{T^2}, \quad (3.31)$$

where $T = n - k$ is the effective number of DOF. With the L-curve method discussed earlier, $\|B\mathbf{x}_k\|_2$ decreases, and the residual norm $\|A\mathbf{x}_k - \mathbf{b}\|_2^2$ increases with increasing values of k . In TSVD, a main difference compared to Tikhonov regularization is that the filter factor is unity if the i 'th singular value is greater than or equal to the k 'th singular value, and zero otherwise. Thus TSVD regularization corresponds to a spectral cut-off filter that simply cuts off the last $n - k$ components, whereas the Tikhonov regularization corresponds to a smooth filter that damps the components corresponding to $\sigma_i < \lambda$.

All aforementioned parameter selection methods can be easily implemented using Hansen's regularization toolbox [\[44\]](#). Note that the quasi-optimality criterion can also be used to compute the truncation parameter in TSVD by choosing k as the smallest index j_q so that:

$$\|\mathbf{x}_{j_q+1} - \mathbf{x}_{j_q}\|_2 = \min_{1 \leq j \leq r} \|\mathbf{x}_{j+1} - \mathbf{x}_j\|, \quad (3.32)$$

where r is the rank of the matrix. Theoretical and numerical investigations on this scheme are more elaborately discussed in [\[1,3,45\]](#).

For calibration models in GRAPPA and SPIRiT which are of a discrete ill-posed nature, two distinct types of decay patterns are generally observed in the singular value profiles. A notable pattern is the type consisting of the singular value profile having two regions: a region with singular values decaying at a faster rate (for large singular values), and a second region which decays at a slower rate (for small singular values). In some datasets, only the first segment consisting of continuously decaying singular values is often seen.

Aliasing artifacts are mostly presented in those datasets for which the Picard's condition is violated in this segment. This is because a large number of singular values that contribute considerably to the signal component are discarded in the calibration.

In the second type of datasets with slower singular value decay rate, sharp L-corners cannot always be identified in Tikhonov and spectral cut-off regularization methods applied to GRAPPA and SPIRiT calibrations. Although the artifacts are always more pronounced in such cases, the mere presence of a sharp L-corner does not guarantee an artifact-free reconstruction. The absence of L-corners may be attributed to overly smooth solutions or low input noise levels. For smooth solutions, the SVD coefficients decay fast to zero, and the solution is dominated by the first few SVD components.

The GCV function may not always exhibit a minimum and can lead to very flat curves with multiple local minima. In the case of under-smoothing, the global minimum can be at the extreme endpoint. This is because the GCV method tends to become unstable when the number of singular values is less or the noise is correlated. For large datasets with uncorrelated errors, the parameter selection using GCV method performs well for Tikhonov as well as spectral cut-off regularization. The parameter selection for GRAPPA and SPIRiT calibration using TSVD regularization is shown in [Figure 3.4](#).

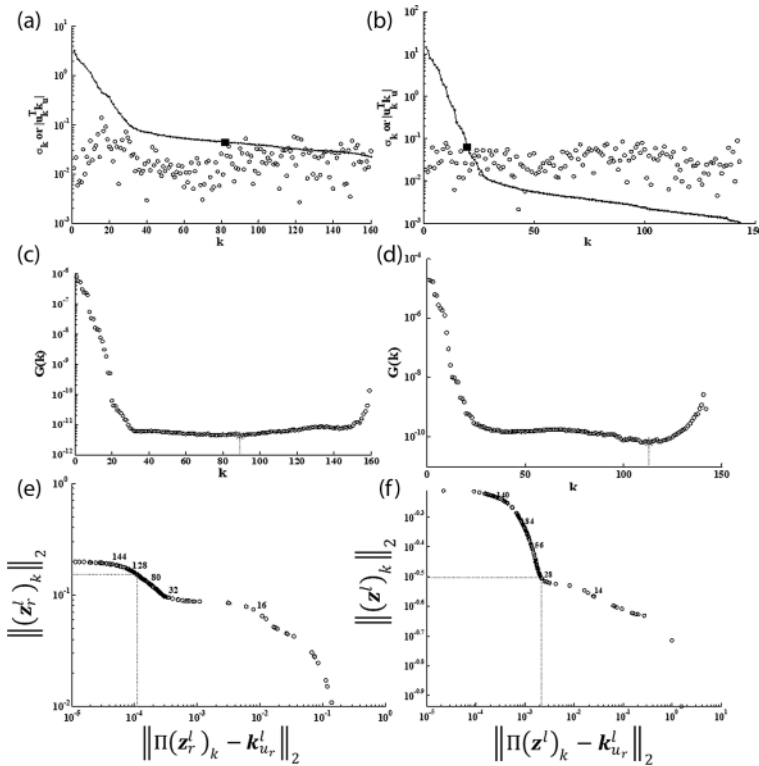


FIGURE 3.4

Singular values of the calibration matrix (black thick line) of (A) GRAPPA and (B) SPIRiT and the corresponding Fourier coefficients in circles. The optimum k is chosen at the point where Picard's condition is violated for the first time. GCV function for TSVD regularization in (C) GRAPPA and (D) SPIRiT calibration. The dotted line indicates truncation parameter (k) chosen at the GCV minimum. L-curves for TSVD regularization in (E) GRAPPA and (F) SPIRiT calibration. The dotted line indicates truncation parameter (k) computed using Regularization toolbox in MATLAB.

3.4 Parameter Selection Strategies for Non-quadratic Regularization

In quadratic regularization methods, the regularization term is formed using the squared norm of the desired solution; that is, $R(\mathbf{x}) = \|\mathbf{B}\mathbf{x}\|_2^2$. Use of such a quadratic regularization term leads to a linear formulation of the solution wherein a closed-form solution can be obtained for a given value of the regularization parameter λ . Despite their computational advantage, quadratic regularizers are limiting in the sense that they do not take into account the outliers in the true image while suppressing the noise component. Furthermore, they blur the discontinuities in the recovered model by applying a homogeneous smoothing to the desired model. For the purpose of addressing this issue, attention has been largely focused on the class of non-quadratic regularizers. This class of regularizers relies on the sparse representation of the solution in some transform domain. Typically, the transform domains can be either the finite-difference and multi-scale domains. In the multi-scale approaches, the observations are expanded on an orthogonal basis such as wavelets, and a back transformation is applied to get the smoothed data.

Effective application of the regularization methods calls for appropriate selection of the regularization parameter to establish a balance between the fidelity and preservation of image details. However, since a closed-form solution does not exist for the non-quadratic regularizers, the extension of the standard parameter selection strategies to non-quadratic regularizers is not straightforward. Particularly, the extension of the L-curve method for non-quadratic penalty is difficult due to the computational complexity associated with solving the minimization problem several times for different λ values. In some cases, it is also difficult to find the maximum value of the curvature of the L-curve, resulting in L-curves without well-defined L-corners.

Similar to the L-curve, the GCV function also cannot be derived explicitly if the regularization term is non-quadratic. As a result, the parameter selection problem becomes a bilevel optimization model which is very difficult to solve compared to the quadratic case [6]. This difficulty is overcome by introduction of a non-linear GCV function [28,46,47] derived by computing the Jacobian matrix consisting of the partial derivatives of Equation (2.6) with respect to the observed data $\mathbf{b}$. In addition, several modifications of statistical methods such as SURE and Bayesian methods have also been proposed for application to non-quadratic settings.

3.4.1 Parameter Selection for Wavelet Regularization

The wavelet regularization or wavelet threshold method is based on the wavelet denoising theory, in which the noise $\mathbf{n}$ is additive and complex Gaussian with zero mean and variance σ_0^2 . As the wavelet transform is a linear transform, the wavelet coefficients of the noisy signal $\mathbf{b}$ can be expressed as the sum of wavelet coefficients representing the true signal $\bar{\mathbf{b}}$ and that representative of noise. The wavelet transform possesses the property of concentrating the signal energy on some large wavelet coefficients and distributing the noise energy throughout the rest of the wavelet domain coefficients. That is, the coefficients of the wavelet transform are usually sparse. Therefore, most of the coefficients in a noiseless wavelet transform are effectively zero. This implies that the problem of recovering $\bar{\mathbf{b}}$ can be reformulated as recovering the wavelet coefficients of $\mathbf{b}$ which are relatively stronger than the Gaussian white noise. Thus the recovery of $\mathbf{b}$ can be achieved by considering the small magnitude wavelet coefficients as pure noise and setting it to zero. Therefore, the coefficient is compared with a threshold value in order to determine whether a particular coefficient constitutes a desirable part of the original signal or not.

The thresholded wavelet coefficients are computed according to the hard- or soft-thresholding rule. Whereas the hard-thresholding rule is usually referred to as wavelet thresholding, as it truncates the wavelet coefficients, the soft-thresholding rule is often called the wavelet shrinkage, as it shrinks the coefficients with high amplitude toward zero. The particular selection methods include Visushrink, SUREshrink, NeighBlock, SUREblock, False discovery rate, Bayesian methods, Ogden's method, cross-validation and GCV. Amongst the various selection strategies, an optimal one is that corresponding to the minimum mean squared error (MSE).

For the wavelet coefficients with j decomposition levels, thresholding is usually applied only to the detail coefficients rather than to the approximation coefficients because the latter represent the 'low-frequency' terms that carry signal information and are less affected by noise. The thresholding extracts the significant coefficients by setting those coefficients whose absolute value is below a certain threshold level λ to zero. In general, the threshold can be made a function of decomposition/resolution level j and the index k , denoted by $\lambda = \lambda(j, k)$. When it is taken as a function of j , the threshold is called level-dependent.

Denoting the orthogonal basis $\mathcal{B} = \{\mathcal{L}_m\}; 0 \leq m \leq N$, the projection onto the orthogonal basis becomes $\mathbf{b}_B(m) = \langle \mathbf{b}, \mathcal{L}_m \rangle$, $\bar{\mathbf{b}}_B(m) = \langle \bar{\mathbf{b}}, \mathcal{L}_m \rangle$ and $\mathbf{n}_B(m) = \langle \mathbf{n}, \mathcal{L}_m \rangle$. Using the orthogonal basis, $\mathbf{b}_B$ can be expressed as:

$$\mathbf{b}_B(m) = \bar{\mathbf{b}}_B(m) + \mathbf{n}_B(m). \quad (3.33)$$

Due to its simple additive nature, the characteristic of noise is not influenced by the choice of this basis [48]. With $\bar{\mathbf{b}}_{B\lambda}$ denoting the vector containing t coefficients of the noise-free data and Δ_λ a vector whose length $N - t$ is set equal to the number of coefficients that are set to zero, the observed signal in the orthogonal basis can be represented as:

$$\mathbf{b} = \begin{bmatrix} S_\lambda & T_\lambda \end{bmatrix} \begin{bmatrix} \bar{\mathbf{b}}_{B\lambda} \\ \Delta_\lambda \end{bmatrix} + \mathbf{n}, \quad (3.34)$$

where the columns of S_λ are selectively formed using $\{\mathcal{L}_{i_j}\}$ for which the absolute values of coefficients are larger than λ , and the columns of T_λ are basis vectors for which the absolute values of coefficients are smaller than λ [49]. In accordance with the representation in Equation (3.34), an estimate of the coefficients obtained after soft thresholding is given by:

$$\hat{\mathbf{b}}_{B\lambda} = \begin{bmatrix} \bar{\mathbf{b}}_{B\lambda} + S_\lambda^T \mathbf{n} - \lambda \text{sign}(S_\lambda^T \mathbf{b}_B) \\ \mathbf{0}_{(N-t \times 1)} \end{bmatrix}. \quad (3.35)$$

The MSE for this estimate is:

$$\begin{aligned} E &= \frac{1}{N} \left\| \bar{\mathbf{b}}_B - \hat{\mathbf{b}}_{B\lambda} \right\|_2^2 \\ &= \frac{1}{N} \left\| S_\lambda^T \mathbf{n} - \lambda \text{sign}(S_\lambda^T \mathbf{b}_B) \right\|_2^2 + \frac{1}{N} \left\| \Delta_\lambda \right\|_2^2, \end{aligned} \quad (3.36)$$

where $S_\lambda^T \mathbf{n}$ denotes the first t elements of the projected noise component $\mathbf{n}_B$ with an underlying complex Gaussian distribution. Therefore, $\frac{1}{N} \left\| S_\lambda^T \mathbf{n} \right\|_2^2 = \frac{1}{N} \sum_{i=1}^t \mathbf{n}_B^2(i)$ represents the

sample of a t th order Chi-square distribution. As a result, E is a sample of a non-central chi-square random variable with the mean and variance as follows:

$$\mathbb{E}(E) = \frac{t}{N}(\sigma_0^2 + t\lambda^2) + \frac{1}{N}\|\Delta_\lambda\|_2^2 \quad (3.37a)$$

$$\text{var}(E) = \frac{2t}{N^2}(\sigma_0^2) + \frac{4t}{N^2}(\sigma_0^2\lambda^2). \quad (3.37b)$$

Although the threshold value λ , data length N and the estimate of the noise variance σ_0^2 are available, the main challenge arises in the estimation of Δ_λ . To bypass this difficulty, an estimate of MSE can also be obtained as the distance between the coefficients of the observed noisy data and the coefficients of a thresholded version of the same data [50]. An estimate of $\|\Delta_\lambda\|_2^2$ with a given confidence probability p can then be obtained from the table of chi-square distribution such that the observed sample of data error is around its mean with probability p .

3.4.1.1 VisuShrink

The VisuShrink method applies a single universal threshold to all wavelet detail coefficients [51]. This universal threshold is given by:

$$\lambda_{\text{uni}} = \sqrt{(2\log N)\hat{\sigma}_0}, \quad (3.38)$$

where N is the number of data points, and $\hat{\sigma}_0$ is an estimate of the noise level σ_0 . The estimate of the noise level is typically computed as a scaled median absolute deviation (MAD) of the empirical wavelet coefficients:

$$\hat{\sigma}_0 = \kappa \text{MAD} \quad (3.39)$$

where $\kappa \approx 1.48$ denotes the scaling parameter. According to Donoho and Johnstone [51], if the random variable representing the wavelet coefficients is of zero mean and unit variance, then:

$$\mathbb{P}\left\{\max_{1 \leq i \leq N} |b_{Bi}| > \sqrt{2\log N}\right\} \sim \frac{1}{\sqrt{\pi \log N}} \xrightarrow{N \rightarrow \infty} 0, \quad (3.40)$$

where $\mathbb{P}\{\cdot\}$ represents the probability. Since there is a high probability that the coefficients with magnitude less than λ_{uni} represent noise, the upper bound on $|b_{Bi}|$ can serve as a universal threshold. Therefore, every sample in the wavelet transform whose value is less than λ_{uni} is set to zero. Because this choice may yield large values for thresholds due to its dependence on the number of samples (which is more than 10^5 for a typical test image of size 512×512), the VisuShrink threshold is known to yield over-smoothed images for image denoising applications.

3.4.1.2 SUREShrink

The problems caused due to the universal threshold can be reduced by decreasing the value of threshold, that is, by choosing a threshold smaller than the universal threshold.

This can be obtained by using a threshold selection strategy that minimizes the SURE risk [18]. A scheme that minimizes the SURE risk which fits a level-dependent threshold λ_j to the wavelet coefficients was proposed by Donoho and Johnstone [52]. The wavelet coefficients of $\mathbf{b}$ with j decomposition levels can be represented using:

$$\mathbf{b}_{B_{jk}}(m) = \bar{\mathbf{b}}_{B_{jk}}(m) + \mathbf{n}_{B_{jk}}(m), \quad (3.41)$$

where k is the index of the coefficient in the level j . Using the expression for the soft thresholding rule in the SURE risk in Equation (3.20), we have:

$$\|\mathbf{e}_y(\mathbf{y})\|_2^2 = \sum_{i=1}^N \left[\min(|\mathbf{b}_{B_{jk_i}}|, \lambda) \right]^2 \quad (3.42a)$$

and

$$\nabla_{\mathbf{e}_y}(\mathbf{y}) = - \sum_{i=1}^N \mathbf{1}_{[-\lambda, \lambda]}(\mathbf{b}_{B_{jk_i}}), \quad (3.42b)$$

where $\mathbf{y} = \bar{\mathbf{b}}_{B_{jk}}$, and $\mathbf{1}_{[-\lambda, \lambda]}(\mathbf{b}_{B_{jk_i}})$ is the indicator function for set $[-\lambda, \lambda]$. Using Equation (3.42), an explicit expression for the SURE risk can be obtained as follows:

$$\hat{R}_\lambda = N + \sum_{i=1}^N \left[\min(|\mathbf{b}_{B_{jk_i}}|, \lambda) \right]^2 - 2 \#(i : |\mathbf{b}_{B_{jk_i}}| < \lambda), \quad (3.43)$$

where $\#$ denotes the cardinality. The threshold λ_{SURE} is obtained by minimizing the SURE risk.

After computing the threshold using SURE, it should be confirmed that the optimal λ in Equation (3.43) is not greater than the universal threshold. The SURE profile follows the actual loss quite closely. However, the selection of the optimal threshold using SURE for the sparse wavelet coefficients is too naive because, in the case of significant sparsity, the noise contributed to the SURE profile by the large number of zero coordinates is much more significant than the information contributed to the SURE profile by the small number of non-zero coordinates. To overcome this, a hybrid scheme called SUREShrink was proposed [52]. Accordingly, the threshold value in the sparse case should be set to the universal threshold to ensure that a noise sample is correctly thresholded to zero. If the wavelet coefficients are not sparsely represented, SURE is supposed to provide a good estimation of the loss. Given the coefficients $\mathbf{y}$, a measure of the sparsity of the coefficients is computed using the following:

$$\begin{aligned} v_N(\mathbf{y}) &= \frac{N^{-1} \sum_{i=1}^s (y_i - 1)}{N^{-1/2} \log_2^{3/2}(N)} \\ &= N^{-1/2} \frac{\sum_{i=1}^N (y_i - 1)}{\log_2^{3/2}(N)}. \end{aligned} \quad (3.44)$$

Using this measure, $\mathbf{y}$ is considered sparse if $v_N(\mathbf{y}) \leq 1$; it is considered non-sparse otherwise. For the sparse coefficients, the thresholding using SUREShrink gives:

$$\lambda_{\text{SUREShrink}}(\mathbf{y}) = \begin{cases} \sqrt{2 \log N}; & \text{if } v_N(\mathbf{y}) \leq 1 \\ \lambda_{\text{SURE}}; & \text{otherwise.} \end{cases} \quad (3.45)$$

Thus, the SUREShrink method is a hybrid version composed of the universal threshold and the SURE threshold, with the choice depending on the sparsity measure of the particular sub-band. The SURE threshold is data-driven and has been shown to yield a good image denoising performance that comes close to the true minimum MSE of the optimal soft-threshold estimator.

3.4.1.3 NeighBlock

The SUREShrink procedure is achieved by the adaptivity according to the sparsity considerations. However, it does not take into account the local neighborhood of each coefficient. This results in excessive truncation of the wavelet coefficients, making it a biased estimator. One possible way to improve precision is by utilizing information about the neighboring wavelet coefficients. The threshold level can be assigned to each group (block) of wavelet coefficients according to the local properties of each. Each group is composed of a subset of the coefficients in the current resolution level. If the neighboring coefficients contain significant information, then it is likely that the current coefficient is also important and hence a lower threshold can be used.

Based on this concept, Cai and Silverman [53] proposed the NeighBlock procedure for threshold selection. At each resolution level j , the wavelet coefficients are grouped into disjoint blocks (jb) of length $L_0 = \frac{\log N}{2}$. Each block is then extended by an amount of $L_1 = \max(1, \lceil L_0 / 2 \rceil)$ in each direction, forming overlapping blocks (jB) of length $L = L_0 + 2L_1$. The coefficients in each disjoint block jb are estimated using:

$$\hat{\mathbf{b}}_{B_{jk}}^{(jb)} = \max \left(0, \frac{S_{(jB)}^2 - \lambda_c L}{S_{(jB)}^2} \right) \mathbf{b}_{B_{jk}}^{(jb)}, \quad (3.46)$$

where $S_{(jB)}^2$ denotes the sum of squared coefficients of the i th block, and $\lambda_c = 4.505$ is chosen as the solution for $\lambda_c - \log \lambda_c = 3$.

Cai and Silverman [53] also proposed another method called NeighCoeff, which takes into account a smaller local neighborhood of each coefficient that follows the same steps as NeighBlock. In this case, $L_0 = L_1 = 1$, $L = 3$ and $\lambda = \frac{2}{3} \log N$, and the value of the computed threshold is smaller than λ_{uni} . In this approach, each individual coefficient is shrunk by an amount that depends on the coefficient and on its immediate neighbors rather than on the neighboring coefficients outside the block of current interest as in NeighBlock.

3.4.1.4 SUREblock

The local block thresholding method has a fixed block size and threshold, and the same thresholding rule is applied to all resolution levels regardless of the distribution of the wavelet coefficients. To overcome this problem, Cai and Zhou [54] developed a data-driven approach to empirically select the block size and the threshold at the individual

resolution levels. In their approach, referred to as SUREblock, the block size and threshold are chosen by minimizing the SURE risk. In this procedure, a block size L (with $m = N / L$ denoting the number of blocks) and a threshold λ are initially fixed and the observations $b_{B_1}, b_{B_2}, \dots, b_{B_N}$ are divided into blocks of size L . Denoting the observation in the t th block as $\mathbf{b}_{B_t} = (b_{B_{(t-1)L+1}}, \dots, b_{B_{tL}})$, the block-wise James-Stein estimator of the corresponding signal component is given by:

$$\hat{\mathbf{b}}_{B_t}(\lambda, L) = \left(1 - \frac{\lambda}{S_t^2}\right)_+ \mathbf{b}_{B_t}; \quad t = 1, 2, \dots, m, \quad (3.47)$$

where $S_t^2 = \|\mathbf{b}_{B_t}\|^2$. The block size L and threshold level λ are chosen empirically by minimizing the SURE risk. An estimate of the SURE risk of the t th block is obtained as follows:

$$\hat{R}_{b_{B_t}, \lambda, L} = L + \frac{\lambda^2 - 2\lambda(L-2)}{S_t^2} \mathbf{1}_{(S_t^2 > \lambda)} + (S_t^2 - 2L) \mathbf{1}_{(S_t^2 \leq \lambda)}. \quad (3.48)$$

The total risk is then computed using the following:

$$\hat{R}_{b_B, \lambda, L} = \sum_{t=1}^m \hat{R}_{b_{B_t}, \lambda, L}. \quad (3.49)$$

Imposing an additional restriction on the search range, the optimal threshold and block length are chosen as follows:

$$\lambda^*, L^* = \min_{\max\{L-2, 0\} \leq \lambda \leq \lambda^F, 1 \leq L \leq \sqrt{N}} \hat{R}_{b_B, \lambda, L}. \quad (3.50)$$

The empirical selection of the block size and threshold enables values to vary across different resolution levels, and exhibits significant advantages over the wavelet thresholding estimators with fixed block sizes.

3.4.1.5 False Discovery Rate

The false discovery rate proposes a statistical procedure for thresholding of wavelet coefficients. In this method, initially proposed by Benjamini and Hochberg [55] and later adapted to be used with wavelets by Abramovich and Benjamini [56], estimation of the wavelet coefficients representing the true signal is viewed as a multiple hypothesis testing problem. Instead of simply choosing a threshold, as in the SURE risk minimizer, they keep or discard a wavelet coefficient in the decomposition of the noisy data based on a hypothesis test to decide whether the coefficient is non-zero or zero. Multiple hypothesis testing requires a more sophisticated statistical argument compared to a single hypothesis test. In a single hypothesis test, for example, if 1023 coefficients were to be tested independently at a significance level of $P = 0.05$, then approximately $1023 * 0.05 \approx 50$ coefficients would not be zeroed. In a multiple hypothesis setting, as in the false discovery rate approach, there are h hypotheses (coefficients) to be tested. Each hypothesis is of the form $H_{jk} : \mathbf{b}_{B_{jk}} = 0$. Among these hypotheses, h_1 is false and the corresponding coefficients need to be included in the reconstruction of the underlying signal. Among the R coefficients which

are not zeroed by a given thresholding procedure that are included in the reconstruction, S are correctly retained in the model and V are erroneously kept so that $R = S + V$. The false discovery rate of coefficients (FDRC) [56] is defined as the expectation of the proportion of included coefficients that should have been zeroed, given by $Q = V/R$. Thus, FDRC gives the expected proportion of the erroneously included coefficients in the reconstruction. The number of coefficients included in the reconstruction can be maximized subject to controlling the FDRC to some level q , typically 0.01 or 0.05. The procedure for FDRC includes the following steps:

1. For a known marginal distribution of the noise in model $\phi(|b_{B_{jk}}|/\sigma_0)$, calculate the two-sided P -values;

$$P_{jk} = 2\left(1 - \phi(|b_{B_{jk}}|/\sigma_0)\right), \quad (3.51)$$

to test the hypothesis $H_{jk} : \bar{b}_{B_{jk}} = 0$.

2. Order the P_{jk} s according to their size, that is, $p_{(1)} \leq p_{(2)} \leq p_{(3)} \leq \dots p_{(N)}$, where each $p_{(i)}$ corresponds to some coefficient $b_{B_{jk}}$.
3. For the largest i for which $p_{(i)} \leq (i/N)q$ and denoted by i_0 , compute the following:

$$\lambda_{i_0} = \sigma_0 \phi^{-1}(1 - p_{(i_0)}/2). \quad (3.52)$$

4. Threshold all coefficients using λ_{i_0} .

Since a coefficient is included in the model only if the corresponding $p_{(i)} < q$, it has to be at least larger than $\lambda_{\min} = \sigma_0 \phi^{-1}(1 - q/2)$. This implies that the FDRC procedure could be performed only for $|b_{B_{jk}}| \geq \lambda_{\min}$. This brings in large computational savings in sorting the coefficients.

3.4.1.6 Bayes Factor Thresholding

The Bayesian approach uses knowledge of prior distributions for σ_0^2 and $\bar{b}_B$ for estimation of $\bar{b}_{B_{jk}}$ [57]. Because the exponential distribution is the entropy maximizer among all distributions supported on $(0, \infty)$ for a fixed first moment [58], the prior distribution for σ_0^2 is chosen to be:

$$\mathbb{P}\{\sigma_0^2 | \mu\} = \mu e^{-\mu\sigma_0^2}, \quad (3.53)$$

where μ is a hyper-parameter. The prior on the signal component of wavelet coefficients $\bar{b}_B$ is chosen to be flat-tailed t family priors, that is:

$$\mathbb{P}\{\bar{b}_B | \tau\} = t_n(0, \tau), \quad (3.54)$$

where τ is a hyper-parameter. Also with a double exponential marginal likelihood:

$$\mathbb{P}\{b_B | \bar{b}_B\} \sim \frac{1}{2} \sqrt{2\mu} e^{-\sqrt{2\mu}|b_B - \bar{b}_B|}. \quad (3.55)$$

In this approach, a hypothesis $H_0: \hat{\bar{b}}_B = 0$ versus $H_A: \hat{\bar{b}}_B \neq 0$ is first tested using the noisy wavelet coefficients $\mathbf{b}_B$, and a Bayesian approach is used to compare the two hypotheses using the Bayes factor computed as:

$$B_y = \frac{\mathbb{P}\{\mathbf{b}_B|0\}}{\int_{\bar{\mathbf{b}}_B \neq 0} \mathbb{P}\{\mathbf{b}_B|\bar{\mathbf{b}}_B\} \zeta(\bar{\mathbf{b}}_B) d\bar{\mathbf{b}}_B}, \quad (3.56)$$

where $\zeta(\bar{\mathbf{b}}_B)$ describes the spread of $\bar{\mathbf{b}}_B$ chosen from the t family. With π_0 denoting the probability that $\bar{\mathbf{b}}_B = 0$ so that $\pi_0 + \pi_1 = 1$, then $\bar{\mathbf{b}}_B$ is estimated from $\mathbf{b}_B$ for H_0 using:

$$\hat{\bar{\mathbf{b}}}_B = \mathbf{b}_B \mathbf{1}_{\left(\mathbb{P}(H_0|\mathbf{b}_B) < \frac{1}{2}\right)}, \quad (3.57)$$

where the posterior probability of H_0 is:

$$\mathbb{P}(H_0|\mathbf{b}_B) = \left(1 + \frac{\pi_1}{\pi_0} \frac{1}{B_y}\right)^{-1}. \quad (3.58)$$

This thresholding rule is called Bayes factor (BF) thresholding since the posterior ratio is obtained by multiplying the Bayes factor with the prior ratio. Thus, a wavelet coefficient is thresholded if the corresponding posterior ratio is greater than 0.5.

3.4.1.7 BayesShrink

BayesShrink is a threshold selection method which is based on the empirical observation that the wavelet coefficients in a sub-band of a natural image can be represented adequately by a generalized Gaussian distribution (GGD) [59]. Thus, it is possible to approximate the average MSE in a sub-band using the squared error risk applied to each sub-band with application of the GGD prior. The GGD priors applied to wavelet-based image processing applications typically employ a shape parameter β , ranging from 0.5 to 1, as given by:

$$G_{\sigma_{\bar{\mathbf{b}}_B}, \beta}(x) = \frac{1}{2\Gamma(1 + 1/\beta)C(\sigma_{\bar{\mathbf{b}}_B}, \beta)} \exp\left\{-\left|\frac{x - \mu}{C(\sigma_{\bar{\mathbf{b}}_B}, \beta)}\right|^\beta\right\}; -\infty < x < \infty, \sigma_{\bar{\mathbf{b}}_B} > 0, \beta > 0, \quad (3.59)$$

where the μ is the mean, $\sigma_{\bar{\mathbf{b}}_B}$ is the standard deviation of $\bar{\mathbf{b}}_B$, $\Gamma(\cdot)$ is the gamma function and $C(\sigma_{\bar{\mathbf{b}}_B}, \beta) = (\sigma_{\bar{\mathbf{b}}_B}^2 \Gamma(1/\beta) / \Gamma(3/\beta))^{1/2}$ is a scaling function. With the prior $\bar{\mathbf{b}}_B \sim G_{\sigma_{\bar{\mathbf{b}}_B}, \beta}$ and the marginal distribution $\mathbf{b}_B|\bar{\mathbf{b}}_B \sim N(\bar{\mathbf{b}}_B, \sigma_0^2)$, the objective is to find a threshold $\lambda^*(\sigma_{\bar{\mathbf{b}}_B}, \beta)$ that minimizes the Bayes risk:

$$R(\lambda) = \mathbb{E}\left(\hat{\bar{\mathbf{b}}}_B - \bar{\mathbf{b}}_B\right)^2 = \mathbb{E}_{\bar{\mathbf{b}}_B} \mathbb{E}_{\mathbf{b}_B|\bar{\mathbf{b}}_B}\left(\hat{\bar{\mathbf{b}}}_B - \bar{\mathbf{b}}_B\right)^2, \quad (3.60)$$

where $\mathcal{T}_\lambda$ denotes the soft-thresholding operator and $\hat{\bar{\mathbf{b}}}_B = \mathcal{T}_\lambda(\mathbf{b}_B)$. Solution of Equation (3.60) requires an estimate of $\sigma_{\bar{\mathbf{b}}_B}$. This is obtained as follows:

$$\hat{\sigma}_{\bar{b}_B} = \sqrt{\max\left(\hat{\sigma}_{b_B}^2 - \sigma_0^2\right), 0}, \quad (3.61)$$

where $\hat{\sigma}_{b_B}^2 = 1/N^2 \sum_{p,q=1}^N b_{B_{pq}}^2$. For image denoising applications, the BayesShrink threshold is shown to yield better performance in terms of the MSE than that obtained using SUREShrink [59]. Although the threshold computed using this method can be used to improve images corrupted by Gaussian noise in smooth regions of the image, its sensitivity is low in the presence of noise around the edges.

3.4.1.8 Ogden's Methods

Ogden developed two distinct methods for wavelet thresholding that include selection thresholding and data analytic thresholding [60]. Selection thresholding relies on the hypothesis testing of coefficients in each level. For a set of coefficients at a particular level, Ogden describes a test statistic which when exceeds some critical value (threshold) retains the largest coefficient into the reconstruction. If it is not larger when compared to the threshold, then the absolute value of the largest remaining coefficients is set as the threshold. The procedure is then continued until all the coefficients are tested.

The data analytic thresholding is based on the standard change point problem in which the null hypothesis is that all the random variables possess the same mean. Thus, the alternative hypothesis corresponds to a shift in the mean at an unknown location. Since a standard non-parametric test for such problems is the cumulative sums (CUSUM) process, the data analytic thresholding is performed by visual evaluation of the plots of cumulative sums of the squares of the coefficients at a particular level. The mean shifts are then detected by application of various functionals such as the Kolmogorov-Smirnov functional to the CUSUM process, computed using the Brownian bridge stochastic process:

$$B_b\left(\frac{i}{N}\right) = \frac{1}{\sigma_0 \sqrt{2N}} \sum_{j=1}^i (b_{B_j}^2 - \mu_{b_B^2}), \quad (3.62)$$

where $\mu_{b_B^2}$ denotes the mean of $b_{B_j}^2$ [61]. The steps involved in this method can be summarized as follows:

1. Test the null hypothesis.
2. For a significant test statistic in Step 1, the coefficient with the largest absolute value is removed from the given data, and N is set to $N - 1$. Then return to Step 1.
3. If the test statistic in Step 1 is not significant, the threshold is set to the largest absolute value of the coefficients in the data.

This approach proceeds by throwing out all the large coefficients until the remaining data behaves like white noise. Thus, shrinking all the coefficients in the remaining set to zero requires shrinking the large coefficients also by the same amount.

3.4.1.9 Cross-validation

In the cross-validation threshold selection procedure, the observed noisy data $b_i; i = 1, 2, \dots, N$, are divided into two subsequences [62]:

$$b_i^{\text{ODD}} = \frac{b_{2i-1} + b_{2i+1}}{2}; i = 1, \dots, N/2; b_{N+1} = b_{N-1} \quad (3.63a)$$

and

$$b_i^{\text{EVEN}} = \frac{b_{2i} + b_{2i+2}}{2}; i = 1, \dots, N/2; b_{N+2} = b_N. \quad (3.63b)$$

Then the threshold using cross-validation method can be obtained as the minimizer of:

$$\widehat{M}(\lambda) = \sum_{j,k} \left(\mathcal{T}_\lambda(b_{B_{jk}}^{\text{EVEN}}) - b_{B_{jk}}^{\text{ODD}} \right)^2 + \sum_{j,k} \left(\mathcal{T}_\lambda(b_{B_{jk}}^{\text{ODD}}) - b_{B_{jk}}^{\text{EVEN}} \right)^2, \quad (3.64)$$

where $b_{B_{jk}}^{\text{EVEN}}$ and $b_{B_{jk}}^{\text{ODD}}$ are the discrete wavelet transforms of b_i^{EVEN} and b_i^{ODD} . With the cross validation method, one can find thresholds that are very close to the optimal threshold by minimizing the l_2 distance between the true signal and the estimate. However, they tend to overfit the model in some cases [57].

3.4.1.10 Wavelet Thresholding

Weyrich and Warhola [63,64] proposed using GCV for estimating the threshold parameter in wavelet denoising applications by modifying Wahba's definition for spline smoothing [10]. Compared to other heuristic methods of estimating the noise level, the GCV method is proven to be asymptotically optimal in the Euclidean norm. In the case of wavelets, the threshold is chosen as the minimizer of the GCV function:

$$\text{GCV}(\lambda) = \frac{\frac{1}{N} \|b_B - \mathcal{T}_\lambda(b_B)\|_2^2}{\left(\frac{N_0}{N}\right)^2}, \quad (3.65)$$

where N_0 is the number of coefficients thresholded to 0 [63,65,66]. Here, the GCV function depends only on the input and output data. It has been proved that, under certain conditions, the threshold choice is asymptotically optimal [66]. For a large number of data points, the method chooses a threshold for which the solution minimizes the expected MSE computed with respect to the noise-free data.

3.4.2 Methods for Parameter Selection in Total Variation (TV) Regularization

The TV of a signal can be viewed as a measure of the signal variability that takes into account the changes that the signal undergoes. Using the variational formulation, the TV measure over domain $\Omega \subset \mathbb{R}^2$ is defined as follows [67]:

$$\text{TV}(x) = \int_{\Omega} |\nabla x| dx dy = \int_{\Omega} \sqrt{x_x^2 + x_y^2} dx dy \quad (3.66)$$

Approaches adopted to deal with the TV-based penalty fall into one of the three classes: (1) a nonlinear partial differential equation (PDE) approach in which the solution is computed by solving the associated Euler-Lagrange equation (ELE), (2) duality-based approaches and (3) prediction-based methods in which the solution is computed from a discrete version of Equation (3.66). However, the computer-based implementation necessitates some type of discretization, so a choice has to be made in either discretizing the solution derived from the initial continuous domain problem or discretizing the initial problem and then finding a solution using a finite-dimensional optimization algorithm. The methods discussed below are commonly used for selection of the regularization parameter for TV regularization.

3.4.2.1 PDE Approach

PDE-based models provide easy incorporation of geometric regularity on the solutions obtained as denoised images, such as smoothness of boundaries. TV is one of the well-known examples of PDE-based edge-preserving regularization approaches. In the TV model proposed by Rudin, Osher and Fatemi (the ROF model), they solved the following constrained minimization problem [67]:

$$\text{minimize} \int_{\Omega} |\nabla \mathbf{x}| dx dy \quad (3.67)$$

$$\text{s.t.} \int_{\Omega} \frac{1}{2} (A\mathbf{x} - \mathbf{b})^2 dx dy = \sigma_0^2,$$

where σ_0^2 is given. Using a Lagrange multiplier, the unconstrained formulation takes the following form:

$$J(\mathbf{x}) = \lambda \int_{\Omega} |\nabla \mathbf{x}| dx dy + \frac{1}{2} \int_{\Omega} ((A\mathbf{x} - \mathbf{b})^2 - \sigma_0^2) dx dy. \quad (3.68)$$

The corresponding Euler-Lagrange equation is:

$$g(\mathbf{x}) = -\lambda \nabla \left(\frac{\nabla \mathbf{x}}{\sqrt{|\nabla \mathbf{x}|^2 + \beta}} \right) + A^* (A\mathbf{x} - \mathbf{b}) = 0, \quad (3.69)$$

where β is a small positive quantity. The ROF model is solved using a time-marching scheme in which the solution to the unconstrained TV problem is obtained by treating $\mathbf{x}$ as a function of time and space. The procedure involves application of the gradient descent method to solve a parabolic equation with time as an evolution parameter. This requires solving the following:

$$\frac{\partial \mathbf{x}}{\partial t} = -g(\mathbf{x}); \text{ for } t > 0, \quad (3.70)$$

where $\mathbf{x} = \mathbf{b}$; at $t = 0$.

Using gradient descent, the solution for Equation (3.70) is recursively estimated as follows:

$$\mathbf{x}^{k+1} = \mathbf{x}^k - \tau_k g(\mathbf{x}^k), \quad (3.71)$$

where τ_k is the step size. However, computation of $g(\mathbf{x}^k)$ in Equation (3.71) involves an unknown parameter λ that also serves as the regularization parameter in Equation (3.68). In order to compute λ , Equation (3.70) is multiplied by $(A\mathbf{x} - \mathbf{b})$ and integrated by parts over Ω . If the steady state has been reached, then the left side of Equation (3.70) vanishes and the parameter λ is estimated as follows:

$$\lambda^t = \left[-\frac{1}{2\sigma_0^2} \int_{\Omega} |\nabla \mathbf{x}| - \left(\frac{\mathbf{b}_x \mathbf{x}_x}{|\nabla \mathbf{x}|} + \frac{\mathbf{b}_y \mathbf{x}_y}{|\nabla \mathbf{x}|} \right) dx \, dy \right]^{-1}. \quad (3.72)$$

The value of λ is dynamic and converges as $t \rightarrow \infty$. This approach can be theoretically justified using the gradient-projection method of Rosen [68]. For numerical implementation, the parameter should be expressed as a function of k with $t_k = k\Delta t$, $h = 1/N$, $x_i = ih$, and $y_i = jh$; $i, j = 1, 2, \dots, N$. Denoting $\mathbf{x}_{ij}^k = \mathbf{x}(x_i, y_i, t_k)$, $\Delta_{\mp}^x = \mp(\mathbf{x}_{i+1,j} - \mathbf{x}_{ij})$ and $\Delta_{\mp}^y = \mp(\mathbf{x}_{i,j+1} - \mathbf{x}_{ij})$, $|\nabla \mathbf{x}|$ can be expressed discretely using $|\nabla \mathbf{x}|_d = \sqrt{(\Delta_{\mp}^x \mathbf{x}_{ij}^k)^2 + (\Delta_{\mp}^y \mathbf{x}_{ij}^k)^2}$. Using this discrete representation, the expression for regularization parameter in Equation (3.72) takes the following form:

$$\lambda^k = \left[\frac{h}{2\sigma_0^2} \sum_{i,j} |\nabla \mathbf{x}|_d - \frac{(\Delta_{\mp}^x \mathbf{b}_{ij}^0)(\Delta_{\mp}^x \mathbf{x}_{ij}^k)}{|\nabla \mathbf{x}|_d} - \frac{(\Delta_{\mp}^y \mathbf{b}_{ij}^0)(\Delta_{\mp}^y \mathbf{x}_{ij}^k)}{|\nabla \mathbf{x}|_d} \right]^{-1}, \quad (3.73)$$

where $\mathbf{b}_{ij}^0 = \mathbf{b}(x_i, y_i) + \psi(x_i, y_i)$ is the modified initial data chosen so that the constraint in Equation (3.67) is satisfied by choosing ψ to be any non-linear function with zero mean and unit l_2 -norm. Using λ^k computed in Equation (3.73), an updated solution can be obtained with an appropriately chosen step size τ_k . Although the numerical implementation of this steepest descent method is straightforward, steepest descent has rather undesirable asymptotic convergence properties that can makes it very inefficient [69].

3.4.2.2 Duality-Based Approaches

The main motivation behind application of duality-based approaches to TV regularization is to overcome the singularity caused by the non-differentiability of $|\nabla \mathbf{x}|$ in the TV-norm through the use of an additional variable in the gradient domain. In this context, the original variable is termed the primal variable and the newly introduced variable is the dual variable, resulting in a system of equations instead of a single objective function. In TV regularization, Chan, Golub and Mulet [70] first proposed the primal dual idea by treating the normalized gradient term in the ELE as an independent variable. Later, Chambolle [71] proposed a semi-implicit gradient descent algorithm based on the Fenchel dual formulation for solving the TV regularization problem and the primal-dual proximal splitting algorithm for solving generalized non-smooth composite

convex minimization problems [72]. In the follow sub-sections, some recent parameter selection methods used in duality-based TV regularization are discussed.

3.4.2.2.1 Discrepancy Principle

For application of the discrepancy principle, a discrete TV functional is defined by introducing a discrete version of the gradient operator ∇ as a linear operator that maps the image domain $\mathbf{x} \in X$ to the gradient domain $\mathbf{p} \in G$, and the discrete version of the divergence operator as $\text{div} = -\nabla^*$, where ∇^* is the adjoint operator of ∇ [5]. Thus, for every $\mathbf{x} \in X$ and $\mathbf{p} \in G$, $\langle -\text{div } \mathbf{p}, \mathbf{x} \rangle_X = \langle \mathbf{p}, \nabla \mathbf{x} \rangle_G$. To facilitate the solution of this discrete TV model using the duality-based approach, it is convenient to express the TV measure in terms of a set $S \equiv \{\mathbf{p} \in G : \|\mathbf{p}\|_\infty \leq 1\}$ as follows [71]:

$$\text{TV}(\mathbf{x}) = \max_{\mathbf{p} \in G} \{ \langle \text{div } \mathbf{p}, \mathbf{x} \rangle - \delta_S(\mathbf{p}) \} = \max_{\mathbf{p} \in S} \langle \text{div } \mathbf{p}, \mathbf{x} \rangle_X, \quad (3.74)$$

in which a characteristic function δ_S can be defined so that $\delta_S(\mathbf{p}) = 0$ for $\mathbf{p} \in S$ and $\delta_S(\mathbf{p}) = +\infty$ for $\mathbf{p} \notin S$. Since S is closed and convex, $\text{TV}(\mathbf{x})$ becomes the Legendre-Fenchel conjugate of δ_S . Replacing the TV measure in Equation (2.6) using Legendre-Fenchel's duality-based representation of the TV norm, the optimization problem is modified as follows:

$$\min_{\mathbf{x}} \max_{\mathbf{p} \in S} \left\{ \mathcal{J}(\mathbf{x}, \mathbf{p}; \lambda) = \langle \mathbf{x}, \text{div } \mathbf{p} \rangle_X + \frac{\kappa}{2} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2 \right\}. \quad (3.75)$$

Here, κ is equivalent to $1/\lambda$ in Equation (2.6) and computed using the discrepancy principle in each iteration. The minimization problem in Equation (3.75) is then solved by finding a saddle point of the primal-dual function. A pair $(\mathbf{x}^*, \mathbf{p}^*)$ is a saddle point if and only if $\mathbf{x}^*$ and $\mathbf{p}^*$ are the optimal solutions of the problems:

$$\mathbf{x}^* = \min_{\mathbf{x}} \mathcal{J}(\mathbf{x}, \mathbf{p}^*; \kappa) \quad (3.76a)$$

and

$$\mathbf{p}^* = \max_{\mathbf{p} \in S} \mathcal{J}(\mathbf{x}^*, \mathbf{p}; \kappa) \quad (3.76b)$$

In this case, the primal-dual proximal point method [73–76] is used to compute the saddle point of Equation (3.75) by alternately optimizing the primal variable $\mathbf{x}$ and dual variable $\mathbf{p}$. Starting at an initial point $(\mathbf{x}^{(0)}, \mathbf{p}^{(0)})$, the sequence is successively generated by solving:

$$\mathbf{p}^{(k+\frac{1}{2})} = \min_{\mathbf{p} \in S} -\mathcal{J}(\mathbf{x}^{(k)}, \mathbf{p}; \kappa) + \frac{1}{2q} \|\mathbf{p} - \mathbf{p}^{(k)}\|_2^2 \quad (3.77a)$$

$$\mathbf{x}^{(k+1)} = \min_{\mathbf{x}} \mathcal{J}\left(\mathbf{x}, \mathbf{p}^{(k+\frac{1}{2})}; \kappa\right) + \frac{1}{2t} \|\mathbf{x} - \mathbf{x}^{(k)}\|_2^2 \quad (3.77b)$$

$$\mathbf{p}^{(k+1)} = \min_{\mathbf{p} \in S} -\mathcal{J}(\mathbf{x}^{(k+1)}, \mathbf{p}; \kappa) + \frac{1}{2q} \|\mathbf{p} - \mathbf{p}^{(k)}\|_2^2, \quad (3.77c)$$

where q and t are constants which are usually set as $t = 1$ and $q = 1/16$. The sub-problems in Equations (3.77a) and (3.77c) can be solved by defining a projection operator onto a set S in which the projection of a vector v onto S is defined as:

$$\mathcal{P}_S(v) = \min_{p \in S} \|p - v\|_2^2. \quad (3.78)$$

Using this projection operator, the minimization problem in Equation (3.77a) is then equivalent to solving:

$$p^{(k+\frac{1}{2})} = \mathcal{P}_S(p^{(k)} - s\nabla x^{(k)}). \quad (3.79)$$

Along similar lines, the solution to Equation (3.77c) is obtained as:

$$p^{(k+1)} = \mathcal{P}_S(p^{(k)} - s\nabla x^{(k+1)}). \quad (3.80)$$

Also, since the objective function in Equation (3.77b) is quadratic with respect to x , a closed-form solution can be obtained as follows:

$$x^{(k+1)} = \left(\kappa^{(k+1)} t A^T A + I \right)^{-1} \left(\kappa^{(k+1)} t A^T b + u_k \right), \quad (3.81)$$

where $u_k = x^{(k)} - t \operatorname{div}(p^{(k+\frac{1}{2})})$. The regularization parameter in Equation (3.81) is computed using the discrepancy principle, which requires the residual error and the upper bound of the perturbation error for parameter selection. From Equation (3.81), the residual error in $(k+1)$ th iteration is given by:

$$e_r^{(k+1)} = A \left(\kappa^{(k+1)} t A^T A + I \right)^{-1} \left(\kappa^{(k+1)} t A^T b + u_k \right) - b. \quad (3.82)$$

The discrepancy principle-based parameter choice can then be obtained in each iteration by solving $\|e_r^{(k+1)}\|_2^2 = c^2$, where $c^2 = N\sigma_0^2$ is an upper bound of the perturbation error. For a good choice of the upper bound c^2 , an optimal parameter value that minimizes the error in the recovered image can be computed in each iteration. However, since the original image is unknown, the upper bound should be chosen according to the noise level [30,77,78]. In the absence of information about the noise level, it can be estimated using the median rule in Equation (3.39).

3.4.2.2.2 GCV Applied to TV Regularization

For the optimization of the cost function in Equation (3.75), a GCV-based parameter selection was proposed [6]. This is based on the first-order primal-dual approach [72], in which a saddle point of mini-max functions is obtained by successively generating the following sequence:

$$x^{(k+1)} = \min_x \mathcal{J}(x, p^{(k)}; \kappa) + \frac{1}{2t} \|x - x^{(k)}\|_2^2 \quad (3.83a)$$

$$\tilde{\mathbf{x}}^{(k+1)} = \mathbf{x}^{(k+1)} + \theta (\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}) \quad (3.83b)$$

$$\mathbf{p}^{(k+1)} = \max_{\mathbf{p} \in S} \mathcal{J}(\mathbf{x}^{(k+1)}, \mathbf{p}; \kappa) - \frac{1}{2q} \|\mathbf{p} - \mathbf{p}^{(k)}\|_2^2, \quad (3.83c)$$

where θ is the combination parameter usually taken as 1. As the sub-problem in Equation (3.83a) can be reformulated as a quadratic regularization problem, the minimizer of the GCV function is computable similar to the Tikhonov problem in Equation (3.26). To evaluate the GCV function, the objective function in Equation (3.83a) is represented in a more convenient form as follows:

$$\mathbf{x}^{(k+1)} = \min_{\mathbf{x}} \left\{ \frac{t\kappa}{2} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2 + \frac{1}{2} \|\mathbf{x} - \mathbf{u}^{(k)}\|_2^2 \right\}, \quad (3.84)$$

where $\mathbf{u}^{(k)} = \mathbf{x}^{(k)} - t \operatorname{div}(\mathbf{p}^{(k)})$. Denoting $\mathbf{d}^{(k+1)} = \mathbf{x}^{(k+1)} - \mathbf{u}^{(k)}$ and $\beta = \kappa/t$, Equation (3.84) can also be expressed as $\mathbf{d}^{(k+1)} = \min_{\mathbf{d}} \left\| \mathbf{A}\mathbf{d} - (\mathbf{b} - \mathbf{A}\mathbf{u}^{(k)}) \right\|_2^2 + \frac{\beta}{2} \|\mathbf{d}\|_2^2$. In this form, the GCV can be expressed as a function of β :

$$\text{GCV}(\beta) = \frac{\frac{1}{N} \left\| I - (A^T A + \beta I^T I)^{-1} A^T (\mathbf{b} - \mathbf{A}\mathbf{u}^{(k)}) \right\|_2^2}{\left[\frac{1}{N} \operatorname{trace} \left(I - (A^T A + \beta I^T I)^{-1} A^T \right) \right]^2}. \quad (3.85)$$

Thus, in each iteration of the primal-dual method, β can be computed using Equation (3.85). With m denoting the number of iterations, the regularization parameter κ for the original TV problem can then be obtained by averaging the regularization parameters obtained in each primal iteration. The value of the estimated parameter κ depends on the step size t . To ensure convergence of the algorithm, it is necessary to tune the parameters q and t . The TV problem needs to be solved twice: first to determine the parameter κ using the GCV functional and second to solve the TV problem with the estimated value of κ .

3.4.2.2.3 Lagrange Multiplier-Based Method

This method is based on the Lagrangian duality in which the solution to the original minimization problem (the primal problem in Equation (2.6)) is obtained by solving the following dual problem:

$$\max_{\lambda} \min_{\mathbf{x}} \left\{ \mathcal{L}(\mathbf{x}, \lambda) = D(\mathbf{A}\mathbf{x}, \mathbf{b}) + \lambda R(\mathbf{x}) \right\}. \quad (3.86)$$

In the constrained form, this is equivalent to solving:

$$\min_{\mathbf{x}} D(\mathbf{x}) \text{ s.t. } \mathcal{R}(\mathbf{x}) \leq \gamma, \quad (3.87)$$

where $\mathcal{R}(\mathbf{x}) \triangleq \text{TV}(\mathbf{x})$ and γ denotes an estimated value of the regularization function in the exact solution such that $\mathcal{R}(\mathbf{x}) - \gamma = R(\mathbf{x})$ [7]. Thus, the dual problem becomes:

$$\max_{\lambda} \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2 + \lambda (\text{TV}(\mathbf{x}) - \gamma). \quad (3.88)$$

By imposing the first-order optimality condition, that is, $\nabla_{\lambda} \mathcal{L}(x, \lambda) = 0$, the dual problem can be transformed into the following two subproblems:

$$\text{find } \lambda; \text{ s.t. } \text{TV}(\mathbf{x}) - \gamma = 0 \quad (3.89a)$$

and

$$\min_x \frac{1}{2} \|A\mathbf{x} - \mathbf{b}\|_2^2 + \lambda (\text{TV}(\mathbf{x}) - \gamma). \quad (3.89b)$$

Consequently, the nonlinear equation $R(x) = \text{TV}(x) - \gamma = 0$ can be solved by computing a sequence $\{\lambda_k\}$ that converges to the root λ^* . For each λ_k , $\mathbf{x}$ can be computed to obtain the sequence $\{\mathbf{x}_k\}$ that converges to $\mathbf{x}^*$. In particular, for a given initial value of the regularization parameter λ_0 , the parameter λ_k can be computed using the bisection method:

$$\lambda_k = \lambda_{k-1} + \text{sign}(\text{TV}(\mathbf{x}_{k-1}) - \gamma) \frac{\lambda_0}{2^k}; k = 1, 2, \dots \quad (3.90)$$

Using the Tikhonov approximate solution $\tilde{\mathbf{x}}$ as in Equation (2.12), the initial value of the parameter λ_0 can be obtained as:

$$\lambda_0 = \alpha \frac{\|A\tilde{\mathbf{x}} - \mathbf{b}\|_2}{\text{TV}(\tilde{\mathbf{x}})}, \quad (3.91)$$

where α is a suitable scale parameter. A disadvantage of this approach is that the value of γ has to be determined heuristically.

3.4.2.3 Prediction Methods

Prediction methods rely on a form of linear relationship between the regularized solution and data. Knowledge of the noise variance is used to compute a proxy measure for the prediction error (risk) as a function of the regularization parameter. The prediction error is the difference between the estimated solution and the true solution. The optimal parameter is then estimated as the minimizer of the proxy measure.

3.4.2.3.1 Monte-Carlo SURE

For the parameter selection using SURE in Equation (3.20), the divergence of the TV denoising operator with respect to the observed data $\mathbf{b}$ is to be computed. In the case of linear algorithms, the desired divergence becomes equivalent to the trace of the corresponding matrix transformation. However, in the absence of an appropriate functional form representation of the denoising operator or if the operator is nonlinear, as in the case of TV regularization, then the divergence computation is not straightforward. To overcome this difficulty, a Monte-Carlo simulation based on SURE is developed [9]. If the estimated signal $\hat{\mathbf{x}} = A_{\lambda}(\mathbf{b})$, and a zero-mean independent and identically distributed (i.i.d.) random vector $\mathbf{v}$ with unit variance and bounded higher order moments is given, then the second-order Taylor expansion of $A_{\lambda}(\mathbf{b} + \varepsilon \mathbf{v})$ is given by:

$$A_{\lambda}(\mathbf{b} + \varepsilon \mathbf{v}) = A_{\lambda}(\mathbf{b}) + \varepsilon J_{A_{\lambda}}(\mathbf{b}) + \varepsilon^2 \mathbf{r}_{A_{\lambda}}, \quad (3.92)$$

where $J_{A_\lambda}(\mathbf{b})$ is the Jacobian matrix of A_λ evaluated at $\mathbf{b}$, and $\mathbf{r}_{A_\lambda}$ represents the vector containing the Lagrange terms corresponding to each component of A_λ . Using the expansion in Equation (3.92), the divergence of the non-linear operator A_λ can be computed using:

$$\text{div}_v \{A_\lambda(\mathbf{b})\} = \lim_{\varepsilon \rightarrow 0} E_v \left\{ \mathbf{v}^T \left(\frac{A_\lambda(\mathbf{b} + \varepsilon \mathbf{v}) - A_\lambda(\mathbf{b})}{\varepsilon} \right) \right\}, \quad (3.93)$$

where E_v represents the expectation with respect to $\mathbf{v}$, and ε is a small positive value. However, as the limit in Equation (3.99) cannot be realized due to finite machine precision, an approximation is obtained by averaging of SURE over all the pixels. Using the estimated divergence, the SURE threshold is computed as the minimizer of the SURE risk in Equation (3.20). The steps for estimation of the SURE parameter are summarized below:

1. For a given $\lambda = \lambda_0$, evaluate $A_\lambda(\mathbf{b})$.
2. Get $\mathbf{z} = \mathbf{b} + \varepsilon \mathbf{v}$ and evaluate $A_\lambda(\mathbf{z})$ for $\lambda = \lambda_0$.
3. Compute $\text{div} = (1/N\varepsilon) \mathbf{v}^T (A_\lambda(\mathbf{z}) - A_\lambda(\mathbf{b}))$ and the SURE parameter using Equation (3.20).

Since a single realization can yield a sufficiently low variance estimate in image-processing applications, it is sufficient to use only one realization of $\mathbf{v}$ to implement Steps 1–3 in this approach [28,79]. This scheme can also be used to compute the optimal regularization parameter for both the variational and wavelet-based methods.

3.4.2.3.2 Unbiased Predictive Risk Estimator

The major difficulty associated with the extension of a UPRE approach to TV regularization is that the penalty functional in the TV term is non-quadratic, and hence a closed-form solution does not exist. Therefore, a quadratic approximation to the non-quadratic penalty is used. This is achieved as part of the lagged diffusivity algorithm [17] in which the TV term is approximated as $\|\mathbf{x}\|_{\text{TV}} = \|\varphi((D_x \mathbf{x})^2 + (D_y \mathbf{x})^2)\|_1$, where $\varphi(x) = \sqrt{x + \beta}$ and β denotes a small positive quantity. Then the gradient of the TV term $\nabla \|\mathbf{x}\|_{\text{TV}}$ at $\mathbf{x}_\lambda$ is $L(\mathbf{x}_\lambda) \mathbf{x}$, where $L(\mathbf{x}_\lambda)$ is given by:

$$L(\mathbf{x}_\lambda) = D_x^T \text{diag}(\varphi'(\mathbf{x}_\lambda)) D_x + D_y^T \text{diag}(\varphi'(\mathbf{x}_\lambda)) D_y. \quad (3.94)$$

Using $L(\mathbf{x}_\lambda)$, a closed-form solution to $\mathbf{x}_\lambda$ is obtained as follows:

$$\mathbf{x}_\lambda = \left(A^T A + \lambda L(\mathbf{x}_\lambda) \right)^{-1} A^T \mathbf{b}. \quad (3.95)$$

Hence, the UPRE functional for TV regularization can be computed as in Equation (3.16) using $A_\lambda = A(A^T A + \lambda L(\mathbf{x}_\lambda))^{-1} A^T$ [8]. The linear approximation of the solution in the TV case complicates the computation of UPRE in Equation (3.16) due to the computation of $\text{trace}(A_\lambda)$. Therefore, a direct computation of UPRE in the TV case imposes severe limits on the problem size due to computation time and memory requirements. Although the computation of Equation (3.16) is straightforward if the SVD decomposition of A is available, it is too expensive to compute the SVD of A in many large-scale problems. To overcome this difficulty, a low-dimension matrix approximate to the large-scale matrix of A_λ is first

computed using the Lanczos method [27]. The trace of the approximated matrix is then computed by a Monte Carlo test as proposed by Girard [28]. Given an i.i.d random vector $\mathbf{v}$ of zero mean and unit variance, the approximated trace is estimated using:

$$\text{trace}(A_\lambda) \approx E_{\mathbf{v}}(\mathbf{v}^T A_\lambda \mathbf{v}). \quad (3.96)$$

Following estimation of the trace value, the UPRE functional is computed using Equation (3.16). The two methods used to locate the minimizer of the UPRE functional are the exhaustive search and golden section search [80]. In the exhaustive search method, the parameter values are swept through all λ points to find the optimal parameter value corresponding to the minimum of the cost function. The golden section search method progressively reduces the interval for locating the minimum. This is based on the idea that the minimum lies within the interval defined by the two points adjacent to the point with the least value so far evaluated. Compared to the exhaustive search method, the golden section search method is faster as it computes the UPRE functional using a fewer number of search points.

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4

Multi-filter Calibration for Auto-calibrating Parallel MRI

4.1 Problems Associated with Single-Filter Calibration

The signal-to-noise ratio (SNR) of the magnetic resonance (MR) signal typically varies across different k -space points; that is, SNR near the DC component is much higher than SNR at the periphery of the k -space. This leads to the idea of non-stationarity of the k -space location and requires either a spatially adaptive or frequency-dependent regularization to minimize the reconstruction error. Several studies [1–5] have shown that estimation of kernel weights can be improved by treating low- and high-frequency signals differently. All these relate to the fact that the kernel weights are correlated to the k -space location. Therefore, the reconstruction can be improved by varying the regularization parameter in relation to the amplitude variation in the k -space data and varying levels of SNR.

A straightforward means to simulate varying levels of SNR in the calibration model is by addition of Gaussian white noise samples to the auto-calibration signal (ACS) lines. As the noise is added to both the calibration matrix and the measurement vector at the same time, it is more appropriate to use the generalized discrepancy principle (GDP) for selection of the regularization parameter. According to this principle, the filter weights should be chosen so that the residual error is the sum of upper bounds of errors in the measurement vector, calibration matrix and the incompatibility measure. Thus, one can only obtain estimates of error bounds using premises based on Monte Carlo methods.

4.2 Effect of Noise in Generalized Autocalibrating Partially Parallel Acquisitions (GRAPPA) Calibration

Although one would expect higher SNR of ACS lines to boost the accuracy of the kernel estimation, Sodickson et al. [6,7] has indicated that this leads to the paradoxical effect of reducing the SNR in GRAPPA reconstructed images. This effect is attributed to the increase in the condition number of the calibration matrix. Using Tikhonov regularization, as was

shown in Equation (2.12) in [Chapter 2](#), the stabilized filter coefficients in Equation (1.40) can be expressed as follows:

$$\mathbf{z}_r^l = (\Pi^H \Pi + \lambda I)^{-1} \Pi^H \mathbf{k}_{ur}^l \quad (4.1)$$

where parameter λ is the regularization strength. The entries of Π comprising the k -space points in the ACS can be represented as the summation of a signal matrix Π_s and a noise matrix Π_n , that is, $\Pi = \Pi_s + \Pi_n$. From the expansion using random matrix theory involving a Gaussian assumption and considering the fact that $M \gg N \gg 1$, the LS solution for Equation (1.40) asymptotically approximates to [\[6,8\]](#):

$$\mathbf{z}_r^l \approx (\Pi_s^H \Pi_s + \sigma_n^2 I)^{-1} \Pi_s^H \mathbf{k}_{ur}^l \quad (4.2)$$

Given Π_n has independent and identically distributed (IID) entries, its smallest singular value is determined by the noise standard deviation. Furthermore, if matrix Π_s is rank deficient, then its smallest singular value is zero. More information on the singular value decomposition (SVD) of random matrices with IID entries can be found in [\[9–11\]](#). Since the largest singular value of Π_s is proportional to the signal strength of the ACS lines, the condition number of Π can be considered proportional to the SNR. This essentially points to the fact that the minimum possible relative l_2 -norm error (RLNE) [\[12–14\]](#) of the reconstructed image is dominated by random noise for low SNR of ACS, and otherwise by aliasing artifacts.

4.3 Monte Carlo Method for Prior Assessment of the Efficacy of Regularization

This section introduces a Monte Carlo method that determines the effectiveness of the regularized reconstruction before application of the reconstruction procedure. This method includes repeated addition of Gaussian white noise samples to the given noisy calibration data. Before noise addition, a threshold singular value that indicates the transition from the decaying to flat Fourier coefficients is determined. This value is chosen so that it is higher than the norm of perturbation error due to noise addition in the ACS data. Also, in each step of noise addition, one needs to ensure that the calibration at that noise level satisfies the GDP condition, requiring the residual error norm to exceed the norm of perturbation error. The cross-over point is identified as the perturbation step for which both residual and perturbation error norms are nearly equal. As long as the perturbation error norm is less than the norm of residual error, the cross-over point may be reached by successive perturbation of the calibration data. The cross-over information allows one to determine an upper bound for perturbation, enabling computation of an error bound computed as the difference between reconstructed k -spaces obtained from calibrations performed on the original and perturbed ACS at cross-over. An upper limit

on the amount of regularization at each unacquired k -space location is determined by matching this error bound to the perturbation in k -space values. The perturbed k -spaces are reconstructed with Tikhonov filters using different regularization parameter values. The possibility of over-regularization is controlled by proper selection of the Tikhonov filters at each frequency point. This selection is achieved by choosing a filter with the largest λ that perturbs the reference estimate at each unacquired k -space location by an amount equal to the error bound at that location. The amount of perturbation (number of noise additions) required to attain the cross-over is an indication of possible noise reduction in the regularized reconstruction. Therefore, the noise reduction without significant loss of image resolution can be achieved only through the employment of an equal or larger number of Tikhonov filters in the reconstruction. The relative difference between minimum singular values of the unperturbed and cross-over calibration matrices shows a positive correlation to the reduction in reconstruction errors obtained using regularization. The cross-over approach is valid for both GRAPPA and iterative self-consistent parallel imaging reconstruction (SPIRiT) type reconstructions.

4.4 Determination of Cross-over

In some cases, the unregularized solution gives better reconstruction than regularized ones, particularly at low condition numbers. This can be explained using the idea of a cross-over point identified by controlled noise addition to the ACS data. This section outlines a numerical algorithm to check the appropriateness of employing a regularized form of GRAPPA and SPIRiT calibration based on perturbation of singular values resulting from the controlled noise addition.

4.4.1 Perturbation of ACS Data for Determination of Cross-over

As noise is added to the acquired k -space locations, the superscript 0 is used to denote the initial unperturbed, and t the perturbed quantity at the t 'th step. With perturbation using the additive Gaussian noise, we have:

$$\tilde{K}_{ACS}^t = K_{ACS}^0 + E^t \quad (4.3)$$

where K_{ACS}^0 denotes the initial unperturbed ACS data, $\tilde{K}_{ACS}^t$ denotes the perturbed ACS and E^t denotes the perturbation due to additive noise. According to the perturbation theory for truncated SVD, $\|E^t\|_2 < \sigma_k^0$ [15,16]. Since the calibration matrix is directly formed out of ACS, Equation (4.3) implies that the perturbation in the calibration matrix can be equivalently represented as $\tilde{\Pi}^t = \Pi^0 + E^t$. Since the threshold singular value remains unchanged throughout the experiment, the noise variance is chosen so that the singular values are perturbed only by a small amount. With M number of rows of Π^0 and additive Gaussian noise with standard deviation σ_0 , this is ensured by keeping the perturbation error norm

such that $\|E^t\|_2 \geq \sigma_0 \sqrt{M}$ in each step of perturbation. The singular value update in each step is performed by projection of E^t onto the original subspace. Following computation of the calibration weight vector $\mathbf{z}_k^t$ using

$$\mathbf{z}_k^t = \sum_{j=1}^k \frac{\mathbf{u}_j^{0T} \mathbf{k}_u^t}{\check{\sigma}_j^t} \mathbf{v}_j^0, \quad (4.4)$$

the resulting norm of residual error $\|\check{\Pi}^t \mathbf{z}_k^t - \mathbf{k}_u^t\|_2$ is compared to $\|E^t\|_2$. With the incremental noise addition steps, both the error norms increase until they become equal at the cross-over point. Since the value of σ_0 determines the accuracy of cross-over estimation as well as the number of steps T_v to reach cross-over, this choice is determined based on how large $\|E^t\|_2$ should be in comparison to a minimum possible value for the norm as obtained from the difference between the lowest singular values of Π^0 .

4.4.2 First Order Update of Singular Values

The singular values of the perturbed calibration matrix $\check{\Pi}^t$ at the t th step can be computed using the perturbation expansion. The perturbation expansions can provide an approximation to the perturbed data when the perturbation error is known [17]. With the assumption that the singular values are not repeated and the error is sufficiently small, a perturbation expansion for $\check{\sigma}_j^t$ takes the following form [18]:

$$\check{\sigma}_j^t = \sigma_j^0 + \mathbf{u}_j^{0T} E^t \mathbf{v}_j^0 + O(\|E^t\|_2^2) \quad (4.5)$$

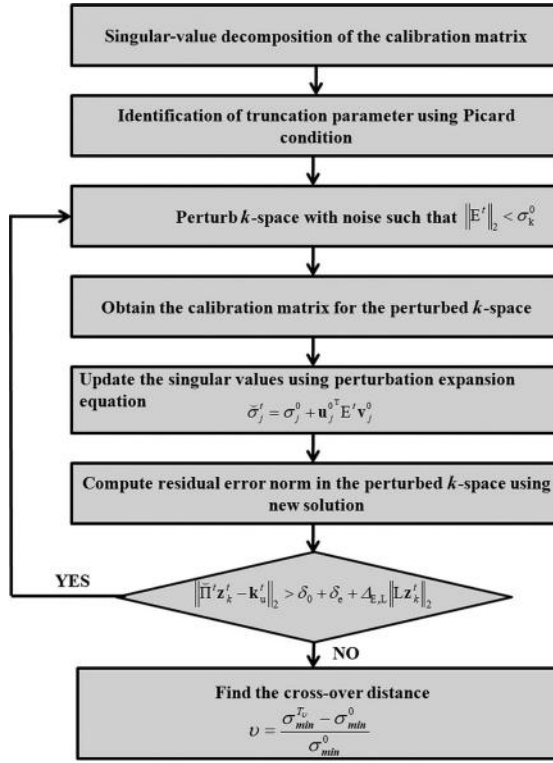
The accuracy of first-order expansion is guaranteed if $\|E^t\|_2^2$ is sufficiently small. If the separation between two consecutive singular values is denoted by δ , then the second-order term is approximately bounded by $\|E^t\|_2^2 / \delta$. In accordance to the theorems by Weyl [19] and Mirsky [20], $\|E^t\|_2$ also serves as an upper bound for the difference between the perturbed and original singular values.

4.4.3 Application of GDP

Application of GDP in each noise perturbation step yields [21]:

$$\|\check{\Pi}^t \mathbf{z}_k^t - \mathbf{k}_u^t\|_2 = \delta_0 + \delta_e + \Delta_{E^t, L} \|\mathbf{L} \mathbf{z}_k^t\|_2, \quad (4.6)$$

where L denotes the regularization matrix, δ_0 is the incompatibility measure (since the system is assumed to be consistent, $\delta_0 = 0$), δ_e is the upper bound for the errors in the observation vector and $\Delta_{E^t, L}$ is an upper bound for $\max_{L \mathbf{z}_k^t \neq 0} \{\|E^t \mathbf{z}_k^t\|_2 / \|\mathbf{L} \mathbf{z}_k^t\|_2\}$ given by the largest generalized singular value of the pair (E^t, L) . In order for the discrepancy principle to hold true for each noise addition step, the induced perturbation error norm (sum of last two terms on the right-hand side [RHS]) should be less than the residual error norm on the left-hand side (LHS).

**FIGURE 4.1**

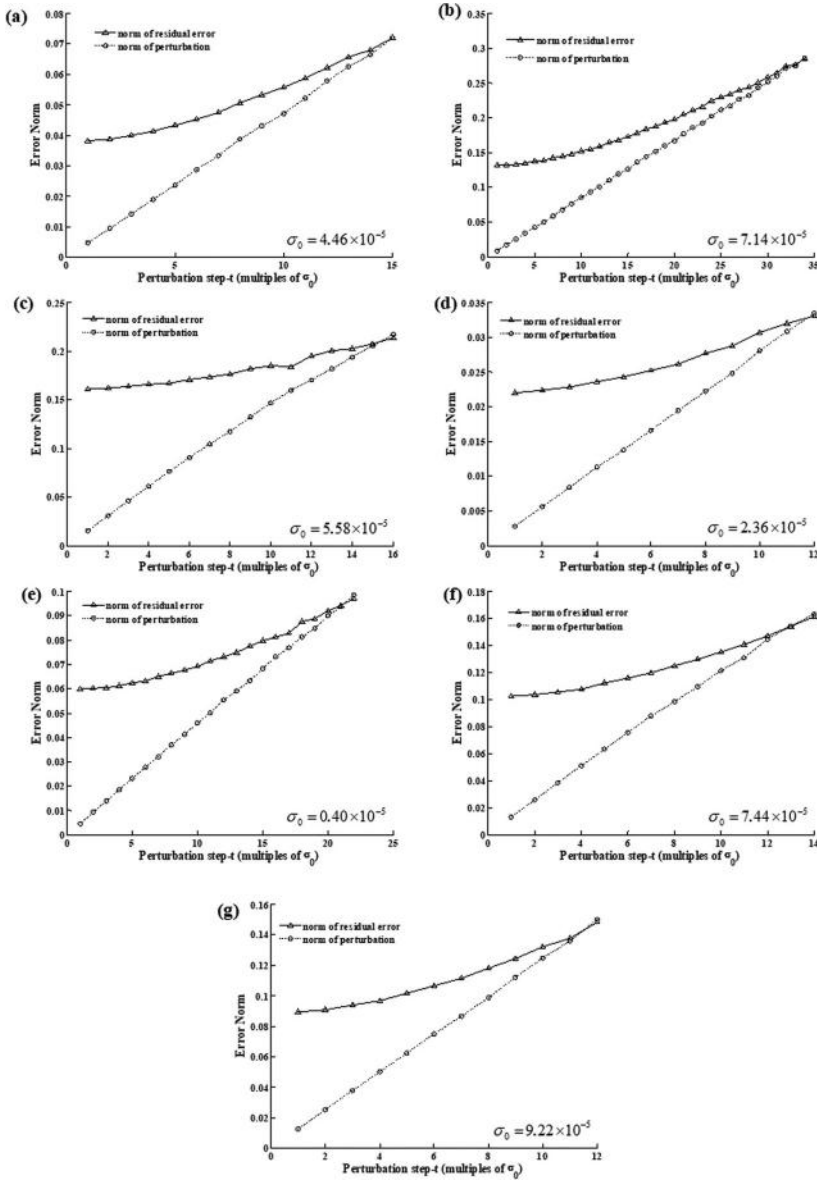
Workflow of cross-over estimation.

4.4.4 Determination of Cross-over

The transition point at which the GDP ceases to be true is called the cross-over. If the GDP criterion $\|\tilde{\Pi}^t \mathbf{z}_k^t - \mathbf{k}_u^t\|_2 > \|E^t\|_2$ is satisfied in a particular step, the cross-over point is then found by repeated noise addition to the calibration data, as outlined in [Sections 4.4.1–4.4.3](#). A detailed workflow is provided in [Figure 4.1](#). After addition of the complex Gaussian noise of variance σ_0^2 to the ACS lines in each step, the singular values of the new calibration matrix $\tilde{\Pi}^1$ are estimated using the first-order perturbation expansion approximation. The test based on application of GDP is then performed by comparison of $\|\tilde{\Pi}^1 \mathbf{z}_k^1 - \mathbf{k}_u^1\|_2$ and $\|E^1\|_2$. The perturbation steps are repeated T_v times until a cross-over point is reached when $\|E^{T_v}\|_2$ just crosses over (exceeds) $\|\tilde{\Pi}^{T_v} \mathbf{z}_k^{T_v} - \mathbf{k}_u^{T_v}\|_2$. The cross-over distance ν is calculated as:

$$\nu = \frac{\tilde{\sigma}_{\min}^{T_v} - \sigma_{\min}^0}{\sigma_{\min}^0} \quad (4.7)$$

Norms of residual and perturbation errors plotted against standard deviation of Gaussian noise added in each perturbation step as multiples of σ_0 are shown in [Figures 4.2 and 4.3](#) respectively. Each panel shows the error norms computed for a different dataset. A brief

**FIGURE 4.2**

(a–g) Norms of residual and perturbation error as functions of perturbation noise level for GRAPPA calibration. Norm of residual error at each perturbation step is indicated using triangular markers and perturbation error norm using circles. Each step (t) represents perturbation of calibration data using additive zero mean Gaussian noise with variance σ^2 .

description of the acquisition details of each dataset is provided in first three columns of [Tables 4.1](#) (GRAPPA) and [4.2](#) (SPIRiT). The residual error norm is typically higher than the perturbation error norm in all datasets at the start of the Monte Carlo test. Although both residual and perturbation errors increase with increasing number of perturbation steps, the residual error norm increases slowly compared to the perturbation error norm $\|E^t\|_2$.

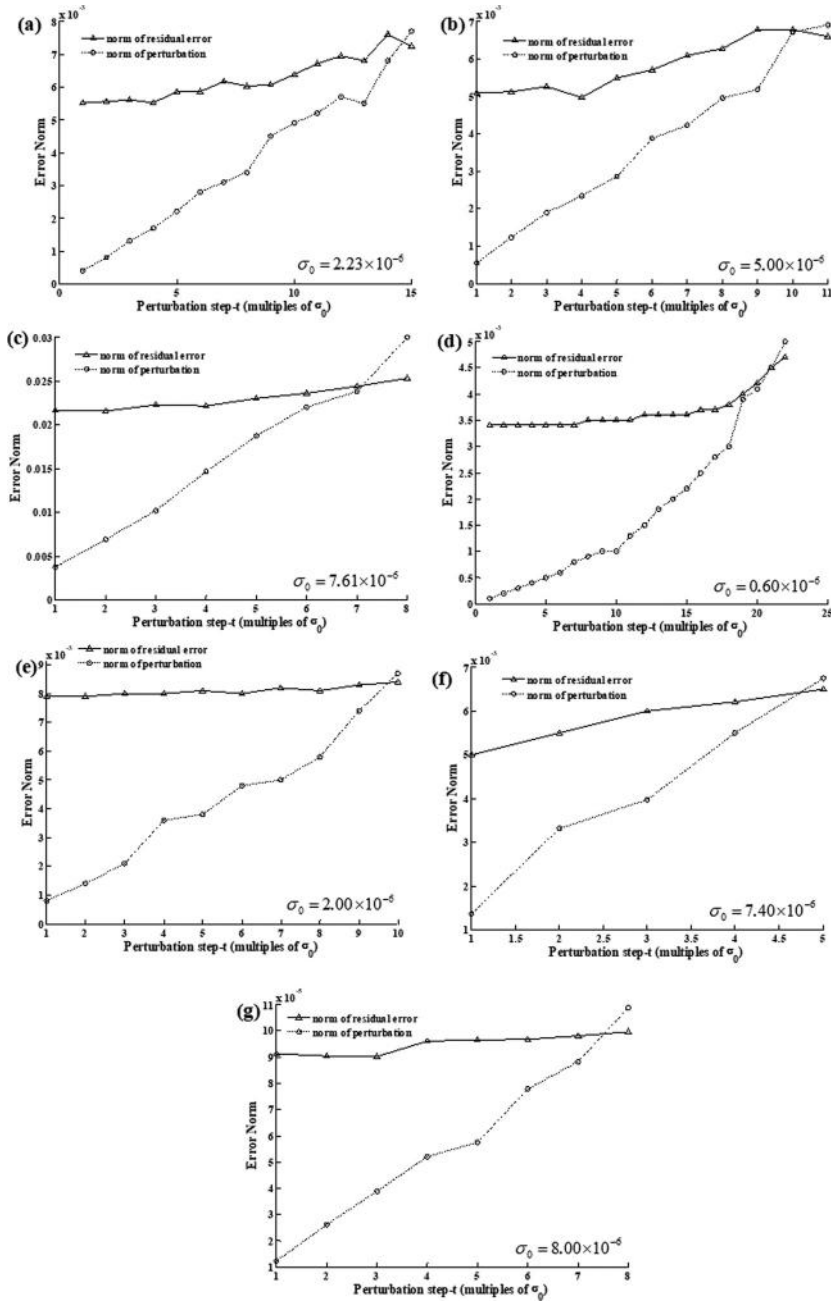


FIGURE 4.3

(a–g) Norms of residual and perturbation error as functions of perturbation noise level for SPIRiT calibration. Norm of residual error at each perturbation step is indicated using triangular markers and perturbation error norm using circles. Each step (t) represents perturbation of calibration data using additive zero mean Gaussian noise with variance σ^2 .

TABLE 4.1Parameters of Cross-Over Determination in GRAPPA Calibration for Datasets I to VIII with $R = 3$

Dataset	Pulse Sequence	Number of Channels	n_{ACS}	M	σ_k^0	$\sigma_{\min}^0$	$\sigma_0 \times 10^{-5}$	T_b	ν
I	FLAIR	16	32	2816	0.1049	0.0171	0.3424	11	1.20
II	T ₂ -weighted spin echo	6	32	3520	0.0307	0.0286	0.5744	22	2.90
III	SWI gradient echo	4	24	4032	0.0482	0.0475	9.5040	12	1.12
IV	BRAVO	12	32	2560	0.1699	0.00978	0.1995	13	2.46
V	BRAVO	12	18	1024	0.1047	0.0135	0.2720	17	2.68
VI	T ₂ -weighted spin echo	4	26	4032	0.2049	0.0311	0.6226	13	1.09
VII	BRAVO	12	28	2048	0.3158	0.0371	0.7428	9	1.74
VIII	BRAVO	12	22	1536	0.2103	0.0212	0.4235	11	1.89

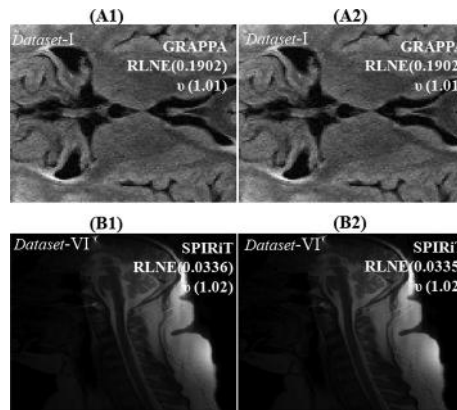
TABLE 4.2Parameters of Cross-Over Determination in SPIRiT Calibration for Datasets I to VII with $R = 3$

Dataset	Pulse Sequence	Number of Channels	n_{ACS}	M	σ_k^0	$\sigma_{\min}^0$	$\sigma_0 \times 10^{-5}$	T_b	ν
I	FLAIR	16	32	784	0.0414	0.0011	2.11	20	1.09
II	T ₂ -weighted spin echo	6	32	294	0.0369	0.0024	4.89	10	2.01
III	SWI gradient echo	4	24	196	0.0435	0.0029	5.86	7	1.17
IV	BRAVO	12	32	588	0.0213	0.0003	6.51	32	2.32
V	BRAVO	12	18	588	0.0202	0.0008	1.52	13	1.78
VI	T ₂ -weighted spin echo	4	26	196	0.0324	0.0012	2.32	15	1.08
VII	BRAVO	12	28	588	0.0055	0.0042	5.00	10	1.77
VIII	BRAVO	12	22	588	0.0301	0.0039	1.55	11	2.01

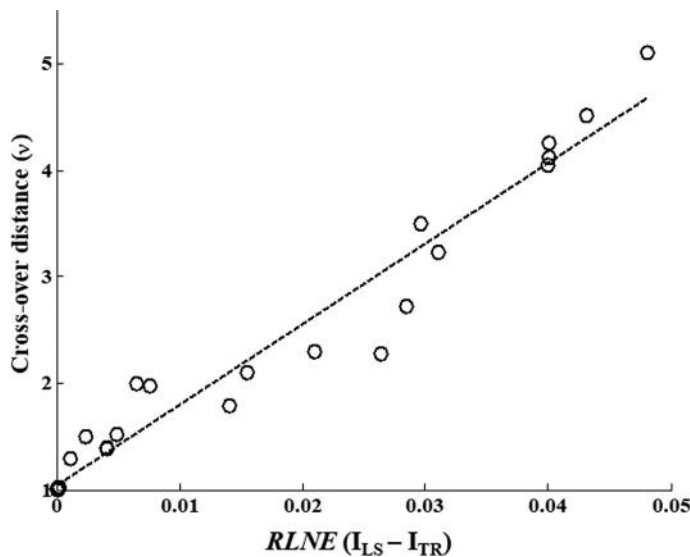
At a particular step, $\|E^t\|_2$ crosses the residual error norm and thereafter exhibits values higher than residual error norm.

For an acceleration factor $R = 3$, the number of ACS lines (n_{ACS}); the number of rows (M) in Π^0 , σ_k^0 , $\sigma_{\min}^0$, σ_0 , T_b ; and the cross-over distance (ν) for GRAPPA and SPIRiT are listed in [Tables 4.1](#) and [4.2](#), respectively. Two types of datasets can be identified from the tables based on the values of their cross-over distances. The regularized reconstructions are not superior to least squares (LS) if the ν -values are closer to unity. In the case of GRAPPA, datasets II, IV, V, VII and VIII exhibit higher ν -values. In SPIRiT, datasets II, IV, V, VII and VIII similarly exhibit ν -values exceeding 1.5, which is indicative of better reconstruction achievable with regularization. [Figure 4.4](#) shows GRAPPA and SPIRiT reconstructions using the first type of datasets, which do not show marked difference between LS and regularization. The corresponding RLNE values and ν are indicated in the insets.

[Figure 4.5](#) shows the scatter plot of ν -values and RLNEs of differences between images reconstructed using LS GRAPPA and Tikhonov regularized GRAPPA for acceleration

**FIGURE 4.4**

LS (A1 and B1) versus regularized (A2 and B2) reconstruction not showing marked differences. The type of reconstruction, RLNE value and cross-over deviation (v) are provided in insets.

**FIGURE 4.5**

Scatter plot of cross-over distance and RLNE of difference between images reconstructed using LS GRAPPA (I_{LS}) and Tikhonov regularized GRAPPA (I_{TR}).

factors $R=2$ and $R=3$. Large v -values correspond to large differences between regularized and unregularized reconstructions, with a correlation coefficient of 0.9789.

The small singular values in the GRAPPA calibration matrix are known to contribute to the noise amplification due to matrix inversion during calibration. If the condition number is too high, the calibration equations become ill-conditioned, and the estimated GRAPPA kernel is dominated by amplified random noise. Consequently, SNR in the corresponding GRAPPA-reconstructed images is reduced. If appropriate regularization is not incorporated, datasets with high SNR ACS lines (having high condition numbers in the calibration) can actually degrade the SNR of GRAPPA-reconstructed images [6,7,22].

With stronger regularization, random noise is suppressed, but image artifacts and blur become more pronounced.

The main point here is that the quality of regularized reconstruction may not always supersede that of LS even under high SNR conditions. Regularization is also required under low SNR conditions to stabilize the solution. For the same dataset, this depends on the exact acquisition conditions, such as number of ACS lines, acceleration factor, number of coils, type of kernels, and so on. Since noise levels may be assumed to be constant throughout the k -space, additional constraints need to be applied for restriction of over-regularization at k -space locations with low signal amplitudes.

The solution norm and residual norm increase with increasing levels of perturbation. Whereas datasets with large ν -values exhibit lower rates of increase in the solution norm, datasets exhibiting ν -values closer to unity exhibit higher rates of increase. The rate of increase can be equivalently described in terms of the norm of the difference between solutions at two consecutive perturbation levels. This can be directly deduced from the theorem (due to P.C. Hansen [23], cf. Theorem 3) and the triangular inequality (as shown in the appendix in Section 4.7). Thus the difference in solution norms at two successive perturbation levels should be less than the sum of the residual errors at the two levels. Since the residual error norms are higher and the norms of the solution difference are lower for large cross-over datasets, the theorem is well satisfied and conforms to the idea that the datasets exhibiting higher cross-over distances are more stable with noise-induced perturbations.

4.5 Multi-filter Calibration Approaches

Most of the existing calibration methods aimed at improving the GRAPPA estimation are based on the assumption of stationarity, which does not consider the data characteristics. In the standard approach, GRAPPA calibration is performed on ACS data that exhibit large amplitude variation, and the estimated filter coefficients are then applied to signals with small amplitude variation for reconstruction. As pointed out by Park et al. [2], the huge differences in signal variations may cause large reconstruction errors. To introduce stationarity of kernel weights, they excluded the centre ACS lines that exhibit large amplitude variation from the calibration matrix. With a similar idea, Yeh et al. [7] utilized the k -space locality principle to develop the parallel magnetic resonance imaging with adaptive radius in k -space (PARS) algorithm for efficient pMRI reconstruction. The k -space locality principle is based on the assumption that only a few nearby points within a small k -space radius contribute to the determination of the synthesized samples. However, the method uses the same assumption as that of standard GRAPPA, and a simple k -space radius constraint may not be sufficient for modeling the non-stationarity of kernel weights.

Similar attempts have been made by Huang et al. [3] in which the ACS data is filtered using a high-pass filter prior to the calibration process. Another approach involves application of additional weights using a bivariate Gaussian function to the regression equations before estimation of the kernel weights [4]. Although such a weighting strategy is considered more general than the selective sample strategy, its efficiency is limited due to an existing shift in the k -space centre or multiple significant components in k -space. Furthermore, algorithms require integration of ACS for final image reconstruction [1–4].

This section focuses on algorithms that do not require ACS integration and takes the local k -space data characteristics into account during the calibration process to compute the non-stationary kernel weights accordingly. Compared to the aforementioned approaches, these rely on the local dependency of the independent variables and can be regarded as a form of localized linear regression. Based on the fact that the non-stationarities can be quantified using localized parameters, several modifications involving multiple sets of kernel weights (filters) have been introduced. This includes modeling non-stationarity of kernel weights (MONKEES) [5], spatially variant GRAPPA (SV-GRAPPA) [24] and frequency-dependent regularization (FDR) [25].

As the k -space magnitudes decrease with increasing distance from the k -space centre, spatially varying weights can be computed by partitioning the k -space into clusters based on the average data magnitude. Utilizing this idea, separate GRAPPA weights are estimated for each cluster in MONKEES GRAPPA. Alternatively, SV-GRAPPA models the non-stationarity in an intermediate space based on the fact that the coil sensitivities are slowly varying along the x -direction, and this variation modulates the GRAPPA calibration weights for estimation of the unknown samples along the x -direction at different PE levels in the intermediate domain. Another straightforward way to account for non-stationarity is to vary the regularization parameter as a function of the k -space location, as in FDR.

4.5.1 MONKEES

In place of regularization, a spatially weighted linear regression model is used in the MONKEES approach [5]. The weights are adapted according to the signal variation in k -space. The kernel weights are estimated using spatially localized linear regression by partitioning the k -space into P clusters based on the average data magnitude [26]. As the k -space magnitudes decrease with increasing distance from the k -space centre, clustering based on distance from the k -space centre is employed. In its most basic form, this is accomplished by partitioning the k -space into low- and high-frequency regions. For each acquired point in k -space, a vector of dimension $(1 \times n_c)$ is formed, with its l th element being the average signal magnitude from l 'th coil. These n_c -dimensional feature vectors are then partitioned into clusters using the k -means algorithm. Separate GRAPPA weights are then estimated for each cluster using geographical weighted regression (GWR) [4]. The geographical weights are obtained using a weight matrix W , whose columns contain weights specific to each cluster. The rows correspond to each acquired point in the ACS. Each matrix element represented by w , is determined based on the distance between feature vectors at that point and the mean vector $\bar{g}_i$ of each cluster. The distance between feature vectors at each acquired ACS location and the cluster mean is given by:

$$d_{\eta,i}(h) = \|\vec{\mu}_{h,\eta} - \bar{g}_i\|_2 \quad (4.8)$$

where h denotes the index of each acquired point in the ACS. Then the diagonal entries of the weighting matrix $W_{\eta}(i)$ are obtained by using the Shepard inverse distance-weighted function as in [27]:

$$w_{\eta,i}(h) = \frac{1/d_{\eta,i}(h)}{\sum_{i=1}^P 1/d_{\eta,i}(h)} ; \text{ for } h = 1, 2, 3, \dots, H, \quad (4.9)$$

where H denotes the number of rows in Π . Then the spatially varying kernel weights are computed using:

$$\mathbf{z}_r^l(i) = \left(\Pi^H W_\eta(i) \Pi \right)^{-1} \Pi^H W_\eta(i) \mathbf{k}_{ur}^l(\eta). \quad (4.10)$$

During estimation, the representative GRAPPA weights corresponding to the cluster yielding the minimum distance between its mean vector and the feature vector constructed from the neighborhood of the estimated k -space location are used for reconstruction. A frequency-discriminated form of k -space partitioning assumes the influence of coil sensitivity in the low-frequency region, and image details in the high-frequency partition. Thus, the kernel weights in the low-frequency partition are highly sensitive to the coil sensitivity profile. Compared with normal GRAPPA, a main advantage of this approach arises from the low residual errors in the low-frequency partition. The relatively high residual from the high-frequency partition affects the image quality only insignificantly. However, the relative reduction in low-frequency errors is largely influenced by the coil sensitivity profile.

Figure 4.6 shows images reconstructed using LS-GRAPPA [28], tailored GRAPPA [29], discrepancy GRAPPA [30], iterative GRAPPA (iGRAPPA) [31], high-pass (hp) [3], virtual

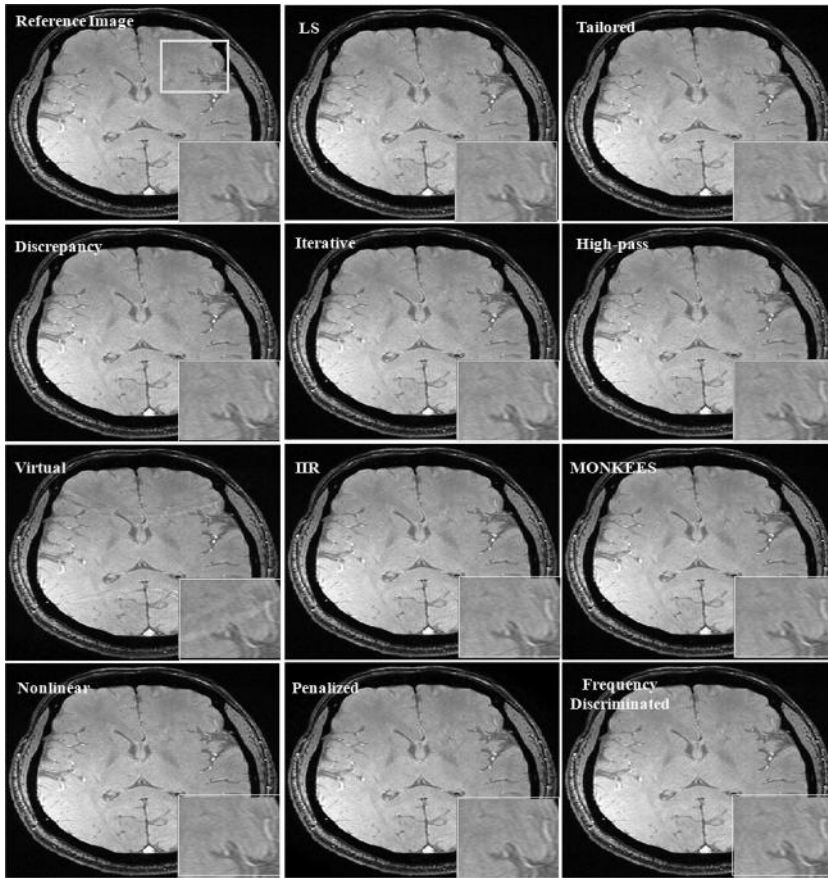
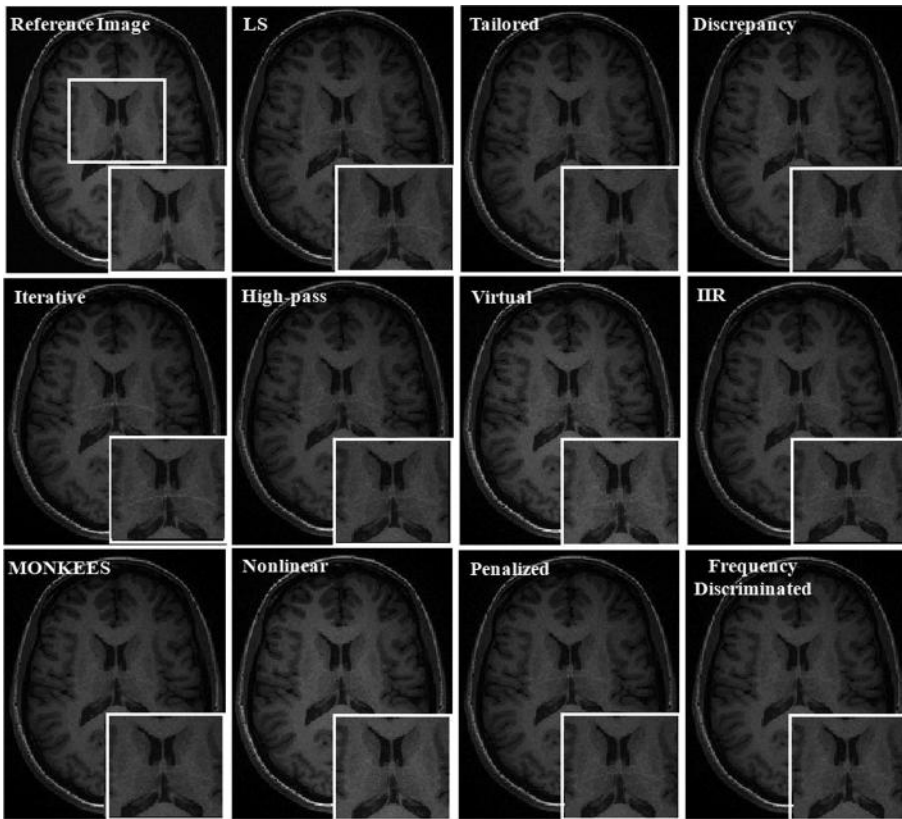


FIGURE 4.6

Reconstructed images from dataset III. For clarity, zoomed versions of the sub-images are shown as insets in the bottom left corner.

**FIGURE 4.7**

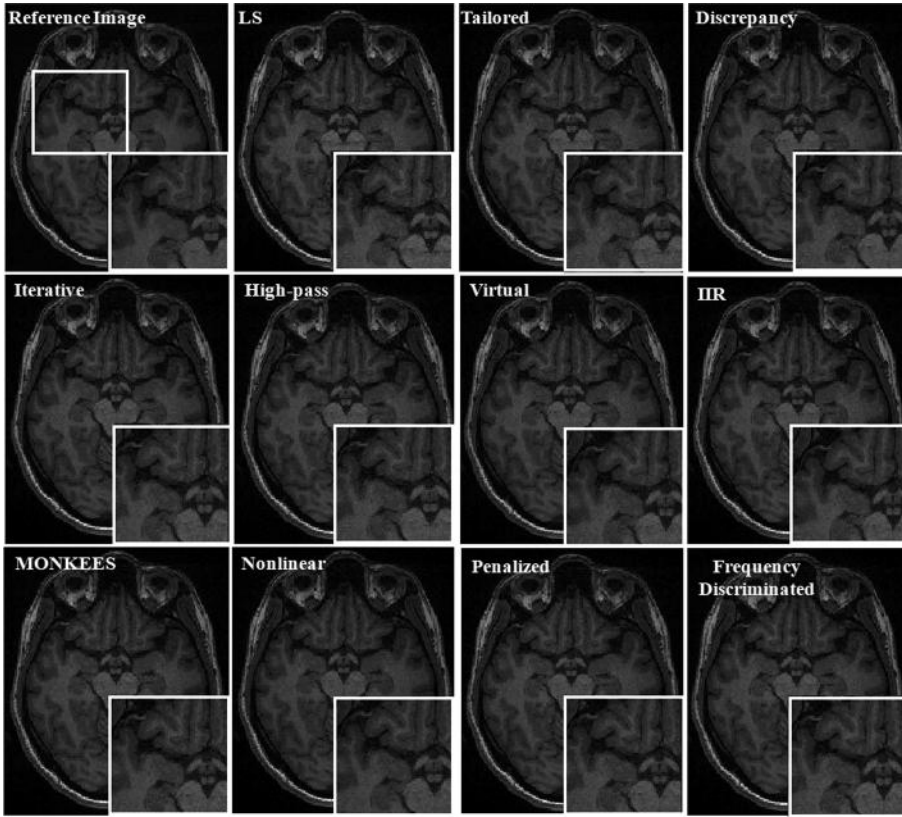
Reconstructed images from dataset V for $R = 3$. For clarity, zoomed versions of the sub-images are shown as insets in the bottom left corner.

GRAPPA [32], IIR-GRAPPA [33], MONKEES, non-linear (NL) GRAPPA [34], penalized GRAPPA [35] and frequency discriminated (FD) [36]. For two-fold acceleration of four-channel images (dataset III), all GRAPPA reconstructions are seen to be artifact-free except virtual GRAPPA. Visual inspection reveals there is very little change between the different variants of GRAPPA.

Figure 4.7 shows GRAPPA reconstructions of dataset V using $R=3$. In the zoomed versions, NL, MONKEES and discrepancy-based methods are shown to yield artifact-suppressed reconstructions. Aliasing artifacts are clearly visible in all other GRAPPA variants.

Figure 4.8 shows reconstructed images for three-fold acceleration of dataset VIII. It is seen that LS and tailored reconstructions exhibit visible artifacts. Other GRAPPA variants are artifact free and exhibit very little changes.

Figure 4.9 illustrates GRAPPA reconstruction in the case of four-fold acceleration of dataset I . To switch the coil number to a higher value ($n_c = 16$), the acquisition was performed in the parallel imaging mode (iPAT) with prospective under-sampling at $R=2$. Additional lines were replaced with zeros to yield a sampling pattern corresponding to $R=4$. NL GRAPPA has the lowest noise amplification and improved white matter smoothness. Visual inspection reveals that all methods introduce aliasing artifacts.

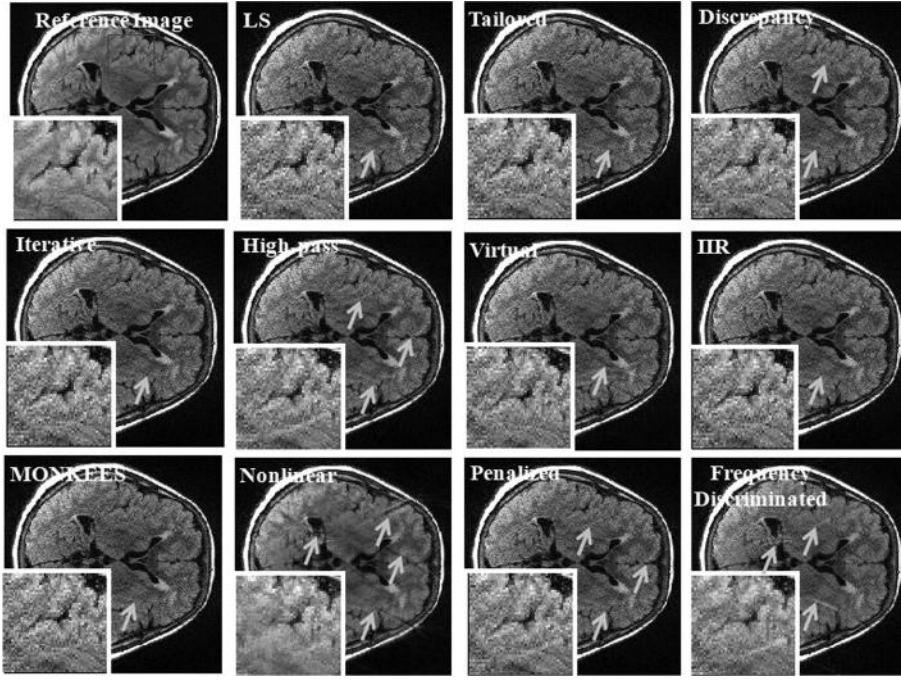
**FIGURE 4.8**

Reconstructed images from dataset VIII for $R = 3$. For clarity, zoomed versions of the sub-images are shown as insets in the bottom left corner.

4.5.2 SV-GRAPPA

As the coil sensitivities $C_c(x, y)$ vary slowly in image space, spatially invariant filter coefficients based on global coil sensitivity characteristics are optimal only when the coil sensitivities are independent of the x -coordinate. However, this assumption is not true for typical receiver coils used in MRI. SV-GRAPPA overcomes this disadvantage by spatially varying the calibration coefficients in accordance to the local coil sensitivity characteristics [24]. In case of pMRI with sampling on the regular 2D Cartesian grid, calibration coil coefficients $\mathbf{z}_r^l(x, k_y)$ can be directly computed from the ACS lines, as in the case of standard GRAPPA [28]. Since the number of ACS data available for each x -position is limited for SV-GRAPPA, the resultant fitting becomes ill determined.

The two types of approaches used to resolve this problem are based on the idea that the coil sensitivities can be described by slowly varying functions in an intermediate domain, and therefore $\mathbf{z}_r^l(x, k_y)$ is also slowly varying relatively to the x -coordinate [24]. In the first approach, the ACS data are divided into overlapping blocks along the x -direction. A set of coefficients are then calculated for each block, and the coefficient sets are interpolated to find $\mathbf{z}_r^l(x, k_y)$ for all x -values. In the second approach, $\mathbf{z}_r^l(x, k_y)$ is expressed using first few terms of the Fourier series as follows:

**FIGURE 4.9**

Reconstructed images of dataset *I* with acceleration $R=4$. Top left panel shows the GRAPPA-reconstructed image using the original acquired data with $R=2$. The region enclosed by the bounding box is chosen to highlight the extent of white-matter smoothing in each reconstruction. For clarity, zoomed versions of the sub-images are shown as insets in the bottom left corner. Arrows indicate regions with aliasing artifacts.

$$\mathbf{z}_r^l(x, k_y') = \sum_m C(m, k_y') \exp(\gamma x m), \quad (4.11)$$

where $\gamma = 2\pi / N_x$, $m = -(N_m - 1)/2, -(N_m - 3)/2, \dots, (N_m - 1)/2$, and N_m is an odd integer. Initially, $C(m, k_y')$ is estimated using the fitting procedure with all available auto-calibrating data (highly over-determined problem). The calibration coefficients $\mathbf{z}_r^l(x, k_y')$ are then computed as in Equation (4.11).

The number of computational operations required to reconstruct the missing k -space lines for SV-GRAPPA is the same as that of standard GRAPPA when the number of reconstruction coefficients is the same for both approaches and the coefficients are known. However, the number of computations required for estimation of coefficients in SV-GRAPPA is dependent on the number of x -blocks and their width for the first approach, and the factor N_m for the second approach. Typically, it is higher than the number of computations required for coefficient estimation in standard GRAPPA.

4.5.3 Reconstruction Using FDR

In general, any form of regularization procedure used to estimate the calibration coefficients provides an unbiased estimation of the unacquired k -space points. Therefore, the extent of regularization is determined a priori using standardized procedures for cross-validation, or other related techniques such as L-curve during the calibration phase. These

are effective only when signal variations in relation to noise level in the accelerated regions are similar to those in the ACS. Also since a single regularized solution is not sufficient to bring about the required trade-off between noise and resolution, any form of deviation from the regularization prior due to signal variations in relation to noise can introduce loss of resolution and over-regularization effects. To overcome the associated difficulties, the range of SNR variations in k -space is modeled by repeated noise perturbation of calibration data and specification of a cross-over error bound. The error bound can be used to control the extent of over-regularization at each frequency location.

As each perturbation step satisfies the Weyl's theorem, the same k -value is used until the cross-over point is reached. Accordingly, the squared sum of the $(n-k)$ singular values of the perturbed matrix should be less than the perturbation error norm at cross-over $\|E^{T_0}\|_2$ [37]. Also since the data matrix has full rank throughout the perturbation steps, the Picard condition is not violated and the magnitudes of all Fourier coefficients lie below the singular values at the cross-over point. This directly follows from a corollary of the Weyl's theorem, according to which the norm of the difference between two matrices, one of which has full rank (n) and the other has reduced rank (k), will be greater than or equal to the squared sum of $(n-k)$ singular values of the full rank matrix. This means that the norm of the difference between the calibration matrices at cross-over and the preceding step will be greater than the squared sum of the $(n-k)$ singular values of the calibration matrix at the cross-over step. In abstract terms, the cross-over distance indicates the admissible extent of SNR variations that the regularizer would tolerate during the reconstruction. One can expect a higher degree of noise reduction with adoption of a regularized form of filter in datasets with $\nu > 1.5$.

In the reconstruction phase, the truncation parameter should be adapted in accordance with the SNR variations in k -space [5,24,38–40]. On the other hand, the first k terms used for filter calibration can be fixed with the inclusion of a spatially dependent filter factor $\alpha_j(k_x, k_y)$, as follows:

$$\mathbf{z}(k_x, k_y) = \sum_{j=1}^k \alpha_j(k_x, k_y) \frac{\mathbf{u}_j^{0T} \mathbf{k}_u^0}{\sigma_j^0} \mathbf{v}_j^0 \quad (4.12)$$

When the same filter is assigned to a set of neighboring locations with inclusion of only the first k terms, a one-to-one correspondence to $\mathbf{z}_k^t$ can be established by comparison of Equation (2.5) in Chapter 2 and Equation (4.12) such that:

$$\alpha_j(k_x, k_y) = \frac{\sigma_j^0}{\tilde{\sigma}_j^t}, \quad (4.13)$$

where t corresponds to the noise perturbation step in which the effect of added noise level $t\sigma_0$ closely approximates the filtering performed at an outer k -space location (k_x, k_y) . With the assumption that the signal levels monotonically decrease with increasing distance from the k -space centre, this is identical to applying the filters $\mathbf{z}_k^t$ with increasing t -values to regions with increasing distance from the k -space centre. Due to perturbation of the calibration data, such an implementation would not be ideal [41]. An improved fit can only be achieved with inclusion of additional terms $j = k+1, k+2, \dots, n$, with a minimum value imposed on the choice of $\alpha_j(k_x, k_y)$ to control the level of smoothing. Analogous to the Tikhonov solution $\mathbf{z}_\lambda$ [42,43] using SVD representation, the filter estimation now takes the following form:

$$\mathbf{z}_\lambda(k_x, k_y) = \sum_{j=1}^k \alpha_j(k_x, k_y) \frac{\mathbf{u}_j^{0T} \mathbf{k}_u^0}{\sigma_j^0} \mathbf{v}_j^0 + \sum_{j=k+1}^n \alpha_j(k_x, k_y) \frac{\mathbf{u}_j^{0T} \mathbf{k}_u^0}{\sigma_j^0} \mathbf{v}_j^0, \quad (4.14)$$

where the regularization parameter λ is such that:

$$\alpha_j(k_x, k_y) = \frac{\sigma_j^{02}}{\sigma_j^{02} + \lambda^2(k_x, k_y)} \quad (4.15)$$

As the regularization parameter estimated using GCV ($\mathbf{z}_{\lambda_{\text{gcv}}}^0$) is always found to be less than that obtained using the L-curve method ($\mathbf{z}_{\lambda_L}^0$), a set of filters $\mathbf{z}_{\lambda_i}^0; \lambda_{\text{gcv}} < \lambda_i < \lambda_L$ for $i = 1, 2, \dots, N$ and $N \geq T_v$ is estimated for each λ_i . The required minimum value for $\alpha_j(k_x, k_y)$ is determined using an absolute measure of error bound ε^v defined as:

$$\varepsilon^v = \left| \mathbf{K}_f^v - \mathbf{K}_f^0 \right|, \quad (4.16)$$

where $\mathbf{K}_f^v$ is the k -space reconstructed at the cross-over point using $\mathbf{z}_{\lambda_L}^v$ after addition of Gaussian noise of variance $\sigma_0^2 T_v$ to all acquired k -space samples, and $\mathbf{K}_f^0$ denotes k -space reconstructed using $\mathbf{z}_{\lambda_{\text{gcv}}}^0$ prior to noise addition. The reconstruction error due to the i 'th filter at a given missing k -space location (k_x, k_y) is given by:

$$\varepsilon_{\lambda_i}^0(k_x, k_y) = \left| \mathbf{k}^{0T}(k_x, k_y) \cdot \Delta \mathbf{z}^0 \right|, \quad (4.17)$$

where $\Delta \mathbf{z}^0 = \mathbf{z}_{\lambda_i}^0 - \mathbf{z}_{\lambda_{\text{gcv}}}^0$ and $\mathbf{k}^0(k_x, k_y)$ denotes the vector comprising the acquired (source) points within the kernel centered at the unacquired k -space location (k_x, k_y) . The frequency-dependent choice of regularization parameter can then be obtained as:

$$\lambda_{\text{opt}}(k_x, k_y) = \min_{\lambda_i} \left(\left| \varepsilon_{\lambda_i}^0(k_x, k_y) - \varepsilon^v(k_x, k_y) \right| \right), \quad (4.18)$$

where $\varepsilon_{\lambda_i}^0(k_x, k_y) < \varepsilon^v(k_x, k_y)$ for $\lambda_i < \lambda_{\text{opt}}(k_x, k_y)$ and $\varepsilon_{\lambda_i}^0(k_x, k_y) > \varepsilon^v(k_x, k_y)$ for $\lambda_i > \lambda_{\text{opt}}(k_x, k_y)$. This shows that, as λ_i is increased from λ_{gcv} , the error on RHS of Equation (4.18) approaches zero at λ_{opt} . Thus $\lambda_{\text{opt}}(k_x, k_y)$ represents the maximum value of regularization parameter or, equivalently, the minimum $\alpha_j(k_x, k_y)$ required to limit the level of smoothing.

The number of filters should be at least equal to T_v for a given choice of σ_0 . Due to the first-order approximation of singular-value perturbation, larger value of σ_0 will result in larger errors, accompanied by reduced accuracy of error bound estimation. This limits the advantage of FDR reconstruction. Similarly, the use of additional filters can lead to reduced reconstruction errors, but with increased cost of computation.

The advantage of producing artifact-suppressed high SNR images with controlled blur increases the compute time of FDR compared to standard GRAPPA/SPIRiT. The computational overheads are mainly due to the error bound calculation and estimation of the unacquired k -space points. Figure 4.10 shows a stage-wise description of FDR GRAPPA/SPIRiT in terms of cross-over determination (Stage I), filter bank synthesis (Stage II) and FDR reconstruction (Stage III). Stage I includes only one SVD step followed by a singular-value update based on first-order perturbation. Therefore, this step is computationally

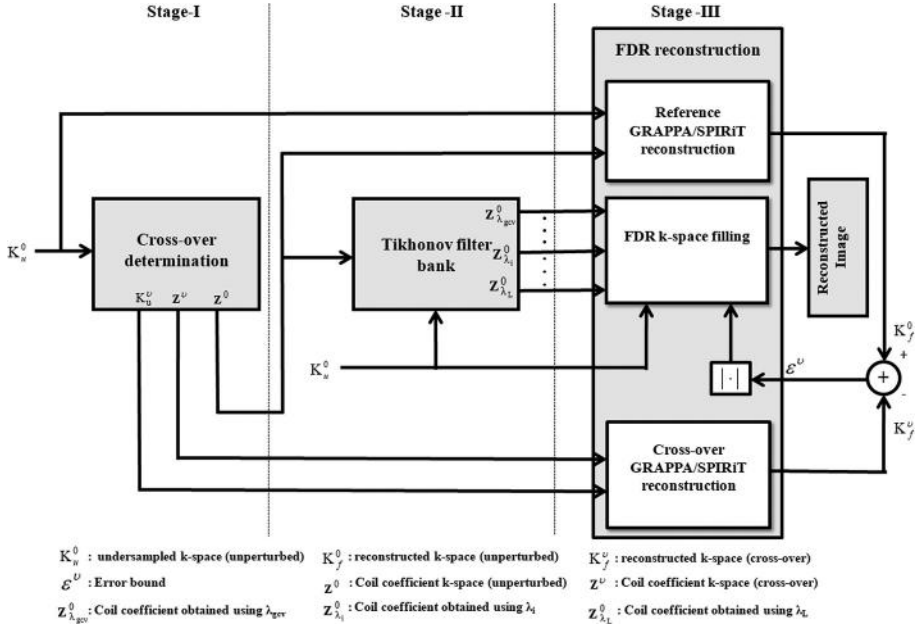


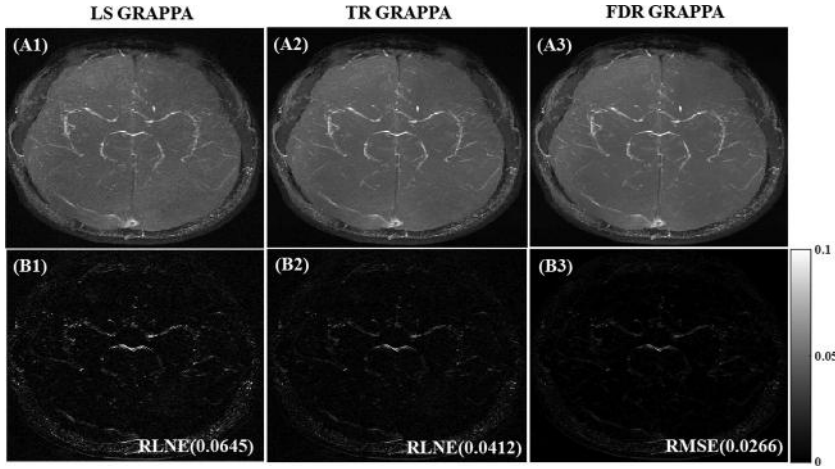
FIGURE 4.10
Block schematic of FDR GRAPPA/SPIRiT.

less expensive. In addition, the filter coefficients to be used in the error bound computation as well as FDR k -space estimation are computed a priori in Stage I. As Stage II does not include any form of matrix inversion or FOR loops, this stage does not add to the computational overhead. Regularized filter coefficients are computed by simple substitution of λ_i 's in Equations (4.14) and (4.15). Mainly, the computational cost is due to Stage III, which includes error bound computation and estimation of the unacquired k -space points. The error bound computation includes two GRAPPA/SPIRiT reconstructions: (1) estimation of original unperturbed k -space using z^0 and (2) estimation of noisy data at cross-over using z^v . The k -space filling operation using FDR results in a three-fold increase in computational cost compared to the standard GRAPPA/SPIRiT.

4.5.3.1 Implementation of FDR Reconstruction

FDR GRAPPA yields improved SNR without substantial reduction of spatial resolution as is shown using three-dimensional (3D) gradient echo (GRE) data. The ground truth data includes 3D fully sampled GRE MRI volumetric data acquired using 12-channel head array coils. After performing retrospective under-sampling along the PE direction, the volumetric data is divided into two-dimensional (2D) slices in the readout direction. Each slice in the channel k -space data ($384 \times 202 \times 32$) is then under-sampled with $R = 2$ and 28 ACS lines.

Reconstructions using Tikhonov regularized and FDR GRAPPA are then performed in each slice. Using 13 out of 26 ACS lines representing the source locations for $R = 2$, M is determined as $202 \times 13 = 2626$. With 12 coils, the number of columns of Π^0 is $12 \times 5 \times 2 = 120$. Performing SVD on Π^0 yields $\sigma_{\min}^0 = 2.3 \times 10^{-4}$. $\|E^t\|_2$ is computed using the difference between

**FIGURE 4.11**

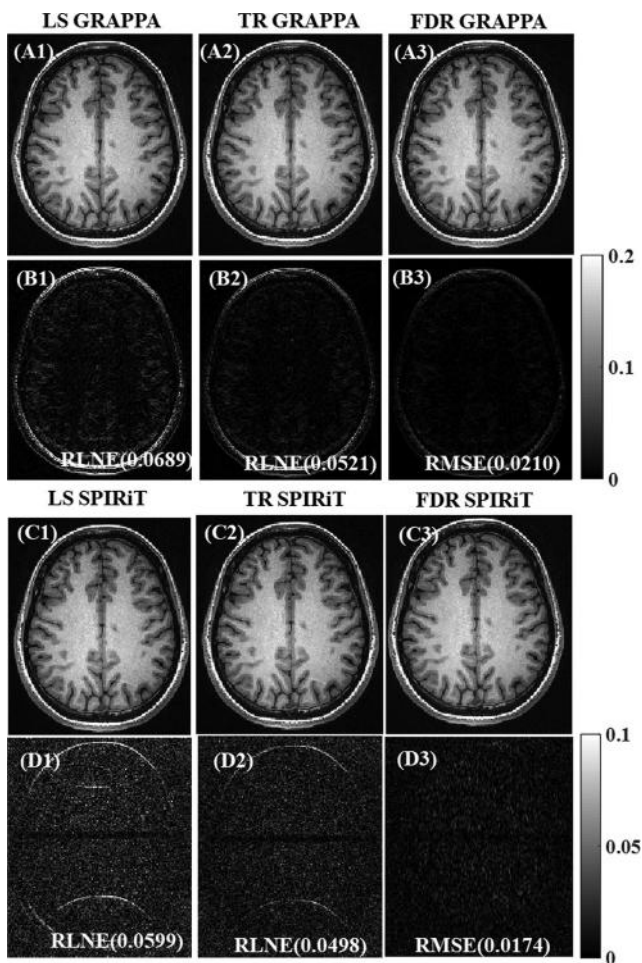
GRAPPA-reconstructed and difference images for SWI-GRE dataset IX. Reconstructed images are shown in the top row, and corresponding difference images are shown in the bottom row. Columns from left to right correspond to LS, Tikhonov (TR) and FDR reconstructions, respectively. The RLNE values are provided in the insets for each type of reconstruction.

the lowest singular values of Π^0 as 3.27×10^{-5} and σ_0 as $3.27 \times 10^{-5} / \sqrt{2626} = 6.38 \times 10^{-7}$. Using this choice, cross-over is achieved in 25 steps; ε^ν at each k -space location is obtained using two GRAPPA reconstructions performed on original data and noisy data at cross-over with $\mathbf{z}_{\lambda_{\text{gcv}}}^0$ and $\mathbf{z}_{\lambda_L}^\nu$, respectively. For each channel, a filter bank consisting of 25 filters is computed for λ_i chosen at equal intervals in the range $\lambda_{\text{gcv}} < \lambda_i < \lambda_L$; k -space filling is then performed for each channel data using the pre-computed Tikhonov filters and ε^ν at each k -space location. Implementation using MATLAB on a PC with an Intel Xeon 2.4 GHz processor and 16 GB of RAM running Windows 7 OS took 32s for a single slice, compared to 11s for TR GRAPPA. The cross-over computation for all slices required the same number of noise perturbation steps and yielded nearly the same cross-over distance of $\nu = 2.74$. A channel-combined sum-of-squares (SoS) image of each slice is then maximum-intensity projected (MIP) to generate the image shown in Figure 4.11.

Column-wise panels from left to right show reconstructions using LS, Tikhonov regularization (TR) and FDR methods, respectively. Difference images show all sources of errors, such as blurring, aliasing and noise.

Figure 4.12 shows GRAPPA and SPIRiT reconstructions ($R=2$) for dataset IV with ν -values of 2.12 and 1.95, respectively. Top and bottom panel pairs show GRAPPA and SPIRiT reconstructions, respectively. FDR-SPIRiT reconstruction took 14 iterations for convergence and required a compute time of 23s compared to 7s for conventional SPIRiT. For each type, reconstructed images are shown in the top row and corresponding difference images in the bottom row. From visual comparison, it is clear that FDR reconstruction yields improved reconstructed images compared to that of TR.

Auto-calibrating pMRI reconstruction methods which rely on sparsity-promoting penalty functions such as L1-SPIRiT [44,45] and correct as DEnoising of Sparse Images from GRAPPA using the Nullspace method (DESIGN) [46] utilize sparsity-promoting regularization. Sparsity is achieved in some transform domains like wavelet or finite difference.

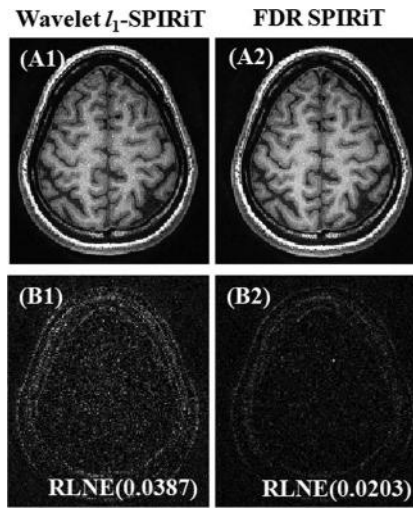
**FIGURE 4.12**

GRAPPA and SPIRiT reconstruction for dataset IV. Top two panels correspond to GRAPPA, and bottom two panels correspond to SPIRiT. In each type, the reconstructed images are shown in the top row, and the difference images are shown in the bottom row. Columns from left to right correspond to LS, Tikhonov (TR) and FDR reconstructions, respectively. RLNE values are provided in the insets.

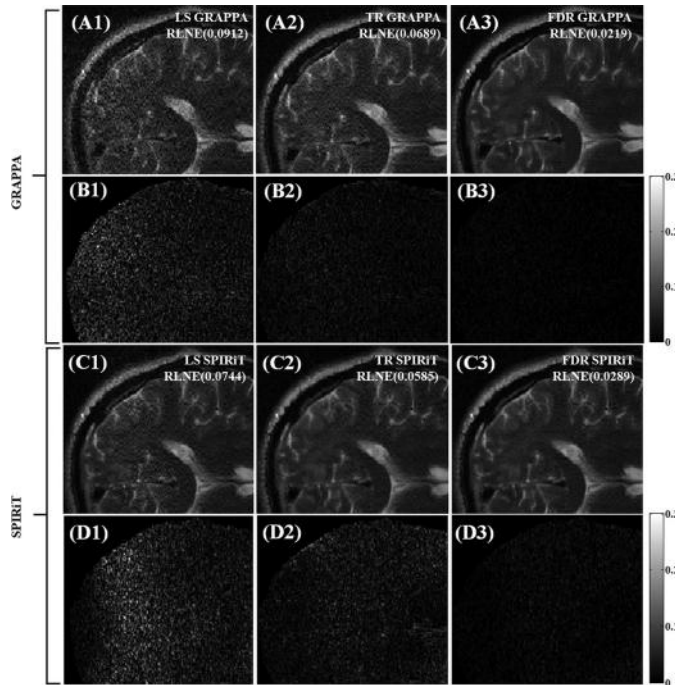
In wavelet-based L1-SPIRiT, reconstruction is performed in the spatial domain using a single calibration filter operator model followed by sparsity-promoting regularization achieved using soft-thresholding operation in the wavelet domain. Here the threshold acts as a regularization parameter.

FDR SPIRiT is compared with L1-SPIRiT for dataset VII using wavelet soft thresholding in [Figure 4.13](#). The threshold for L1-SPIRiT is selected using an ad hoc search procedure to achieve the best possible reconstruction by visual inspection of the difference image. From the difference images, it is evident that the FDR approach provides improved trade-off between resolution and SNR.

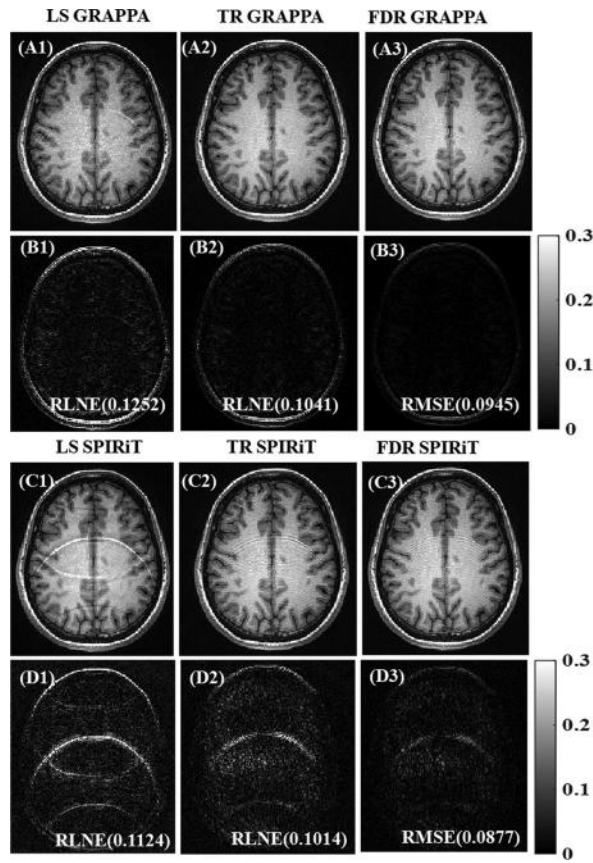
[Figure 4.14](#) shows FDR reconstructions performed with $R = 3$ on dataset II. Top and bottom panel pairs illustrate GRAPPA and SPIRiT reconstructions, with ν -values of

**FIGURE 4.13**

SPIRiT reconstruction applied to dataset VII for a random under-sampling mask with 40% acquired samples. Reconstructed images are shown in the top row, and corresponding difference images are shown in the bottom row. RLNE values are provided in the insets.

**FIGURE 4.14**

GRAPPA and SPIRiT reconstruction in dataset II for an under-sampling factor $R = 3$. Top two panels correspond to GRAPPA, and bottom two panels correspond to SPIRiT. In each type, enlarged views of reconstructed images are shown in the top row, and corresponding difference images are shown in the bottom row. Columns from left to right correspond to LS, Tikhnov (TR) and FDR reconstructions respectively. The type of reconstruction and RLNE values are provided in insets.

**FIGURE 4.15**

GRAPPA and SPIRiT reconstruction applied to dataset IV for $R = 4$. Top two panels correspond to GRAPPA, and bottom two panels correspond to SPIRiT. In each type, the reconstructed images are shown in the top row, and the difference images are shown in the bottom row. Columns from left to right correspond to LS, Tikhonov (TR) and FDR reconstructions, respectively. RLNE values are provided in the insets.

2.9 and 2.01, respectively. Enlarged views of the reconstructed images in each case are shown in the top row, with the corresponding difference images shown below. Column-wise panels indicate LS, TR and FDR reconstructions, respectively. The type of reconstruction and RLNE values are provided in insets. Reconstruction parameters for $R = 3$ are summarized in [Table 4.1](#) for GRAPPA and in [Table 4.2](#) for SPIRiT.

[Figure 4.15](#) shows FDR GRAPPA and SPIRiT reconstruction applied to dataset IV for $R = 4$. Top and bottom panels show reconstructed images using GRAPPA and SPIRiT, respectively. The corresponding difference images are shown below each reconstructed image, with corresponding RLNE values provided in the insets. Column-wise panels correspond to LS, TR and FDR reconstructions, respectively.

The FDR can be applied to datasets with higher cross-over distance, resulting in improved reconstruction quality. Assuming the noise level to be constant over the entire k -space, the SNR decays with an increase in frequency (distance from the k -space centre). By selectively choosing the parameter values using a monotonically increasing function of spatial frequencies, the noise reduction can be effectively controlled in datasets with large cross-over distance. However, this can also lead to loss of resolution.

An optimal trade-off is provided at each k -space location by limiting the value of the regularization parameter. This is achieved by minimizing the discrepancy between an error bound computed based on calibration performed at cross-over, and reconstruction error deviation with respect to a reference filter (such as GCV).

Although GRAPPA was initially developed for Cartesian trajectories, some of its recent variants have been extended to non-Cartesian sampling trajectories, such as radial and spiral [47–50]. While under-sampling is performed in the azimuthal dimension in radial GRAPPA, spiral GRAPPA employs the radial dimension. In non-Cartesian GRAPPA, the k -space samples are reordered to Cartesian in a hybrid (r, θ) space. Unlike the case of Cartesian GRAPPA where the same filter weights can be applied to the entire k -space, the non-Cartesian GRAPPA requires calibration for each segment in the hybrid space. Extension of FDR GRAPPA to each segment of the hybrid space requires the cross-over and error bound to be determined separately for each segment. This also requires computation of the best matching Tikhonov filters at each location within the segments. In FDR SPIRiT, local filters are applied at each k -space location using both acquired samples and the previously estimated unacquired samples. The same filters are used in the successive iterations. Due to the requirement for use of local k -space filters in FDR SPIRiT, a single SPIRiT operator-based model cannot be used. Thus, the only possibility for its extension to non-Cartesian trajectories would be to operate in the segment-wise hybrid space similar to GRAPPA operator for wider radial bands (GROWL) [50].

Despite the three-fold increased processing time of FDR GRAPPA/SPIRiT, it is particularly useful to process 3D under-sampled data for reconstruction of high-resolution MIP images with the additional advantage of not having to repeat the cross-over computations because each slice would inherit approximately the same cross-over distance. For 3D volumetric acquisitions, FDR-GRAPPA can also be used with under-sampling patterns along two PE directions, as in 2D GRAPPA [51] and Controlled Aliasing In Parallel Imaging Results IN Higher Acceleration (CAIPRINHA)-type acquisition [52,53]. Furthermore, computation time for FDR GRAPPA is comparatively lower than some of the other recent GRAPPA variants such as iGRAPPA [31,54], or NL-GRAPPA [34,54].

4.6 Effect of Noise Correlation

Representing the noise pattern in the complex image domain using a complex multivariate (one variable per coil) Additive White Gaussian Noise (AWGN) process, the input noise covariance matrix is given by [55,56]:

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12}^2 & \dots & \sigma_{1n_C}^2 \\ \sigma_{21}^2 & \sigma_2^2 & \dots & \sigma_{2n_C}^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n_C 1}^2 & \sigma_{n_C 2}^2 & \dots & \sigma_{n_C}^2 \end{bmatrix}, \quad (4.19)$$

where $\sigma_{ij}^2 = \rho_{ij}\sigma_i\sigma_j$ and ρ_{ij} is the coefficient of correlation between the i 'th and j 'th coils. The coefficient of correlation is dependent only on the electromagnetic coupling between

coils i and j . On the other hand, the variance of noise for each coil is obtained as a scaled version of the noise variance $\sigma_{K_j}^2$ in k -space [56]:

$$\sigma_j^2 = \frac{1}{N_{pe}N_{fe}}\sigma_{K_j}^2, \quad (4.20)$$

where N_{pe} and N_{fe} represent image dimensions along the phase and frequency-encode directions, respectively. Using Cholesky decomposition $\Sigma = LL^H$, the pre-whitened image data can be computed as:

$$\hat{\mathbf{u}}(x, y) = L^{-1}\mathbf{u}(x, y), \quad (4.21)$$

where $\mathbf{u}(x, y) = [u_1(x, y), u_2(x, y), \dots, u_{n_c}(x, y)]^T$, with each element representing the image intensities at (x, y) of coils-1 to n_c . Datasets with different levels of noise correlation are generated by setting pre-assigned values for the coefficient of correlation, and constant noise variance in the covariance structure. Cross-over values are then computed by addition of uncorrelated noise to (1) dataset before pre-whitening (U) and (2) dataset after pre-whitening ($\hat{U}$). The cross-over distance is found to decrease with increasing noise levels. For the same dataset, cross-over distances are almost the same for both types of cross-over computations. Thus, pre-whitening does not significantly affect cross-over computation. Simulations are performed by assuming the same value of coefficient of correlation across any pair of channels. Table 4.3 summarizes the input noise variance and coefficient of correlation used for simulation of noisy calibration data. For an a priori known amount of correlation, the table shows cross-over distances computed using Gaussian noise added to calibration data with and without pre-whitening. The table clearly shows that the maximum deviation in cross-over distance computed without pre-whitening is within 5% of that obtained with pre-whitening.

TABLE 4.3

Cross-Over Distances Computed Using Gaussian Noise Added to Calibration Data with and without Pre-whitening

Noisy Calibration with Known Covariance Structure				
σ_{in}	ρ_{in}	$\sigma_0 \times 10^{-4}$	U (with Pre-Whitening)	U (without Pre-Whitening)
0.0001	0.1	1.38	1.85	1.89
	0.3	1.38	1.94	1.99
	0.7	1.38	2.07	2.15
0.0005	0.1	3.04	1.39	1.34
	0.3	3.04	1.45	1.38
	0.7	3.04	1.60	1.66
0.0010	0.1	4.38	1.16	1.18
	0.3	4.38	1.20	1.19
	0.7	4.38	1.52	1.55

Appendix

The stability of the solution with noise-induced perturbation can be analyzed using the norm of solution difference at two consecutive perturbation levels. Let the solutions at two consecutive perturbation levels t_1 and t_2 be denoted by $\mathbf{z}^{t_1}$ and $\mathbf{z}^{t_2}$. The norm of solution difference with respect to the initial solution can be represented as:

$$\|\mathbf{z}^{t_1} - \mathbf{z}^{t_2}\| = \left\| (\mathbf{z}^{t_1} - \mathbf{z}_0) - (\mathbf{z}^{t_2} - \mathbf{z}_0) \right\| \quad (4.22)$$

Applying the triangular inequality principle, we have:

$$\|\mathbf{z}^{t_1} - \mathbf{z}^{t_2}\| \leq \|\mathbf{z}^{t_1} - \mathbf{z}_0\| + \|\mathbf{z}^{t_2} - \mathbf{z}_0\| \quad (4.23)$$

Using P. C. Hansen's [23] Theorem 3, at perturbation level t_1 , we have:

$$\|\mathbf{z}_0 - \mathbf{z}^{t_1}\| \leq f(\check{\Pi}^{t_1} \mathbf{z}^{t_1} - \mathbf{k}_u^{t_1}) \quad (4.24)$$

where $f(\cdot)$ represents a monotonically increasing function of the argument. Similarly, at perturbation level t_2 , we have:

$$\|\mathbf{z}_0 - \mathbf{z}^{t_2}\| \leq f(\check{\Pi}^{t_2} \mathbf{z}^{t_2} - \mathbf{k}_u^{t_2}) \quad (4.25)$$

Therefore, the norm of solution difference becomes:

$$\|\mathbf{z}^{t_1} - \mathbf{z}^{t_2}\| \leq f\left(\check{\Pi}^{t_1} \mathbf{z}^{t_1} - \mathbf{k}_u^{t_1}\right) + f\left(\check{\Pi}^{t_2} \mathbf{z}^{t_2} - \mathbf{k}_u^{t_2}\right) \quad (4.26)$$

This shows that the norm of solution difference is a function of the residual norms at the perturbation levels t_1 and t_2 .

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5

Parameter Adaptation for Wavelet Regularization in Parallel MRI

5.1 Image Representation Using Wavelet Basis

Similar to the Fourier basis, the wavelet basis reveals the signal regularity based on the amplitude of transform domain coefficients, and their structured nature facilitates fast computation. A main difference is that the wavelets are well localized, and only a few coefficients are required to represent local transient structures. Also in contrast to the Fourier basis, a wavelet basis provides a sparse representation of piecewise regular signals that include transients and singularities.

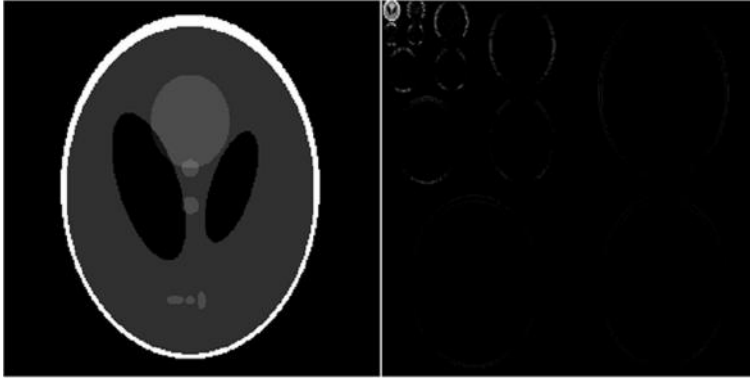
Two-dimensional wavelet basis is discretized to define orthonormal basis of images, including N^2 pixels. Wavelet orthonormal basis of images can be constructed from wavelet orthonormal basis of one-dimensional signals. To construct an orthonormal basis of the space $\mathbb{L}^2(\mathbb{R}^2)$ of finite energy functions $f(x) = f(x_1, x_2)$, three mother wavelets $\psi^1(x)$, $\psi^2(x)$ and $\psi^3(x)$, with $x = (x_1, x_2) \in \mathbb{R}^2$, are dilated by 2^j and translated by $2^j n$ with $n = (n_1, n_2) \in \mathbb{Z}^2$. In general, for a mother wavelet with directionality s , the resultant orthonormal basis is given by:

$$\psi_{j,n}^s(x) = \frac{1}{2^j} \psi^s\left(\frac{x - 2^j n}{2^j}\right), \text{ for } j \in \mathbb{Z}, n \in \mathbb{Z}^2, 1 \leq s \leq 3. \quad (5.1)$$

Figure 5.1 shows the array of N^2 discretized wavelet coefficients computed for an image with N^2 pixels. Each scale 2^j corresponds to a subimage, with positions of coefficients above a threshold: $|\langle f, \psi_{j,n}^s \rangle| \geq \alpha$ shown in white. Similar to the one-dimensional case, a two-dimensional wavelet coefficient has large amplitude in the neighborhood of edges and irregular textures.

5.2 Structure of Wavelet Coefficients

Wavelet transform exploits the underlying structure of wavelet coefficients in the given data. Applications related to wavelet-based recovery use the fact that a large number of wavelet coefficients can be set to zero without any significant change in the reconstruction accuracy.

**FIGURE 5.1**

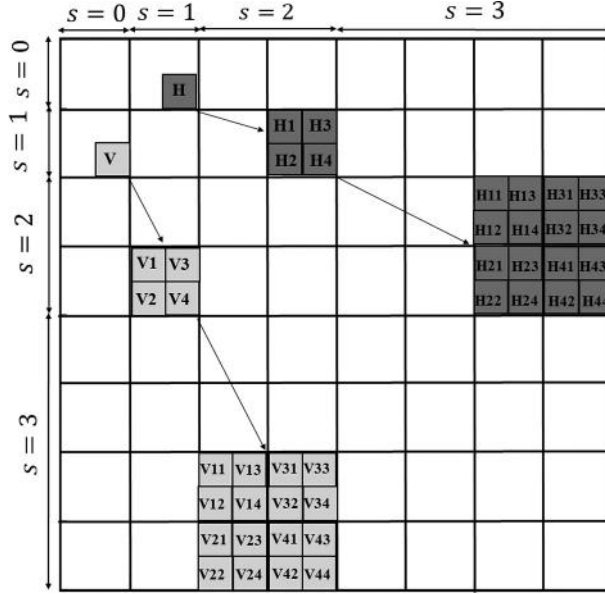
Shepp-Logan phantom of size 256×256 and array of N^2 orthogonal wavelet coefficients $|f, \psi_{j,n}^s| \geq \alpha$ for $s = 1, 2, 3$, and 4 scales 2^j .

Implementation of the discrete wavelet transform of an image using a bank of high-pass and low-pass filters, with decimation performed after each filtering stage, results in a quadtree structure of the wavelet coefficients [1]. In the quadtree structure, each wavelet coefficient serves as a ‘parent’ for four ‘children’ coefficients. The root nodes correspond to the wavelet coefficients at the coarsest scale, and leaf nodes correspond to the finest scale. For a natural image, the negligible wavelet coefficients tend to be clustered together. The concept of a zero tree implies that a tree or subtree of the negligible wavelet coefficients can be collectively neglected [2]. This is based on the fact that if a wavelet coefficient at a particular scale is negligible, then its children are also negligible. Wavelet coefficients of an image in the form of a tree structure are shown in Figure 5.2. The root and leaf nodes are represented by coefficients at scale $s = 1$ and $s = L$, respectively. The top left block corresponding to $s = 0$ represents the scaling coefficients that contain information about the coarse-scale component. At scales $1 \leq s \leq L - 1$, each wavelet coefficient has four children coefficients at the next successive scale $s + 1$. Greedy compressed sensing (CS) algorithms such as Iterative signal recovery from incomplete and inaccurate samples (CoSaMP) [3] and iterative hard thresholding (IHT) [4] employ the wavelet tree structure.

5.2.1 Statistics of Wavelet Coefficients

Wavelet coefficients can be modeled either as jointly Gaussian [5–8] or as non-Gaussian but independent observations [9–12]. Linear correlations between wavelet coefficients can be efficiently expressed using jointly Gaussian models. Compared to the Gaussian models, non-Gaussian models are found to be more suitable because the wavelet coefficient histograms peak at zero with a heavy tail compared to the Gaussian. The use of independent non-Gaussian models can be justified based on the interpretation of the wavelet transform as a decorrelator, making each wavelet coefficient statistically independent of the rest of the coefficients. However, in practice, the wavelet transform cannot serve as a perfect decorrelator because of the inherent dependency structure existing between the wavelet coefficients. This is further analyzed using the secondary properties of the wavelet transform, such as clustering and persistence.

Clustering refers to the property that if a particular wavelet coefficient is large/small, then adjacent coefficients also tend to be large/small [13]. Alternatively, persistence refers

**FIGURE 5.2**

Wavelet decomposition of an image, with the tree structure depicted across scales. The wavelet transform is performed with three wavelet decomposition levels, and two wavelet trees. The top-left block $s=0$ represents scaling coefficients, and other regions represent wavelet coefficients.

to the property by which large/small values of wavelet coefficients exhibit the tendency to spread across scales [14,15]. Although modeling of the joint probability density function of the wavelet coefficients enables characterization of the dependencies, it is practically impossible to estimate the complete joint probability density. Therefore, it is simpler to model the wavelet coefficients as statistically independent but neglect the dependencies between wavelet coefficients. A more precise model would be to establish a compromise between the two extreme cases by incorporating only the key dependencies. The non-Gaussian nature of the wavelet coefficients can be utilized to model the marginal probability of each coefficient as a mixture density with a hidden state variable. The key dependencies between the wavelet coefficients can then be characterized by modeling the Markovian dependencies between the hidden state variables. These types of model are generally referred to as hidden Markov models (HMMs) [16–18].

Because the independent mixture model ignores the inter-coefficient dependencies, the state variables remain unconnected. While the hidden Markov chain model links the state variables horizontally within each scale, the hidden Markov tree (HMT) model links the state variables vertically across the scales. In the HMT model, each wavelet coefficient is drawn from one of the two zero-mean Gaussian distributions representing the two hidden states [19]. The low and high states are defined by assigning small and high values, respectively, for the variance of the Gaussian distribution. While a large wavelet coefficient has a high probability to be in the high state, a coefficient with a relatively small value is more likely to be in the low state. The dependence of the probability of a particular state on the state of associated parent coefficient yields a Markov representation across scales.

Using the Markov transition property represented by a 2×2 matrix P , each $P(i, j)$ indicates the probability that children coefficients are in state j given that the associated parent coefficient is in state i ; $i=1, j=1$ represents the low state and $i=2, j=2$ represents the high state.

Typically, the form of P such that $P(1,1)=1-\epsilon$ and $P(1,2)=\epsilon$, where $0 < \epsilon \ll 1$, indicates the probability that if a parent coefficient is small, the associated children are also likely to be small. Note that P usually varies between different scales and across different wavelet quadrees. For the root nodes, P is a 1×2 vector, which represents an initial-state distribution.

5.3 CS Using Wavelet Transform Coefficients

Recovery of the underlying signal based on CS relies on the sparsity pertaining to the signal of interest and incoherence to the sensing modality [20]. To recover the signal from the compressive measurements, it is therefore required to find the sparsest one that agrees with the measurements. Sparse representations can be obtained in a basis that utilizes some form of regularity of the input signals, thereby resulting in a large number of small-amplitude coefficients. Typically, for most natural signals, the wavelet coefficients w are compressible, indicating that it is possible to set a large fraction of the coefficients to zero. If the wavelet coefficients w with m large coefficients set to zero is denoted by w_e , the resultant error $\|w - w_e\|_2$ is proportional to $(m+1)^{-1/2}$ [21,22]. This compressibility property enables the wavelet to recover images from a small number of projection measurements. With m significant transform domain coefficients, and the rest $N-m$ coefficients being negligibly small, $w = w_{N-m} + w_e$, where w_{N-m} represents the original w with the smallest $N-m$ coefficients set to zero. Using this notation, Equation (1.43) takes the following form:

$$y = \phi \Psi w = \phi \Psi w_{N-m} + \phi \Psi w_e. \quad (5.2)$$

Since ϕ is the sensing matrix, it represents the way one samples the signal. For example, in the case of subsampling, ϕ is the identity matrix with corresponding rows removed. As each element of w_e can be modeled as a zero-mean Gaussian with small variance, each element of $n_e = \phi \Psi w_e$ can also be modeled by a zero-mean Gaussian with appropriate variance. Therefore, with the assumption that the CS measurements are corrupted with zero-mean Gaussian noise n_0 , Equation (5.2) becomes:

$$y = \phi \Psi w_{N-m} + n_e + n_0 = \phi \Psi w_{N-m} + n. \quad (5.3)$$

Given the measurements and the sensing matrix ϕ , the objective of the wavelet-based CS reconstruction problem is to estimate the values and the locations of the nonzero elements in the transform coefficients. In order for the CS algorithms to perform well, the sensing matrix should either satisfy the incoherence properties, restricted isometry properties (RIPs) or null space properties (NSPs), as discussed in [Section 1.6.3](#).

An important characteristic of CS-based recovery algorithms is that they do not utilize any particular structure in the signal apart from sparsity. For example, the wavelet coefficients of piecewise smooth signals are not only sparse; they also cluster around a connected subtree [19,23]. Consequently, CS algorithms that exploit structured sparsity can be used to further reduce the required number of sampling measurements by exploring the tree structure [24–26]. By introducing dependencies between locations of the signal coefficients utilizing the structure within the transform coefficients, Baraniuk et al. [25] demonstrated that it is possible to improve the performance of CS reconstruction. These ideas have been extended to magnetic resonance (MR) image reconstruction by Chen and Huang [27].

5.3.1 Structured Sparsity Model

The structured sparsity model is based on the information that the support of the large coefficients in the sparse representation of images often has an underlying inter-dependency structure. To explain the structured sparsity model, consider a signal $\mathbf{s}$ whose coefficients decay according to the power law:

$$|\mathbf{s}_D(i)| \leq Gi^{-1/r}; \quad i = 1, \dots, N^2, \quad (5.4)$$

where index D denotes the sorted coefficients, when sorted in the descending order of magnitude. Due to the rapid decay of the coefficients, such signals can be well approximated by S_k -sparse signals. With $\mathbf{s}_{S_k} \in \Sigma_{S_k}$ representing the best S_k -term approximation of $\mathbf{s}$, the l_p -norm error in this approximation can be expressed as:

$$\min_{\bar{\mathbf{s}} \in \Sigma_{S_k}} \|\mathbf{s} - \bar{\mathbf{s}}\|_p = \|\mathbf{s} - \mathbf{s}_{S_k}\|_p. \quad (5.5)$$

A structured sparsity model provides the S_k -sparse signal $\mathbf{s}$ with additional structure and allows only certain S_k -dimensional subspaces in Σ_{S_k} [28,29]. With m_{S_k} allowable S_k -dimensional subspaces, the set containing all the corresponding supports can be represented as $\{\Omega_1, \dots, \Omega_{m_{S_k}}\}$, where $|\Omega_m| = S_k$ for each $m = 1, \dots, m_{S_k}$. Using this representation, a structured sparsity model can be defined as:

$$\mathcal{M}_{S_k} = \bigcup_{m=1}^{m_{S_k}} \chi_m \text{ s.t. } \chi_m : \left\{ \mathbf{s} : \mathbf{s}|_{\Omega_m} \in \mathbb{R}^{S_k}, \mathbf{s}|_{\Omega_m^c} = 0 \right\}, \quad (5.6)$$

such that each subspace contains all signals $\mathbf{s}$ with $\text{supp}(\mathbf{s}) \subseteq \Omega_m$. Consequently, signals from $\mathcal{M}_{S_k}$ are called S_k -structured sparse with $m_k \leq \binom{N}{S_k}$. Based on the two secondary properties of the wavelet transform coefficients, two different structured sparsity models have been developed to recover the underlying signal. The first model is based on the fact that the large wavelet coefficients of piecewise smooth signals and images possess a connected tree structure [19]. The second model utilizes the fact that, generally, the largest set of sparse coefficients cluster together into blocks [30–32].

5.3.1.1 Model-Based RIP

If the acquired signal $\mathbf{y}$ is known to be S_k -structured sparse, then stable recovery from the compressive measurements can be achieved by relaxing the RIP constraint (refer to Section 1.6.3) on the CS measurement matrix Φ [28,29]. Accordingly, an $M \times N$ matrix Φ has the $\mathcal{M}_{S_k}$ restricted isometry property ($\mathcal{M}_{S_k}$ -RIP) with constant $\delta_{\mathcal{M}_{S_k}}^\Phi$ if:

$$\left(1 - \delta_{\mathcal{M}_{S_k}}^\Phi\right) \|\mathbf{s}\|_2^2 \leq \|\Phi \mathbf{s}\|_2^2 \leq \left(1 + \delta_{\mathcal{M}_{S_k}}^\Phi\right) \|\mathbf{s}\|_2^2, \quad \text{for all } \mathbf{s} \in \mathcal{M}_{S_k}. \quad (5.7)$$

The number of measurements M necessary for a random CS matrix to have the $\mathcal{M}_{S_k}$ -RIP with a given probability is:

$$M \geq \frac{2}{c(\delta_{\mathcal{M}_{S_k}}^\Phi)^2} \left(\ln(2m_k) + S_k \ln \frac{12}{\delta_{\mathcal{M}_{S_k}}^\Phi} + t \right), \quad (5.8)$$

where c and t are positive constants such that $t > 0$. The number of rows needed for a random matrix to have the $\mathcal{M}_{S_k}$ -RIP can be significantly lower than the number of rows required for the standard RIP since the number of subspaces m_k is considerably smaller than in standard CS. However, as the structured sparse concepts cannot be immediately extended to structured compressible signals, a generalization of the $\mathcal{M}_{S_k}$ -RIP is required for quantification of the stability of recovery for structured compressible signals [25]. This technique used to measure the robustness of structured compressible signal recovery is referred to as the restricted amplification property (RAmP). The key quantity involved in this property is the amplification of the structured sparse approximation residual through ϕ . This can be explained using the set of residual subspaces of size S_k defined as:

$$\mathcal{R}_{l,S_k}(\mathcal{M}) = \left\{ \mathbf{u} \in \mathbb{R}^N : \mathbf{u} = \mathbb{M}(\mathbf{s}, jS_k) - \mathbb{M}(\mathbf{s}, (j-1)S_k) \right\}; \mathbf{s} \in \mathbb{R}^N, l = 1, \dots, \lfloor N/S_k \rfloor, \quad (5.9)$$

where $\mathbb{M}(\mathbf{s}, S_k) = \min_{\tilde{\mathbf{s}} \in \mathcal{M}_{S_k}} \|\mathbf{s} - \tilde{\mathbf{s}}\|_p$. A matrix ϕ has the (ϵ_k, r) -RAmP for the residual subspaces $\mathcal{R}_{l,S_k}$ of model $\mathcal{M}$ [25] if:

$$\|\phi \mathbf{u}\|_2^2 \leq (1 + \epsilon_k) l^{2r} \|\mathbf{u}\|_2^2, \quad (5.10)$$

for any $\mathbf{u} \in \mathcal{R}_{l,S_k}$ and r is the regularity parameter that controls the growth rate of the amplification of $\mathbf{u} \in \mathcal{R}_{l,S_k}$ as a function of l . The choice of r is made so that the growth in amplification with l balances the decay of the norm in each residual subspace $\mathcal{R}_{l,S_k}$. Accordingly, the number of compressive measurements M needed for a random measurement matrix ϕ to have the RAmP with probability $1 - e^{-t}$ is given by:

$$M \geq \max_{1 \leq l \leq \lfloor N/S_k \rfloor} \frac{2S_k + 4 \ln R_l N + 2t}{\left(l^r \sqrt{1 + \epsilon_k} - 1 \right)^2} \quad (5.11)$$

where R_l is the number of subspaces of dimension S_k in $\mathcal{R}_{l,S_k}(\mathcal{M})$.

5.3.1.2 Model-Based Signal Recovery

The model-based signal recovery can be implemented by integrating structured sparsity models into CoSaMP [3] and IHT [4] algorithms. This is achieved by replacing the S_k -term sparse approximation step with a best S_k -term structured sparse approximation using two concrete structured sparsity models. Whereas the first model relies on the fact that large wavelet coefficients obtained from piecewise smooth signals exhibit a rooted and connected tree structure [19], the second model uses the fact that large wavelet coefficients often cluster together into blocks [30–32]. With application of structured sparsity, there is only a search over the m_k subspaces of $\mathcal{M}_{S_k}$ rather than $\binom{N}{S_k}$ subspaces of Σ_{S_k} in each iteration of the model-based recovery algorithm. This indicates that the same quality robust signal recovery using standard CS algorithms can be achieved using fewer measurements in model-based CS.

5.3.1.2.1 Tree-Based Recovery from Compressive Measurements

For tree-sparse signals, one can find a sub-Gaussian random matrix with the $\mathcal{T}_{S_k}$ -RIP property with constant $\delta_{\mathcal{T}_{S_k}}^\phi$ and probability $1 - e^{-t}$ if the number of measurements is such that:

$$M \geq \frac{2}{c(\delta_{T_{S_k}}^\phi)^2} \left(S_k \ln \left(\frac{48}{\delta_{T_{S_k}}^\phi} \right) + \ln \frac{512}{S_k e^2} + t \right). \quad (5.12)$$

From Equation (5.12), it is evident that the number of measurements required for obtaining stable recovery of tree-sparse signals is linear in S_k , without the dependence on N , as in the case of conventional CS recovery model. The number of subspaces in each residual set for the approximation class in the case of tree-compressible signals is given by:

$$R_j \leq \frac{(2e)^{S_k(2l+1)}}{(S_k l + S_k + 1)(S_k l + 1)}. \quad (5.13)$$

The matrix ϕ has the (ϵ_k, r) -RAmP for the structured sparsity model $\mathcal{T}$ with $r > 0$ with probability $(1 - e^{-t})$ if:

$$M \geq \frac{2 \left(10S_k + 2 \ln \frac{N}{S_k(S_k + 1)(2S_k + 1)} + t \right)}{(\sqrt{1 + \epsilon_k} - 1)^2}. \quad (5.14)$$

This gives a simplified bound on the required number of measurements $M = \mathcal{O}(S_k)$, which is a substantial improvement over that required by the conventional CS recovery methods with $M = \mathcal{O}(S_k \log N / S_k)$. The signal can be recovered using a model-based CoSaMP [25] algorithm. In each iteration, the structured sparse approximation is performed using a condensing sort and select algorithm (CSSA) [33–35], which is an efficient algorithm to solve $\mathbb{M}(\mathbf{s}, S_k) = \min_{\mathbf{s} \in T_{S_k}} \|\mathbf{s} - \bar{\mathbf{s}}\|_p$. In the case of general wavelet coefficients, $\mathbb{M}$ is computed by condensing the non-monotonic segments of the tree branches. To obtain the condensed version, an iterative sort-and-average scheme is used during a greedy search through the nodes. The algorithm computes the average magnitude of the wavelet coefficients for each sub-tree rooted at each node, and the largest average among all the sub-trees is taken as the energy for that node. The algorithm then searches for the unselected node which has the largest energy. It then adds that sub-tree corresponding to the node's energy to the estimated support as a single super-node to provide a condensed representation of the corresponding sub-tree [35].

5.3.1.2.2 Block-Based Recovery from Compressive Measurements

For a block-sparse signal, the locations corresponding to the largest coefficients cluster in blocks under a specific sorting order. A signal ensemble received using multiple coils can be reshaped as a single vector by concatenation, and the concatenated vector can be rearranged to introduce block sparsity. The block-sparse structure can yield signal recovery from a reduced number of CS measurements, for the single channel [30,31,36] as well as the signal ensemble case [32]. For description of the signal recovery using block sparsity, consider a class of signal vectors $\mathcal{S}$ such that $\mathbf{s} \in \mathbb{R}^J$, where J and N are integers. This signal can be reshaped into a matrix X of size $J \times N$. The class $\mathcal{S}_{S_k}$ contains $\mathcal{S} = \binom{N}{S_k}$ sub-spaces of dimension $J S_k$. Note that the set of S_k -block sparse signals:

$$\mathcal{S}_{S_k} = \left\{ X = [\mathbf{s}_1 \dots \mathbf{s}_N] \in \mathbb{R}^{J \times N} \text{ s.t. } \mathbf{s}_n = 0 \text{ for } n \notin \Omega, \Omega \subseteq \{1, \dots, N\}, |\Omega| = S_k \right\}, \quad (5.15)$$

has sparsity $S_k J$, which is dependent on the size of the block (J). This formulation can be extended to each length N signals of the block with common support. A sub-Gaussian random matrix ϕ possesses the δ_{S_k} -RIP property with constant $\delta_{S_k}^\phi$ and probability $(1 - e^{-t})$ if the number of measurements is:

$$M \geq \frac{2}{c(\delta_{S_k}^\phi)^2} \left(S_k \ln \left(\frac{48}{\delta_{S_k}^\phi} \right) + \ln \frac{512}{S_k e^2} + t \right). \quad (5.16)$$

Therefore, the number of measurements required is $M = \mathcal{O}(JS_k + S_k \log N / S_k)$, which is less than $M = \mathcal{O}(JS_k \log N / S_k)$ as required by the standard CS recovery methods. The signal recovery can be obtained via model-based CoSaMP recovery [25] with the block-based approximation of the signal X as follows:

$$X_{S_k}^\mathcal{S} = \min_{\bar{X} \in \mathbb{R}^{J \times N}} \|X - \bar{X}\|_{2,2} \text{ s.t. } \|\bar{X}\|_{2,0} \leq S_k, \quad (5.17)$$

where the (p, q) mixed norm of the matrix X is defined as $\|X\|_{p,q} = \left(\sum_{n=1}^N \|s_n\|_p^q \right)^{1/q}$. The approximation in Equation (5.17) can be obtained by performing column-wise hard thresholding of X . Because the block-based approximation step involves sorting of only fewer coefficients, the block-based recovery is much faster than CoSaMP and requires fewer iterations to converge.

5.3.2 Wavelet Sparsity Model

With reference to the earlier description of CS-MRI reconstruction provided in [Section 2.9](#), the wavelet coefficients obtained using Landweber update is given by:

$$\hat{\mathcal{W}}^{(k)} = \mathcal{W}^{(k-1)} + \mathcal{E}_{\text{res}}^{(k)}, \quad (5.18)$$

where $\mathcal{E}_{\text{res}}^{(k)} = \Psi(F_u'(K_u^\epsilon - F_u(\Psi' \mathcal{W}^{(k-1)})))$ represents the wavelet coefficients of the consistency error, and $\hat{\mathcal{W}}^{(k)}$ and $\mathcal{W}^{(k)}$ denote the wavelet coefficients in the k 'th iteration before and after soft thresholding, respectively. The second term on the right-hand side (RHS) of Equation (5.18), that is, $\mathcal{E}_{\text{res}}$, is given the superscript k since the consistency error in each iteration is derived from the thresholded wavelet coefficients $\mathcal{W}^{(k-1)}$ of the preceding iteration. Using soft thresholding, the updated wavelet coefficient is obtained as:

$$\mathcal{W}^{(k)} = \mathcal{T}_\alpha^{(k-1)}(\hat{\mathcal{W}}^{(k)}), \quad (5.19)$$

where $\mathcal{T}_\alpha$ indicates a component-wise application of the soft-thresholding operator defined as:

$$\mathcal{T}_\alpha(\hat{\mathcal{W}}_i) = \max\{|\hat{\mathcal{W}}_i| - \alpha, 0\} \frac{(\hat{\mathcal{W}}_i - \alpha)}{|\hat{\mathcal{W}}_i - \alpha|}, \quad (5.20)$$

where $\hat{\mathcal{W}}_i$ denotes the i th component of $\hat{\mathcal{W}}$. The soft-thresholding operator sets the coefficients with values less than the threshold to 0, whereas those above the threshold are shrunk by subtracting the threshold value from the non-zero coefficients based on

the assumption that the threshold corresponds to the noise variance of the realizations of the wavelet coefficients. The assumption that consistency error is equivalent to some sort of noise in the wavelet coefficients is used to adapt the threshold in [37]. Such a noise-driven strategy assumes that the noise (ϵ) is stationary and Gaussian. With σ denoting the standard deviation of $\mathcal{E}_{\text{res}}^{(k)}$ and τ a constant $3 \leq \tau \leq 5$, the threshold $\alpha = \tau\sigma$ can be directly determined using the consistency error, that is, $\mathcal{E}_{\text{res}}^{(\infty)} = \Psi F_u' \epsilon$, at convergence. This may tempt one to choose $\mathcal{E}_{\text{res}}^{(k)}$ for estimating the standard deviation for threshold selection. Such estimation may not be reliable as $\mathcal{E}_{\text{res}}^{(k)}$ contains structure along with noise.

For wavelet sparsity-based formulations, the reconstructed image quality is highly dependent on the choice of the regularization parameter (threshold). As discussed in Chapter 3, threshold selection is typically carried out using exact mini-max [38], VisuShrink [38], Stein's unbiased risk estimation (SURE) shrink [39], cross-validation [40–42], false discovery rate [43,44], Bayesian methods [45,46] and Ogden's data analytic thresholding method [47]. The rest of this chapter discusses the dependence of threshold selection methods and the nature of their influence on the speed of convergence and image quality in CS-MRI.

5.4 Influence of Threshold on Speed of Convergence and Need for Iteration-Dependent Threshold Adaptation

In all methods based on an iterated soft-thresholding algorithm (ISTA), the wavelet threshold determines both steady-state errors as well as the rate of convergence. A higher threshold yields faster convergence at the cost of increased steady-state errors, and vice versa. As the upper bound of the error levels is found to vary with iteration in iterative approaches for sparsity-promoting regularization, the parameter should be varied in accordance with the variation in the upper bound in each iteration. This in turn gives rise to an iteration-dependent parameter selection.

The key idea behind such iteration-dependent parameter selection strategies is to solve the optimization problem with a large value of α at the start of the iterations and gradually decrease α in the succeeding iterations until the target regularization is reached. For a sufficiently large α , the solution coincides with the sparsest one resulting in a very large data fidelity. However, as the sparse coefficients are unreliable at early-stage iterations of the algorithm, a large value of α can be used initially to favor the penalty term. Then α is monotonically decreased following a linear pattern.

The idea of iterative thresholding with a varying threshold that requires knowledge of a decay factor was pioneered by Starck et al. [48,49] and Elad et al. [50]. In their approach, the initial threshold was set to a large value and was made to decrease in each iteration following a linear or exponential pattern. However, the fast convergence offered by such varying thresholds may not always lead to the minimum possible steady-state error. This type of threshold update scheme was initially derived from the homotopy continuation or path following [51–53]. The exact homotopy path-following methods were initially proposed in the statistics literature for computing the complete regularization path when varying the parameter β from large to small [51,52,54].

5.4.1 Selection of Initial Threshold

The selection of the initial threshold value is one of the challenging factors in threshold adaptation. Typically, the threshold selection is carried out to obtain the minimum possible relative l_2 -norm error (RLNE) [55–57], if the ground truth were known. For practical implementations in which the ground truth reference is not available, several parameter selection strategies have been proposed. For a given initial threshold chosen by any parameter selection method, the adaptation scheme is formulated to iteratively update the threshold and thus enable accelerated convergence to the minimum possible RLNE.

Since the universal VisuShrink approach for selecting the initial threshold always yields a high initial value, it gives better results when used with adaptation. The universal VisuShrink threshold is dependent, however, on the standard deviation of noise present in the input data and on the input data size. A higher sampling ratio requires a higher threshold for optimum reconstruction. The choice of a higher initial threshold can be justified based on the higher value of expectation of the minimum energy reconstruction and absolute wavelet coefficients of the zero-filled image. For initial threshold values larger than the VisuShrink threshold, the reconstructed image quality does not vary significantly with adaptation. On the other hand, thresholds much less than that given by VisuShrink (closer to zero) may lead to poor reconstruction quality. Thus, even at low noise levels, selection of the initial threshold in the proximity of the VisuShrink threshold ensures the best possible performance.

5.5 Parallelism to the Generalized Discrepancy Principle (GDP)

A key point of discussion is the iteration-dependent threshold adaptation in the parlance of the GDP, capable of providing low steady-state reconstruction error and fast convergence simultaneously. Typically, for a consistent linear inverse problem with an ill-posed system matrix H and observation vector y , application of the GDP yields the regularized solution x_{opt} that satisfies:

$$\|Hx_{\text{opt}} - y\|_2 \geq \|e\|_2 + \|E\|_2, \quad (5.21)$$

where e and E denote errors in y and H , respectively [58]. Extension of Equation (5.21) for adaptation of wavelet threshold in ISTA or thresholded Landweber (TL) algorithm is achieved by drawing a parallel between the fidelity term on the left-hand side (LHS) and the perturbation errors on the right-hand side (RHS) versus the difference between wavelet coefficients before and after soft thresholding. Since soft thresholding introduces sparsity, we refer to this difference quantity as the sparse approximation error. With l_1 -norms of the consistency and sparse approximation errors replacing the LHS and RHS, Condition (5.21) is now applied in the synthesis model for wavelet-based reconstruction [59,60]. Monotonically decreasing parameter values resulting in low steady-state errors can be obtained by reducing the absolute difference between l_1 -norms of the consistency and sparse approximation errors. In further discussions, this difference is referred to as the discrepancy level.

The significance of the discrepancy level in image reconstruction from randomly under-sampled k -space can be attributed to the fact that the noise-like artifacts in the

reconstructed image occurs due to perturbation in the regression matrix resulting from a mismatch between the adopted basis F_u and the fully sampled Fourier operator F [61,62]. The perturbations resulting in both noise and deviation of the operator from the adopted basis are accounted for using the sparse approximation error defined as:

$$\mathcal{E}_n^{(k-1)} = \hat{\mathcal{W}}^{(k-1)} - \mathcal{W}^{(k-1)}. \quad (5.22)$$

This is also evident from the plot of $\|\mathcal{E}_n^{(1)}\|_1$ against the sampling ratio in Figure 5.3 using retrospectively under-sampled phantom with added noise ϵ . Here the added noise represents perturbation in acquired k -space samples and sampling ratio determines the deviation of adopted basis F_u from the fully sampled Fourier operator F . The plot in Figure 5.3 corresponds to $\|\mathcal{E}_n^{(1)}\|_1$ in the absence of noise and three input noise levels $\sigma=0.001, 0.01$ and 0.02 , respectively. Gaussian noise is first added to the real and imaginary part of k -space at the acquired sampling locations. For a given noise level, the l_1 -norm of sparse approximation error increases with decrease in sampling ratio. Similarly for a given sampling ratio, $\|\mathcal{E}_n^{(1)}\|_1$ increases with additive noise level. This confirms that errors in both the noise and operator are encoded in the sparse approximation error.

Using the expression for $\mathcal{W}^{(k-1)}$ from Equation (5.22), Equation (5.18) in terms of $\mathcal{E}_{\text{res}}^{(k)}$ takes the form:

$$\hat{\mathcal{W}}^{(k)} - \hat{\mathcal{W}}^{(k-1)} = \mathcal{E}_{\text{res}}^{(k)} - \mathcal{E}_n^{(k-1)}. \quad (5.23)$$

Taking l_1 -norm on both sides of Equation (5.23) gives:

$$\|\hat{\mathcal{W}}^{(k)} - \hat{\mathcal{W}}^{(k-1)}\|_1 = \|\mathcal{E}_{\text{res}}^{(k)} - \mathcal{E}_n^{(k-1)}\|_1. \quad (5.24)$$

When the amount of noise present in the data approaches zero, the LHS of Equation (5.24) as well as the threshold α also approach zero if the transform operator is invertible (due to Daubechies et al. [63], cf. Theorem 4.1). Also using backward triangle inequality, Equation (5.24) becomes:

$$\|\mathcal{E}_{\text{res}}^{(k)} - \mathcal{E}_n^{(k-1)}\|_1 \geq \|\mathcal{E}_{\text{res}}^{(k)}\|_1 - \|\mathcal{E}_n^{(k-1)}\|_1. \quad (5.25)$$

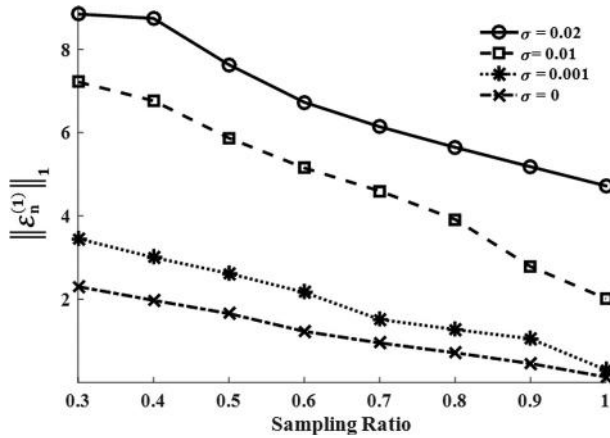


FIGURE 5.3

l_1 -norm of the sparse approximation error plotted as a function of sampling ratio.

If noise is absent in the data, then the LHS of Equation (5.25) approaches zero, and the discrepancy level defined as:

$$\mathfrak{d}_1^{(k)} \triangleq \left\| \mathcal{E}_{\text{res}}^{(k)} \right\|_1 - \left\| \mathcal{E}_n^{(k-1)} \right\|_1 \quad (5.26)$$

also tends to zero. Even by omitting the absolute value on the RHS, it can be shown that $\mathfrak{d}_1^{(k)} \geq 0$. To show this, it is necessary to start from the expression for the soft-thresholded wavelet coefficients given by:

$$\mathcal{W}^{(k-1)} = \begin{cases} \hat{\mathcal{W}}^{(k-1)} - \alpha^{(k-2)}, & \text{if } \hat{\mathcal{W}}^{(k-1)} \geq \alpha^{(k-2)} \\ \hat{\mathcal{W}}^{(k-1)} + \alpha^{(k-2)}, & \text{if } \hat{\mathcal{W}}^{(k-1)} \leq -\alpha^{(k-2)} \\ 0, & \text{otherwise.} \end{cases} \quad (5.27)$$

Then the thresholded wavelet coefficients $\mathcal{W}^{(k-1)}$ in the $(k-1)$ 'th iteration are obtained by thresholding $\hat{\mathcal{W}}^{(k-1)}$ with the threshold $\alpha^{(k-2)}$. From Equation (5.27):

$$\left| \mathcal{W}^{(k-1)} \right| = \begin{cases} \left| \hat{\mathcal{W}}^{(k-1)} \right| - \alpha^{(k-2)}, & \text{if } \left| \hat{\mathcal{W}}^{(k-1)} \right| \geq \alpha^{(k-2)} \\ 0, & \text{otherwise.} \end{cases} \quad (5.28)$$

Using Equation (5.28), the l_1 -norm of $\mathcal{W}^{(k-1)}$ takes the following form:

$$\left\| \mathcal{W}^{(k-1)} \right\|_1 = \left\| \hat{\mathcal{W}}^{(k-1)} \right\|_{1(\geq \alpha^{(k-2)})} - N_{(k-2)} \alpha^{(k-2)}, \quad (5.29)$$

where $\left\| \hat{\mathcal{W}}^{(k-1)} \right\|_{1(\geq \alpha^{(k-2)})}$ represents the l_1 -norm of the $N_{(k-2)}$ wavelet coefficients satisfying $|\hat{\mathcal{W}}^{(k-1)}| \geq \alpha^{(k-2)}$. Expressing Equation (5.18) in terms of $\mathcal{E}_{\text{res}}^{(k)}$ and using backwards triangle inequality, the l_1 -norm of $\mathcal{E}_{\text{res}}^{(k)}$ can be expressed as:

$$\left\| \mathcal{E}_{\text{res}}^{(k)} \right\|_1 \geq \left\| \hat{\mathcal{W}}^{(k)} \right\|_1 - \left\| \mathcal{W}^{(k-1)} \right\|_1. \quad (5.30)$$

Substitution of $\left\| \mathcal{W}^{(k-1)} \right\|_1$ from Equation (5.29) into Equation (5.30) gives:

$$\left\| \mathcal{E}_{\text{res}}^{(k)} \right\|_1 \geq \left\| \hat{\mathcal{W}}^{(k)} \right\|_1 - \left\| \hat{\mathcal{W}}^{(k-1)} \right\|_{1(\geq \alpha^{(k-2)})} + N_{(k-2)} \alpha^{(k-2)}. \quad (5.31)$$

For further simplification, it is possible to approximate $\left\| \mathcal{E}_n^{(k-1)} \right\|_1$ as $N_{(k-2)} \alpha^{(k-2)}$ because the histogram of $|\mathcal{E}_n^{(k-1)}|$ can be approximated as an impulse at $\alpha^{(k-2)}$, and the number of wavelet coefficients satisfying $|\hat{\mathcal{W}}^{(k-1)}| < \alpha^{(k-2)}$ is much less than $N_{(k-2)}$. Although $\left\| \hat{\mathcal{W}}^{(k)} \right\|_1 < \left\| \hat{\mathcal{W}}^{(k-1)} \right\|_1$, the selection of threshold such that $\alpha^{(k-1)} < \alpha^{(k-2)}$ ensures that the noisy component dominates

the absolute wavelet coefficients satisfying $|\hat{\mathcal{W}}^{(k)}| \geq \alpha^{(k-1)}$ and $|\hat{\mathcal{W}}^{(k-1)}| \geq \alpha^{(k-2)}$. The following two propositions are used to further simplify the first two terms on the RHS of Equation (5.31): (1) since $\|\hat{\mathcal{W}}^{(k)}\|_{1(\geq \alpha^{(k-1)})} > \|\hat{\mathcal{W}}^{(k-1)}\|_{1(\geq \alpha^{(k-2)})}$, the signal component of $\hat{\mathcal{W}}^{(k)}$ is stronger compared to that of $\hat{\mathcal{W}}^{(k-1)}$, and (2) since $\|\hat{\mathcal{W}}^{(k)}\|_{1(< \alpha^{(k-1)})} < \|\hat{\mathcal{W}}^{(k-1)}\|_{1(< \alpha^{(k-2)})}$, the noise component of $\hat{\mathcal{W}}^{(k)}$ is weaker compared to that of $\hat{\mathcal{W}}^{(k-1)}$. As the signal components contribute more to the sum of absolute wavelet coefficients and $\|\hat{\mathcal{W}}^{(k)}\|_{1(\geq \alpha^{(k-1)})} > \|\hat{\mathcal{W}}^{(k-1)}\|_{1(\geq \alpha^{(k-2)})}$ due to (1), it easily follows that $\|\hat{\mathcal{W}}^{(k)}\|_1 > \|\hat{\mathcal{W}}^{(k-1)}\|_{1(\geq \alpha^{(k-2)})}$ and hence:

$$\|\mathcal{E}_{\text{res}}^{(k)}\|_1 \geq \|\mathcal{E}_n^{(k-1)}\|_1. \quad (5.32)$$

The resemblance in the type of errors in Equations (5.32) and (5.21) and the dependence of the difference between the LHS and the RHS of Equation (5.32) on the noise level enables one to establish a parallel to the GDP.

5.6 Adaptive Thresholded Landweber

A higher value for the threshold can result in accelerated convergence at the cost of increased steady-state errors, and vice versa. Conversely, a monotonically decreasing threshold with a high initial value can provide a low steady-state error. This is achieved by decreasing the discrepancy level in each iteration. As l_1 -norm can be represented as the sum of absolute values of all the coefficients, Equation (5.32) takes the form:

$$\sum_i |\mathcal{E}_{\text{res}}^{(k)}(i)| > \sum_i |\mathcal{E}_n^{(k-1)}(i)|. \quad (5.33)$$

Also, from the impulse shape of the histogram of $|\mathcal{E}_n^{(k-1)}|$, the inequality in Equation (5.33) can be recast in terms of the probability of the absolute error coefficients to exceed the threshold, as follows:

$$\sum_i \mathbb{P}\left\{|\mathcal{E}_{\text{res}}^{(k)}(i)| \geq \alpha^{(k-1)}\right\} > \sum_i \mathbb{P}\left\{|\mathcal{E}_n^{(k-1)}(i)| \geq \alpha^{(k-1)}\right\}. \quad (5.34)$$

As our aim is to decrease the steady-state errors at a faster rate, the adaptation should ensure $\alpha^{(k)} < \alpha^{(k-1)}$. This implies $\mathbb{P}(X \geq \alpha^{(k)}) > \mathbb{P}(X \geq \alpha^{(k-1)})$, and hence Equation (5.34) becomes:

$$\sum_i \mathbb{P}\left\{|\mathcal{E}_{\text{res}}^{(k)}(i)| \geq \alpha^{(k)}\right\} > \sum_i \mathbb{P}\left\{|\mathcal{E}_n^{(k-1)}(i)| \geq \alpha^{(k-1)}\right\}. \quad (5.35)$$

Using Markov inequality, the LHS and RHS of Equation (5.35) can be expressed as:

$$\sum_i \mathbb{P}\left\{|\mathcal{E}_{\text{res}}^{(k)}(i)| \geq \alpha^{(k)}\right\} \leq \frac{\mathbb{E}\left(|\mathcal{E}_{\text{res}}^{(k)}|\right)}{\alpha^{(k)}}, \quad (5.36a)$$

and

$$\sum_i \mathbb{P}\left\{\left|\mathcal{E}_n^{(k-1)}(i)\right| \geq \alpha^{(k-1)}\right\} \leq \frac{\mathbb{E}\left(\left|\mathcal{E}_n^{(k-1)}\right|\right)}{\alpha^{(k-1)}}. \quad (5.36b)$$

Since at convergence, $\mathbb{E}\left(\left|\mathcal{E}_{\text{res}}^{(k)}\right|\right) > \mathbb{E}\left(\left|\mathcal{E}_n^{(k-1)}\right|\right)$ and $\alpha^{(k)} < \alpha^{(k-1)}$, Equation (5.36) becomes:

$$\frac{\mathbb{E}\left(\left|\mathcal{E}_{\text{res}}^{(k)}\right|\right)}{\alpha^{(k)}} > \frac{\mathbb{E}\left(\left|\mathcal{E}_n^{(k-1)}\right|\right)}{\alpha^{(k-1)}}. \quad (5.37)$$

By taking the difference between the LHS and RHS of Equation (5.37) as an increasing function Φ of $\mathfrak{d}_1^{(k)}$ that satisfies a predefined set of conditions for all iterations, and approaches zero as $\mathfrak{d}_1^{(k)} \rightarrow 0$, $\alpha^{(k)}$ can be obtained using [50]:

$$\alpha^{(k)} = \frac{\mathbb{E}\left(\left|\mathcal{E}_{\text{res}}^{(k)}\right|\right)}{\Phi\left(\mathfrak{d}_1^{(k)}\right) + \left(\mathbb{E}\left(\left|\mathcal{E}_n^{(k-1)}\right|\right) / \alpha^{(k-1)}\right)}. \quad (5.38)$$

The function $\Phi(\cdot)$ is selected using the conditions derived by utilizing the monotonically decreasing nature of $\mathfrak{d}_1^{(k)}$, $\mathbb{E}\left(\left|\mathcal{E}_{\text{res}}^{(k)}\right|\right)$, $\mathbb{E}\left(\left|\mathcal{E}_n^{(k)}\right|\right)$ and $\alpha^{(k)}$ with k as a result of adaptation. As shown in the appendix in [Section A5.1](#), the lower and upper limits for $\Phi\left(\mathfrak{d}_1^{(k)}\right)$ are:

$$\Phi\left(\mathfrak{d}_1^{(k)}\right) > \frac{1}{\alpha^{(k-1)}} \left(\mathbb{E}\left(\left|\mathcal{E}_{\text{res}}^{(k)}\right|\right) - \mathbb{E}\left(\left|\mathcal{E}_n^{(k-1)}\right|\right) \right), \quad (5.39)$$

and

$$\Phi\left(\mathfrak{d}_1^{(k)}\right) < \frac{\mathbb{E}\left(\left|\mathcal{E}_{\text{res}}^{(k)}\right|\right)}{\alpha^{(k)}} - \frac{\mathbb{E}\left(\left|\mathcal{E}_n^{(k)}\right|\right)}{\alpha^{(k-1)}}. \quad (5.40)$$

The expectations on the RHS of Equations (5.39) and (5.40) can be obtained using the histograms of $|\mathcal{E}_{\text{res}}^{(k)}|$, $|\mathcal{E}_n^{(k)}|$ and $|\mathcal{E}_n^{(k-1)}|$. Thus, the function $\Phi(\cdot)$ is chosen so that it has an increasing function of $\mathfrak{d}_1^{(k)}$ with $\Phi(\mathfrak{d}_1^{(k)}) \rightarrow 0$ as $\mathfrak{d}_1^{(k)} \rightarrow 0$, and numerically satisfies the limits in Equations (5.39) and (5.40) for $\mathfrak{d}_1^{(k)}$. Three sample functions $\mathfrak{d}_1^{(k)}$, $\log(1 + \mathfrak{d}_1^{(k)})$ and $1 - \exp(-\mathfrak{d}_1^{(k)})$ are used for $\Phi(\cdot)$ that satisfy the conditions for all datasets used. To generalize the adaptation functions to other datasets, the functions are scaled using a scale factor c so that the scaled functions are given by $c\mathfrak{d}_1^{(k)}$, $\log(1 + c\mathfrak{d}_1^{(k)})$ or $1 - \exp(-c\mathfrak{d}_1^{(k)})$. At the start of the iterations, it is required to search for a suitable value of c for which the limits are satisfied within the first few iterations. The lower limit of the threshold as shown in the appendix in [Section A5.2](#), confirms the convergence of the threshold α as $k \rightarrow \infty$. Practically, the iterations can be terminated much earlier as too many iterations result in only a marginal improvement in RLNE with no further improvements in visual quality. For this purpose, a stopping criterion is defined that monitors the relative decrease in l_2 -norms of images reconstructed in two successive iterations using:

$$\frac{\|U^{(k)} - U^{(k-1)}\|_2}{\|U^{(k-1)}\|_2} \leq \text{tol}, \quad (5.41)$$

where tol represents the tolerance level.

5.6.1 Level-Dependent Adaptive Thresholding

Since the high-frequency bands are more affected by noise, wavelet coefficients are noisier at higher decomposition levels. This indicates that higher threshold values are required at higher decomposition levels. Typically, the coarsest level of decomposition J is chosen so that $N / 2^J \ll N$ and $N / 2^J > 1$, where N denotes the number of data samples. The detail coefficients of a two-dimensional (2D) orthogonal wavelet transform can then be represented as HH_j , HL_j , LH_j , where $j = 1, 2, \dots, J$, each with size $N / 2^j \times N / 2^j$. A level-dependent threshold can be applied to all the wavelet coefficients at that level to obtain the corresponding reconstruction. Typically, thresholding is applied to the detail coefficients, whereas the approximation coefficients remain unaltered because the latter carries a larger proportion of signal components and hence is less prone to noise. Using $\mathcal{E}_{\text{res } j}^{(k)}$ and $\mathcal{E}_{n j}^{(k-1)}$ to represent the consistency and sparse approximation errors at level j , the level-dependent discrepancy level is:

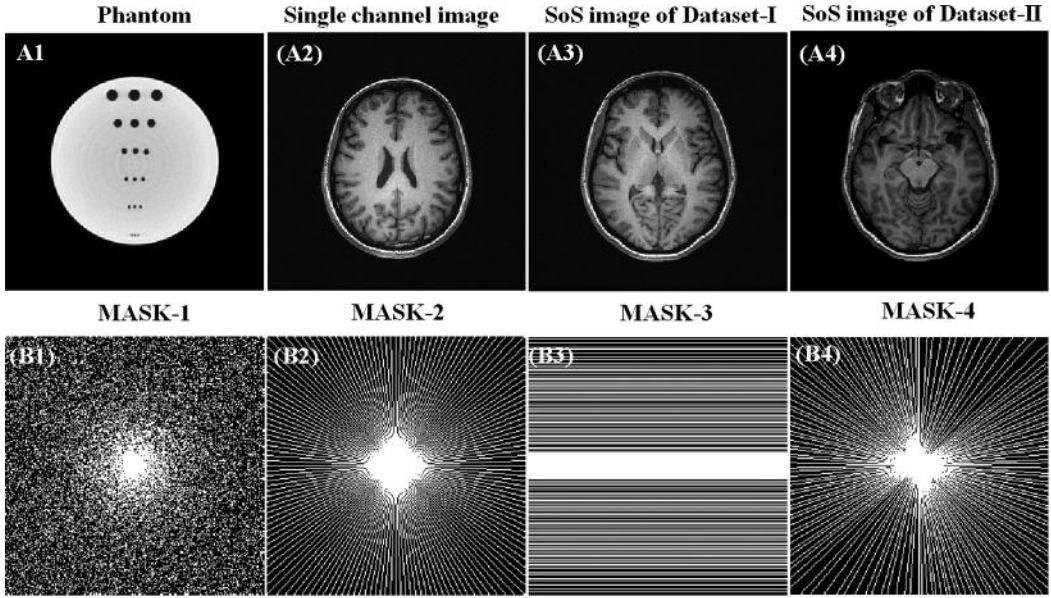
$$\mathfrak{d}_{1,j}^{(k)} = \left\| \mathbb{E} \left(\mathcal{E}_{\text{res } j}^{(k)} \right) \right\|_1 - \left\| \mathbb{E} \left(\mathcal{E}_{n j}^{(k-1)} \right) \right\|_1. \quad (5.42)$$

The level-dependent adaptation yields lower steady-state reconstruction errors compared to global adaptation.

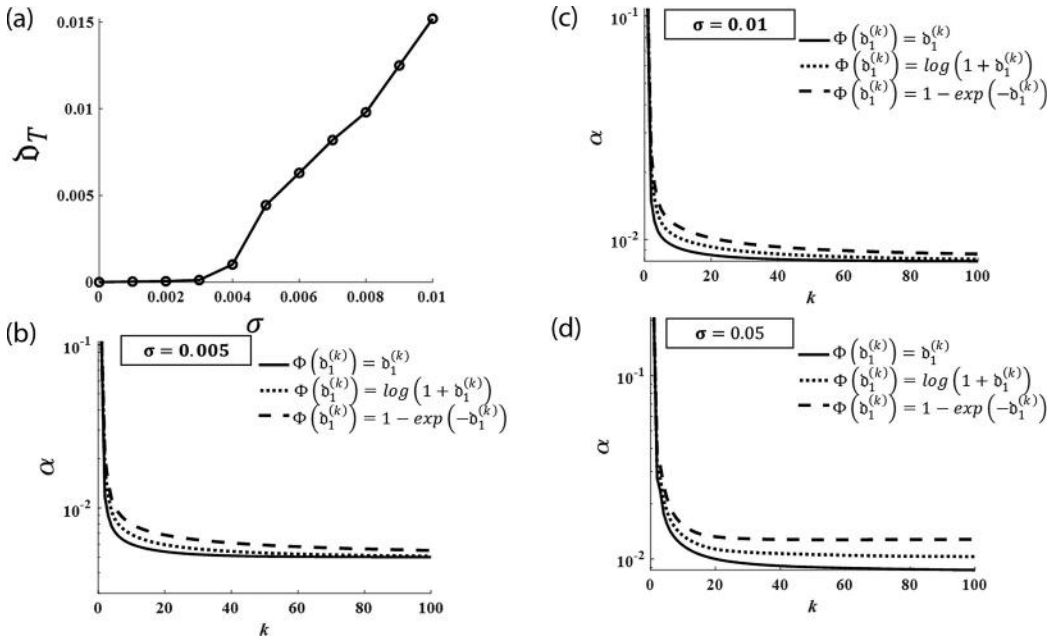
5.6.2 Numerical Simulation of Wavelet Adaptive Shrinkage CS Reconstruction Problem

In this section, the adaptive TL algorithm is applied to reconstruct the noisy versions of the Shepp-Logan phantom to illustrate (1) the dependence of $\mathfrak{d}_T = \mathfrak{d}_1^{(\infty)}$ on the noise level and (2) the effect of change in the adaptation function on reconstruction errors. For (1), adaptive TL reconstructions are performed on the phantom (shown in [Figure 5.4](#)), with added noise randomly under-sampled using MASK-1 (shown in [Figure 5.4](#)).

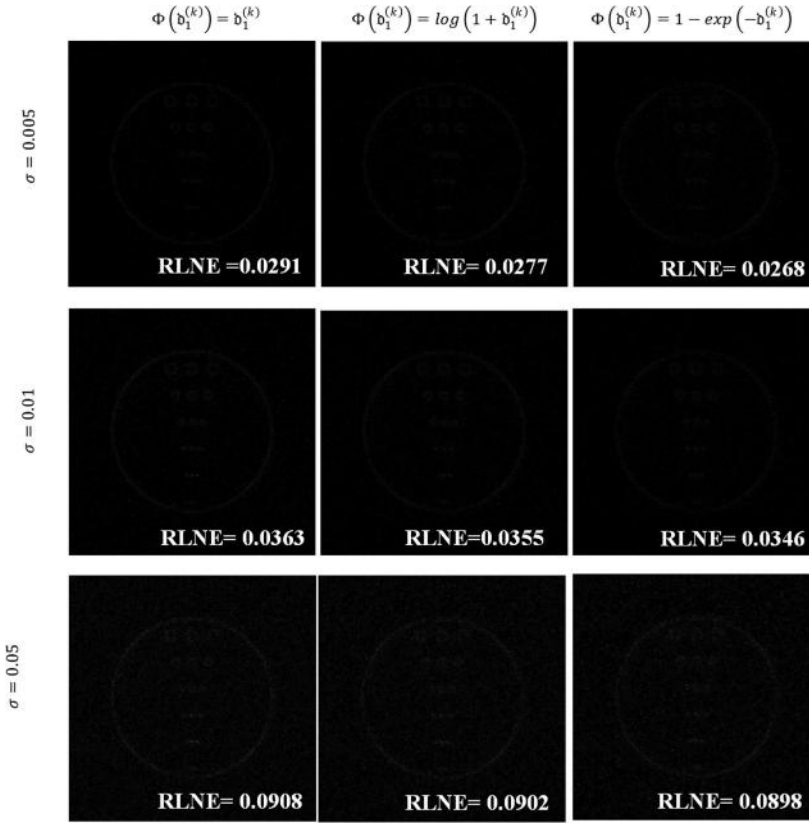
[Figure 5.5a](#) shows discrepancy level $\mathfrak{d}_T$ at convergence plotted as a function of the noise standard deviation. For (2), adaptive TL reconstructions are performed using three adaptive functions, including $\mathfrak{d}_1^{(k)}$, $\log(1 + \mathfrak{d}_1^{(k)})$ and $1 - \exp(-\mathfrak{d}_1^{(k)})$. Plots of threshold versus number of iterations k are shown for three input noise levels with standard deviation $\sigma = 0.005, 0.01$ and 0.05 in [Figure 5.5b–d](#). Adaptive TL reconstructions are performed using three sample functions $\Phi(\cdot)$. From the plots, it is evident that the threshold at convergence increases with increase in the input noise level. Furthermore, the threshold variation for all exemplar adaptive functions looks identical. The difference images obtained using all exemplar functions are shown in [Figure 5.6](#). Thresholds obtained using the adaptive algorithm are found to decrease rapidly and converge to a steady-state value. The initial sharp decrease is due to large differences between l_1 -norms of the consistency and sparse approximation errors.

**FIGURE 5.4**

(A1) Phantom image, (A2) single-channel image, (A3,A4), SoS images of datasets I and II, (B1–B4) sampling masks.

**FIGURE 5.5**

(a) Discrepancy level at convergence plotted as a function of noise standard deviation. (b–d) Threshold versus iteration number for three noise levels (provided in inset).

**FIGURE 5.6**

Difference images of adaptive TL reconstructions for noisy versions of phantom using MASK-1.

5.6.3 Illustration Using Single-Channel MRI

For the sake of illustration, the fully sampled k -space of a channel combined image is retrospectively under-sampled using MASK-1 (shown in Figure 5.4). The variable density sampling mask is generated using a power law decaying function $g(\kappa) = (1 - |\kappa|/|\kappa_m|)^q + \beta$, for $q = 2$, real constant $\beta \in [-1, 1]$, and $|\kappa| = \kappa_x^2 + \kappa_y^2$, where κ_m corresponds to the maximum spatial frequency of the acquired k -space [64,65]. Values of $g(\kappa_i)$ lie in the interval $[0, 1]$ as described in [61]. All reconstructions shown in this section are performed using Daubechies wavelets with four decomposition levels and adaptation function $\Phi(b_1^{(k)}) = b_1^{(k)}$. The adaptive TL algorithm is compared with block-matching and three-dimensional filtering (BM3D)-MRI [66,67] which utilizes the self-repetitive structure in images and enforces nonlocal sparsity as provided in the MATLAB script, shared at http://web.itu.edu.tr/eksioglu/pubs/BM3D_MRI.htm.

Despite the fact that nonlocal sparse regularization enables preservation of finer details, it has the drawback of generating noise-like artifacts within piece-wise smooth regions [68], as shown in Figure 5.7. The advantages of both techniques can be obtained by combining BM3D with adaptive TL using the fast composite splitting denoising (FCSD) method [69] shown in Figure 5.7 (A5). For this purpose, the original problem is first decomposed into l_1 and patch-based non-local sparsity regularization, respectively, wherein the reconstructed image is computed from the weighted average of solutions from two sub-problems in an iterative framework, as summarized in Algorithm (5.1).

Algorithm 5.1 FCSD

Initialization

Initialize $r^{(0)} = \mathbf{0}$, $U^0 = \mathbf{0}$, $t^{(0)} = 1$, $\alpha, \gamma, \text{tol} = 1.0e - 4$.

Input zero-filled k -space (K_u)

Main Iteration

1) Iterate for $k = 1, \dots, \text{MaxIter}$:

$$U_g = r^{(k)} + \text{FT}'(K_u - \text{FT}(U^{(k-1)}))$$

$$U_1 = \min_U \left\{ 2\alpha \|\Psi U^{(k-1)}\|_1 + \left(\|U^{(k-1)} - U_g\|_2 \right)^2 / 2 \right\}$$

$$U_2 = \min_U \left\{ 2\gamma \|U^{(k-1)}\|_{NL} + \left(\|U^{(k-1)} - U_g\|_2 \right)^2 / 2 \right\}$$

$$U^k = (U_1 + U_2) / 2$$

$$t^{(k+1)} = \frac{1 + \sqrt{1 + 4t^{(k)2}}}{2}$$

$$r^{(k+1)} = U^k + \frac{(t^{(k)} - 1)}{t^{(k+1)}} (U^{(k)} - U^{(k-1)})$$

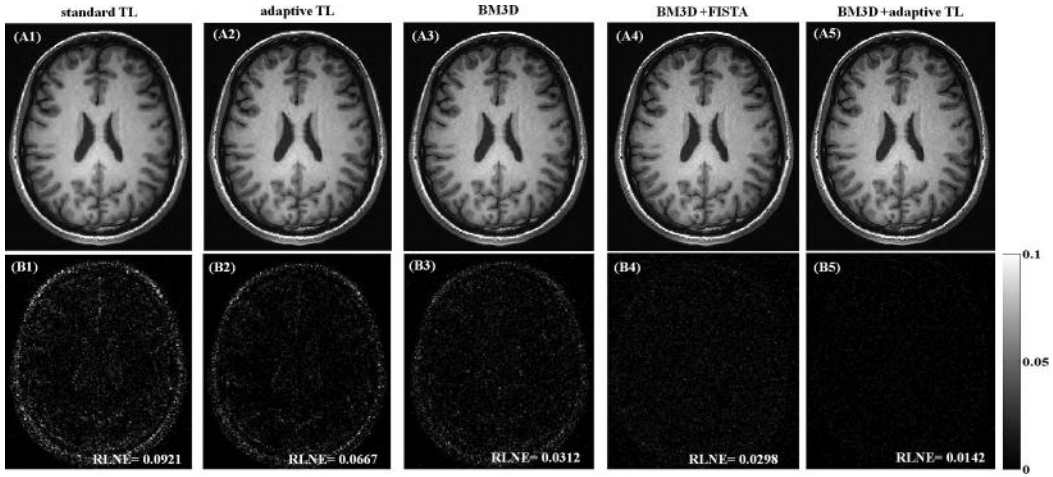
2) Repeat step 1 until $\frac{\|U^{(k)} - U^{(k-1)}\|_2}{\|U^{(k-1)}\|_2} \leq \text{tol}$.

end

NL: Non-local wavelet norm is the l_1 -norm of the wavelet coefficients obtained by stacking similar patches in a 3D pattern. The wavelet coefficients are obtained by computing the two-dimensional (2D) wavelet transform on each patch and then performing one-dimensional (1D) wavelet transform along the third direction.

The top and bottom panels in [Figure 5.7](#) show the reconstructed and difference images using MASK-1 (shown in [Figure 5.4](#)). Columns from left to right represent reconstructions obtained using standard TL, adaptive TL, BM3D, FCSD (BM3D, fast iterated soft-thresholding algorithm [FISTA]) and FCSD (BM3D, Adaptive TL). For each type of reconstruction, RLNE values are indicated in insets. The difference images are provided to show all sources of errors, including blurring, aliasing, and noise.

In the FCSD implementation, the parameter γ denotes the denoising threshold of the BM3D denoiser. In the BM3D implementation, a diminishing threshold is used with the

**FIGURE 5.7**

Comparison of reconstructions using standard TL, adaptive TL, BM3D, FCSD (TL, BM3D) and FCSD (adaptive TL, BM3D) using MASK-1, (A1–A5) reconstructed images, (B1–B5) difference images.

parameter decreasing on a logarithmic scale, starting from a higher initial parameter to a lower value [66]. The initial γ value can be manually tuned around the default parameter suggested by the original BM3D implementation [67] to make the RLNE as low as possible. The changes in initial γ do not significantly affect the performance. However, as indicated in the original implementation of BM3D-MRI [66], the RLNE begins to slowly increase if γ is further reduced to smaller values.

5.6.4 Application to pMRI

The following three types of threshold adaptation implementations derived from Equation (5.38) are used in parallel MRI: (1) $\mathcal{E}_{\text{res}}^{(k)}$ and $\mathcal{E}_n^{(k-1)}$ computed from wavelet coefficients of combined image (Method I), (2) sum-of-squares (SoS) of channel-wise errors (Method II) and (3) covariance matrix of channel-wise errors (Method III).

5.6.4.1 Update Calculation Using Error Information from Combined Image (Method I)

In each iteration, the wavelet coefficients of consistency and sparse approximation errors calculated from the SoS combined image are used to obtain the update, as in the single channel case. Individual channels are reconstructed using the updated threshold.

5.6.4.2 Update Calculation Using SoS of Channel-wise Errors (Method II)

Errors in the image domain calculated from individual channels are combined using:

$$E_{\text{res}}^{(k)} = \sqrt{\sum_{c=1}^{nC} \left(\Psi' \left(\mathcal{E}_{\text{res},c}^{(k)} \right) \right)^2} \quad (5.43)$$

and

$$E_n^{(k-1)} = \sqrt{\sum_{c=1}^{nC} \left(\Psi' \left(\mathcal{E}_{n,c}^{(k-1)} \right) \right)^2},$$

where n_c is the number of channels, and $\mathcal{E}_{X,c}^{(k)}$ denotes the respective channel-wise wavelet-based error coefficients. With $\epsilon_{\text{res}}^{(k)} = \Psi(E_{\text{res}}^{(k)})$ and $\epsilon_n^{(k-1)} = \Psi(E_n^{(k-1)})$ representing the respective wavelet coefficients, it is possible to relate the probability of absolute error exceeding the threshold using the following inequality:

$$\sum_i \mathbb{P} \left\{ \left| \epsilon_{\text{res}}^{(k)} \right| - \mathbb{E} \left(\left| \epsilon_{\text{res}}^{(k)} \right| \right) \geq \alpha^{(k)} \right\} > \sum_i \mathbb{P} \left\{ \left| \epsilon_n^{(k-1)} \right| - \mathbb{E} \left(\left| \epsilon_n^{(k-1)} \right| \right) \geq \alpha^{(k-1)} \right\} \quad (5.44)$$

Application of Tchebycheff inequality to Equation (5.44) gives:

$$\mathbb{P} \left\{ \left| \epsilon_{\text{res}}^{(k)} \right| - \mathbb{E} \left(\left| \epsilon_{\text{res}}^{(k)} \right| \right) \geq \alpha^{(k)} \right\} \leq \frac{\left(\text{var} \left(\left| \epsilon_{\text{res}}^{(k)} \right| \right) \right)^2}{\left(\alpha^{(k)} \right)^2} \quad (5.45a)$$

and

$$\mathbb{P} \left\{ \left| \epsilon_n^{(k-1)} \right| - \mathbb{E} \left(\left| \epsilon_n^{(k-1)} \right| \right) \geq \alpha^{(k-1)} \right\} \leq \frac{\left(\text{var} \left(\left| \epsilon_n^{(k-1)} \right| \right) \right)^2}{\left(\alpha^{(k-1)} \right)^2}, \quad (5.45b)$$

where $\text{var}(\cdot)$ denotes the variance of the respective errors. Using Equations (5.38), (5.45a) and (5.45b), the new threshold can be computed as:

$$\alpha^{(k)} = \left[\frac{\left(\text{var} \left(\left| \epsilon_{\text{res}}^{(k)} \right| \right) \right)^2}{\Phi \left(\mathfrak{d}_1^{(k)} \right) + \left(\left(\text{var} \left(\left| \epsilon_n^{(k-1)} \right| \right) \right)^2 / \left(\alpha^{(k-1)} \right)^2} \right)} \right]^{\frac{1}{2}}. \quad (5.46)$$

The updated threshold is applied to the individual channels for reconstruction.

5.6.4.3 Update Calculation Using Covariance Matrix (Method III)

The channel-wise consistency ($\mathcal{E}_{\text{res},c}^{(k)}$) and sparse approximation errors ($\mathcal{E}_{n,c}^{(k-1)}$) are first computed from wavelet coefficients of individual channel images. With the assumption that the channel-wise errors possess different variances and are correlated, another form of the Tchebycheff inequality is utilized to obtain the threshold update. Using $\xi_X^{(k)}$ to denote the sum averaged wavelet error coefficients, application of the Tchebycheff inequality gives:

$$\mathbb{P} \left\{ \left| \xi_{\text{res}}^{(k)} \right| - \mathbb{E} \left(\left| \xi_{\text{res}}^{(k)} \right| \right) \geq \alpha^{(k)} \right\} \leq \frac{\sum_i \sum_j \sigma_{ij}^{(k)}(\text{res})}{\left(\alpha^{(k)} \right)^2} \quad (5.47)$$

and

$$\mathbb{P} \left\{ \left| \xi_n^{(k-1)} - \mathbb{E} \left(\xi_n^{(k-1)} \right) \right| \geq \alpha^{(k-1)} \right\} \leq \frac{\sum_i \sum_j \sigma_{ij(n)}^{(k-1)}}{(\alpha^{(k-1)})^2},$$

where σ_{ij}^2 denote covariances of absolute wavelet coefficients of the respective errors from the channel pair (i, j) . Using Equation (5.47), the updated threshold is obtained as:

$$\alpha^{(k)} = \left[\frac{\sum_i \sum_j \sigma_{ij(\text{res})}^{(k)}}{\Phi(\mathfrak{d}_1^{(k)}) + \left(\sum_i \sum_j \sigma_{ij(n)}^{(k-1)} / \alpha^{(k-1)2} \right)} \right]^{\frac{1}{2}}. \quad (5.48)$$

5.6.4.4 Illustration Using In Vivo Data

The differences between each algorithm lie in the threshold adaptation step that include $\mathfrak{d}_1^{(k)}$ computation and $\alpha^{(k)}$ update. The steps followed in the implementation of standard TL are provided in Algorithm (5.2) for comparison. In Algorithm (5.3), the wavelet coefficients of errors computed from the SoS combined image are used to obtain the update as in the single-channel case. In Algorithm (5.4), the consistency and sparse approximation errors of individual channels are combined using SoS. In Algorithm (5.5), the two errors are derived from the covariance of the respective errors. To enable further acceleration, FISTA steps are included in all algorithms; $\tilde{\mathcal{W}}^{(k)}$ is used to denote the FISTA updated wavelet coefficients. The initialization steps are same for all algorithms and include setting $\mathcal{W}^{(0)} = \tilde{\mathcal{W}}^{(0)} = \mathbf{0}$, $t^{(0)} = 1$, $\text{tol} = 1.0e-4$, K_u as the zero-filled k -space, and the initial threshold $\alpha^{(0)}$ is chosen as outlined in [Section 5.4.1](#).

Algorithm 5.2 Standard TL with FISTA

Iterate for $k = 1, \dots, \text{MaxIter}$:

1) Iterate for $c = 1, \dots, n_c$:

(i) Calculate $\mathcal{E}_{\text{res}}^{(k)}$ as in (5.18)

(ii) Find $\hat{\mathcal{W}}^{(k)} = \tilde{\mathcal{W}}^{(k-1)} + \mathcal{E}_{\text{res}}^{(k)}$

(iii) Soft Threshold the wavelet Coefficients

$$\mathcal{W}^{(k)} = \mathcal{T}_\alpha \left\{ \hat{\mathcal{W}}^{(k)} \right\}$$

(iv) Find $\mathcal{U}_c^{(k)} = \Psi' \mathcal{W}^{(k)}$

$$(v) \text{ Find } t^{(k)} = \frac{1 + \sqrt{1 + 4t^{(k-1)2}}}{2}$$

$$(vi) \tilde{\mathcal{W}}^{(k)} = \mathcal{W}^{(k)} + \frac{t^{(k-1)} - 1}{t^{(k)}} \left(\mathcal{W}^{(k)} - \mathcal{W}^{(k-1)} \right)$$

end

2) Compute the SoS image $U^{(k)} = \sqrt{\sum_{c=1}^{n_C} (U_c^{(k)})^2}$

3) Repeat steps (1)–(2) until $\frac{\|U^{(k)} - U^{(k-1)}\|_2}{\|U^{(k-1)}\|_2} \leq tol$

end

Algorithm 5.3 Adaptive TL with FISTA (Method I)

Iterate for $k = 1, \dots, \text{MaxIter}$:

1) *Iterate for $c = 1, \dots, n_C$:*

i) Follow steps (1)–(2) as in (A5.2)

2) *Calculate $\mathcal{E}_n^{(k-1)}$ and $\mathcal{E}_{\text{res}}^{(k)}$ from combined image*

3) *Calculate the expectations of $|\mathcal{E}_{\text{res}}^{(k)}|$ and $|\mathcal{E}_n^{(k-1)}|$ from respective histograms*

4) *Calculate $\mathfrak{d}_1^{(k)} = \|\mathcal{E}_{\text{res}}^{(k)}\|_1 - \|\mathcal{E}_n^{(k-1)}\|_1$*

5) *Update α^k using (5.38)*

6) Repeat steps (1) to (5) until $\frac{\|U^{(k)} - U^{(k-1)}\|_2}{\|U^{(k-1)}\|_2} \leq tol$

end

Algorithm 5.4 Adaptive TL with FISTA (Method II)

Iterate for $k = 1, \dots, \text{MaxIter}$:

1) *Iterate for $c = 1, \dots, n_C$:*

i) Follow steps (1)–(2) as in (A5.2)

2) *Compute $\epsilon_{\text{res}}^{(k)}$ and $\epsilon_n^{(k)}$ from the image domain errors obtained from (5.25)*

3) *Calculate the expectations of $|\epsilon_{\text{res}}^{(k)}|$ and $|\epsilon_n^{(k-1)}|$ from respective histogram*

4) *Calculate $\mathfrak{d}_1^{(k)} = \|\epsilon_{\text{res}}^{(k)}\|_1 - \|\epsilon_n^{(k-1)}\|_1$*

5) *Update the threshold using (5.46)*

6) Repeat steps (1) to (5) until $\frac{\|U^{(k)} - U^{(k-1)}\|_2}{\|U^{(k-1)}\|_2} \leq \text{tol.}$

end

Algorithm 5.5 Adaptive TL with FISTA (Method III)

Iterate for $k = 1, \dots, \text{MaxIter}$:

1) *Iterate for* $c = 1, \dots, n_c$:

i) *Follow steps (1)–(2) as in (A5.2)*

2) Calculate $\xi_{\text{res}}^{(k)} = \left(\frac{1}{n_c}\right) \sum_{c=1}^{n_c} \mathcal{E}_{\text{res},c}^{(k)}$ and $\xi_n^{(k)} = \left(\frac{1}{n_c}\right) \sum_{c=1}^{n_c} \mathcal{E}_{n,c}^{(k)}$

3) *Calculate the covariance measures between the all*

channel combinations $\left(\sigma_{ij(\text{res})}^{(k)} \text{ and } \sigma_{ijn}^{(k-1)}\right)$ *from respective histograms*

4) Calculate $\mathfrak{d}_1^{(k)} = \|\xi_{\text{res}}^{(k)}\|_1 - \|\xi_n^{(k-1)}\|_1$

5) *Update the threshold using (5.48)*

6) Repeat steps (1) to (5) until $\frac{\|U^{(k)} - U^{(k-1)}\|_2}{\|U^{(k-1)}\|_2} \leq \text{tol.}$

end

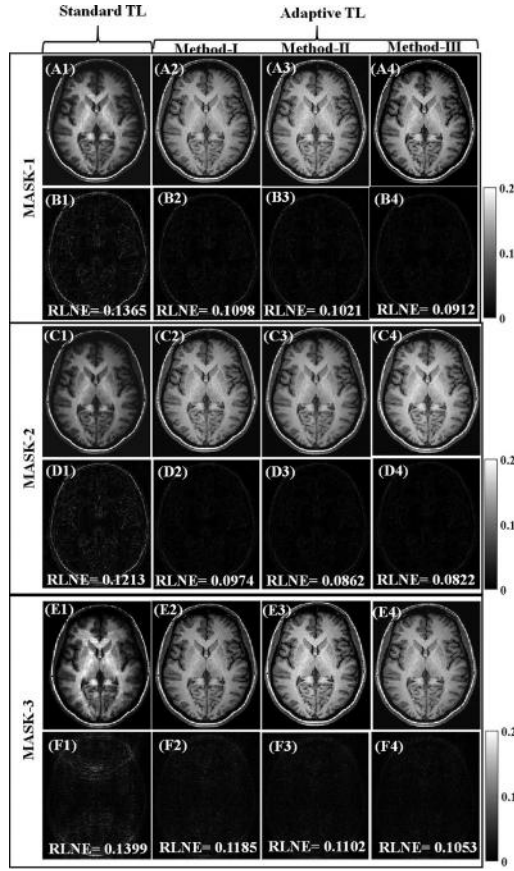
Reconstructions are performed using an adaptive TL algorithm in which threshold updates are computed using: (1) errors from combined k -space (Method I), (2) combined errors from individual channel k -spaces (Method II) and (3) error covariance matrix (Method III). Details of acquisition of the in vivo data are summarized in the first eight columns of [Table 5.1](#). Threshold values used for standard TL and initial threshold values for adaptive TL are summarized in columns 9 and 10. Reconstruction errors for adaptive TL reconstruction and standard TL using MASK-1 (shown in [Figure 5.4](#)) are listed in the last four columns.

[Figure 5.8](#) shows reconstructions of dataset I obtained using standard TL (column 1) and adaptive TL Methods I, II and III, respectively. The top two panels show the reconstructed and difference images obtained with MASK-1 (shown in [Figure 5.4](#)). The bottom four panels correspond to (1) radial sampling scheme using MASK-2 (shown in [Figure 5.4](#)), and (2) random PE under sampling of Cartesian k -space trajectories based on the golden ratio using MASK-3 (shown in [Figure 5.4](#)) [70]. For the radial sampling scheme, 80 spokes are used, with 256 samples along each spoke. The sampling scheme using random phase encodes with 128 PE lines; 28 lines from the central k -space region are equally spaced and the rest are randomly spaced. RLNE values provided in the insets conform to the improved quality of reconstruction obtained using the adaptive TL method as confirmed by visual inspection of the difference images.

TABLE 5.1
Acquisition Parameters, Initial Thresholds, and RLNE for Datasets I-III

Dataset	Pulse Sequence	Number of Channels	TE (ms)	TR (ms)	FOV (mm)	Matrix Size	Scanner	Initial Threshold $\alpha^{(1)}$		RLNE			
								^a Standard TL	Adaptive TL	Standard TL	Adaptive TL (Method-I)	Adaptive TL (Method-II)	Adaptive TL (Method-III)
I	BRAVO	12	3	8	240	256 × 256	GE 3T	0.008	0.20	0.1365	0.1098	0.1021	0.0912
II	BRAVO	12	3.6	9.2	240	256 × 256	GE 3T	0.010	0.17	0.1821	0.1625	0.1606	0.1588
IV	T2 weighted spin echo	6	92	5500	230	256 × 256	Siemens 1.5 T	0.015	0.09	0.1341	0.1076	0.0781	0.0698

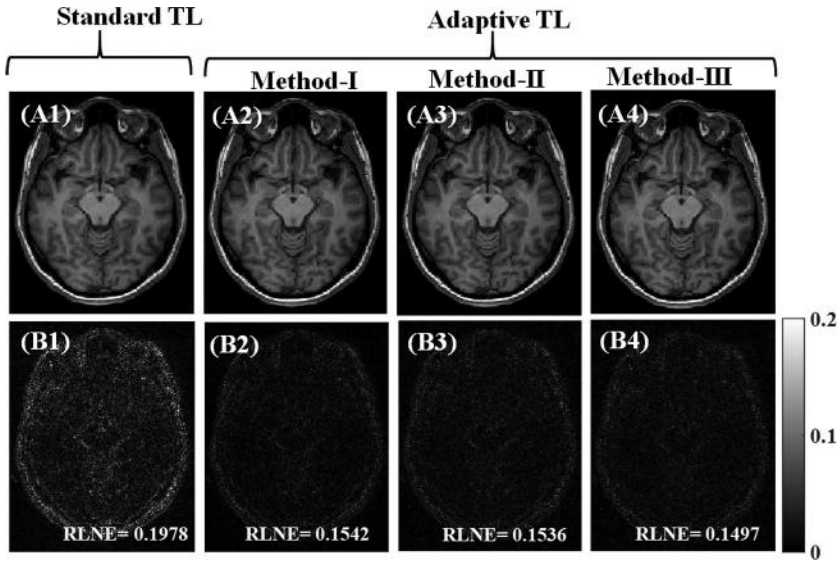
^a $\alpha^{(1)}$ does not change with iterations in standard TL.

**FIGURE 5.8**

Multi-channel reconstructions for *Dataset-I* using standard TL (Column-1) and adaptive TL methods-I, II and III. Top, middle and bottom panels show ((A1)-(A4), (C1)-(C4), and (E1)-(E4)) reconstructed and ((B1)-(B4), (D1)-(D4), and (F1)-(F4)) difference images for retrospective under-sampling using MASKs-1, 2 and 3 respectively. RLNE values are inserted in the insets.

Figure 5.9 shows adaptive TL reconstruction for dataset II using a radial sampling scheme with angular separation based on the golden ratio [71] using MASK-4 (shown in Figure 5.4). The trade-off of signal-to-noise ratio (SNR) versus resolution is more optimal for adaptive TL Method III. The difference images shown in the bottom panel illustrate localized errors and the effect on resolution.

Comparing the three variants of adaptive TL algorithms applied to parallel MRI, the fastest one is that corresponding to threshold adaptation using the SoS of consistency and sparse approximation errors. This is based on the computation of channel-wise errors as well as SoS error images for updating the threshold. The threshold update approach using error information from the combined channel data requires an additional loop for coil combination. This leads to an increase in computation time compared to adaptation using SoS error images. Although the scheme based on covariance update provides the best image quality, it has the highest computation time compared to the above two pMRI variants due to the additional error covariance computation in each iteration.

**FIGURE 5.9**

Reconstructions using (A1) standard TL, (A2) Adaptive TL-Method I, (A3) Adaptive TL-Method II and (A4) Adaptive TL-Method III for dataset-II. (B1)-(B4) Difference images.

5.6.4.5 Illustration Using Synthetic Data

MR coil images are synthesized by element-wise multiplication of a sample proton density map with simulated coil sensitivity maps. The complex coil sensitivities are synthesized using the Biot-Savart law applied over a 2D-plane, with the phantom located at the centre of the field of view (FOV). The coils are placed at equal distances from the centre of FOV. The absolute sum-of-squared coil image is identical to the reference proton density map. Figure 5.10 shows plots of reconstruction time as a function of the number of coils for (A) adaptive TL stand-alone modes I, II and III and (B) FISTA combined with adaptive TL modes I, II, and III. For ease of comparison, the performance curve for FISTA is shown in both panels.

A phantom image involving thin structures is used to evaluate the SNR resolution trade-off in Figure 5.11. The difference images are used to show all sources of errors, including blurring, aliasing and noise. The improved resolution and reduction in localized errors is clearly evident from the difference images.

Wavelet threshold is iteratively adapted based on a discrepancy level computed from the consistency and sparse approximation errors deduced with analogy to the residual and perturbation errors in the GDP. The threshold adaptation is shown to result in reduced reconstruction errors along with faster convergence. Accelerated convergence is ensured when the threshold is adapted so that its value in the succeeding iteration is always less than the current value. With incorporation of threshold adaptation steps, the order of complexity increases by only N^2 flops per iteration for an image of size $N \times N$.

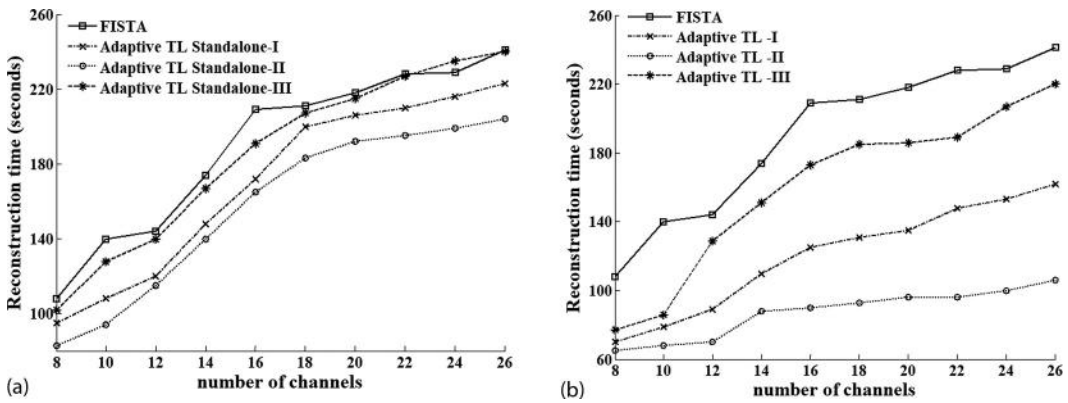


FIGURE 5.10

Reconstruction time as a function of the number of coils for (a) adaptive TL stand-alone modes I, II and III and (b) FISTA combined with adaptive TL modes I, II and III. For ease of comparison, the performance curve for FISTA is shown in both panels.

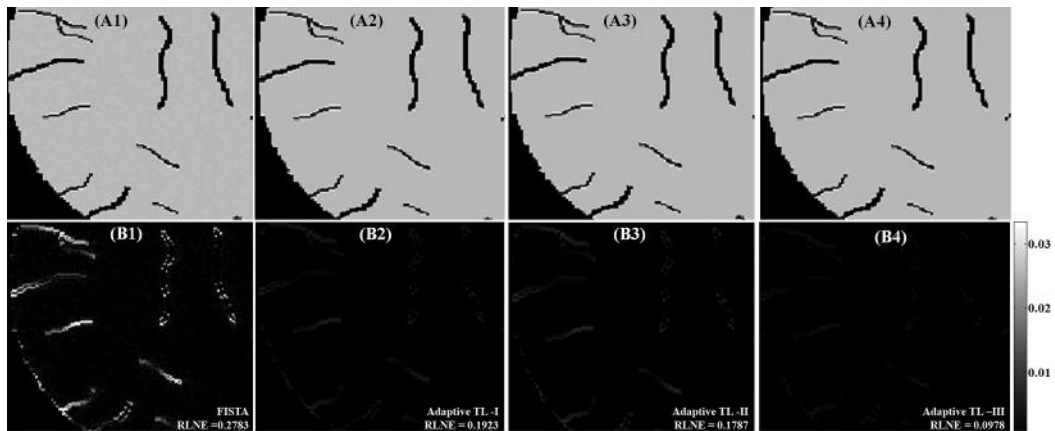


FIGURE 5.11

Reconstructions using (A1) standard TL, (A2) Adaptive TL-Method I, (A3) Adaptive TL-Method II and (A4) Adaptive TL-Method III. (B1)-(B4) Difference images.

The threshold adaptation shows the combined advantages of the reweighting strategy and FISTA. The main advantage of using the reweighting is a fractional improvement in the peak signal-to-noise ratio (PSNR), without improvements in convergence rates. In fact, any change in PSNR after 30 dB is insignificant (in image processing). However, the method gains attention only when it is accompanied by improvements in either reduction in complexity or computation time.

In addition to the reduction in reconstruction errors, all three variants of adaptive TL are shown to preserve image resolution compared to the standard ISTA algorithm. Thus, an improved trade-off between SNR and image resolution can be achieved with adaptation compared to other variants without involving adaptation.

Appendix

A5.1 Conditions for Selection of $\Phi(\cdot)$

The conditions for choosing an adaptation function $\Phi(\mathfrak{d}_1^{(n)})$ can be obtained by using the monotonically decreasing nature of $\mathfrak{d}_1^{(n)}$, $\mathbb{E}(|\mathcal{E}_{\text{res}}^{(n)}|)$, $\mathbb{E}(|\mathcal{E}_n^{(n)}|)$ and $\alpha^{(n)}$ with $k = n$. Using the fact that $\alpha^{(n)} < \alpha^{(n-1)}$ and Equation (5.38), we have:

$$\frac{\Phi(\mathfrak{d}_1^{(n)})}{\mathbb{E}(|\mathcal{E}_{\text{res}}^{(n)}|)} + \frac{\mathbb{E}(|\mathcal{E}_n^{(n-1)}|)}{\alpha^{(n-1)}\mathbb{E}(|\mathcal{E}_{\text{res}}^{(n)}|)} > \frac{1}{\alpha^{(n-1)}}. \quad (5.49)$$

From Equation (5.49), we have:

$$\Phi(\mathfrak{d}_1^{(n)}) > \frac{1}{\alpha^{(n-1)}} \left(\mathbb{E}(|\mathcal{E}_{\text{res}}^{(n)}|) - \mathbb{E}(|\mathcal{E}_n^{(n-1)}|) \right). \quad (5.50)$$

From the fact that $\mathbb{E}(|\mathcal{E}_{\text{res}}^{(n-1)}|) > \mathbb{E}(|\mathcal{E}_{\text{res}}^{(n)}|)$ and $\alpha^{(n)} < \alpha^{(n-1)}$ with adaptation, the denominators of Equation (5.38) should satisfy:

$$\Phi(\mathfrak{d}_1^{(n)}) + \frac{\mathbb{E}(|\mathcal{E}_n^{(n-1)}|)}{\alpha^{(n-1)}} > \Phi(\mathfrak{d}_1^{(n-1)}) + \frac{\mathbb{E}(|\mathcal{E}_n^{(n-2)}|)}{\alpha^{(n-2)}}. \quad (5.51)$$

This is the same as:

$$\Phi(\mathfrak{d}_1^{(n)}) > \Phi(\mathfrak{d}_1^{(n-1)}) - \left[\frac{\mathbb{E}(|\mathcal{E}_n^{(n-1)}|)}{\alpha^{(n-1)}} - \frac{\mathbb{E}(|\mathcal{E}_n^{(n-2)}|)}{\alpha^{(n-2)}} \right] \quad (5.52)$$

Substituting $\Phi(\mathfrak{d}_1^{(n-1)}) = \frac{\mathbb{E}(|\mathcal{E}_{\text{res}}^{(n-1)}|)}{\alpha^{(n-1)}} - \frac{\mathbb{E}(|\mathcal{E}_n^{(n-2)}|)}{\alpha^{(n-2)}}$ in Equation (5.52) gives:

$$\Phi(\mathfrak{d}_1^{(n)}) > \frac{\mathbb{E}(|\mathcal{E}_{\text{res}}^{(n-1)}|)}{\alpha^{(n-1)}} - \frac{\mathbb{E}(|\mathcal{E}_n^{(n-1)}|)}{\alpha^{(n-1)}} \quad (5.53)$$

Adding and subtracting $\frac{\mathbb{E}(|\mathcal{E}_{\text{res}}^{(n)}|)}{\alpha^{(n-1)}}$ in Equation (5.53) gives:

$$\Phi(\mathfrak{d}_1^{(n)}) > \frac{1}{\alpha^{(n-1)}} \left(\mathbb{E}(|\mathcal{E}_{\text{res}}^{(n)}|) - \mathbb{E}(|\mathcal{E}_n^{(n-1)}|) \right) + \frac{1}{\alpha^{(n)}} \left(\mathbb{E}(|\mathcal{E}_{\text{res}}^{(n-1)}|) - \mathbb{E}(|\mathcal{E}_{\text{res}}^{(n)}|) \right) \quad (5.54)$$

As $\mathbb{E}(|\mathcal{E}_{\text{res}}^{(n-1)}|) > \mathbb{E}(|\mathcal{E}_{\text{res}}^{(n)}|)$, the second term on the RHS of Equation (5.54) is always positive. Equation (5.54) reveals that the RHS of Equation (5.53) consists of the sum of a positive quantity and the RHS of Equation (5.50); thus, Equation (5.53) places a stronger

condition compared to Equation (5.50). Since $\mathbb{E}(|\mathcal{E}_n^{(n)}|) < \mathbb{E}(|\mathcal{E}_n^{(n-1)}|)$, replacing $\mathbb{E}(|\mathcal{E}_n^{(n-1)}|)$ in Equation (5.38) with $\mathbb{E}(|\mathcal{E}_n^{(n)}|)$ gives:

$$\alpha^{(n)} < \frac{\mathbb{E}(|\mathcal{E}_{\text{res}}^{(n)}|)}{\Phi(\mathfrak{d}_1^{(n)}) + \left(\mathbb{E}(|\mathcal{E}_n^{(n)}|) / \alpha^{(n-1)}\right)} \quad (5.55)$$

Rearranging Equation (5.55), we have:

$$\Phi(\mathfrak{d}_1^{(n)}) < \frac{\mathbb{E}(|\mathcal{E}_{\text{res}}^{(n)}|)}{\alpha^{(n)}} - \frac{\mathbb{E}(|\mathcal{E}_n^{(n)}|)}{\alpha^{(n-1)}} \quad (5.56)$$

Replacing n with $n+k$, and using the monotonically decreasing nature of $\mathbb{E}(|\mathcal{E}_n^{(n)}|)$ and $\alpha^{(n)}$, limits similar to Equations (5.50) and (5.56) can be obtained for $\Phi(\mathfrak{d}_1^{(n+k)})$ for $k=1,2,3,\dots$.

A5.2 Stability of the Adaptation Method

Using the limiting conditions derived for $\Phi(\mathfrak{d}_1^{(k)})$, the adapted threshold in each step is lower-bounded by:

$$\frac{\mathbb{E}(|\mathcal{E}_{\text{res}}^{(k)}|) - \mathbb{E}(|\mathcal{E}_n^{(k-1)}|)}{\Phi(\mathfrak{d}_1^{(k)})} < \alpha^{(k-1)} \quad (5.57)$$

From the definition of soft thresholding, $\mathcal{E}_{\text{res}}^{(k)} = \hat{\mathcal{W}}^{(k)} - \mathcal{W}^{(k-1)}$ and $\mathcal{E}_n^{(k-1)} = \hat{\mathcal{W}}^{(k-1)} - \mathcal{W}^{(k-1)}$ can be expressed as:

$$\mathcal{E}_{\text{res}}^{(k)} = \begin{cases} \hat{\mathcal{W}}^{(k)} - (\hat{\mathcal{W}}^{(k-1)} - \alpha^{(k-2)}), & \text{if } \hat{\mathcal{W}}^{(k-1)} \geq \alpha^{(k-2)} \\ \hat{\mathcal{W}}^{(k)} - (\hat{\mathcal{W}}^{(k-1)} + \alpha^{(k-2)}), & \text{if } \hat{\mathcal{W}}^{(k-1)} \leq -\alpha^{(k-2)} \\ \hat{\mathcal{W}}^{(k)}, & \text{otherwise.} \end{cases} \quad (5.58)$$

and

$$\mathcal{E}_n^{(k-1)} = \begin{cases} \alpha^{(k-2)}, & \text{if } \hat{\mathcal{W}}^{(k-1)} \geq \alpha^{(k-2)} \\ -\alpha^{(k-2)}, & \text{if } \hat{\mathcal{W}}^{(k-1)} \leq -\alpha^{(k-2)} \\ \hat{\mathcal{W}}^{(k-1)}, & \text{otherwise.} \end{cases} \quad (5.59)$$

Since $|\hat{\mathcal{W}}^{(k)} - \hat{\mathcal{W}}^{(k-1)}| \rightarrow 0$ as $k \rightarrow \infty$ (due to Daubechies et al. [63], cf. Lemma 3.8), approximation of the expectations of errors can be obtained as:

$$\mathbb{E}(|\mathcal{E}_{\text{res}}^{(k)}|) \rightarrow \alpha^{(k-2)} + \mathbb{E}(|\hat{\mathcal{W}}^{(k)}|)_{<\alpha^{(k-2)}} \quad (5.60)$$

where $\mathbb{E}(|\hat{\mathcal{W}}^{(k)}|)_{<\alpha^{(k-2)}}$ represents the expectation of absolute wavelet coefficients that satisfies $|\hat{\mathcal{W}}^{(k)}| < \alpha^{(k-2)}$. Similarly:

$$\mathbb{E}(|\mathcal{E}_n^{(k-1)}|) \rightarrow \alpha^{(k-2)} + \mathbb{E}(|\hat{\mathcal{W}}^{(k-1)}|)_{<\alpha^{(k-2)}}, \quad (5.61)$$

where $\mathbb{E}(|\hat{\mathcal{W}}^{(k-1)}|)_{<\alpha^{(k-2)}}$ represents the expectation of absolute wavelet coefficients satisfying $|\hat{\mathcal{W}}^{(k-1)}| < \alpha^{(k-2)}$. Thus, at convergence:

$$\mathbb{E}(|\mathcal{E}_{\text{res}}^{(k)}|) - \mathbb{E}(|\mathcal{E}_n^{(k-1)}|) = \mathbb{E}(|\hat{\mathcal{W}}^{(k)}|)_{<\alpha^{(k-2)}} - \mathbb{E}(|\hat{\mathcal{W}}^{(k-1)}|)_{<\alpha^{(k-2)}}. \quad (5.62)$$

As $k \rightarrow \infty$, substitution of Equation (5.62) into the LHS of Equation (5.57) gives:

$$\frac{\mathbb{E}(|\hat{\mathcal{W}}^{(k)}|)_{<\alpha_T} - \mathbb{E}(|\hat{\mathcal{W}}^{(k-1)}|)_{<\alpha_T}}{\Phi(\delta_{1T})} < \alpha_T, \quad (5.63)$$

where $\alpha_T = (\alpha^{(k)})_{k \rightarrow \infty}$ and $\delta_{1T} = (\delta_1^{(k)})_{k \rightarrow \infty}$. The fact that the numerator of Equation (5.63) approaches zero in the limiting case as $k \rightarrow \infty$ indicates that the threshold never reduces to zero. In accordance with Theorem 4.1 due to Daubechies et al. [63], this also conforms to the fact that $\alpha_T \rightarrow 0$ as the noise level approaches zero.

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Parameter Adaptation for Total Variation–Based Regularization in Parallel MRI

6.1 Total Variation–Based Image Recovery

In comparison to the frequency localization transforms for sparse domain representation [1–5], total variation (TV)–based sparsity promoting regularization [6–8] offers potentially superior edge preservation and structure recovery while suppressing the noise present in the image [9]. The responsible property is the piece-wise-smooth nature of the TV regularization term. However, the major difficulty hindering TV-based reconstruction in finding a fast numerical solution is the non-differentiability of its cost term.

Numerous algorithms have been proposed to address the non-differentiability problem. Using an approximation to the TV functional as in the Rudin, Osher and Fatemi (ROF) model, an artificial time marching scheme was introduced to solve the Euler-Lagrange equation (ELE) [6]. To overcome the slow convergence, Vogel and Oman proposed [10] a fixed-point iteration scheme in which the ELE equation is linearized by lagging the diffusion coefficient, and the update obtained by solving a sparse linear equation. In an attempt to improve the speed, Chan, Golub and Mulet [11] proposed the primal-dual idea by treating the normalized gradient term in the ELE as an independent variable, with a pair of nonlinear equations solved using Newton’s method. Later, Chambolle [12] proposed a semi-implicit gradient descent algorithm based on the Fenchel dual formulation for the TV denoising problem, and primal-dual proximal splitting algorithm for solving generalized non-smooth composite convex minimization problems [13]. An alternating-direction method based on variable splitting was also proposed by Wang et al. [14] for efficient TV regularized image recovery. An alternative variable-splitting method using Bregman iterations was proposed by Goldstein and Osher [15] to address l_1 -regularized optimization problems. Implementation of TV-based minimization using alternating direction method of multipliers (ADMM) was applied to image restoration by Ng et al. [16] for Gaussian noise-corrupted images. In multiplicative noise models, ADMM for TV regularization was used with cost functions based on penalized likelihood by Carlván and Blanc-Féraud [17], and Kullback-Leibler (KL) divergence by Teuber et al. [18]. An overview of these different approaches is provided in Burger et al. [19] and Glowinski et al. [20].

Selecting an optimal setting for the regularization parameter in TV optimization plays a central role in determining its reconstruction accuracy. Several parameter selection strategies have been applied to TV optimization, such as the discrepancy principle [21–24], the generalized cross validation (GCV) method [25], the unbiased predictive risk estimator (UPRE) [26], the variational Bayesian approach [27,28], and the majorization-minimization (MM) approach [29] as discussed in [Chapter 3](#).

6.2 Parameter Selection Using Continuation Strategies

The regularization parameter (β) can be either constant or iteration dependent. The issues pertaining to a constant single-parameter-based recovery can be overcome using an iteration-dependent threshold. The key idea of all continuation strategies is to solve the optimization problem with a large value of β at the start of the iterations and gradually decrease β in the succeeding iterations until the target regularization is reached. For a sufficiently large β , the solution coincides with the sparsest one, resulting in a very large data fidelity. However, as the sparse coefficients are unreliable at early stage iterations of the algorithm, a large value of β can be used initially to favor the penalty term. Then β is monotonically decreased following a linear pattern.

The idea of iterative thresholding with varying threshold was pioneered by Starck et al. [30,31] and Elad et al. [32] and requires knowledge of a decay factor. In their approach, the initial threshold was set to a large value, typically $\|F_u \Psi^* K_u\|_\infty$, and β was made to decrease in each iteration following a linear or exponential pattern. Although such a varying threshold yields fast convergence, there is no guarantee that it converges to a low steady-state error. This type of threshold update scheme was initially derived from the homotopy continuation or path following [33–35]. The exact homotopy path-following methods were initially developed in the statistics literature to compute the complete regularization path when varying the parameter β from large to small [33,34,36,37].

A continuation strategy was subsequently used to overcome the speed difficulties associated with a small regularization parameter in gradient projection for sparse reconstruction (GPSR) [38]. It was shown to yield a fast convergence using a large parameter initially, which they called a warm start, and the performance was compared with a constant parameter called a cold start [38]. They used a simple heuristic that starts at a predetermined value and then decreases by a constant factor in five steps until the desired value of the parameter is obtained. An adaptive continuation method was proposed [39] with reference to the sparse reconstruction by separable approximation (SpaRSA) algorithm and has been shown to yield faster convergence for large initial values of β . However, this adaptation scheme involves a constant parameter to be set by the user. Another continuation-based approach that solves a sequence of problems was proposed by Fadili et al. [40] in which linearly or exponentially decreasing values of β were used to solve the optimization problem.

Following this, the convergence and recovery guarantees of iterated soft-thresholding algorithm (ISTA) with varying thresholds have been investigated [41] using an exponential update schedule for β and adaptive line search for the step size. The researchers show that the method converges only if $F_u^* \Psi$ satisfies the restricted isometry property. For a given $\beta^{(0)}$, they use a decreasing geometric sequence for the values of β until the target value is reached. The parameter is updated as a scaled version of the parameter in the previous iteration. However, the aforementioned update schemes cannot be regarded as data-driven and involve heuristics in one way or the other.

Hale et al. applied the continuation strategy to the fixed-point algorithm for achieving fast convergence by choosing an initial threshold as $\leq \|F_u \Psi^* K_u\|_\infty$. The continuously decreasing values of thresholds can be justified as larger values of threshold yields sparser solutions, while smaller thresholds favor less sparse solutions. For a given current value of threshold denoted by β^k , and an approximate solution $\tilde{x}^k = \tilde{x}(\beta^k)$ with the gradient of the fidelity term denoted by $\tilde{g}^k = g(\tilde{x}^k)$, one can specify a constant $\vartheta \geq 1$ such that:

$$\frac{\|\tilde{g}^k\|_\infty}{\|\tilde{g}^{k+1}\|_\infty} \geq \vartheta. \quad (6.1)$$

This leads to the simple relationship between two consecutive thresholds such that:

$$\beta^{k+1} \geq \frac{\beta^k}{\vartheta}. \quad (6.2)$$

This relationship is utilized in the iterations for achieving solution using the continuation algorithm. According to Equation (6.2), the threshold in the succeeding iteration is a scaled version of the previous threshold. Since ϑ is arbitrary in Equation (6.1), it is required to develop a data-driven approach to estimate ϑ in each step, and hence β . The procedure used is an extension of the scheme discussed in [Chapter 5](#) for adapting the wavelet threshold to the TV iterative shrinkage (TVIS) framework recently proposed by Michailovich [42]. In TVIS, the method of iterative shrinkage is directly applied to the TV functional rather than its smoothed versions. In further discussions, the threshold adaptation scheme introduced into TVIS is referred to as the adaptive TVIS (ATVIS).

6.3 TV Iterative Shrinkage Based Reconstruction Model

Recalling the sparsity-based image reconstruction model in [Section \(2.9\)](#), the optimum solution is determined by minimizing:

$$\min_U \frac{1}{2} \|K_u - F_u U\|_2^2 + \beta \|\Psi U\|_1, \quad (6.3)$$

where β represents the trade-off between sparsity and data fidelity. The total variation can be defined isotropically or anisotropically as:

$$\|\Psi U\|_1 = \begin{cases} \left\| \sqrt{(\nabla_x U_{ij})^2 + (\nabla_y U_{ij})^2} \right\|_1; & \text{(Isotropic)} \end{cases} \quad (6.4a)$$

$$\left\| |\nabla_x U_{ij}| + |\nabla_y U_{ij}| \right\|_1; \quad \text{(Anisotropic)}, \quad (6.4b)$$

where ∇_x and ∇_y represent the forward finite difference operators on the horizontal and vertical coordinates. The definition of gradient ∇ depends on the boundary conditions. With symmetric boundary conditions (SBC), we have:

$$(\nabla U)_{n,m} = \begin{cases} (U_x)_{n,m} = U_{n,m} - U_{n-1,m} & \text{with } U_{-1,m} = U_{0,m} \\ (U_y)_{n,m} = U_{n,m} - U_{n,m-1} & \text{with } U_{n,-1} = U_{n,0}, \end{cases} \quad (6.5)$$

where $n = 0, 1, \dots, N-1$ and $m = 0, 1, \dots, M-1$. In the case of periodic boundary conditions (PBC), the gradient is defined as:

$$(\nabla U)_{n,m} = \begin{cases} (U_x)_{n,m} = U_{n,m} - U_{n-1,m} & \text{with } U_{-1,m} = U_{N-1,m} \\ (U_y)_{n,m} = U_{n,m} - U_{n,m-1} & \text{with } U_{n,-1} = U_{n,M-1}. \end{cases} \quad (6.6)$$

In Equations (6.5) and (6.6), U_x and U_y denote the partial differences of U taken in the column and row direction, respectively. Now the discrete versions of the TV functionals in Equation (6.4a) and (6.4b) can be defined as:

$$\text{TV}_i(U) = \left\langle \mathbf{1}, \sqrt{|U_x|^2 + |U_y|^2} \right\rangle \quad (6.7a)$$

$$\text{TV}_a(U) = \left\langle \mathbf{1}, |U_x| + |U_y| \right\rangle, \quad (6.7b)$$

where TV_i and TV_a denote the isotropic and anisotropic versions, respectively, of the TV functional and $\mathbf{1}$ denotes an $N \times M$ matrix of ones. A connection between the two versions of the TV functional in Equation (6.7a) and (6.7b) is established using the identity [42]:

$$\sqrt{|U_x|^2 + |U_y|^2} = \frac{\int_0^{\pi/2} \left(|U_x \cos \theta_k^L + U_y \sin \theta_k^L| + |U_y \cos \theta_k^L - U_x \sin \theta_k^L| \right) d\theta}{\int_0^{\pi/2} (\cos \theta_k^L + \sin \theta_k^L) d\theta}, \quad (6.8)$$

where $k = 0, 1, \dots, L-1$ and $\theta_0^L, \theta_1^L, \dots, \theta_{L-1}^L$ are uniformly distributed in $[0, \pi/2)$ for a fixed L . Replacing the integral in Equation (6.8) using the Riemannian approximation, we have:

$$\text{TV}_i(U) \cong d_L \left\langle \mathbf{1}, \sum_{k=0}^{L-1} |U_x \cos \theta_k^L + U_y \sin \theta_k^L| + |U_y \cos \theta_k^L - U_x \sin \theta_k^L| \right\rangle, \quad (6.9)$$

where $d_L = \left[\sum_{k=0}^{L-1} (\cos \theta_k^L + \sin \theta_k^L) \right]^{-1}$. Accordingly, the TV functionals in Equation (6.7a) and (6.7b) can be expressed as:

$$\text{TV}_L = \begin{cases} \text{TV}_a(U); & \text{if } L = 1 \\ \text{TV}_i(U); & \text{if } L \rightarrow \infty. \end{cases} \quad (6.10)$$

Using discrete samples of θ_k^L in Equation (6.9), the approximative TV functional can be computed by using the multi-directional gradients (MDGs) with L directions [42]. The transformation of the approximative TV functional (TV_L) into the anisotropic or isotropic versions of TV is determined by the value of L . Setting $L = 1$ transforms the approximative TV functional into anisotropic TV, whereas the TV_L assumes isotropic TV as $L \rightarrow \infty$. For obtaining the MDG, the rotated gradients $\nabla_{\theta_k^L} U$ are first computed as:

$$\nabla_{\theta_k^L} U = \begin{pmatrix} U_x \cos \theta_k^L + U_y \sin \theta_k^L \\ U_y \cos \theta_k^L - U_x \sin \theta_k^L \end{pmatrix}. \quad (6.11)$$

where $L \geq 1$, and $k = 0, 1, \dots, L-1$. Given a set of L different angles $\bar{\theta}_L = \{\theta_k^L\}_{k=0}^{L-1}$, the MDG is defined as a map $\nabla_{\bar{\theta}_L} : \mathbb{R}^{N \times M} \rightarrow (\mathbb{R}^{N \times M})^{2L}$:

$$\nabla_{\bar{\theta}_L} : U \mapsto \begin{pmatrix} \nabla_{\theta_0^L} U \\ \nabla_{\theta_1^L} U \\ \vdots \\ \nabla_{\theta_{L-1}^L} U \end{pmatrix}. \quad (6.12)$$

Then, the MDG $\mathbf{d} = (\nabla_{\theta_0^L} U \quad \nabla_{\theta_1^L} U \quad \dots \quad \nabla_{\theta_{L-1}^L} U)^T$. Therefore, the l_1 -norm of the gradient space variables can be obtained as:

$$\|\mathbf{d}\|_1 = \sum_{n,m} \sum_{k=0}^{L-1} \left(|U_x \cos \theta_k^L + U_y \sin \theta_k^L| + |U_y \cos \theta_k^L - U_x \sin \theta_k^L| \right)_{n,m}. \quad (6.13)$$

The corresponding divergence operator $\text{div}_{\bar{\theta}_L} : (\mathbb{R}^{N \times M})^{2L} \rightarrow \mathbb{R}^{N \times M}$ can be computed as:

$$\text{div}_{\bar{\theta}_L} : \mathbf{u} \mapsto -\text{div} \begin{pmatrix} \sum_{k=0}^{L-1} (u_x^k \cos \theta_k^L - u_y^k \sin \theta_k^L) \\ \sum_{k=0}^{L-1} (u_y^k \cos \theta_k^L + u_x^k \sin \theta_k^L) \end{pmatrix}, \quad (6.14)$$

where the elements of the vector $\mathbf{u} = (u_x^0, u_y^0, u_x^1, u_y^1, \dots, u_x^{L-1}, u_y^{L-1})$. A solution to the optimization problem in Equation (6.3) is provided in the following section.

6.3.1 Derivative Shrinkage

An iterative shrinkage approach to TV is based on the assumption that both the original and observed images belong to a bounded norm vector space $\mathbb{U}$. The bounded norm vector space requires all the vectors in $\mathbb{U}$ to be bounded and hence possesses a finite l_2 -norm. Furthermore, elements of $\mathbb{U}$ are also constrained to have zero mean value, which leads to a formal definition of $\mathbb{U}$ as:

$$\mathbb{U} = \{U \in \mathbb{R}^{N \times M} \mid \langle \mathbf{1}, U \rangle = 0, \|U\|_2 < \infty\}. \quad (6.15)$$

Image recovery in TV minimization is achieved using an operator $\mathcal{X}(U)$ given by:

$$\mathcal{X}(U) = -\nabla \left(\mathcal{F}^{-1} \left(\mathcal{F}(U) \odot W_i^* \right) \right) \quad (6.16)$$

where $\odot$ denotes the Hadamard product, $\text{div}(\cdot)$ the discrete divergence operator, $\mathcal{F}$ represents the discrete cosine transform operator in the case of SBC, and the discrete Fourier

transform operator otherwise. Also, W_i denotes the integration filter coefficients defined as $(W_i)_{k,l} = (2\cos(a\pi k/M) + 2\cos(a\pi l/N) - 4)^{-1}$, with $a=1$ for SBC and 2 for PBC, respectively [42]. The adjoint operator $\mathcal{X}^*$ is defined as $\mathcal{X}^*(\mathbf{d}) = \frac{1}{L} \mathcal{F}^{-1}(\mathcal{F}(\text{div}_{\bar{\partial}_L}(\mathbf{d})) \odot W_i)$. Using this operator, the optimization problem in Equation (6.3) can now be represented in the derivative space as:

$$\min_{\mathbf{d}} \left\{ J(\mathbf{d}) = \frac{1}{2} \|K_u - F_u \mathcal{X}^*(\mathbf{d})\|_2^2 + \beta \|\mathbf{d}\|_1 \right\}, \quad (6.17)$$

where the optimization problem in the image space $U \in \mathbb{U}$ is now replaced by the minimization of its corresponding gradient image $\mathbf{d} \in \mathbb{D}$. This is similar to the l_1 -norm constrained reconstruction problem in [43], enabling the reconstruction using iterative soft thresholding. Therefore, the solution to the TV optimization problem can be implemented using the following steps incorporating iterative soft thresholding. In the first step, the Landweber update for the gradient image is obtained as:

$$\hat{\mathbf{d}}^{(k+1)} = \mathbf{d}^{(k)} + \mathcal{X}'_{F_u} (K_u - F_u \mathcal{X}^* \mathbf{d}^{(k)}), \quad (6.18)$$

where $\hat{\mathbf{d}}$ and $\mathbf{d}$ denote the gradient images before and after thresholding. As the second term in the right-hand side (RHS) of Equation (6.18) represents the gradient image ϵ_{res} of the consistency error, the Landweber update now becomes:

$$\hat{\mathbf{d}}^{(k+1)} = \mathbf{d}^{(k)} + \epsilon_{\text{res}}^{(k+1)}. \quad (6.19)$$

At the start of the iterations, the MDG $\mathbf{d}^{(0)}$ is initialized to a zero-filled array of dimension $N \times M \times 2L$. Application of the soft-thresholding function to $\hat{\mathbf{d}}^{(k+1)}$ gives:

$$\mathbf{d}^{(k+1)} = \mathcal{T}_\beta \left(\hat{\mathbf{d}}^{(k+1)} \right), \quad (6.20)$$

where $\mathcal{T}_\beta$ denotes a component-wise application of the soft-thresholding function applied to complex data, defined by:

$$(d^{(k+1)})_i = \begin{cases} (\hat{d}^{(k+1)})_i - \beta \frac{(\hat{d}^{(k+1)})_i}{|(\hat{d}^{(k+1)})_i|}, & \text{if } |(\hat{d}^{(k+1)})_i| > \beta \\ 0, & \text{if } |(\hat{d}^{(k+1)})_i| \leq \beta. \end{cases} \quad (6.21)$$

6.3.2 Selection of Initial Threshold

Since the threshold adaptation scheme requires a monotonically decreasing pattern, a higher initial threshold is required to achieve faster reduction to the smallest possible relative l_2 -norm error (RLNE) [44–46]. Therefore, the threshold selection procedure provided by Michailovich [42] cannot yield best results when used with the threshold adaptation scheme. Alternatively, the universal threshold proposed by Sardy and Monajemi [47] using finite differences is found to be more suitable in this case. The universal threshold is estimated using:

$$\beta^{(0)} = \hat{\sigma} \mathcal{G}_{\mu, \gamma}^{-1} \left(1 - 2 / \sqrt{\log P_M} \right), \quad (6.22)$$

where $P_M = 2N(N-1)$, $\mathcal{G}_{\mu, \gamma}^{-1}(p) = \mu - \gamma \log(-\log p)$, $\mu = \exp(\hat{a}^{\mu_0} + \hat{b}^{\mu_0} \log \log N)$, $\gamma = \exp(\hat{a}^{\gamma_0} + \hat{b}^{\gamma_0} \log \log N)$, with the values of the coefficients as $\hat{a}^{\mu_0} = -0.395$, $\hat{b}^{\mu_0} = 0.552$, $\hat{a}^{\gamma_0} = -1.512$ and $\hat{b}^{\gamma_0} = -0.247$ [47]. The standard deviation of the noise used in the threshold determination is a scaled median absolute deviation of the finite differences given by [48]:

$$\hat{\sigma} = (1.4826 / \sqrt{2}) \cdot \text{median} \left(\left| \left(\hat{\mathbf{d}}^{(1)} \right) - \text{median} \left(\hat{\mathbf{d}}^{(1)} \right) \right| \right). \quad (6.23)$$

6.4 Adaptive Derivative Shrinkage

Image recovery using the iterative shrinkage approach requires knowledge of an optimal regularization parameter that depends on the presence of noise, image size, image type, and so on. Although the reconstructed image quality is well preserved near the optimal parameter value, it deteriorates rapidly away from this value. The procedures for finding the optimal parameter value in the absence of information on noise level and its statistics can pose difficulties in optimal reconstruction. As a consequence, inferences obtained using different selection strategies can lead to suboptimal reconstructions. These difficulties can be overcome with the use of an adaptive parameter.

An adaptive threshold can be realized by reducing the difference between the TV norms of the gradient domain consistency error and the sparse approximation error. The sparse approximation error is defined in the derivative domain and accounts for the perturbations resulting from noise as well as the deviation of the adopted basis from the true basis. It is given by:

$$\epsilon_n^{(k)} = \hat{\mathbf{d}}^{(k)} - \mathbf{d}^{(k)}. \quad (6.24)$$

In the adaptive strategy, a parallel is drawn between (1) the residual error versus the consistency error, and (2) the perturbation error versus the difference between coefficients before and after soft thresholding in the gradient domain (i.e., the sparse approximation error). The adaptation relies on the fact that the absolute difference between TV norms of the consistency and sparse approximation errors, referred to as the discrepancy level, approaches zero in the absence of noise and shows up as a small noise-dependent positive quantity at convergence. An adaptive strategy is enforced by using a discrepancy-level dependent function to update the threshold. The dependence between the extent of denoising and data consistency is exploited to adapt the threshold toward accelerated reduction of the steady-state error with a reduced loss of resolution. Analogous to the formulation in [Chapter 5](#), the discrepancy level, defined as the difference between the TV norms of the consistency and sparse approximation errors, can be represented as:

$$\mathfrak{d}_i^{(k+1)} \triangleq \left\| \left\| \epsilon_{\text{res}}^{(k+1)} \right\|_1 - \left\| \epsilon_n^{(k)} \right\|_1 \right\|. \quad (6.25)$$

The discrepancy level exhibits noise dependence characteristics, as shown in Theorem 1 in Section 6.6.1. The relationship between the l_1 -norms of consistency and sparse approximation errors at convergence can be expressed as:

$$\left\| \epsilon_{\text{res}}^{(k+1)} \right\|_1 \geq \left\| \epsilon_n^{(k)} \right\|_1. \quad (6.26)$$

The proof for Equation (6.26) follows Theorem 2, shown in Section 6.6.2. The above representation bears a close resemblance to the generalized discrepancy principle (GDP), except that both errors are expressed in terms of the l_1 -norm. However, since ϵ denotes the gradient of the respective errors, the l_1 -norms in Equation (6.26) correspond to their TV norms. Since the TV norm is the sum of absolute gradient values, Equation (6.26) becomes:

$$\sum_i \left| \epsilon_{\text{res}}^{(k+1)}(i) \right| \geq \sum_i \left| \epsilon_n^{(k)}(i) \right|. \quad (6.27)$$

At convergence, the expectations of the errors satisfy the inequality:

$$\mathbb{E} \left(\left| \epsilon_{\text{res}}^{(k+1)} \right| \right) \geq \mathbb{E} \left(\left| \epsilon_n^{(k)} \right| \right), \quad (6.28)$$

as shown in Section 6.6.3. Also, since the threshold adaptation requires $\beta^{(k+1)} < \beta^{(k)}$, as dictated by the fixed-point continuation methods [49], we have:

$$\frac{\mathbb{E} \left(\left| \epsilon_{\text{res}}^{(k+1)} \right| \right)}{\beta^{(k+1)}} > \frac{\mathbb{E} \left(\left| \epsilon_n^{(k)} \right| \right)}{\beta^{(k)}}. \quad (6.29)$$

Taking the difference between the left-hand side (LHS) and the right-hand side (RHS) as an increasing function Φ of $\mathfrak{d}_1^{(k+1)}$ satisfying the conditions specified in [Chapter 5](#), the threshold update scheme takes the following form:

$$\beta^{(k+1)} = \frac{\mathbb{E} \left(\left| \epsilon_{\text{res}}^{(k+1)} \right| \right)}{\Phi \left(\mathfrak{d}_1^{(k+1)} \right) + \left(\mathbb{E} \left(\left| \epsilon_n^{(k)} \right| \right) / \beta^{(k)} \right)}. \quad (6.30)$$

The expectations can be computed from the histograms of $|\epsilon_{\text{res}}^{(k+1)}|$ and $|\epsilon_n^{(k)}|$. In order to yield decreasing values of $\{\beta^k\}$, the relationship in Equation (6.30) implies:

$$\frac{\beta^{(k)}}{\beta^{(k+1)}} = \frac{\beta^{(k)} \Phi \left(\mathfrak{d}_1^{(k+1)} \right) + \mathbb{E} \left(\left| \epsilon_n^{(k)} \right| \right)}{\mathbb{E} \left(\left| \epsilon_{\text{res}}^{(k+1)} \right| \right)} > 1. \quad (6.31)$$

This means that:

$$\beta^{(k)} > \frac{\mathbb{E} \left(\left| \epsilon_{\text{res}}^{(k+1)} \right| \right) - \mathbb{E} \left(\left| \epsilon_n^{(k)} \right| \right)}{\Phi \left(\mathfrak{d}_1^{(k+1)} \right)}. \quad (6.32)$$

Replacing the expectations of errors in terms of their respective l_1 -norms, that is, $\mathbb{E}(|\epsilon_{\text{res}}^{(k+1)}|) = \|\epsilon_{\text{res}}^{(k+1)}\|_1/d_n$ and $\mathbb{E}(|\epsilon_n^{(k)}|) = \|\epsilon_n^{(k)}\|_1/d_n$, the numerator of Equation (6.32) becomes $\mathfrak{d}_1^{(k+1)}/d_n$, where $d_n = 2LMN$. Therefore, decreasing values of thresholds can be obtained if:

$$\beta^{(k)} > \frac{\mathfrak{d}_1^{(k+1)}}{d_n \Phi(\mathfrak{d}_1^{(k+1)})}. \quad (6.33)$$

Thus, if $\Phi(\cdot)$ is chosen as an increasing function of $\mathfrak{d}_1^{(k+1)}$ with $\Phi(\mathfrak{d}_1^{(k+1)}) \rightarrow 0$ as $\mathfrak{d}_1^{(k+1)} \rightarrow 0$, it is first required to numerically satisfy the limits for $\mathfrak{d}_1^{(k+1)}$ specified in [50]. Three sample adaptation functions $\mathfrak{d}_1^{(k+1)}$, $\log(1 + \mathfrak{d}_1^{(k+1)})$ and $1 - \exp(-\mathfrak{d}_1^{(k+1)})$ for $\Phi(\cdot)$ satisfy the limits for all datasets used. For other datasets, it may be useful to scale the functions as $\Phi(c\mathfrak{d}_1^{(k+1)})$ and search for a suitable value of c for which the limits are satisfied within the first few iterations. The convergence of the adaptive threshold sequence is shown in Sections 6.6.4 and 6.6.5.

6.5 Algorithmic Implementation for Parallel MRI (pMRI)

In each iteration, the derivative space errors, calculated from the sum-of-squares (SoS) image is utilized to obtain the threshold update as in the single-channel case. The updated threshold is then applied to the individual channels for reconstruction. The steps for threshold adaptation are summarized below, with TVIS steps provided in Algorithm 6.1. In Algorithm 6.2, the errors in derivative space, calculated from the SoS combined image, are utilized to obtain the update. The initialization steps are common to all algorithms and include setting $\mathbf{d}^{(0)} = \tilde{\mathbf{d}}^{(0)} = 0$, $t^{(0)} = 1$, $\text{tol} = 1.0e-4$, and K_u as the zero-filled k -space with n_c number of channels. The initial threshold $\beta^{(0)}$ is determined as outlined in Section 6.3.2.

Algorithm 6.1 Standard TVIS with FISTA

Iterate for $k = 1, \dots, \text{MaxIter}$:

1) Iterate for $c = 1, \dots, n_c$:

(i) Calculate $\epsilon_{\text{res}}^{(k)}$ as in (6.18)

(ii) Find $\hat{\mathbf{d}}^{(k)} = \tilde{\mathbf{d}}^{(k-1)} + \epsilon_{\text{res}}^{(k)}$

(iii) Soft Threshold the gradient domain Coefficients

$$\mathbf{d}^{(k)} = \mathcal{T}_\beta(\hat{\mathbf{d}}^{(k)})$$

(iv) Find $U_c^{(k)} = \mathcal{X}^*(\mathbf{d}^{(k)})$

(v) Find $t^{(k)} = \frac{1 + \sqrt{1 + 4t^{(k-1)2}}}{2}$

(vi) $\tilde{\mathbf{d}}^{(k)} = \mathbf{d}^{(k)} + \frac{t^{(k-1)} - 1}{t^{(k)}}(\mathbf{d}^{(k)} - \mathbf{d}^{(k-1)})$

end

- 2) Compute the SoS image $U^{(k)} = \sqrt{\sum_{c=1}^{n_c} (U_c^{(k)})^2}$
- 3) Repeat steps (1)–(2) until $\frac{\|U^{(k)} - U^{(k-1)}\|_2}{\|U^{(k-1)}\|_2} \leq \text{tol.}$
- end

Algorithm 6.2 ATVIS with FISTA

Iterate for $k = 1, \dots, \text{MaxIter}$:

- 1) Follow steps (1)–(2) as in (A6.1)
- 2) Calculate $\epsilon_n^{(k-1)}$ and $\epsilon_{\text{res}}^{(k)}$ from combined image
- 3) Calculate the expectations of $|\epsilon_{\text{res}}^{(k)}|$ and $|\epsilon_n^{(k-1)}|$ from respective histograms
- 4) Calculate $d_1^{(k)} = \|\epsilon_{\text{res}}^{(k)}\|_1 - \|\epsilon_n^{(k-1)}\|_1$
- 5) Update β^k using (6.30)
- 6) Repeat steps (1) to (5) until $\frac{\|U^{(k)} - U^{(k-1)}\|_2}{\|U^{(k-1)}\|_2} \leq \text{tol.}$
- end

The improved reconstruction performance of the anisotropic version of ATVIS ($L=1$) in comparison to constant threshold TVIS is illustrated using a noise-free and noisy versions of phantom image I, under-sampled with the sampling mask, as shown in Figure 6.1. Each phantom image is reconstructed with TVIS and ATVIS. Plots of the threshold and RLNE versus iteration number are shown in Figure 6.2. For all noise conditions, the TVIS reconstruction using a constant threshold yields slower and suboptimal reconstruction. The difference images of phantom-I, with their RLNE values (indicated in insets), are

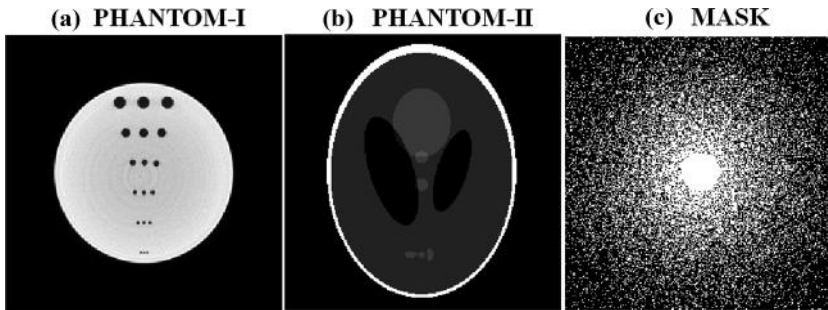
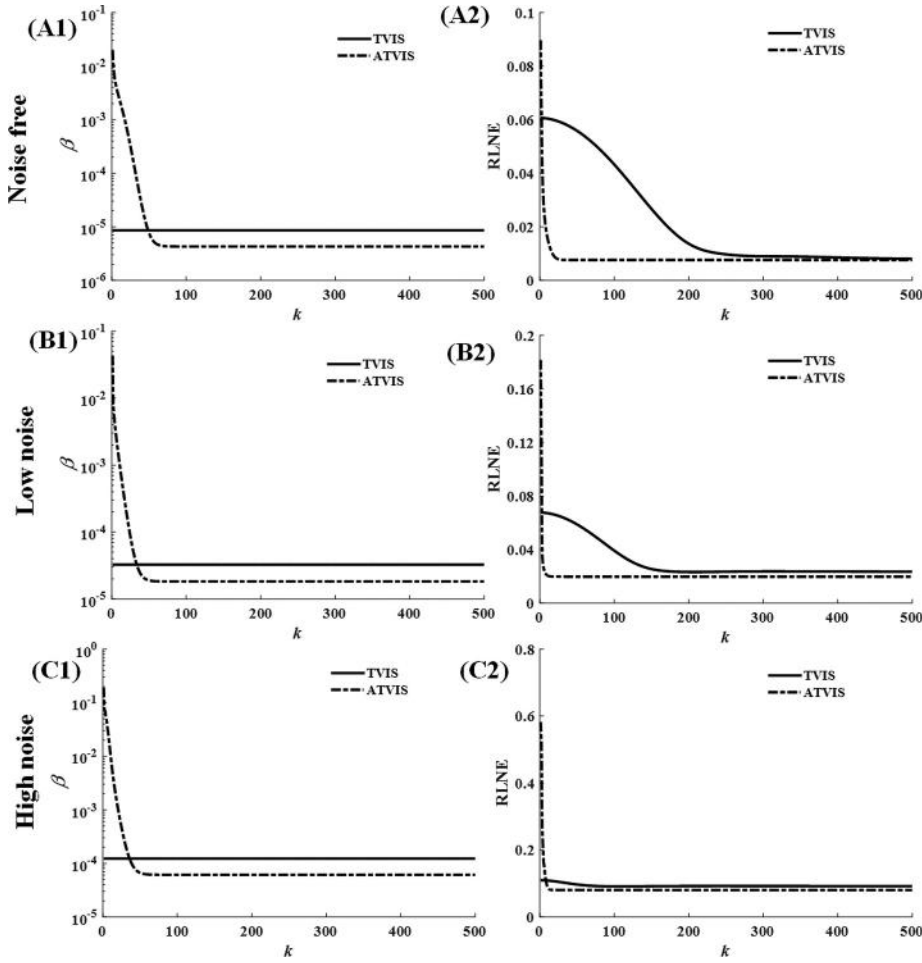


FIGURE 6.1

(a and b) Phantom images, (c) The sampling mask generated using 30% acquired samples, with 1.55% of samples fully acquired from the central k -space.

**FIGURE 6.2**

Plots of threshold (left) and RLNE (right) as functions of iteration number at three different noise levels.

shown in Figure 6.3. For all methods, a common stopping criterion (as shown in Algorithm 6.1, Step 3) is employed. The algorithms are implemented using MATLAB (Mathworks, Natick, Massachusetts) on a personal computer (PC) with Intel Xeon 2.4 GHz processor and 16 GB of RAM running Windows 7. The adaptation function used is $\Phi(\mathfrak{d}_1^{(k)}) = \mathfrak{d}_1^{(k)}$. The threshold for restoration using TVIS is computed from the threshold selection criterion defined in [42]. However, the initial threshold in the adaptive case is computed using the procedure defined in Section 6.3.2. For faster implementation, over-relaxation is also included in both variants of TVIS algorithm as in fast iterated soft-thresholding algorithm (FISTA) [51,52].

In order to compare the performance of anisotropic and isotropic TV adaptation using the approximative TV functional, reconstruction experiments are performed on phantom II with $L = 1, 3$ and 5, and $\sigma = .02$, with and without adaptation. The difference images corresponding to the restored images are shown in Figure 6.4, with RLNE values inserted in the insets. The TVIS provides improvement in performance with increasing values of L and is in agreement with the experimental results reported in [42]. In both extremes, the RLNE values indicate that adaptive shrinkage using ATVIS show improved performance.

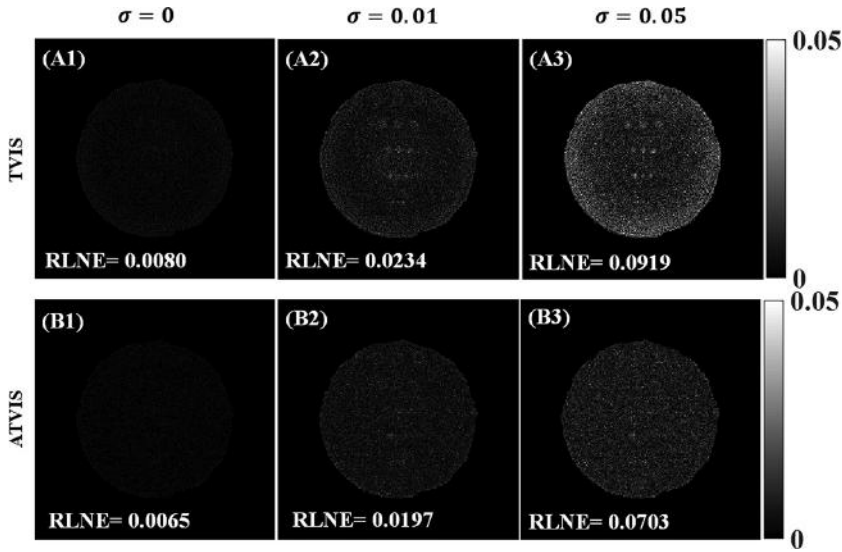


FIGURE 6.3

Difference images of TVIS and ATVIS reconstructions using phantom I for three different input noise levels.

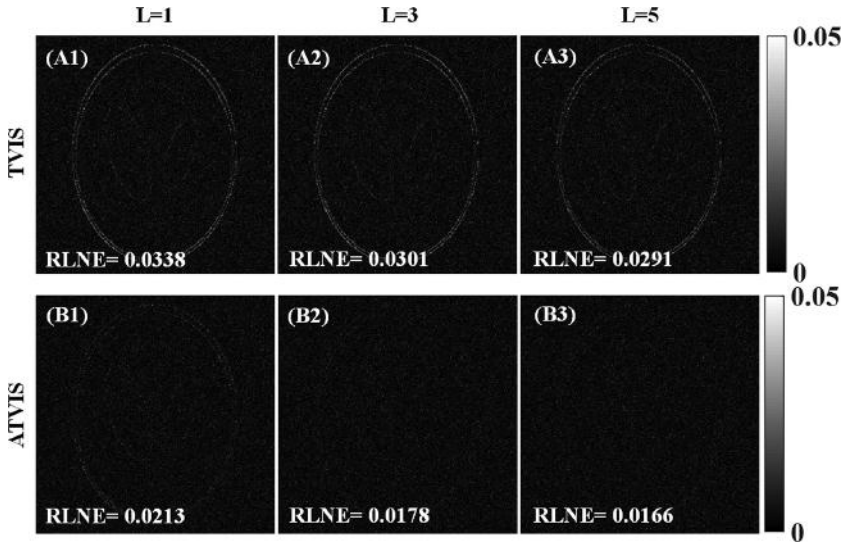
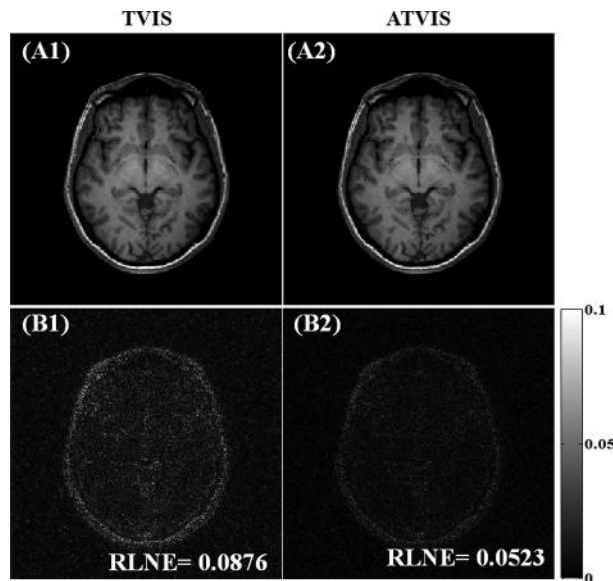


FIGURE 6.4

Difference images of TVIS and ATVIS reconstructions for noisy versions of phantom II with $L = 1, 3$, and 5 .

Visual inspection of difference images also supports this claim. In all further discussions, the implementation of ATVIS uses the anisotropic TV formulation ($L = 1$).

Fully sampled k -space of a channel-combined image is retrospectively under-sampled using a sampling mask, as shown in Figure 6.1. The sampling mask is generated using 30% acquired samples, with 1.55% of samples fully acquired from the central k -space. The ATVIS reconstruction is compared with constant threshold TVIS. The top and bottom panels in Figure 6.5 show the reconstructed and difference images. Columns from left to

**FIGURE 6.5**

Comparison of single-channel reconstructions using (A1) TVIS, and (A2) ATVIS, (B1–B2) corresponding difference images.

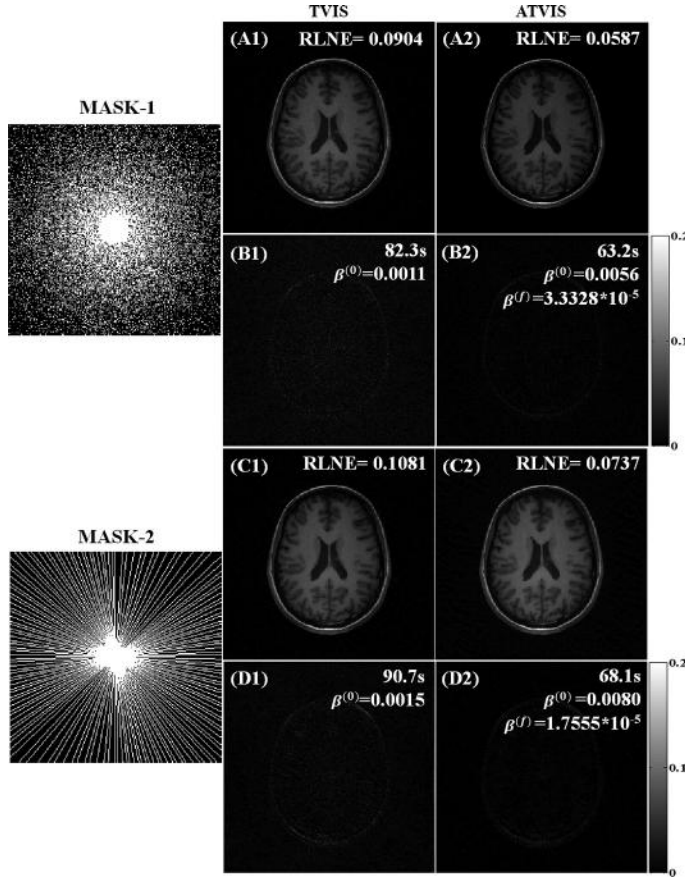
right show reconstructions using constant threshold TVIS and ATVIS, respectively, with their corresponding RLNE values provided in the insets.

The application of the adaptation strategy to parallel MRI is illustrated using multi-channel in-vivo data. Raw k -space data were acquired from a GE Discovery 3T 750W clinical MR scanner. *Datasets I* and *II* are images obtained using GE's brain volume imaging acquisition (BRAVO) sequence.

Figure 6.6 show reconstructions for *dataset I* using TVIS, and ATVIS methods. The top two panels of Figure 6.6 show the reconstructed and difference images obtained with a random sampling mask (MASK-1). The lower two panels show reconstructions pertaining to a radial sampling scheme (MASK-2) with angular separation based on the golden ratio [53]. For the radial sampling scheme, 80 spokes are used, with 256 samples along each spoke. RLNE values indicate the improved quality of reconstruction obtained with the ATVIS method. This is confirmed by visual inspection of the difference images.

Figure 6.7 illustrates reconstructions for *dataset II* using the TVIS and ATVIS methods. The top two panels show the reconstructed and difference images obtained with a radial sampling scheme with uniform angular separation (MASK-3). Again, 80 spokes with 256 samples along each spoke are used. The lower two panels show reconstructions using a Cartesian sampled k -space, retrospectively under-sampled along the phase-encode direction based on the golden ratio (MASK-4) [54]. In the random phase-encode under-sampling scheme, 120 phase encode lines are acquired. Of these, 32 lines from the central region of k -space are equally spaced, and the rest are randomly spaced. RLNE values as well as the initial and final thresholds are shown in the insets. Visual inspection and RLNE values attest to the improved quality of ATVIS reconstruction.

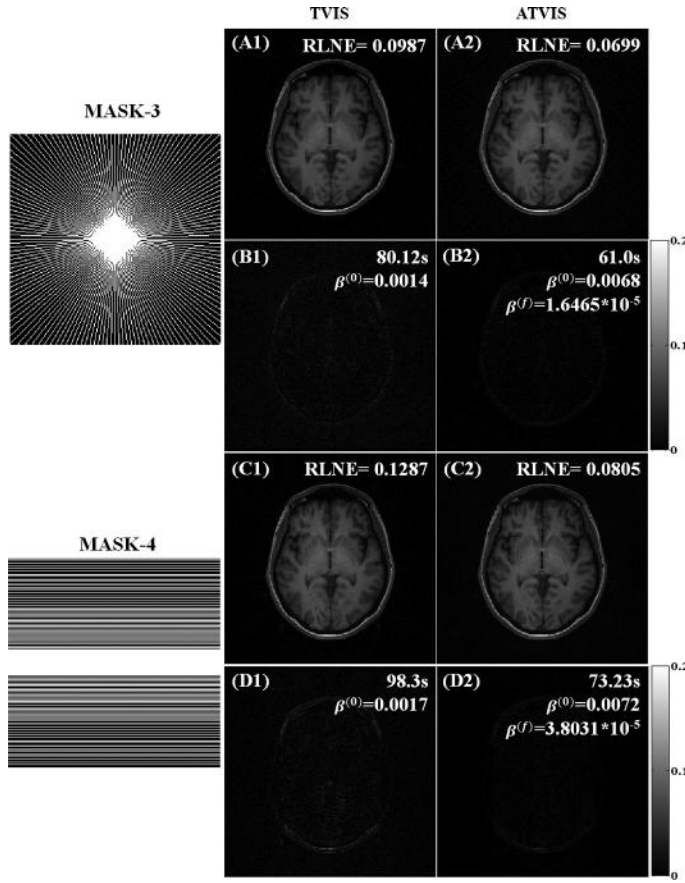
The adaptive scheme is applied to dynamic MRI in which several k -spaces are collected at multiple time points. As the accelerated dynamic MR images are more susceptible to artifacts, approaches that provide higher levels of sparsity, such as adaptive dictionary

**FIGURE 6.6**

Multi-channel reconstructions for *dataset I* using (A1 and C1) TVIS, and (A2 and C2) ATVIS. Top and bottom panels show reconstructed and difference images for retrospective under-sampling using MASK-1 and MASK-2, respectively. The insets in the difference image depict the reconstruction time, initial threshold ($\beta^{(0)}$) and final threshold ($\beta^{(f)}$) values. For TVIS, the initial and final thresholds are the same.

learning (DL), are preferred over wavelet or TV-based schemes. In DL with temporal gradients (DLTG) [55], an additional sparsifying transform in the temporal dimension is used along with DL for improved dynamic MR image reconstruction of cardiac cine data.

The implementation of DLTG is made up of three parts in an iterative framework, as outlined in Algorithm 6.3. The input to the algorithm is a complex valued single-slice cardiac cine data denoted by $\mathbf{X}^{(3D)}$ and its concatenated column vector denoted by $\mathbf{x}$. The corresponding zero-filled sequence is denoted by $\mathbf{x}_z$. In DL, $\mathbf{x}_{T,i}$ denotes the training patches represented as column vectors of size m , and Γ_T is the matrix formed with columns $\gamma_{T,i}$, each containing the sparse representation of $\mathbf{x}_{T,i}$. The dictionary is denoted as $\mathbf{D}$, with real and imaginary parts denoted as $\Re(\mathbf{x})$ and $\Im(\mathbf{x})$, and their corresponding sparse coding as $\Gamma_{\Re}$ and $\Gamma_{\Im}$, respectively. With R_i denoting an operator that extracts a column vector corresponding to a three-dimensional (3D) patch in the dataset starting from pixel location i , the columns of $\Gamma_{\Re}$ and $\Gamma_{\Im}$ (represented as $\gamma_{\Re,i}$ and $\gamma_{\Im,i}$) are obtained by the coding of patch $R_i \Re(\mathbf{x})$ and $R_i \Im(\mathbf{x})$, respectively.

**FIGURE 6.7**

Multi-channel reconstructions for *dataset II* using (A1 and C1) TVIS, and (A2 and C2) ATVIS. Top and bottom panels show reconstructed and difference images for retrospective under-sampling using MASK-3, and MASK-4, respectively. The insets in the difference image depict the reconstruction time, initial threshold ($\beta^{(0)}$) and final threshold ($\beta^{(T)}$) values. For TVIS, the initial and final thresholds are the same.

Algorithm 6.3 DLTG

Initialization

Input the zero-filled sequence $\mathbf{x}_z$.

Initialize $r_1 = 0$, $r_2 = 0$, $\mathbf{x}^{(0,0)} = \mathbf{x}_z$, $\Gamma_{\mathcal{R}} = \Gamma_{\mathcal{I}} = 0$, $tol = 1.0e - 4$.

Main Iteration

1) Iterate for $r_1 = 1, \dots, MaxIter1$

(i) Dictionary Training

$$\{D, \Gamma_T\} \leftarrow \min_{D, \Gamma_T} \|\gamma_{T,i}\|_0 \text{ s.t. } \|R_i \mathbf{x}_T^{(r_1, r_2)} - D \gamma_{T,i}\|_2^2 < \epsilon$$

(ii) *Sparse Coding of Real and Imaginary parts separately*

$$\Gamma_{\Re} \leftarrow \min_{\Gamma_{\Re}} \|\gamma_{R,i}\|_0 \text{ s.t. } R_i \Re(\mathbf{x}^{(r_1, r_2)}) - D\gamma_{\Re,i} \Big|_2^2 < \epsilon$$

$$\Gamma_{\Im} \leftarrow \min_{\Gamma_{\Im}} \|\gamma_{I,i}\|_0 \text{ s.t. } R_i \Im(\mathbf{x}^{(r_1, r_2)}) - D\gamma_{\Im,i} \Big|_2^2 < \epsilon$$

(iii) *DL Sparse Approximation*

$$U' \leftarrow \frac{1}{n} \sum_{i=1}^P R_i^T D (\gamma_{\Re,i} + \gamma_{\Im,i})$$

(iv) *Iterate for $r_2 = 1, \dots, \text{MaxIter1}$:*

(a) *Enforce data consistency*

(b) *Enforce TG Sparsity*

$$\mathbf{x}' = \min_{\mathbf{x}'} \left\| \mathbf{x}^{(r_1, r_2)} - \mathbf{x}' \right\|_2^2 + \beta^{(r_2)} \left\| \nabla_t |\mathbf{x}'| \right\|_1$$

end

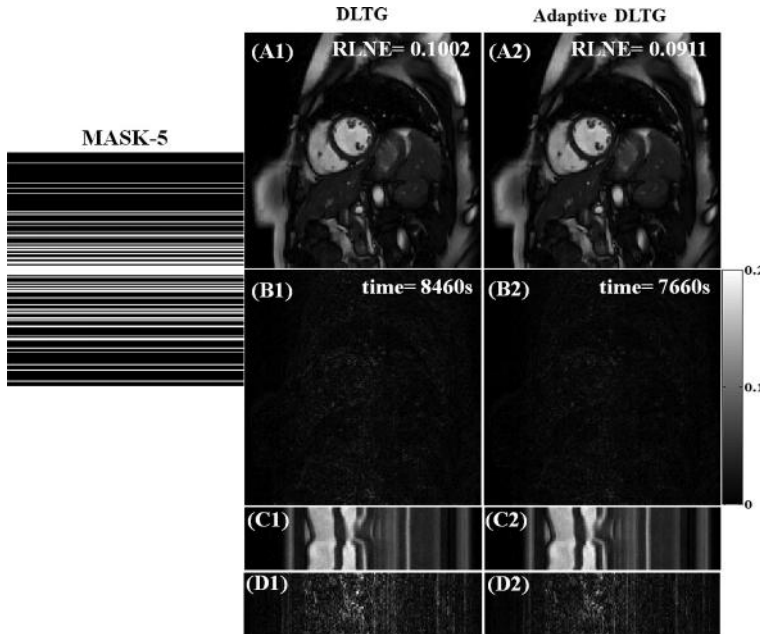
$$(v) \text{ Repeat steps (i)–(iv) until } \frac{\left\| \mathbf{x}^{(r_1, r_2)} - \mathbf{x}^{(pr_1, pr_2)} \right\|_2}{\left\| \mathbf{x}^{(pr_1, pr_2)} \right\|_2} \leq \text{tol},$$

where $\mathbf{x}^{(pr_1, pr_2)}$ is the previous iterate of the solution.

end

The adaptive implementation of DLTG follows the threshold β update scheme in Equation (6.30). The use of temporal gradient sparsity in DLTG affects the convergence rate of the algorithm [55]. Setting the threshold parameter to a high initial value and reducing it with succeeding iterations result in faster convergence and lower steady-state errors compared to DLTG with a constant parameter. In this way, the adaptation scheme applied to temporal gradient thresholding can accelerate the convergence while yielding lower steady-state reconstruction errors. Figure 6.8 depicts reconstructions of dynamic data (borrowed from <http://caballerojose.com/code.html>) containing 30 temporal frames of size 256×256 . The reconstructions are performed using both constant parameter and adaptive parameter schemes for enforcing sparsity along the temporal dimension. The top two panels show the reconstructed and difference images obtained with the sampling scheme labeled MASK-5. The lower two panels include the temporal profile of row 128 in the original dataset and their respective difference images.

In estimating the missing samples in compressed sensing MRI, regularization provides a mechanism for balancing the two errors. The presence of noise or artifacts in the resultant image is attributed to the increase in either of these two errors. Thus, low consistency errors obtained by relaxing the difference domain sparsity result in a noisy reconstruction.

**FIGURE 6.8**

Dynamic MRI reconstructions for cardiac cine data using (A1) DLTG, and (A2) adaptive DLTG. The top and bottom panels show reconstructed and difference images for retrospective under-sampling using MASK-5. The lower two panels show the temporal profile of row 128 in the original dataset and the respective difference images amplified by 6.

Conversely, relaxing the model, along with excess truncation in the difference domain, can result in artifact-prone reconstruction. In summary, the method works iteratively to improve image quality by continuously reducing the consistency and sparse approximation errors. The speed of reconstruction depends on how fast the errors decrease in relation to each other.

As the TVIS algorithm is a single-step shrinkage scheme, its rate of convergence can be improved by incorporation of the two-step iterative shrinkage (IS) strategies, such as Two-step iterative shrinkage/thresholding algorithms (TwIST) [56] and FISTA [51,52] algorithms. Since multiple-coil reconstructions are involved in a single iteration, the time advantage of the adaptation schemes becomes more relevant when used for multi-channel reconstructions. Compute times observed in the multi-channel reconstructions demonstrate that the computational cost of the ATVIS algorithm is 30% lower than that of FISTA.

The adaptation scheme is more suited to practical implementation in pMRI and dynamic MRI because the adaptation does not require computation of the exact upper bound on the sparse approximation error. Computation of the exact upper bound requires accurate knowledge of the measurement or input noise, as in the discrepancy principle. On the other hand, other TV-based parameter adaptation schemes such as the Adaptive parameter estimation (APE-ADMM) [24] require prior knowledge of noise, thereby limiting their practical use. The adaptation scheme is shown to result in reduction of reconstruction error along with acceleration of the compute time. Further improvements in image quality can be achieved using higher-order TV-based implementation using the total generalized variation (TGV) [57,58]. TGV was introduced to control or eliminate the staircasing effect

that often manifests in reconstructions using first-order TV implementations. The stair-casing artifacts occur because the piece-wise constant assumption required for efficient TV recovery in MR image reconstruction is not satisfied due to the inhomogeneities of the exciting **B1** field of high field systems (3.0 T and above) and of the receive coils. The adaptation strategy can be easily extended to higher-order TV implementation. The adaptive thresholding scheme for the TVIS algorithm accelerates the sparse image reconstruction for compressed sensing MRI. The acceleration is attained by a fast minimization of the absolute difference between TV norms of the consistency and sparse approximation errors in the derivative space.

Appendix

A6.1 Noise Dependence Property of Discrepancy Level

Theorem 1: When noise variance decreases to zero, the discrepancy level $\mathfrak{d}_1^{(k+1)}$ reduces to zero.

Proof. Using the definition of $\epsilon_{\text{res}}^{(k+1)}$ in Equation (6.18), we have:

$$\begin{aligned} \left\| \epsilon_{\text{res}}^{(k+1)} \right\|_1 &= \left\| \mathcal{X}F_u' \left(K_u - F_u \mathcal{X}^* \mathbf{d}^{(k)} \right) \right\|_1 \\ &\leq \left\| \mathcal{X}F_u' \right\|_1 \left\| F_u \mathcal{X}^* \mathbf{d}^{(k)} - K_u \right\|_1 \end{aligned} \quad (6.34)$$

Applying limit to Equation (6.34), yields:

$$\begin{aligned} \lim_{k \rightarrow \infty} \left\| \epsilon_{\text{res}}^{(k+1)} \right\|_1 &\leq \lim_{k \rightarrow \infty} \left\| \mathcal{X}F_u' \right\|_1 \left\| F_u \mathcal{X}^* \mathbf{d}^{(k)} - K_u \right\|_1 \\ &\leq \left\| \mathcal{X}F_u' \right\|_1 \left\| F_u \mathcal{X}^* \mathbf{d} - K_u \right\|_1 \\ &\leq \eta \sqrt{2LMN} \left\| \mathcal{X}F_u' \right\|_1, \end{aligned} \quad (6.35)$$

where $\left\| F_u \mathcal{X}^* \mathbf{d} - K_u \right\|_2^2 \leq \eta^2$ is an upper bound that depends on the noise level σ . When noise variance = 0, $\eta^2 = \tau MN \sigma^2 = 0$, where τ is a constant. Hence $\left\| \epsilon_{\text{res}}^{(k+1)} \right\|_1 = 0$, and $\left\| \epsilon_{\text{res}}^{(k)} \right\|_1 = 0$ at convergence. Consequently $\mathfrak{d}_1^{(k+1)} = 0$ at convergence.

A6.2 Relationship between l_1 -Norms of Consistency Error and Sparse Approximation Error

The following Lemmas are used to prove the relationship between the l_1 -norms of consistency error and sparse approximation error.

Lemma 1: Denoting $|\mathbf{b}| = (|b_1|, |b_2|, \dots, |b_n|)$, where $\mathbf{b}$ is any vector, and $n_{>\beta}(\hat{\mathbf{d}}^{(k)})$ the number of coefficients of $\hat{\mathbf{d}}^{(k)} > \beta$:

$$\|\mathbf{d}^{(k)}\|_1 = \|\hat{\mathbf{d}}^{(k)}\|_{1(>\beta)} - \beta n_{>\beta}(\hat{\mathbf{d}}^{(k)}),$$

Proof. The proof of this lemma is straightforward and can be obtained from the definition of the l_1 -norm:

$$\|\mathbf{d}^{(k)}\|_1 = \sum_{i=1}^{d_n} |d_i^{(k)}|, \quad (6.36)$$

where $d_n = 2LMN$. The coefficients $d_i^{(k)}$ can be split into two limits with reference to the threshold value, as follows:

$$\|\mathbf{d}^{(k)}\|_1 = \sum_{\forall i \text{ s.t. } |d_i^{(k)}| > \beta}^{d_n} |d_i^{(k)}| + \sum_{\forall i \text{ s.t. } |d_i^{(k)}| \leq \beta}^{d_n} |d_i^{(k)}|. \quad (6.37)$$

Using the definition of soft thresholding, Equation (6.37) becomes:

$$\begin{aligned} \|\mathbf{d}^{(k)}\|_1 &= \sum_{\forall i \text{ s.t. } |d_i^{(k)}| > \beta}^{d_n} \left| \hat{d}_i^{(k)} - \beta \frac{(\hat{\mathbf{d}}^{(k)})_i}{\left| (\hat{\mathbf{d}}^{(k)})_i \right|} \right| \\ &= \|\hat{\mathbf{d}}^{(k)}\|_{1(>\beta)} - \beta n_{>\beta}(\hat{\mathbf{d}}^{(k)}). \end{aligned} \quad (6.38)$$

Lemma 2: With $n_{>\beta}(\hat{\mathbf{d}}^{(k)})$ denoting the number of coefficients of $\hat{\mathbf{d}}^{(k)} > \beta$, we have:

$$\|\boldsymbol{\epsilon}_n^{(k)}\|_1 = \beta n_{>\beta}(\hat{\mathbf{d}}^{(k)}) + \|\hat{\mathbf{d}}^{(k)}\|_{1(\leq \beta)}.$$

Proof. The definition of the sparse approximation error can also be defined using soft thresholding, as follow:

$$\boldsymbol{\epsilon}_n^{(k)} = \begin{cases} \beta \frac{(\hat{\mathbf{d}}^{(k)})_i}{\left| (\hat{\mathbf{d}}^{(k)})_i \right|}; & \text{if } |d_i^{(k)}| > \beta \\ \hat{d}_i^{(k)}; & \text{if } |d_i^{(k)}| \leq \beta. \end{cases} \quad (6.39)$$

By definition:

$$\begin{aligned} \left\| \epsilon_n^{(k)} \right\|_1 &= \sum_{i=1}^{d_n} \left| \epsilon_n^{(k)} \right|_i. \\ &= \sum_{\forall i \text{ s.t. } \left| \hat{d}_i^k \right| > \beta} \left| \epsilon_n^{(k)} \right|_i + \sum_{\forall i \text{ s.t. } \left| \hat{d}_i^k \right| \leq \beta} \left| \epsilon_n^{(k)} \right|_i \end{aligned} \quad (6.40)$$

Using Equation (6.40), we have:

$$\begin{aligned} \left\| \epsilon_n^{(k)} \right\|_1 &= \sum_{\forall i \text{ s.t. } \left| \hat{d}_i^k \right| > \beta} \beta + \sum_{\forall i \text{ s.t. } \left| \hat{d}_i^k \right| \leq \beta} \left| \hat{d}_i^{(k)} \right| \\ &= \beta n_{>\beta} \left(\hat{\mathbf{d}}^{(k)} \right) + \left\| \hat{\mathbf{d}}^{(k)} \right\|_{1(\leq \beta)}. \end{aligned} \quad (6.41)$$

Lemma 3: Defining $G = F_u \mathcal{X}^*$, $\|GG^*\| < 1$ [42], then the series $\sum_{k=0}^{\infty} \left\| \hat{\mathbf{d}}^{(k+1)} - \hat{\mathbf{d}}^{(k)} \right\|_2^2$ is convergent [43].

Proof. To prove that the series $\sum_{k=0}^{\infty} \left\| \hat{\mathbf{d}}^{k+1} - \hat{\mathbf{d}}^k \right\|_2^2$ is convergent, first define for $G = F_u \mathcal{X}^*$:

$$\phi(\mathbf{d}, \mathbf{a}) = \left\| \mathbf{d} - \mathbf{a} \right\|_2^2 - \left\| G\mathbf{d} - G\mathbf{a} \right\|_2^2, \quad (6.42)$$

where $\mathbf{a}$ is any auxiliary vector in the sparse domain. Since $\|G^*\| < 1$, $I - GG^*$ is a strictly positive operator, hence $\phi(\mathbf{d}, \mathbf{a})$ is strictly convex in $\mathbf{d}$ for any choice of $\mathbf{a}$. Then $\phi(\mathbf{d}, \mathbf{a})$ is added to the cost function $J(\mathbf{d})$ to obtain the surrogate functional:

$$J^{\text{SUR}}(\mathbf{d}) = J(\mathbf{d}) - \left\| G\mathbf{d} - G\mathbf{a} \right\|_2^2 + \left\| \mathbf{d} - \mathbf{a} \right\|_2^2. \quad (6.43)$$

Since $\phi(\mathbf{d}, \mathbf{a})$ is strictly convex and $J(\mathbf{d})$ is convex in $\mathbf{d}$, $J^{\text{SUR}}(\mathbf{d})$ is strictly convex in $\mathbf{d}$ and has a unique minimizer for any choice of $\mathbf{a}$. One can then try to find the minimizer of $J(\mathbf{d})$ by an iterative process. The iterative procedure starts with an arbitrary vector $\hat{\mathbf{d}}^0$; determine the minimizer $\mathbf{d}^1$ computed by soft thresholding $\hat{\mathbf{d}}^1$ for $\mathbf{a} = \hat{\mathbf{d}}^0$. The soft-thresholded coefficients in each successive iterate $\mathbf{d}^k$ is the minimizer for $\mathbf{d}$ of the surrogate functional in Equation (6.43) anchored at the previous iterate; that is, $\mathbf{a} = \hat{\mathbf{d}}^{k-1}$.

Claim 3.1: $J(\hat{\mathbf{d}}^k)$ is a non-increasing sequence.

Proof. To prove this, consider the difference of the l_2 -norms:

$$\begin{aligned} \left\| \mathbf{h} \right\|_2^2 - \left\| G^* \mathbf{h} \right\|_2^2 &= \langle \mathbf{h}, \mathbf{h} \rangle - \langle G^* \mathbf{h}, G^* \mathbf{h} \rangle \\ &= \langle \mathbf{h}, \mathbf{h} \rangle - \langle GG^* \mathbf{h}, \mathbf{h} \rangle, \\ &= \langle (I - GG^*) \mathbf{h}, \mathbf{h} \rangle. \end{aligned} \quad (6.44)$$

where $\mathbf{h}$ is a vector in the sparse domain. Defining $L = \sqrt{I - GG^*}$, we have:

$$\begin{aligned}\|L\mathbf{h}\|_2^2 &= \langle L\mathbf{h}, L\mathbf{h} \rangle \\ &= \left\langle \sqrt{I - GG^*} \mathbf{h}, \sqrt{I - GG^*} \mathbf{h} \right\rangle. \\ &= \left\langle (I - GG^*) \mathbf{h}, \mathbf{h} \right\rangle.\end{aligned}\tag{6.45}$$

From Equations (6.44) and (6.45), we have:

$$\|\mathbf{h}\|_2^2 - \|G^* \mathbf{h}\|_2^2 = \|L\mathbf{h}\|_2^2.\tag{6.46}$$

Using Equation (6.46), the surrogate function in Equation (6.43) becomes:

$$J^{\text{SUR}}(\mathbf{d}) = J(\mathbf{d}) + \|L(\mathbf{d} - \mathbf{a})\|_2^2.\tag{6.47}$$

To show that the sequence $J(\hat{\mathbf{d}}^k)$ is non-increasing, consider the $J^{\text{SUR}}(\mathbf{d})$ with $\mathbf{d} = \hat{\mathbf{d}}^{k+1}$ and $\mathbf{a} = \hat{\mathbf{d}}^k$:

$$J(\hat{\mathbf{d}}^{k+1}) + \|L(\hat{\mathbf{d}}^{k+1} - \hat{\mathbf{d}}^k)\|_2^2 = J^{\text{SUR}}(\hat{\mathbf{d}}^{k+1}, \hat{\mathbf{d}}^k)\tag{6.48}$$

Since, $\hat{\mathbf{d}}^{k+1}$ is a minimizer of the function, replacing $\hat{\mathbf{d}}^{k+1}$ with $\hat{\mathbf{d}}^k$ in the RHS of Equation (6.48) yields:

$$J(\hat{\mathbf{d}}^{k+1}) + \|L(\hat{\mathbf{d}}^{k+1} - \hat{\mathbf{d}}^k)\|_2^2 \leq J^{\text{SUR}}(\hat{\mathbf{d}}^k, \hat{\mathbf{d}}^k).\tag{6.49}$$

Therefore:

$$J(\hat{\mathbf{d}}^{k+1}) \leq J(\hat{\mathbf{d}}^k).\tag{6.50}$$

Thus, the sequence $J(\hat{\mathbf{d}}^k)$ is a non-increasing sequence.

From the above proof, we have:

$$\|L(\hat{\mathbf{d}}^{k+1} - \hat{\mathbf{d}}^k)\|_2^2 \leq J(\hat{\mathbf{d}}^k) - J(\hat{\mathbf{d}}^{k+1}).\tag{6.51}$$

From the strict positive definiteness of L , for any $N \in \mathbb{N}$, we have:

$$\sum_{k=0}^N \|\hat{\mathbf{d}}^{k+1} - \hat{\mathbf{d}}^k\|_2^2 \leq \frac{1}{A} \sum_{k=0}^N \|L(\hat{\mathbf{d}}^{k+1} - \hat{\mathbf{d}}^k)\|_2^2,\tag{6.52}$$

where A is a strictly positive bound for the spectrum of L^*L . From Equation (6.51), we have:

$$\sum_{k=0}^N \left\| L(\hat{\mathbf{d}}^{k+1} - \hat{\mathbf{d}}^k) \right\|_2^2 \leq \sum_{k=0}^N \left[J(\hat{\mathbf{d}}^k) - J(\hat{\mathbf{d}}^{k+1}) \right]. \quad (6.53)$$

Equation (6.53) simplifies to:

$$\sum_{k=0}^N \left\| L(\hat{\mathbf{d}}^{k+1} - \hat{\mathbf{d}}^k) \right\|_2^2 \leq J(\hat{\mathbf{d}}^0) - J(\hat{\mathbf{d}}^{N+1}) \leq J(\hat{\mathbf{d}}^0). \quad (6.54)$$

This follows that $\sum_{k=0}^N \left\| \hat{\mathbf{d}}^{k+1} - \hat{\mathbf{d}}^k \right\|_2^2$ is a bounded family in N , so that the infinite series $\sum_{k=0}^{\infty} \left\| \hat{\mathbf{d}}^{k+1} - \hat{\mathbf{d}}^k \right\|_2^2$ converges. As an immediate consequence, $\left\| \hat{\mathbf{d}}^{k+1} - \hat{\mathbf{d}}^k \right\|_2^2 \rightarrow 0$ as $k \rightarrow \infty$. Since $\|\mathbf{d}\|_1 \leq \sqrt{d_n} \|\mathbf{d}\|_2$, it is evident that $\left\| \hat{\mathbf{d}}^{k+1} - \hat{\mathbf{d}}^k \right\|_1 \rightarrow 0$ as $k \rightarrow \infty$.

Theorem 2: The relationship between the l_1 -norms of consistency and sparse approximation errors at convergence can be expressed as:

$$\left\| \boldsymbol{\epsilon}_{\text{res}}^{(k+1)} \right\|_1 \geq \left\| \boldsymbol{\epsilon}_n^{(k)} \right\|_1.$$

Proof. Consider the difference:

$$\boldsymbol{\epsilon}_{\text{res}}^{(k+1)} - \boldsymbol{\epsilon}_n^{(k)} = \hat{\mathbf{d}}^{(k+1)} - \hat{\mathbf{d}}^{(k)}. \quad (6.55)$$

Consider the l_1 -norm of consistency error:

$$\left\| \boldsymbol{\epsilon}_{\text{res}}^{(k+1)} \right\|_1 = \left\| \hat{\mathbf{d}}^{(k+1)} - \mathbf{d}^{(k)} \right\|_1. \quad (6.56)$$

Since $\left\| \hat{\mathbf{d}}^{(k+1)} - \mathbf{d}^{(k)} \right\|_1 \geq \left\| \hat{\mathbf{d}}^{(k+1)} \right\|_1 - \left\| \mathbf{d}^{(k)} \right\|_1$, Equation (6.56) becomes:

$$\left\| \boldsymbol{\epsilon}_{\text{res}}^{(k+1)} \right\|_1 \geq \left\| \hat{\mathbf{d}}^{k+1} \right\|_1 - \left\| \mathbf{d}^{(k)} \right\|_1. \quad (6.57)$$

$$\geq \left\| \hat{\mathbf{d}}^{k+1} \right\|_1 - \left\| \mathbf{d}^{(k)} \right\|_1.$$

Using Lemma 1, Equation (6.57) becomes:

$$\left\| \boldsymbol{\epsilon}_{\text{res}}^{(k+1)} \right\|_1 \geq \left\| \hat{\mathbf{d}}^{(k+1)} \right\|_1 - \left\| \hat{\mathbf{d}}^{(k)} \right\|_{>\beta} + \beta n_{>\beta}(\hat{\mathbf{d}}^{(k)}). \quad (6.58)$$

Adding and subtracting $\|\hat{\mathbf{d}}^{(k)}\|_{1(\leq\beta)}$ to the RHS yields:

$$\|\epsilon_{\text{res}}^{(k+1)}\|_1 \geq \|\hat{\mathbf{d}}^{(k+1)}\|_1 - \|\hat{\mathbf{d}}^{(k)}\|_1 + \beta n_{>\beta}(\hat{\mathbf{d}}^{(k)}) + \|\hat{\mathbf{d}}^{(k)}\|_{1(\leq\beta)}. \quad (6.59)$$

Using Lemma 2, Equation (6.59) becomes:

$$\|\epsilon_{\text{res}}^{(k+1)}\|_1 \geq \|\hat{\mathbf{d}}^{(k+1)}\|_1 - \|\hat{\mathbf{d}}^{(k)}\|_1 + \|\epsilon_n^{(k)}\|_1. \quad (6.60)$$

At convergence, according to Lemma 3, $\|\hat{\mathbf{d}}_1^{(k+1)}\| - \|\hat{\mathbf{d}}_1^{(k)}\| \rightarrow 0$, and Equation (6.60) yields [59]. Compressed Sensing Parallel MRI with Adaptive Shrinkage TV Regularization. arXiv preprint arXiv:1809.06665. 2018 Sep 18.:

$$\|\epsilon_{\text{res}}^{(k+1)}\|_1 \geq \|\epsilon_n^{(k)}\|_1. \quad (6.61)$$

A6.3 Relationship between the Expectation of Consistency and Sparse Approximation Errors

Proposition 1: At convergence, the expectations of the errors can be related as follows:

$$\mathbb{E}\left(\left|\epsilon_{\text{res}}^{(k+1)}\right|\right) \geq \mathbb{E}\left(\left|\epsilon_n^{(k)}\right|\right)$$

Proof. To prove this, consider the absolute error coefficients:

$$\left|\epsilon_{\text{res}}^{(k+1)}\right| = \left|\hat{\mathbf{d}}^{(k+1)} - \mathbf{d}^{(k)}\right|. \quad (6.62)$$

Substituting $\mathbf{d}^{(k)} = \hat{\mathbf{d}}^{(k)} - \epsilon_n^{(k)}$, we have:

$$\left|\epsilon_{\text{res}}^{(k+1)}\right| = \left|\hat{\mathbf{d}}^{(k+1)} - \hat{\mathbf{d}}^{(k)} + \epsilon_n^{(k)}\right|. \quad (6.63)$$

Using triangle inequality, we have

$$\left|\epsilon_{\text{res}}^{(k+1)}\right| \geq \left|\left|\hat{\mathbf{d}}^{(k+1)} - \hat{\mathbf{d}}^{(k)}\right| - \left|\epsilon_n^{(k)}\right|\right| \quad (6.64)$$

Applying the limit $k \rightarrow \infty$ in Equation (6.64) implies that the first term in the RHS $\left|\hat{\mathbf{d}}^{(k+1)} - \hat{\mathbf{d}}^{(k)}\right| \rightarrow 0$ at convergence, as in Lemma 3, whereas the LHS and the second term do not. Hence:

$$\lim_{k \rightarrow \infty} \left|\epsilon_{\text{res}}^{(k+1)}\right| \geq \lim_{k \rightarrow \infty} \left|\epsilon_n^{(k)}\right|. \quad (6.65)$$

As the expectation $\mathbb{E}(\cdot)$ preserves monotonicity [60], we have:

$$\mathbb{E}\left(\left|\epsilon_{\text{res}}^{(k+1)}\right|\right) \geq \mathbb{E}\left(\left|\epsilon_n^{(k)}\right|\right), \quad (6.66)$$

A6.4 Convergence of Threshold Sequence

Lemma 5: The adaptive threshold sequence $\{\beta^k\}$ converges as $k \rightarrow \infty$.

Proof. Since the threshold adaptation yields $\beta^{(k+1)} < \beta^{(k)}$, as dictated by the fixed-point continuation methods [49], usage of the threshold adaptation equation should satisfy:

$$\frac{\Phi\left(\mathfrak{d}_1^{(k+1)}\right)}{\mathbb{E}\left(\left|\epsilon_{\text{res}}^{(k+1)}\right|\right)} + \frac{\mathbb{E}\left(\left|\epsilon_n^{(k)}\right|\right)}{\beta^{(k)}\mathbb{E}\left(\left|\epsilon_{\text{res}}^{(k+1)}\right|\right)} > \frac{1}{\beta^{(k)}}. \quad (6.67)$$

From Equation (6.67), we have:

$$\Phi\left(\mathfrak{d}_1^{(k+1)}\right) > \frac{1}{\beta^{(k)}}\left(\mathbb{E}\left(\left|\epsilon_{\text{res}}^{(k+1)}\right|\right) - \mathbb{E}\left(\left|\epsilon_n^{(k)}\right|\right)\right) \quad (6.68)$$

Since $\mathbb{E}\left(\left|\epsilon_{\text{res}}^{(k+1)}\right|\right) \geq \mathbb{E}\left(\left|\epsilon_n^{(k)}\right|\right)$ (see Proposition 1), and the requirement for $\beta^{(k+1)} < \beta^{(k)}$, the denominators of the respective threshold adaptation equations should satisfy the following:

$$\Phi\left(\mathfrak{d}_1^{(k+1)}\right) + \frac{\mathbb{E}\left(\left|\epsilon_n^{(k)}\right|\right)}{\beta^{(k)}} > \Phi\left(\mathfrak{d}_1^{(k)}\right) + \frac{\mathbb{E}\left(\left|\epsilon_n^{(k-1)}\right|\right)}{\beta^{(k-1)}}. \quad (6.69)$$

Rearranging Equation (6.69), we have:

$$\Phi\left(\mathfrak{d}_1^{(k+1)}\right) > \Phi\left(\mathfrak{d}_1^{(k)}\right) - \left[\frac{\mathbb{E}\left(\left|\epsilon_n^{(k)}\right|\right)}{\beta^{(k)}} - \frac{\mathbb{E}\left(\left|\epsilon_n^{(k-1)}\right|\right)}{\beta^{(k-1)}} \right] \quad (6.70)$$

Substitution of $\Phi\left(\mathfrak{d}_1^{(k)}\right) = \frac{\mathbb{E}\left(\left|\epsilon_{\text{res}}^{(k)}\right|\right)}{\beta^{(k)}} - \frac{\mathbb{E}\left(\left|\epsilon_n^{(k-1)}\right|\right)}{\beta^{(k-1)}}$ into Equation (6.70) gives:

$$\Phi\left(\mathfrak{d}_1^{(k+1)}\right) > \frac{\mathbb{E}\left(\left|\epsilon_{\text{res}}^{(k)}\right|\right)}{\beta^{(k)}} - \frac{\mathbb{E}\left(\left|\epsilon_n^{(k)}\right|\right)}{\beta^{(k)}} \quad (6.71)$$

From Equation (6.71), we have:

$$\frac{\mathbb{E}\left(\left|\epsilon_{\text{res}}^{(k+1)}\right|\right) - \mathbb{E}\left(\left|\epsilon_n^{(k)}\right|\right)}{\Phi\left(\mathfrak{d}_1^{(k)}\right)} < \beta^{(k)}. \quad (6.72)$$

Using the definition of soft thresholding, $\epsilon_{\text{res}}^{(k+1)}$ and $\epsilon_n^{(k)}$ can be expressed as:

$$\left(\epsilon_{\text{res}}^{(k+1)}\right)_i = \begin{cases} \left(\hat{d}^{(k+1)}\right)_i - \left(\left(\hat{d}^{(k)}\right)_i - \beta^{(k-1)} \frac{\left(\hat{d}^{(k)}\right)_i}{\left|\left(\hat{d}^{(k)}\right)_i\right|}\right), & \text{if } \left|\left(\hat{d}^{(k)}\right)_i\right| \geq \beta^{(k-1)} \\ \left(\hat{d}^{(k+1)}\right)_i, & \text{otherwise.} \end{cases} \quad (6.73)$$

and

$$\left(\epsilon_n^{(k)}\right)_i = \begin{cases} \beta^{(k-1)} \frac{\left(\hat{d}^{(k)}\right)_i}{\left|\left(\hat{d}^{(k)}\right)_i\right|} & \text{if } \left|\left(\hat{d}^{(k)}\right)_i\right| \geq \beta^{(k-1)} \\ \left(\hat{d}^{(k)}\right)_i, & \text{otherwise.} \end{cases} \quad (6.74)$$

Since $\left|\hat{d}^{(k+1)} - \hat{d}^{(k)}\right| \rightarrow 0$ as $k \rightarrow \infty$ (due to Lemma 3), the expectations of errors can be approximated as:

$$\mathbb{E}\left(\left|\epsilon_{\text{res}}^{(k+1)}\right|\right) \rightarrow \beta^{(k-1)} + \mathbb{E}\left(\left|\hat{d}^{(k+1)}\right|\right)_{<\beta^{(k-1)}}, \quad (6.75)$$

where $\mathbb{E}\left(\left|\hat{d}^{(k+1)}\right|\right)_{<\beta^{(k-1)}}$ denotes the expectation of absolute transform domain coefficients satisfying $\left|\hat{d}^{(k+1)}\right| < \beta^{(k-1)}$. Similarly,

$$\mathbb{E}\left(\left|\epsilon_n^{(k)}\right|\right) \rightarrow \beta^{(k-1)} + \mathbb{E}\left(\left|\hat{d}^{(k)}\right|\right)_{<\beta^{(k-1)}}, \quad (6.76)$$

where $\mathbb{E}\left(\left|\hat{d}^{(k)}\right|\right)_{<\beta^{(k-1)}}$ denotes the expectation of absolute transform domain coefficients satisfying $\left|\hat{d}^{(k)}\right| < \beta^{(k-1)}$. Thus, at convergence:

$$\mathbb{E}\left(\left|\epsilon_{\text{res}}^{(k+1)}\right|\right) - \mathbb{E}\left(\left|\epsilon_n^{(k)}\right|\right) = \mathbb{E}\left(\left|\hat{d}^{(k+1)}\right|\right)_{<\beta^{(k-1)}} - \mathbb{E}\left(\left|\hat{d}^{(k)}\right|\right)_{<\beta^{(k-1)}} \quad (6.77)$$

As $k \rightarrow \infty$, substitution of Equation (6.77) into the LHS of Equation (6.72) gives:

$$\frac{\mathbb{E}\left(\left|\hat{d}^{(k+1)}\right|\right)_{<\beta^*} - \mathbb{E}\left(\left|\hat{d}^{(k)}\right|\right)_{<\beta^*}}{\Phi\left(\mathfrak{d}_1^{(k+1)}\right)} < \beta^*, \quad (6.78)$$

where $\beta^* = \left(\beta^{(k)}\right)_{k \rightarrow \infty}$ and $\mathfrak{d}_{\text{IT}} = \left(\mathfrak{d}_1^{(k)}\right)_{k \rightarrow \infty}$. The fact that the numerator of Equation (6.78) approaches zero in the limiting case as $k \rightarrow \infty$ indicates that the threshold never drops

down to zero, and a lower bound exists for $\{\beta^{(k)}\}$. A lower bound exists for $\{\beta^{(k)}\}$, and $\beta^{(k+1)} < \beta^{(k)}$ implies that the adaptive threshold sequence $\{\beta^{(k)}\}$ converges as $k \rightarrow \infty$.

A6.5 Convergence of ATVIS Algorithm

The proof for convergence of the adaptive algorithm uses the following two theorems.

Theorem 3: The monotone line search condition (MLSC) is satisfied with adaptation, whenever $\rho \leq \frac{1-\sigma}{l}$, where l is the Lipschitz constant of the gradient of $\frac{1}{2}\|K_u - F_u \mathcal{X}^* \mathbf{d}\|_2^2$ and σ is a constant in the interval $(0, 1)$.

Proof. The monotone line search condition [61] is said to be satisfied if:

$$J(\mathbf{d}^{k+1}) \leq J(\mathbf{d}^k) - \frac{\sigma}{2\rho} \|\mathbf{d}^{k+1} - \mathbf{d}^k\|_2^2. \quad (6.79)$$

The general iterative shrinkage and thresholding algorithm solves Equation (6.17) by generating a sequence $\{\mathbf{d}^k\}$ using:

$$\mathbf{d}^{k+1} = \min_{\mathbf{d}} \frac{1}{2} \|G\mathbf{d}^k - K_u\|_2^2 + \langle G^*G\mathbf{d}^k - G^*K_u, \mathbf{d} - \mathbf{d}^k \rangle + \beta^k \|\mathbf{d}_1\|_1 + \frac{1}{2\rho} \|\mathbf{d}^k - \mathbf{d}\|_2^2, \quad (6.80)$$

where $G = F_u \mathcal{X}^*$ as in Lemma 3. Since $\mathbf{d}^{k+1}$ is a minimizer, replacing $\mathbf{d}$ with $\hat{\mathbf{d}}^{k+1}$ with $\hat{\mathbf{d}}^k$ in the RHS of Equation (6.80) separately yields:

$$\begin{aligned} & \frac{1}{2} \|G\mathbf{d}^k - K_u\|_2^2 + \langle G^*G\mathbf{d}^k - G^*K_u, \mathbf{d}^{k+1} - \mathbf{d}^k \rangle + \beta^k \|\mathbf{d}^{k+1}\|_1 + \frac{1}{2\rho} \|\mathbf{d}^k - \mathbf{d}^{k+1}\|_2^2 \\ & \leq \frac{1}{2} \|G\mathbf{d}^k - K_u\|_2^2 + \langle G^*G\mathbf{d}^k - G^*K_u, \mathbf{d}^k - \mathbf{d}^k \rangle + \beta^{k-1} \|\mathbf{d}^k\|_1 + \frac{1}{2\rho} \|\mathbf{d}^k - \mathbf{d}^k\|_2^2, \end{aligned} \quad (6.81)$$

From Equation (6.81), it can be deduced that:

$$\langle G^*G\mathbf{d}^k - G^*K_u, \mathbf{d}^{k+1} - \mathbf{d}^k \rangle + \beta^k \|\mathbf{d}^{k+1}\|_1 + \frac{1}{2\rho} \|\mathbf{d}^k - \mathbf{d}^{k+1}\|_2^2 \leq \beta^{k-1} \|\mathbf{d}^k\|_1. \quad (6.82)$$

Since $\frac{1}{2}\|G\mathbf{d}^{k+1} - K_u\|_2^2$ is the Lipschitz differentiable, we have:

$$\frac{1}{2} \|G\mathbf{d}^{k+1} - K_u\|_2^2 \leq \frac{1}{2} \|G\mathbf{d}^k - K_u\|_2^2 + \langle G^*G\mathbf{d}^k - G^*K_u, \mathbf{d}^{k+1} - \mathbf{d}^k \rangle + \frac{l}{2} \|\mathbf{d}^k - \mathbf{d}^{k+1}\|_2^2. \quad (6.83)$$

Rearranging Equation (6.83), we have:

$$\frac{1}{2} \|G\mathbf{d}^{k+1} - K_u\|_2^2 - \frac{1}{2} \|G\mathbf{d}^k - K_u\|_2^2 - \frac{l}{2} \|\mathbf{d}^k - \mathbf{d}^{k+1}\|_2^2 \leq \langle G^*G\mathbf{d}^k - G^*K_u, \mathbf{d}^{k+1} - \mathbf{d}^k \rangle. \quad (6.84)$$

Using Equation (6.84), Equation (6.82) becomes:

$$\frac{1}{2} \|G\mathbf{d}^{k+1} - K_u\|_2^2 - \frac{1}{2} \|G\mathbf{d}^k - K_u\|_2^2 - \frac{l}{2} \|\mathbf{d}^k - \mathbf{d}^{k+1}\|_2^2 + \beta^k \|\mathbf{d}^{k+1}\|_1 + \frac{1}{2\rho} \|\mathbf{d}^k - \mathbf{d}^{k+1}\|_2^2 \leq \beta^{k-1} \|\mathbf{d}^k\|_1. \quad (6.85)$$

Rearranging the terms in Equation (6.85), we have:

$$\frac{1}{2} \|G\mathbf{d}^{k+1} - K_u\|_2^2 + \beta^k \|\mathbf{d}^{k+1}\|_1 \leq \frac{1}{2} \|G\mathbf{d}^k - K_u\|_2^2 + \beta^{k-1} \|\mathbf{d}^k\|_1 + \left(\frac{l}{2} - \frac{1}{2\rho}\right) \|\mathbf{d}^k - \mathbf{d}^{k+1}\|_2^2. \quad (6.86)$$

This implies that:

$$J(\mathbf{d}^{k+1}) \leq J(\mathbf{d}^k) + \left(\frac{l}{2} - \frac{1}{2\rho}\right) \|\mathbf{d}^k - \mathbf{d}^{k+1}\|_2^2. \quad (6.87)$$

Therefore, the MLSC is satisfied whenever:

$$\frac{1-\rho l}{2\rho} \geq \frac{\sigma}{2\rho} \Rightarrow \rho \leq \frac{1-\sigma}{l}. \quad (6.88)$$

Theorem 4: If the MLSC is satisfied, then all limit points of the sequence $\{\mathbf{d}^k\}$ generated by Equation (6.80) are critical points of Equation (6.17).

Proof. The existence of limit points for the sequence $\{\mathbf{d}^k\}$ is provided in Theorem 4 by Gong et al. [61]. If the MLSC is satisfied, we have:

$$J(\mathbf{d}^{k+1}) \leq J(\mathbf{d}^k) \quad \forall k \geq 0, \quad (6.89)$$

which implies that the sequence $\{J(\mathbf{d}^k)\}$ is monotonically decreasing. Let $\mathbf{d}^*$ be a limit point of the sequence $\mathbf{d}^k$; that is, there exists a subset of N of $\mathbb{N}$ such that:

$$\lim_{k \in N \rightarrow \infty} \mathbf{d}^k = \mathbf{d}^*. \quad (6.90)$$

Using the facts that $J(\cdot)$ is lower bounded and $\{J(\mathbf{d}^k)\}$ is monotonically decreasing, $\lim_{k \rightarrow \infty} J(\mathbf{d}^k)$ exists. Observing that $J(\cdot)$ is continuous, we have:

$$\lim_{k \rightarrow \infty} J(\mathbf{d}^k) = \lim_{k \in N \rightarrow \infty} J(\mathbf{d}^k) = J\left(\lim_{k \in N \rightarrow \infty} \mathbf{d}^k\right) = J(\mathbf{d}^*). \quad (6.91)$$

Taking limits on both sides of Equation (6.79) yields:

$$\lim_{k \in N \rightarrow \infty} \|\mathbf{d}^{k+1} - \mathbf{d}^k\|_2^2 = 0. \quad (6.92)$$

Because the minimizer $\mathbf{d}^{k+1}$ is also a critical point of the problem in Equation (6.17), by the definition of sub-differential, we have:

$$0 \in \nabla \left(\frac{1}{2} \|F_u \mathcal{X}^* \mathbf{d}^{k+1} - K_u\|_2^2 \right) + \frac{\rho}{2} \nabla \|\mathbf{d}^k - \mathbf{d}^{k+1}\|_2^2 + \beta^k \partial \|\mathbf{d}^{k+1}\|_1 \quad \forall k. \quad (6.93)$$

Hence:

$$0 \in \lim_{k \in N \rightarrow \infty} \nabla \left(\frac{1}{2} \|F_u \mathcal{X}^* \mathbf{d}^{k+1} - K_u\|_2^2 \right) + \frac{\rho}{2} \lim_{k \in N \rightarrow \infty} \nabla \|\mathbf{d}^k - \mathbf{d}^{k+1}\|_2^2 + \lim_{k \in N \rightarrow \infty} \beta^k \lim_{k \in N \rightarrow \infty} \partial \|\mathbf{d}^{k+1}\|_1. \quad (6.94)$$

This implies that:

$$\begin{aligned} 0 &\in \nabla \left(\frac{1}{2} \|F_u \mathcal{X}^* \mathbf{d}^* - K_u\|_2^2 \right) + \beta^* \partial \|\mathbf{d}^*\|_1. \\ &\Rightarrow 0 \in \partial J(\mathbf{d}^*), \end{aligned} \quad (6.95)$$

where $\mathbf{d}^*$ is a critical point of the problem in Equation (6.17), and β^* is the limit point of the sequence $\{\beta^k\}$ from Lemma 5. This implies that Equation (6.17) has an optimal solution with threshold adaptation.

Lemma 6: The threshold adaptation algorithm yields accelerated convergence.

Proof. In order to prove the accelerated convergence, consider the scaled form of the cost function $J(\mathbf{d}) = \frac{1}{2\beta} \|K_u - F_u \mathcal{X}^*(\mathbf{d})\|_2^2 + \|\mathbf{d}\|_1$. Then, in accordance with Theorem 2 due to Kowalski [62], the cost functions at $(k+1)^{th}$ and k^{th} iterations can be related as follows:

$$J(\mathbf{d}^{(k+1)}) \leq J(\mathbf{d}^{(k)}) - \left(\frac{1}{\beta} - \frac{l}{2\beta} \right) \|\mathbf{d}^{(k)} - \mathbf{d}^{(k+1)}\|_2^2, \quad (6.96)$$

where l is the Lipschitz constant of the gradient of $\frac{1}{2} \|F_u \mathcal{X}^* \mathbf{d}^{(k)} - K_u\|_2^2$. With threshold adaptation, Equation (6.96) is modified as:

$$J(\mathbf{d}^{(k+1)}) \leq J(\mathbf{d}^{(k)}) - \left(\frac{1}{\beta^{(k)}} - \frac{l}{2\beta^{(k)}} \right) \|\mathbf{d}^{(k)} - \mathbf{d}^{(k+1)}\|_2^2 \quad (6.97a)$$

$$J(\mathbf{d}^{(k+2)}) \leq J(\mathbf{d}^{(k+1)}) - \left(\frac{1}{\beta^{(k+1)}} - \frac{l}{2\beta^{(k+1)}} \right) \|\mathbf{d}^{(k+1)} - \mathbf{d}^{(k+2)}\|_2^2 \quad (6.97b)$$

Using Equation (6.97a) and (6.97b) at convergence, we have:

$$\left(\frac{1}{\beta^{(k)}} - \frac{l}{2\beta^{(k)}} \right) \|\mathbf{d}^{(k)} - \mathbf{d}^{(k+1)}\|_2^2 \geq \left(\frac{1}{\beta^{(k+1)}} - \frac{l}{2\beta^{(k+1)}} \right) \|\mathbf{d}^{(k+1)} - \mathbf{d}^{(k+2)}\|_2^2 \quad (6.98)$$

This implies that:

$$\|\mathbf{d}^{(k+1)} - \mathbf{d}^{(k+2)}\|_2^2 \leq \frac{\beta^{(k+1)}}{\beta^{(k)}} \|\mathbf{d}^{(k)} - \mathbf{d}^{(k+1)}\|_2^2 \quad (6.99)$$

In the case of a constant threshold, this becomes equivalent to $\|d^{(k+1)} - d^{(k+2)}\|_2^2 \leq \|d^{(k)} - d^{(k+1)}\|_2^2$. According to Equation (6.99), the norm of the difference between successive iterates is $\frac{\beta^{(k+1)}}{\beta^{(k)}} < 1$ times less than the norm of the difference between previous iterates. This ensures accelerated convergence.

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Combination of Parallel Magnetic Resonance Imaging and Compressed Sensing Using L1-SPIRiT

7.1 Combination of Parallel Magnetic Resonance Imaging and Compressed Sensing

Longer scan times during magnetic resonance (MR) data acquisition demands the development of sophisticated techniques that can accelerate the scanning process by reducing the number of k -space measurements without compromising the image quality. In parallel magnetic resonance imaging (pMRI), the data redundancy provided by the spatial sensitivity of different coils, together with gradient encoding, is exploited to allow for a significant reduction of the number of necessary k -space samples [1–6]. On the other hand, the image recovery using compressed sensing (CS) principles is achieved with the assumption that the signal has a sparse representation with incoherent measurements [7–9]. As the two schemes rely on two different auxiliary information, many recent attempts have been made to combine these two techniques (CS-pMRI), resulting in a better recovery than each of the methods used individually [10–16].

CS-pMRI methods can be broadly classified into two categories: (1) those performing a joint optimization procedure for image reconstruction and (2) those in which a CS-based reconstruction is treated as a separate stage, either before or after the pMRI reconstruction stage. The examples of type 1 include Sparse sensitivity encoding (SENSE) [14] and its equivalents [17,18]. In addition, Weller et al. [19] proposed a Generalized autocalibrating partially parallel acquisitions (GRAPPA)-based CS-pMRI in which the l_1 -norm minimization problem and l_2 -norm fidelity term are minimized together during GRAPPA reconstruction. The method can be realized by performing the CS-SENSE algorithm in which a reconstruction is first obtained by using SparseMRI [7] on a set of aliased images with a localized field of view (FOV) for each coil, and the final reconstruction is then obtained from the aliased images using Cartesian SENSE [12].

The efficient implementation of any CS reconstruction is directly dependent on the following factors: (1) sparsity of representation, (2) incoherent sampling and (3) reconstruction that enforces non-linear sparsity. Thus, any attempt to combine CS and pMRI requires a modified sparsity model that can be used for multiple coil images, an incoherent sampling scheme that does not harm the conditioning of pMRI reconstruction, and a parallel imaging reconstruction scheme that uses information from the multiple coils as well as the sparsity of the underlying data in some transform domain. In other words, the implementation of CS-pMRI methods require a parallel imaging reconstruction framework that works well for random k -space trajectories enabling the introduction of sparsity constraints into the reconstruction problem.

From the joint optimization problem in Equation (2.86) in [Chapter 2](#), it is evident that SPIRiT is an auto-calibrating method that allows the incorporation of sparsity constraints along with parallel imaging. Although the use of random sampling brings about the incoherence property required for compressed sensing, it is not optimized for parallel imaging. The larger gaps between k -space data samples reduce the reconstruction conditioning and increase noise. It has been found that Poisson-disc distribution [20,21] sampling provides a high degree of incoherence as well as a uniform distance between samples. This also introduces flexibility for fractional and anisotropic acceleration by the use of ellipsoids rather than discs.

7.2 L1-SPIRiT

Although SPIRiT was initially proposed as a generalized iterative version of GRAPPA, the method gained more significance with the advent of l_1 -norm based iterative self-consistent parallel imaging reconstruction (L1-SPIRiT). L1-SPIRiT is an iterative method that utilizes l_1 -norm penalty and k -space data self-consistency [10,11]. It is implemented using an iterative GRAPPA-like approach [7] that enforces both calibration consistency and data acquisition consistency along with additional sparsity constraint. To utilize the correlation between the sparse domain coefficients of the individual coil images, a joint sparsity functional is utilized in L1-SPIRiT. For introduction of joint sparsity-based penalization, the vector of transform domain coefficients is represented as [22]:

$$\omega(r) = [\omega_1(r), \omega_2(r), \dots, \omega_{n_C}(r)], \quad (7.1)$$

where r denotes the position across all n_C coils. The l_1 -norm is generalized to be the sum of magnitudes as $\sum_r \|\omega(r)\|_2$. The joint sparsity-based L1-SPIRiT optimization function can be expressed as:

$$\begin{aligned} \min \sum_r \|\omega(r)\|_2, \\ \text{s.t. } k = Zk, k_{NC} = Dk, \omega = \Psi F_u' k. \end{aligned} \quad (7.2)$$

In the joint sparse model, the existence of a large coefficient in any one of the coils preserves the coefficients in the rest of the coils from being suppressed by the non-linear reconstruction. Since SPIRiT exploits the correlation among both acquired and unacquired k -space samples, the estimation of the missing samples is first performed using the calibration consistency projection using:

$$K^{f(k)} = Z \hat{K}^{f(k)}, \quad (7.3)$$

where $\hat{K}^{f(k)}$ is the k -space before filtering. At the start of the iterations, $\hat{K}^{f(k)}$ is same as the zero-filled k -space K_u . The data consistency error at the acquired k -space locations after k -space filling, using the calibration consistency projection in the k th iteration, is given by:

$$E_{\text{res}}^{(k)} = K_u - K^{f(k)}, \quad (7.4)$$

where K_u denotes the zero filled k -space, and K^f is the filter reconstructed k -space. In L1-SPIRiT, the consistency corrected k -space $K^{f(k)} + E_{\text{res}}^{(k)}$ is soft-thresholded in the transform domain for removing noise without affecting the image resolution. The soft-thresholded transform domain coefficients are given by:

$$\omega^{(k)} = \mathcal{S}_v \left(\Psi F_u' \left(K^{f(k)} + E_{\text{res}}^{(k)} \right) \right), \quad (7.5)$$

where $\mathcal{S}$ denotes the joint soft-thresholding operator, F_u denotes the multi-coil under-sampling Fourier operator obtained by setting the unacquired frequency points as zeros, v denotes the threshold, and Ψ and Ψ' represent the multi-coil sparsifying transform operator and its adjoint operator, respectively. The incorporation of l_1 -norm regularization requires the data acquisition consistency correction step to be performed at the acquired k -space locations before the start of the next iteration, as follows [10]:

$$\hat{K}^{f(k+1)} = F_u \left(\Psi' \left(\omega^{(k)} \right) \right) + K^z - F_u \left(\Psi' \left(\omega^{(k)} \right) \right). \quad (7.6)$$

Algorithm 7.1 summarizes the steps involved in L1-SPIRiT implementation.

Algorithm 7.1 L1-SPIRiT

Iterate for $k = 1, \dots, \text{MaxIter}$:

- (i) *Compute $K^{f(k)}$ using calibration consistency projection in (7.3)*
- (ii) *Compute the data consistency error as in (7.4)*
- (iii) *Obtain the consistency corrected k -space as $K^{f(k)} + E_{\text{res}}^{(k)}$*
- (iv) *Soft threshold the transform domain as in (7.5)*
- (v) *Compute the consistency corrected k -space as in (7.6)*
- (vi) *Compute the SoS image $U^{(k)} = \sqrt{\sum_{c=1}^{n_c} \left(U_c^{(k)} \right)^2}$*
- (vii) *Repeat steps (i)–(vi) until $\frac{\|U^{(k)} - U^{(k-1)}\|_2}{\|U^{(k-1)}\|_2} \leq \text{tolerance}$.*

end

The noise performance of SPIRiT-based reconstruction is found to be better compared to GRAPPA. Further improvements in noise performance can be achieved using optimization of the kernel chosen for calibration and reconstruction. One typical example is the approach scheme similar to that in [23] for achieving optimal kernel calibration. Another approach to optimize the kernel for noise is to include noise covariance matrices in the SPIRiT reconstruction, as provided in [1].

7.2.1 Reconstruction Steps for Non-Cartesian SPIRiT

The steps for non-Cartesian k -space-based reconstruction are given below:

1. Model an interpolation kernel based on a specific maximum aliasing amplitude, image weighting, kernel width, and grid over-sampling [6].
2. Compute the calibration filter coefficients by performing a calibration step that supports the over-sampled grid size.
3. Reconstruct the Cartesian k -space data by performing a conjugate gradient (CG) algorithm and obtain the corresponding images by taking the inverse fast Fourier transform (IFFT).
4. Crop to the desired grid size.
5. Multiply the images with the IFFT of the interpolation kernel to obtain the apodized images.

For solving the SPIRiT optimization, two types of iterative methods are generally followed: projection over convex sets (POCS) and CG descent. POCS finds a solution by keeping the acquired k -space data untouched, whereas CG finds a numerical solution of the optimization problem since the gradient of the objective function is known analytically. The CG-based implementation includes calculation of the gradient of the objective function with data consistency and calibration consistency terms. The estimate of k -space in the current iteration on the over-sampled grid is split into two parallel paths. In the first path, the gradient of the data consistency term is calculated by interpolation of the current estimate onto a non-Cartesian grid, and then the error with the acquired data is computed by taking the difference and interpolating the error back onto a Cartesian grid. The second path includes the computation of the gradient of the calibration consistency by carrying out the SPIRiT convolution, and the result is convolved with the adjoint of the SPIRiT convolution operator, and scaling with the trade-off parameter. The CG direction is found using the computed gradient and subtracting from the previous estimate of k -space. Once the iterations converge, an IFFT is performed for each coil, and the images are cropped according to the FOV and apodized by the inverse Fourier transform of the k -space convolution interpolation kernel.

The data consistency operator performs a convolution interpolation from an over-sampled Cartesian grid onto a non-Cartesian grid. The corresponding adjoint operator performs a convolution interpolation from a non-Cartesian grid to an over-sampled Cartesian grid. The calibration consistency operator carries out the k -space convolutions with the SPIRiT kernels computed using SPIRiT calibration. The adjoint operator also performs similar convolutions with reordered, flipped (in both the horizontal and vertical directions), and conjugated SPIRiT kernels. The main advantage of using a k -space implementation is that it does not require a Fourier transform during the iterations. However, it has some associated disadvantages. For satisfying the bandpass ripple requirement, the interpolation kernel should be inevitably larger than what is commonly used for gridding for achieving the same interpolation error as well as grid over-sampling. The calibration consistency projection performed using convolution also needs to be performed on an over-sampled grid size, with considerably large calibration kernel sizes. The aforementioned considerations increase the computational complexity in each iteration of the reconstruction phase. On the other hand, in image domain-based implementations, large calibration kernels do not increase the computational complexity since the convolution is performed as multiplications in the image space. The image domain-based reconstruction algorithm is similar

to the k -space implementation, except for some factors. The data consistency and its adjoint operators perform the inverse-gridding and gridding algorithms, whereas the calibration consistency and its adjoint operators perform image multiplications instead of k -space convolutions. Another difference is that after the CG iterations converge, the result is the reconstructed images and no further processing is required.

Even though k -space-based and image domain approaches can be applied to SPIRiT reconstruction, the image domain approach is preferred as the gridding algorithm is unchanged, and the SPIRiT kernel size can be chosen to be the same as that of Cartesian SPIRiT, regardless of the gridding parameters. However, in the k -space approach, the gridding brings in a passband ripple parameter in the kernel design, and the trade-offs are less obvious because the choice of grid over-sampling affects the SPIRiT kernel size [6]. The image-based regularization is more efficient compared to the k -space-based regularization.

Since the SPIRiT reconstruction is an iterative approach, the computational complexity increases compared to direct reconstruction, typically in the case of uniform under-sampling. In the implementation of Cartesian POCS algorithm, an operation similar to a single GRAPPA reconstruction is present in each iteration. On the other hand, the CG Cartesian algorithm requires two such operations per iteration. However, the latter approach is more flexible and more stable, requiring almost half the number of iterations compared to the POCS algorithm. Therefore, the CG approach is preferred over POCS despite the conceptual and implementation-wise simplicity of POCS. For implementation of the non-Cartesian algorithm, two GRAPPA-type operations along with two gridding operations (for each coil) are used per iteration. However, the advantage of SPIRiT is that it can be applied to any arbitrary sampling patterns. In standard GRAPPA-based reconstruction, one needs to perform calibration for numerous sampling patterns. This becomes computationally cumbersome as there may be numerous individual patterns for random under-sampling schemes. SPIRiT reconstruction can be performed with negligible reconstruction times compared to standard GRAPPA.

As in the case of CG SENSE, SPIRiT can be applied to data sampled with arbitrary k -space trajectories, and the reconstructed output is always Cartesian, without requiring an additional gridding step. This flexibility offered by the SPIRiT framework is considered one of its main advantages. Since SPIRiT kernel convolution is performed over a small area of k -space, this has the effect of locally averaging the data and removing noise after several iterations. However, it may be costly to implement using the k -space domain convolution for reconstruction if the kernel sizes are large. In the image domain reconstruction, multiplication by the IFFT of the SPIRiT kernel is fast even for large kernel sizes, but there is the additional burden of calculating the non-uniform FFT (NUFFT) for each iteration of the non-Cartesian reconstruction. However, the use of multiple graphics processing unit (GPUs) and parallel computing have been found to yield clinically feasible runtimes [11].

7.3 Computational Complexity in L1-SPIRiT

With n_C channels and v number of voxels per channel, the computational complexity of the L1-SPIRiT algorithm is:

$$O\left(C_C n_C^3 + T(C_{SD} + C_F) n_C v \log v + C_S n_C^2 v\right) \quad (7.7)$$

where T denote the number of iterations, and C_C , C_{SD} , C_F and C_S are constants that indicate the relative computational cost of the SPIRiT calibration, sparse domain transforms, Fourier transforms, and SPIRiT interpolation and are heavily dependent on input data size.

Finding the solution of the calibration by performing a least-norm least-squares fit of the SPIRiT model leads to $O(n_C^4)$ and is computationally very challenging as the number of coils increases. By reusing a single Cholesky factorization for all channels, the complexity can be reduced to $O(n_C^3)$ [16]. The $n_C v \log v$ term represents the Fourier and sparse domain transforms, and the $n_C^2 v$ term represents the image-domain implementation of the k -space SPIRiT interpolation. The linearity in the latter term is because the k -space convolution is implemented as multiplication in the image domain.

Since clinical use often expects the MR images to be available immediately after the scanning process, faster implementation of reconstruction algorithms becomes inevitable for clinical applications. Accelerated reconstructions using MRI have recently been performed by implementing and optimizing them for massively parallel processors. Since nested data parallelism is present in the MR image reconstruction operations, Fourier and wavelet transforms are performed over k -dimensional (k -D) slices through the N -dimensional reconstruction volume, with $k < N$. These operations are performed for all k -D slices, resulting in another source of parallelism that can be exploited for accelerating the reconstruction. Such k -D operations are parallelizable, but they involve substantial synchronization and data sharing.

Several functionally equivalent implementations can be identified by parallelizing at different levels of the hierarchy. All these implementations will produce identical results, but with different performance characteristics. Also, the performances may vary with implementations; that is, the performance for a given implementation is dependent on the size of the images and coil arrays. Generally, the optimal implementation is a trade-off between effective use of the memory hierarchy and amortization of parallelization overheads.

7.4 Faster Non-Cartesian SPIRiT Using Augmented Lagrangian with Variable Splitting

Although the k -space implementation of SPIRiT eliminates the need for iteratively computing the Fourier transforms, the image domain formulation is preferred for larger images because the SPIRiT convolutions become multiplications in the image domain. It has been shown in the image domain implementation of the SPIRiT kernels [24] that the number of iterations required to achieve a desired reconstruction quality can be reduced considerably using variable splitting, alternating minimization, and preconditioning [25,26].

Consequently, all faster implementation methods are based on the image domain reconstruction that attempts to estimate the coil stacked image $\mathbf{U} \in \mathbb{R}^{MN \times n_C}$ obtained by arranging the vectorized channel images into a matrix with n_C columns. Using the notation for a vectorized form of coil images as $\mathbf{u} = \text{col}(\mathbf{U}) \in \mathbb{R}^{n_C MN \times 1}$, the SPIRiT self-consistency in the image domain can be expressed as $\mathbf{u} = \mathcal{Z}\mathbf{u}$. Here, the operator $\mathcal{Z}$ consists of $n_C \times n_C$ blocks of diagonal matrices, each of size $MN \times MN$. The image domain formulation of L1-SPIRiT involves minimizing the cost function:

$$\min_{\mathbf{U}} \frac{1}{2n_c \text{MN}} \|\mathbf{A}\mathbf{U} - \mathbf{D}\|_F^2 + \frac{\lambda}{2} \|(\mathcal{Z} - \mathbf{I})\mathbf{u}\|_2^2 + R(\mathbf{U}), \quad (7.8)$$

where $\mathbf{D} \in \mathbb{C}^{Q \times nc}$ denotes the Q number of known k -space samples of each coil arranged into the form of a matrix, $\mathbf{A} \in \mathbb{C}^{\text{MN} \times \text{MN}}$ denotes the discrete-space Fourier transform (DSFT) operator that maps a uniform image grid onto a non-uniform k -space grid, and R denotes the multi-channel regularizer of the following form:

$$R(\mathbf{U}) = \gamma \Phi(\mathbf{R}_1 \mathbf{U}, \mathbf{R}_2 \mathbf{U} \cdots \mathbf{R}_L \mathbf{U}), \quad (7.9)$$

where $\mathbf{R}_1, \mathbf{R}_2 \cdots \mathbf{R}_L$ are linear transform operators with circulant Gram matrices, $\Phi(\cdot): \mathbb{C}^{ncL} \rightarrow \mathbb{R}_{\geq 0}$ denotes a convex function and γ controls the extent of regularization. For example, in the case of a total variation (TV) penalty, the operators $\mathbf{R}_1, \mathbf{R}_2 \cdots \mathbf{R}_L$ correspond to the finite differences with L neighboring locations. In particular, for the anisotropic and isotropic versions of the TV functional, $\Phi(\cdot)$ corresponds to the $l_{1,2}$ and $l_{2,2}$ mixed norms, respectively.

7.4.1 Regularized Non-Cartesian SPIRiT Using Split Bregman Technique

Implementation of the minimization problem in Equation (7.8) using the split Bregman technique is achieved using an auxiliary variable $\mathbf{W} = [\mathbf{R}_1 \mathbf{U}, \mathbf{R}_2 \mathbf{U} \cdots \mathbf{R}_L \mathbf{U}]$, a dual variable $\mathbf{B}_W \in \mathbb{C}^{1 \times ncL}$ and an extra tuning parameter μ . Thus, Equation (7.8) is transformed into the constrained minimization problem:

$$\begin{aligned} \min_{\mathbf{U}, \mathbf{W}} \frac{1}{2n_c \text{MN}} \|\mathbf{A}\mathbf{U} - \mathbf{D}\|_F^2 + \frac{\lambda}{2} \|(\mathcal{Z} - \mathbf{I})\text{col}(\mathbf{U})\|_2^2 + \gamma \Phi(\mathbf{W}), \\ \text{s.t. } \mathbf{W} = [\mathbf{R}_1 \mathbf{U}, \mathbf{R}_2 \mathbf{U} \cdots \mathbf{R}_L \mathbf{U}]. \end{aligned} \quad (7.10)$$

The algorithmic steps include solving the sub-problems:

$$\begin{aligned} \mathbf{U}^{(k)} \leftarrow \min_{\mathbf{U}} \frac{1}{2n_c \text{MN}} \|\mathbf{A}\mathbf{U} - \mathbf{D}\|_F^2 + \frac{\lambda}{2} \|(\mathcal{Z} - \mathbf{I})\text{col}(\mathbf{U})\|_2^2 \\ + \frac{\mu}{2} \|\mathbf{W}^{(k-1)} - [\mathbf{R}_1 \mathbf{U}, \mathbf{R}_2 \mathbf{U} \cdots \mathbf{R}_L \mathbf{U}] - \mathbf{B}_W^{(k-1)}\|_F^2. \end{aligned} \quad (7.11)$$

$$\begin{aligned} \mathbf{W}^{(k)} \leftarrow \min_{\mathbf{W}} \gamma \Phi(\mathbf{W}) \\ + \frac{\mu}{2} \left\| \mathbf{W} - [\mathbf{R}_1 \mathbf{U}^{(k)}, \mathbf{R}_2 \mathbf{U}^{(k)} \cdots \mathbf{R}_L \mathbf{U}^{(k)}] - \mathbf{B}_W^{(k-1)} \right\|_F^2. \end{aligned} \quad (7.12)$$

$$\mathbf{B}_W^{(k)} \leftarrow \mathbf{B}_W^{(k-1)} + \left([\mathbf{R}_1 \mathbf{U}^{(k)}, \mathbf{R}_2 \mathbf{U}^{(k)} \cdots \mathbf{R}_L \mathbf{U}^{(k)}] - \mathbf{W}^{(k)} \right). \quad (7.13)$$

Since the cost function in Equation (7.10) is strictly convex, adjusting the value of μ only affects the convergence rate and not the steady-state error. When the potential function $\Phi(\cdot)$ corresponds to the $l_{1,2}$ mixed norm or the TV norm, the solution to the sub-problem in Equation (7.12) can be directly obtained using iterated soft-thresholding algorithm (ISTA). For other forms of penalizing functions, the proximal gradient algorithm or the method

of non-linear CG can be applied to obtain an approximate solution. Since the operators A and $\mathcal{Z}$ have different form of structures that are not suitable for preconditioning, it is hard to solve the Least Squares (LS) sub-problem in Equation (7.11) using the standard iterative methods such as the CG algorithm due to its slow convergence. To address this difficulty, the SPIRiT objective function is recast into a form suitable for applying the alternating direction method of multipliers (ADMM) method.

7.4.2 Iterative Non-Cartesian SPIRiT Using ADMM

The main aim of applying ADMM is to simplify the sub-problems for solving Equation (7.10) to speed up the convergence. The modification of the augmented Lagrangian problem in the framework of ADMM requires introduction of a new auxiliary variable $\mathbf{u}$ along with the dual variable $\mathbf{b}_u$. This modifies the cost function in Equation (7.10) to the following constrained form:

$$\begin{aligned} \min_{\mathbf{u}, \mathbf{W}} \quad & \frac{1}{2n_c \text{MN}} \|A\mathbf{U} - \mathbf{D}\|_F^2 + \frac{\lambda}{2} \|(\mathcal{Z} - I)\mathbf{u}\|_2^2 + \gamma \Phi(\mathbf{W}), \\ \text{s.t.} \quad & \mathbf{u} = \text{col}(\mathbf{U}), \mathbf{W} = [\mathbf{R}_1 \mathbf{U}, \mathbf{R}_2 \mathbf{U} \dots \mathbf{R}_L \mathbf{U}]. \end{aligned} \quad (7.14)$$

Solved iteratively using the alternating minimization steps, we have:

$$\begin{aligned} \mathbf{U}^{(k)} \leftarrow \min_{\mathbf{u}} \quad & \frac{1}{2n_c \text{MN}} \|A\mathbf{U} - \mathbf{D}\|_F^2 + \frac{\mu_u}{2} \|\text{col}(\mathbf{U}) - \mathbf{u}^{(k-1)} + \mathbf{b}_u^{(k-1)}\|_2^2 \\ & + \frac{\mu_W}{2} \|\mathbf{W}^{(k-1)} - [\mathbf{R}_1 \mathbf{U}, \mathbf{R}_2 \mathbf{U} \dots \mathbf{R}_L \mathbf{U}] - \mathbf{B}_W^{(k-1)}\|_F^2. \end{aligned} \quad (7.15)$$

$$\mathbf{u}^{(k)} \leftarrow \min_{\mathbf{u}} \quad \frac{\lambda}{2} \|(\mathcal{Z} - I)\mathbf{u}\|_2^2 + \frac{\mu_u}{2} \|\text{col}(\mathbf{U}^{(k)}) - \mathbf{u} + \mathbf{b}_u^{(k-1)}\|_2^2. \quad (7.16)$$

$$\mathbf{b}_u^{(k)} \leftarrow \mathbf{b}_u^{(k-1)} + (\text{col}(\mathbf{U}^{(k)}) - \mathbf{u}^{(k)}). \quad (7.17)$$

$$\begin{aligned} \mathbf{W}^{(k)} \leftarrow \min_{\mathbf{W}} \quad & \gamma \Phi(\mathbf{W}) \\ & + \frac{\mu_W}{2} \|\mathbf{W} - [\mathbf{R}_1 \mathbf{U}^{(k)}, \mathbf{R}_2 \mathbf{U}^{(k)} \dots \mathbf{R}_L \mathbf{U}^{(k)}] - \mathbf{B}_W^{(k-1)}\|_F^2. \end{aligned} \quad (7.18)$$

$$\mathbf{B}_W^{(k)} \leftarrow \mathbf{B}_W^{(k-1)} + ([\mathbf{R}_1 \mathbf{U}^{(k)}, \mathbf{R}_2 \mathbf{U}^{(k)} \dots \mathbf{R}_L \mathbf{U}^{(k)}] - \mathbf{W}^{(k)}). \quad (7.19)$$

With initial estimates of the coil images $\mathbf{U}^{(0)}$, the auxiliary variables are initially set to $\mathbf{W}^{(0)} \leftarrow [\mathbf{R}_1 \mathbf{U}^{(0)}, \mathbf{R}_2 \mathbf{U}^{(0)} \dots \mathbf{R}_L \mathbf{U}^{(0)}]$ and $\mathbf{u}^{(0)} = \text{col}(\mathbf{U}^{(0)})$. Apart from separating the solvers involving the matrices A and $\mathcal{Z}$, computational efficiency is better due to the parallelism in the sub-problems. Due to the decoupling of sub-problems involving A and $\mathcal{Z}$, Equations (7.15) and (7.16) can be easily solved using preconditioned conjugate gradients.

In the absence of preconditioning, the convergence of LS sub-problems in Equations (7.15) and (7.16) becomes slower. Pre-conditioning exploits the matrix structure

to make the inversion more efficient. Solving Equation (7.15), for example, involves inverting a positive definite Gram matrix $(1/MN)A'A + \mu_W \sum_{l=1}^L R_l'R_l + \mu_u I$. Because the Gram matrices for sparsifying operators representing TV functional and shift invariant wavelets are circulant, an effective pre-conditioner having a circulant structure can be identified for efficiently carrying out the LS inversion in this case. In particular, the multi-level block-Toeplitz structure of the Gram matrix for the DSFT operator $T \triangleq A'A$ endows the pre-conditioner also to have a similar structure to improve the efficiency of inversion. Thus, a multi-level block-circulant pre-conditioner C is derived by minimizing $\|C - T_F\|$. For arbitrary $N \times N$ circulant matrices with elements c_n and t_n , minimizing the Frobenius norm has been shown to result in the following relationship between the two element pairs [27]:

$$c_n = \frac{n}{N} t_{-(N-n)} + \frac{N-n}{N} t_n \quad (7.20)$$

Starting from the lowest level blocks, the pre-conditioner blocks are recursively computed using the following relationship:

$$C_n = \frac{n}{N} c(T_{-(N-n)}) + \frac{N-n}{N} c(T_n) \quad (7.21)$$

If the original block-Toeplitz matrix is positive definite, the resulting pre-conditioner will also be positive definite. With addition of $\mu_u I$, a suitable pre-conditioner with circulant structure to solve Equation (7.15) would therefore be $c((1/MN)A'A + \mu_u I) + \mu_W \sum_{l=1}^L R_l'R_l$. Further, the Gram matrix $(Z - I)'(Z - I)$ in the sub-problem in Equation (7.16) has a diagonal block structure that can be transformed by reordering the elements to a block diagonal structure having block size $n_C \times n_C$ via permutations. Therefore, inversion of $\Gamma = \lambda(Z - I)'(Z - I) + \mu_u I$ required to solve the linear sub-problem in Equation (7.16) simply involves inversion of each $n_C \times n_C$ diagonal block.

The choice of values for μ_W and μ_u determines the convergence characteristics of the ADMM approach. While the choice of larger values causes the objective function to decrease slowly, setting the tuning parameter to lower values affects the convergence speed of the dual variable. The convergence behavior of each sub-problem also depends on the conditioning of their respective system matrices. For example, the system matrix for the LS update of $\mathbf{U}$ becomes poorly conditioned for large values of μ_W and μ_u . In contrast, the system matrix for the LS update of $\mathbf{u}$ tends to become poorly conditioned by reducing μ_u . Therefore, a minimum value for μ_u can be obtained by setting an upper limit on the condition number κ_u of the system matrix used for the LS update of $\mathbf{u}$. Similarly, setting an upper limit on the condition number κ_U of the system matrix used for the LS update of $\mathbf{U}$, the values of μ_W and μ_u can be tuned with the goal of providing the required conditioning. In summary, the ADMM parameter choice requires an initial setting of the maximum condition numbers κ_U and κ_u , and the thresholding fraction τ_W to set the penalty parameter μ_W that thresholds no more than τ_W fraction of the total number of sparse transform coefficients. The parameter selection procedure includes the following steps:

1. Compute the largest ($\sigma_{\max}$) and smallest ($\sigma_{\min}$) singular values of the $n_C \times n_C$ blocks of the Gram matrix $(Z - I)'(Z - I)$.
2. Find the smallest μ_u such that $(\lambda\sigma_{\max} + \mu_u) / (\lambda\sigma_{\min} + \mu_u) \leq \kappa_u$.

3. Compute the largest and smallest singular values of the circulant pre-conditioners for the circulant Gram matrices $A'A$ and $\sum_{l=1}^L R_l'R_l$.
4. Adjust μ_W and μ_u so as to satisfy the following inequality:

$$\left(\frac{\sigma_{\max,A}}{MN} + \mu_u + \mu_W \sigma_{\max,R} \right) / \left(\frac{\sigma_{\min,A}}{MN} + \mu_u + \mu_W \sigma_{\min,R} \right) < \kappa_U.$$

7.4.3 Fast Iterative Cartesian SPIRiT Using Variable Splitting

Although ADMM makes the convergence faster and can achieve high spatial resolution in the Cartesian SPIRiT problem, its efficiency in terms of complexity and convergence speed decreases with an increase in the number of channels [28]. In this case, a more efficient algorithm is obtained when the cost function of the Cartesian SPIRiT problem consisting of a data fidelity term, a self-calibration term and regularization terms is transformed into one with only the data fidelity and the regularization terms. This is then solved using the split Bregman formulation in which the operator splitting (OS) technique is employed to split the minimization problem into a gradient descent minimization and a denoising problem with two regularization terms. Although the steady-state error is almost similar to that obtained with ADMM, the convergence speed is higher for reconstructions involving more channels.

Considering $\mathcal{G}$ as an interpolation operator in k -space that convolves the entire k -space with the SPIRiT kernel and K as the concatenated k -spaces of all channels, the k -space-based self-consistency term is expressed as $K = \mathcal{G}K$. Likewise, the data consistency is expressed in terms of a linear operator D that serves to select only the acquired k -space data $\mathbf{k}_u \in \mathbb{C}^{Q_{nc} \times 1}$ as $\mathbf{k}_u = DK$. With the self- and data consistency terms forming the constraints, the SPIRiT reconstruction problem can be recast into the following form:

$$\begin{aligned} \min_K & \|\Psi F_u' K\|_{2,1}, \\ \text{s.t. } & K = \mathcal{G}K, \mathbf{k}_u = DK. \end{aligned} \quad (7.22)$$

where Ψ is coil-by-coil the sparsifying transform operator and F_u' is the adjoint of the coil-by-coil under-sampled Fourier operator. Because pMRI data are joint sparse in both wavelet and finite differences domains, it is necessary to apply the regularization term using joint l_1 -norm regularization and the joint total variation (JTV) regularization term involving all coil images together. In the unconstrained form, this is equivalent to:

$$\min_K \frac{\lambda_1}{2} \|DK - \mathbf{k}_u\|_2^2 + \frac{\lambda_2}{2} \|(\mathcal{G} - I)K\|_2^2 + \gamma_1 \|\Psi F_u' K\|_{2,1} + \gamma_2 \|\Psi F_u' K\|_{JTV} \quad (7.23)$$

The trick is to combine the first terms of the cost by introduction of some new operators and variables. These are the vectors representing the unacquired k -space, denoted as $\tilde{\mathbf{k}}_u$, and the operators D, D', D_c and D_c' , where the primes represent the adjoint operation. The suffix C denotes the complement operation that corresponds to selection of the unacquired points. Also the adjoint operators D' and D_c' represent operations that reassign the acquired and unacquired samples to their appropriate locations in k -space. Using these new operators and variables, the SPIRiT cost function can be recast into the following form:

$$\min_{\tilde{\mathbf{k}}_u} \frac{1}{2} \|(\mathcal{G} - I)D_C' \tilde{\mathbf{k}}_u - \mathbf{b}\|_2^2 + \gamma_1 \|\Psi F_u' (D' \mathbf{k} + D_C' \tilde{\mathbf{k}}_u)\|_{2,1} + \gamma_{21} \|F_u' (D' \mathbf{k} + D_C' \tilde{\mathbf{k}}_u)\|_{JTV} \quad (7.24)$$

where $\mathbf{b} = -(\mathcal{G} - I)D' \mathbf{k}$. Letting $p = D' \mathbf{k} + D_C' \tilde{\mathbf{k}}_u$ and $q = D' \mathbf{k} + D_C' \mathbf{k}_u$, the cost function becomes:

$$\min_p \frac{1}{2} \|p - q\|_2^2 + \frac{\gamma_1}{L} \|\Psi F_u' p\|_{2,1} + \frac{\gamma_2}{L} \|F_u' p\|_{JTV}, \quad (7.25)$$

where L denotes the Lipschitz constant for the gradient term $\frac{1}{2} \|(\mathcal{G} - I)D_C' \tilde{\mathbf{k}}_u - \mathbf{b}\|_2^2$. Further transforming the cost function using image domain variables $u = F_u' p$ and $v = F_u' q$ representing the vectorized coil images stacked column-wise, the SPIRiT reconstruction becomes equivalent to a denoising problem given by:

$$\min_u \frac{1}{2} \|u - v\|_2^2 + \alpha_1 \|\Psi u\|_{2,1} + \alpha_2 \|u\|_{JTV}, \quad (7.26)$$

where $\alpha_1 = \gamma_1 / L$, $\alpha_2 = \gamma_2 / L$, and $v \in \mathbb{C}^{MN \times n_C}$ denotes the noisy multi-coil images. Using the earlier operator notation for sparsifying transforms and the operator for joint sparsity defined as $\|u\|_{JTV} = \|\mathbf{R}_{hv} u\|_1 = \sum_{r=1}^{MN} \sqrt{\sum_{c=1}^{n_C} [(\mathbf{R}_h u)_{rc}]^2 + [(\mathbf{R}_v u)_{rc}]^2}$, the reconstruction problem becomes:

$$\min_u \frac{1}{2} \|u - v\|_2^2 + \alpha_1 \|\Psi u\|_{2,1} + \alpha_2 \|\mathbf{R}_{hv} u\|_1 \quad (7.27)$$

With the introduction of auxiliary variables $d = \mathbf{R}_{hv} u$, $z = \Psi u$ and corresponding dual variables b_d, b_z , the solution can be obtained using the following steps of split Bregman iterations:

$$d^{k+1} = \min_d \frac{\beta_2}{2} \|d - \mathbf{R}_{hv} u^k - b_d^k\|_2^2 + \|d\|_1, \quad (7.28)$$

$$z^{k+1} = \min_z \frac{\beta_1}{2} \|z - \Psi u^k - b_z^k\|_2^2 + \|z\|_1, \quad (7.29)$$

$$u^{k+1} = \min_u \frac{1}{2} \|u - v\|_2^2 + \frac{\alpha_1}{\beta_1} \|z^k - \Psi u - b_z^k\|_2^2 + \frac{\alpha_2}{\beta_2} \|d^k - \mathbf{R}_{hv} u - b_d^k\|_2^2 \quad (7.30)$$

$$b_d^{k+1} = b_d^k + (\mathbf{R}_{hv} u^{k+1} - d^{k+1}), \quad (7.31)$$

$$b_z^{k+1} = b_z^k + (\Psi u^{k+1} - z^{k+1}). \quad (7.32)$$

The sub-problem in Equation (7.28) is solved using the two-dimensional shrinkage operation, defined as:

$$\text{Shrink2D}(d, T) = \max(|d| - T, 0) \cdot \frac{d}{\text{repmat}\left(\sqrt{\sum_c \left[|(d_h)_c|^2 + |(d_v)_c|^2\right]}, [1, n_c]\right)}, \quad (7.33)$$

where $\text{repmat}(\cdot)$ denotes the MATLAB function for replication of arrays. Using Equation (7.33), the solution to Equation (7.28) can be expressed as:

$$d^{k+1} = \text{Shrink2D}\left(\mathbf{R}_{hv} u^k + b_d^k, \frac{1}{\beta_2}\right). \quad (7.34)$$

Similarly, the solution to Equation (7.29) is determined using the joint one-dimensional shrinkage operator defined as:

$$\text{ShrinkJ}(z, T) = \max(|z| - T, 0) \cdot \frac{z}{\text{repmat}\left(\sqrt{\sum_c |z_c|^2}, [1, n_c]\right)}. \quad (7.35)$$

As for Equation (7.30), it is easier to solve the coil variables separately as they are independent for different coils. Recasting Equation (7.30) in terms of the individual coil image u_c leads to the following closed-form solution:

$$u_c^{k+1} = \left(I + \alpha_1 \beta_1 \Psi^T \Psi + \alpha_2 \beta_2 \mathbf{R}_{hv}^T \mathbf{R}_{hv}\right)^{-1} \cdot \left(v_c + \alpha_1 \beta_1 \Psi^T \left(z_c^k - b_{zc}^k\right) + \alpha_2 \beta_2 \mathbf{R}_{hv}^T \left(d_c^k - b_{dc}^k\right)\right). \quad (7.36)$$

where $\Psi^T \Psi = I$ and $\mathbf{R}_{hv}^T \mathbf{R}_{hv} = \mathbf{R}_h^T \mathbf{R}_h + \mathbf{R}_v^T \mathbf{R}_v$. The operator in Equation (7.36) can be easily realized by considering the fact that $\mathbf{R}_{hv}^T \mathbf{R}_{hv}$ becomes block circulant under periodic boundary conditions, and can be diagonalized by application of the two-dimensional discrete Fourier transform operator. This leads to the closed-form solution:

$$u_c^{k+1} = F' \left(\frac{F \left(v_c + \alpha_1 \beta_1 \Psi^T \left(z_c^k - b_{zc}^k \right) + \alpha_2 \beta_2 \mathbf{R}_{hv}^T \left(d_c^k - b_{dc}^k \right) \right)}{I + \alpha_1 \beta_1 I + \alpha_2 \beta_2 F \mathbf{R}_{hv}^T \mathbf{R}_{hv} F'} \right). \quad (7.37)$$

Thus, the entire procedure uses the FFT operation and component-wise operations to solve the SPIRiT problem as the equivalent of a denoising problem. This form of SPIRiT implementation is summarized in Algorithm (7.2).

Algorithm 7.2 Operator splitting algorithm for L1-SPIRiT

Iterate for $k = 1, \dots, \text{MaxIter}$:

(i) Set $u^1 = 0, d^1 = 0, z^1 = 0, b_d^1 = 0, b_z^1 = 0$

(ii) for $k = 1, \dots, \text{MaxIter}$

(iii) $d^{k+1} = \text{Shrink}2D\left(R_{hv}u^k + b_d^k, \frac{1}{\beta_2}\right)$

(iv) $z^{k+1} = \text{Shrink}J\left(\Psi u^k + b_z^k, \frac{1}{\beta_1}\right)$

(v) for $c = 1, \dots, n_C$:

(vi) Compute u_c^{k+1} using (7.36)

end for

(vii) $b_d^{k+1} = b_d^k + (R_{hv}u^{k+1} - d^{k+1})$

(viii) $b_z^{k+1} = b_z^k + (\Psi u^{k+1} - z^{k+1})$

end for

7.5 Challenges in the Implementation of L1-SPIRiT

Despite the easy incorporation of additional constraints such as sparsity enforcement of the data, the SPIRiT reconstruction model possesses some potential difficulties, for example: (1) the mechanism for selection of the appropriate parameter values for the optimization is often carried out empirically, leading to a rigorous optimization; (2) it lacks a proper stopping criterion to terminate the conjugate gradient descent, as the use of too many iterations may fit noise into the reconstructed images leading to sub-optimal images with no clinical relevance; and (3) computation is complex. However, the difficulty associated with computational complexity is not an obstacle as faster realization of SPIRiT can be achieved by the use of multiple GPUs and parallel computing, as outlined in [16]. Therefore, it is evident that parameter selection and the decision on when to terminate the algorithm are the main factors that hinder the clinical usefulness of the SPIRiT algorithm.

As the implementation of SPIRiT kernel convolutions in k -space are computationally expensive due to the iterative steps, an image domain approach is preferred over the k -space approach [6]. With both type of reconstruction approaches (POCs and CG), the image quality can be degraded if too many or too few iterations are used [29]. As a rule for practical implementations, an early termination of 10–12 iterations is preferred in the case of CG iterations. However, no such stopping rule suggestions are available for POCs-based iterations. Generally, the relative decrease in l_2 -norms of images reconstructed in two successive iterations computed as in Equation (5.41) in Chapter 5 is used as a stopping criterion. Since Equation (5.41) has no bearing on the reconstruction error (computed if the ground truth were known), we cannot bank on such a criterion to provide the optimal reconstruction.

7.5.1 Effect of Incorrect Parameter Choice on Reconstruction Error

L1-SPIRiT reconstruction of an in vivo dataset (dataset II) using a POCs-based iterative scheme is used to illustrate the influence of parameter values used for both sparsity enforcement and filter calibration on the relative l_2 -norm error (RLNE) [30–32]. The top row panel in Figure 7.1 corresponds to an optimal sparsity parameter with a varying filter parameter, whereas the bottom row panel corresponds to an optimal filter parameter with a varying sparsity parameter. Plots of the RLNE for different parameter values are shown in Figure 7.1.

The minimum possible error is obtained when the optimum parameters are chosen. Parameter values larger or smaller than the optimum value result in increased minimum RLNE. The reconstruction error exhibits a semi-convergence behavior, resulting in increased errors with increased number of iterations. Changes in parameter values can result in significant variations in the performance characteristics of the algorithm. Therefore, determining the optimum parameter is crucial for practical applications. Even with an appropriate tuning of the filter and sparsity parameters, there is no guarantee that the minimum RLNE will occur in the first 10–12 iterations. Thus, the termination criterion remains unpredictable without the knowledge of a reference or ground truth image. This clearly indicates that an ad hoc criterion may not be always suitable to arrive at the most useful solution and calls for introduction of an appropriate stopping rule by efficient manipulation of parameters.

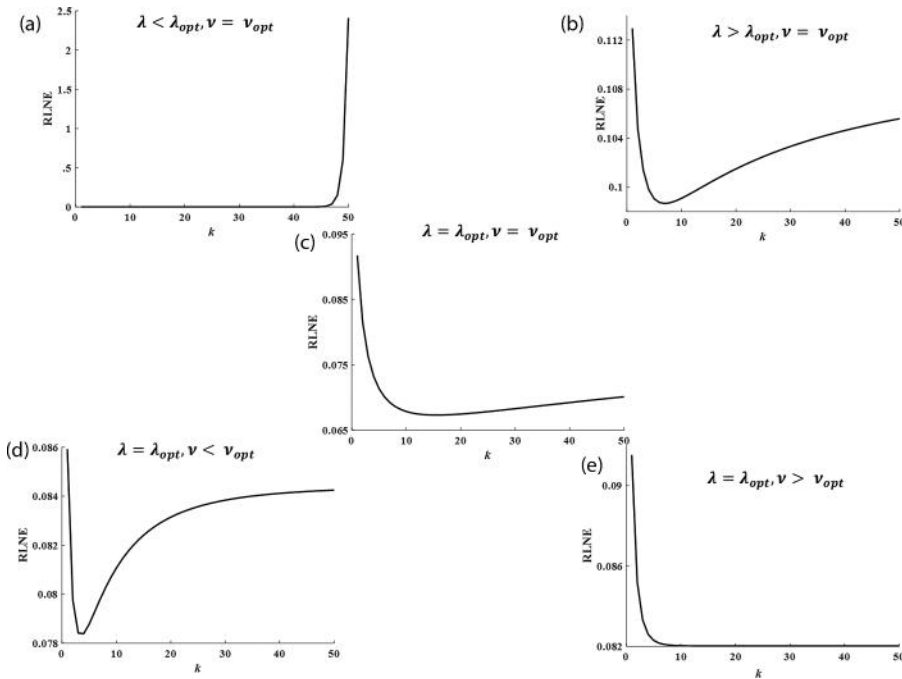


FIGURE 7.1

(a)–(e) Plots of RLNE for different parameter values as indicated in the insets.

7.6 Improved Calibration Framework for L1-SPIRiT

An improved calibration framework for SPIRiT is proposed in [33] that utilizes the non-linear transform to reduce errors occurring due to the presence of noise in the auto-calibration signal (ACS) region and discrepancy-based regularization to adaptively generate the linear weighting coefficients. Since SPIRiT is an iterative method, the noise-induced errors in the calibration coefficients brought into k -space may deteriorate the image quality.

From the calibration model of SPIRiT presented in Section 1.6.2.3 in Chapter 1, it is evident that the bias in the calibration coefficients is not a linear function of the noise present in the ACS data, and one cannot completely remove this bias even with an infinite number of ACS [34]. An attempt to eliminate such errors for GRAPPA-based reconstruction was put forward by Chang et al.; it is called non-linear GRAPPA in which a non-linear transform is introduced into the GRAPPA calibration framework [34]. Similar to their approach, a non-linear transform is applied to SPIRiT calibration through the use of a polynomial mapping to the ACS data for reducing the effect of non-linear noise-related errors [33]. Following the notations in Section 1.6.2.3, the application of polynomial mapping to the calibration modifies the computation of calibration coefficients as follows:

$$\mathbf{k}_u^l = \varphi\left(\Pi\hat{\mathbf{z}}^l\right), \quad (7.38)$$

where φ denotes the polynomial mapping, and $\hat{\mathbf{z}}$ is the updated calibration coefficients in the feature space. It is evident that the observation vector formed using Equation (7.38) is also a linear combination in this high-dimensional feature space, enabling the computation of the corresponding filter coefficients using:

$$\hat{\mathbf{z}}^l = \left[\varphi^H\left(\Pi\hat{\mathbf{z}}^l\right)\varphi^H\left(\Pi\hat{\mathbf{z}}^l\right) \right]^{-1} \varphi^H\left(\Pi\hat{\mathbf{z}}^l\right)\mathbf{k}_u^l. \quad (7.39)$$

The newly formed calibration coefficients $\hat{\mathbf{z}}^l$ have dimensions much higher than the original coefficients and prevent the direct application to SPIRiT. Therefore, a subset of coefficients that has the same dimension of $\hat{\mathbf{z}}^l$ is generated. Since the first-order coefficients yield coefficients with exactly the same dimension as that of $\hat{\mathbf{z}}^l$, such coefficients are picked out as the new set of filter coefficients for SPIRiT optimization.

7.6.1 Modification of Polynomial Mapping

Since the SPIRiT calibration kernel chooses an entire neighborhood compared to GRAPPA, the polynomial mapping is modified before applying to the non-linear calibration of SPIRiT. The polynomial mapping is modified such that all first-order terms and high-order self-terms are kept while truncating all the high-order cross-terms. For a vector $= [a_1, a_2, \dots, a_K]$, the modified polynomial mapping is obtained as:

$$\varphi(a) = [1, a_1, a_2, \dots, a_K, a_1^2, \dots, a_K^2, a_1^3, \dots, a_K^3] \quad (7.40)$$

The products of the high-order terms in the polynomial formulation are eliminated with the assumption that their corresponding coefficients denote the hidden noise. The performance of the polynomial mapping is highly dependent on the number of high-order terms. Therefore, with only few high-order terms, the relationship between the unacquired k -space and the calibration kernel cannot be described precisely in the presence of noise. On the other hand, increasing the number of high-order terms results in overfitting, with large deviations in the first-order term coefficients. Then, the coefficients become too sensitive to the noise in the calibration data. Therefore, only the first-order and second-order terms are kept in this polynomial mapping to achieve an acceptable reconstruction performance without increase in computational complexity.

7.6.2 Regularization Parameter Choice

The accuracy of the calibration coefficients as well as the final reconstruction quality depends on the choice of regularization parameter. If the SPIRiT reconstruction is performed using coil coefficients computed without incorporating regularization, then the SPIRiT iterations may not converge to a stable solution. For example, by application of polynomial mapping in non-linear calibration, a certain degree of regularization is already incorporated. Even then, there is no guarantee that the solution converges to a stable solution. Therefore, regularization is indispensable for obtaining stable reconstructions using SPIRiT. Tikhonov regularization, in its standard form, can be used in this case. The L-curve method or the discrepancy principle can be used to automatically choose the regularization parameter. However, the polynomial mapping in Equation (7.37) simultaneously keeps the magnitude of the first-order and second-order terms in different orders, and in most cases the plot no longer exhibits a sharp L-corner. This is the main reason for applying the discrepancy principle to select the optimal regularization parameter for the non-linear calibration. The parameter computed using the discrepancy principle may lead to tremendous deviations in the final results due to the presence of outliers in the intermediate coefficients.

7.7 Automatic Parameter Selection for L1-SPIRiT Using Monte Carlo SURE

In Monte Carlo Stein's unbiased risk estimation (SURE) (see [Section 3.4.2.3.1](#) in [Chapter 3](#) [35]), SURE-based iterative reconstruction originally developed for single-channel MRI is extended to the L1-SPIRiT reconstruction using a complex-valued Monte Carlo-based approach [24]. As the true values at unacquired k -space locations are unknown, a Monte Carlo-based approach is used to approximate the mean squared error (MSE) resulting in a Monte Carlo estimate [36]. The authors use POCS-based implementation with a four-level 'db4' wavelet as the sparsifying transform consisting of 20 L1-SPIRiT iterations. Since Monte Carlo SURE is a black-box approach and utilizes only the output of the reconstruction, this implementation can also be generalized to other parallel imaging reconstruction and denoising methods.

7.8 Continuation-Based Threshold Adaptation in L1-SPIRiT

Compared to the adaptation schemes in CS-MR image recovery [37], the transform domain threshold adaptation scheme extended to the SPIRiT iterative framework is intended to avoid errors introduced in the reconstructed images due to fitting of noise into the reconstructed data and thereby enable quick convergence. Another difference is that the adaptation scheme is applied to a combination of partially parallel imaging and CS. Consequently, the current application involves an additional filter reconstruction step prior to the linear filtering and soft-thresholding operations in CS recovery.

Since l_1 -norm minimization step is performed on the calibration consistency projected k -space rather than the l_1 -norm minimized previous iterate in the case of CS-based recovery, it is more appropriate to use the errors derived from the calibration consistency projection to update the threshold used in the l_1 -norm regularization. Similar to the sparse approximation error in the case of CS-based recovery [37], the filter approximation error is defined as the difference between k -spaces before and after calibration consistency projection:

$$E_n^{(k)} = \hat{K}^{f(k)} - K^{f(k)}. \quad (7.41)$$

Since a joint sparsity operation is involved in L1-SPIRiT, the channel combined sparse domain error is obtained as:

$$\varepsilon_{\text{res}}^{(k)} = \sum_c |\Psi E_{\text{res},c}^{(k)}| \text{ and } \varepsilon_n^{(k)} = \sum_c |\Psi E_{n,c}^{(k)}|, \quad (7.42)$$

where c denotes the channel. Using $v^{(0)}$, chosen as described in Section 5.4.1 in Chapter 5, Algorithm 7.3 summarizes the threshold adaptation steps in L1-SPIRiT using the sparse domain errors.

Algorithm 7.3 Adaptive L1-SPIRiT

Iterate for $k = 1, \dots, \text{MaxIter}$:

(i) Compute $K^{f(k)}$ using calibration consistency projection

(ii) Perform (7.3)–(7.6)

(iii) Compute discrepancy level $\mathfrak{d}_1^{(k)}$ as

$$\mathfrak{d}_1^{(k)} = \left| \|\varepsilon_{\text{res}}^{(k)}\|_1 - \|\varepsilon_n^{(k)}\|_1 \right|$$

(iv) Update threshold (v) using

$$v^{(k+1)} = \frac{\mathbb{E}\left(\left|\varepsilon_{\text{res}}^{(k)}\right|\right)}{\Phi\left(\mathfrak{d}_1^{(k)}\right) + \left(\mathbb{E}\left(\left|\varepsilon_n^{(k)}\right|\right) / v^{(k)}\right)},$$

where $\mathbb{E}(\cdot)$ denote the expectation of the respective errors

and $\Phi(\mathfrak{d}_1^{(k)})$ is an increasing function of $\mathfrak{d}_1^{(k)}$

(v) Compute the SoSimage $U^{(k)} = \sqrt{\sum_{c=1}^{n_c} (U_c^{(k)})^2}$

(vi) Repeat steps (i)–(v) until $\frac{\|U^{(k)} - U^{(k-1)}\|_2}{\|U^{(k-1)}\|_2} \leq \text{tolerance}$.

end

7.8.1 L1-SPIRiT Examples

This section illustrates the application of Algorithm 7.3 for pMRI reconstruction. The L-curve parameter is chosen for the filter calibration step, as this choice can yield better results compared to its alternatives. All reconstructions are performed using MATLAB on a PC with an Intel Xeon 2.4 GHz processor and 16 GB of RAM running Windows 7, and using adaptation function $\Phi(\mathfrak{d}_1^{(k)}) = \mathfrak{d}_1^{(k)}$. The sum-of-squares (SoS) images and the sampling masks used are shown in Figure 7.2.

Figure 7.3 illustrates the reconstructed images for dataset I using L1-SPIRiT and adaptive L1-SPIRiT with random sampling mask (MASK-1). The top two panels show the reconstructed and difference images obtained with L1-SPIRiT reconstruction performed with the optimum parameter values. The optimal values in standard L1-SPIRiT are chosen empirically to obtain the minimum possible RLNE. The bottom four panels show adaptive L1-SPIRiT reconstructions performed with the filter parameter chosen from the L-curve and the parameter for standard L1-SPIRiT. RLNE values shown in insets confirm that the quality of reconstruction obtained using adaptive L1-SPIRiT is superior to that of L1-SPIRiT. It is more significant that the image quality is preserved with increased number of iterations in the adaptive case.

Figure 7.4 shows reconstructions for dataset II using standard L1-SPIRiT and adaptive L1-SPIRiT, respectively, with a radial sampling scheme (MASK-2). For the radial sampling scheme, 80 spokes are used, with 256 samples along each spoke. The top two panels show the reconstructed and difference images obtained with standard L1-SPIRiT reconstruction performed with empirically chosen parameter values. The bottom four panels show images reconstructed using adaptive L1-SPIRiT, with the filter parameter chosen from the

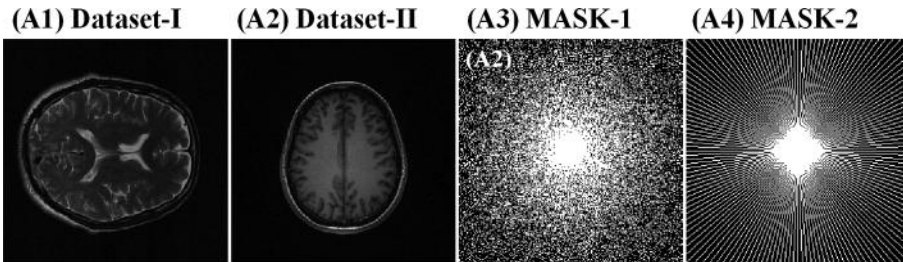
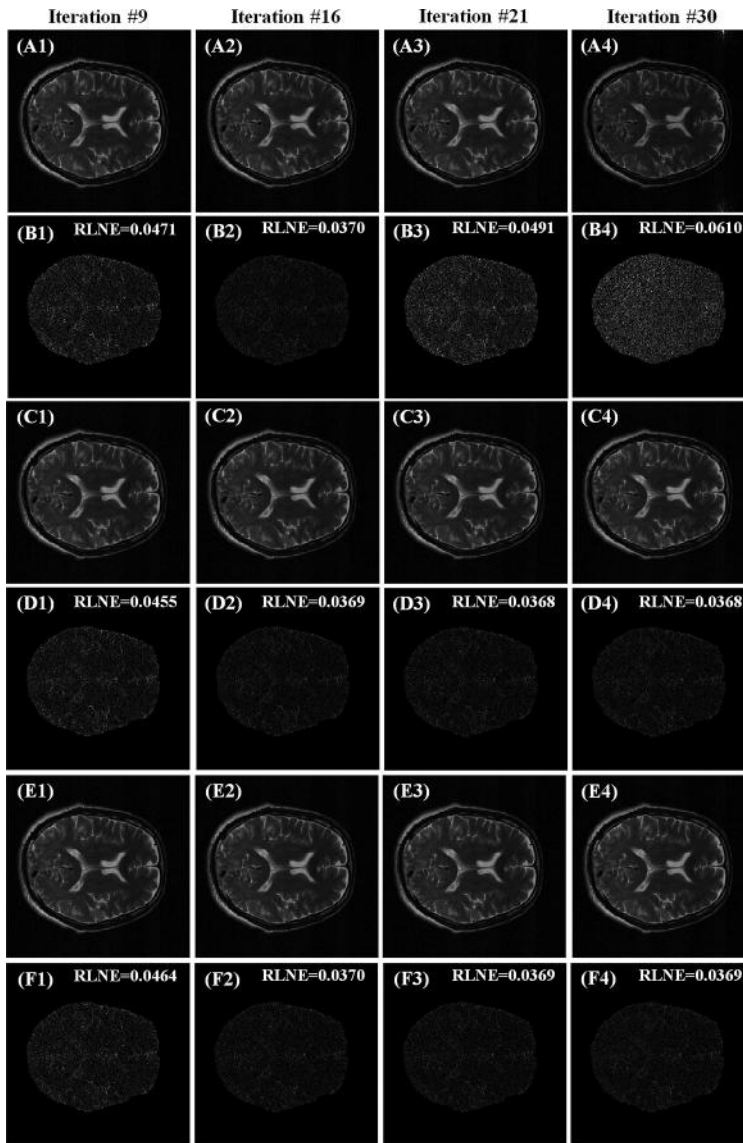


FIGURE 7.2

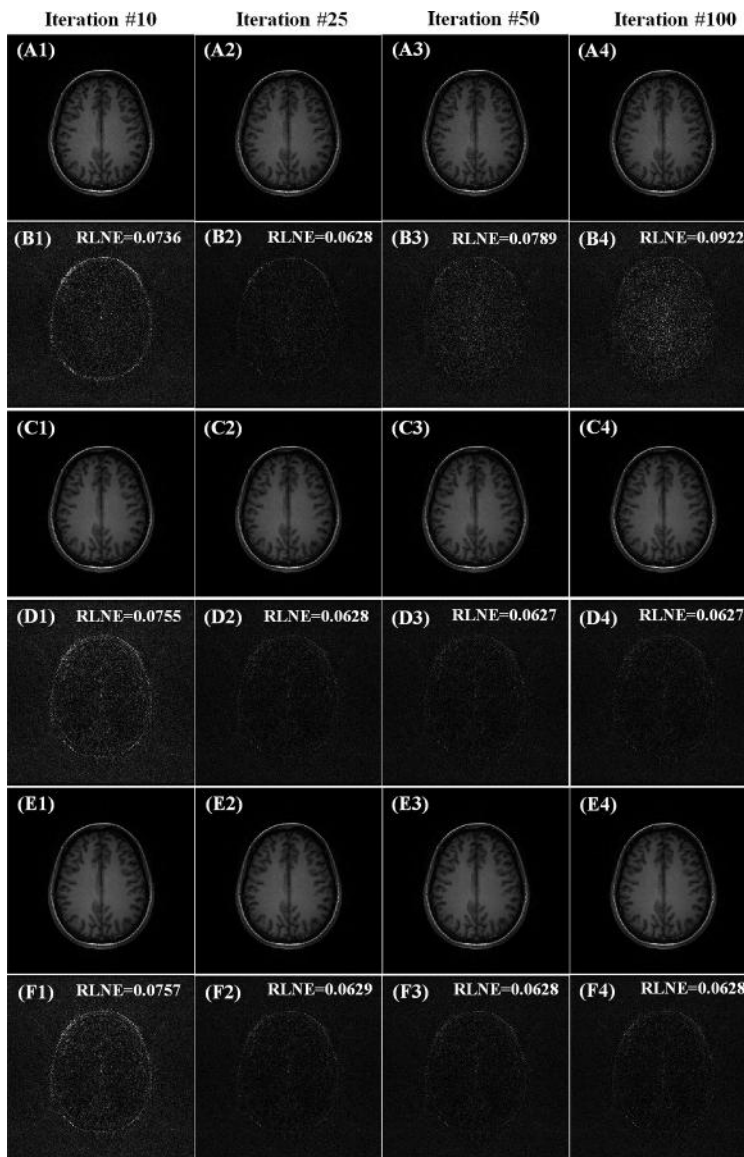
(A1), (A2) sum-of-square (SoS) images of multi-channel datasets 1 and 2, respectively; (A3) sampling mask using 30% acquired samples and 1.55% of samples fully acquired from the central k -space; (A4) radial sampling mask with 80 spokes.

**FIGURE 7.3**

Comparison of L1-SPIRiT reconstructions for dataset I using MASK1. The top two panels show L1-SPIRiT reconstructed and difference images with empirically chosen parameter values kept constant. The bottom four panels show adaptive L1-SPIRiT reconstructions with the filter parameter chosen from the L-curve and the empirically chosen parameter as in standard L1-SPIRiT.

L-curve and the empirically chosen parameter for standard L1-SPIRiT. As in the previous case, the image quality deteriorates with an increased number of iterations for the standard L1-SPIRiT. In the adaptive case, the image quality is maintained with an increased number of iterations, which is indicative of convergence of the algorithm. This is as opposed to semi-convergence in the case of standard L1-SPIRiT.

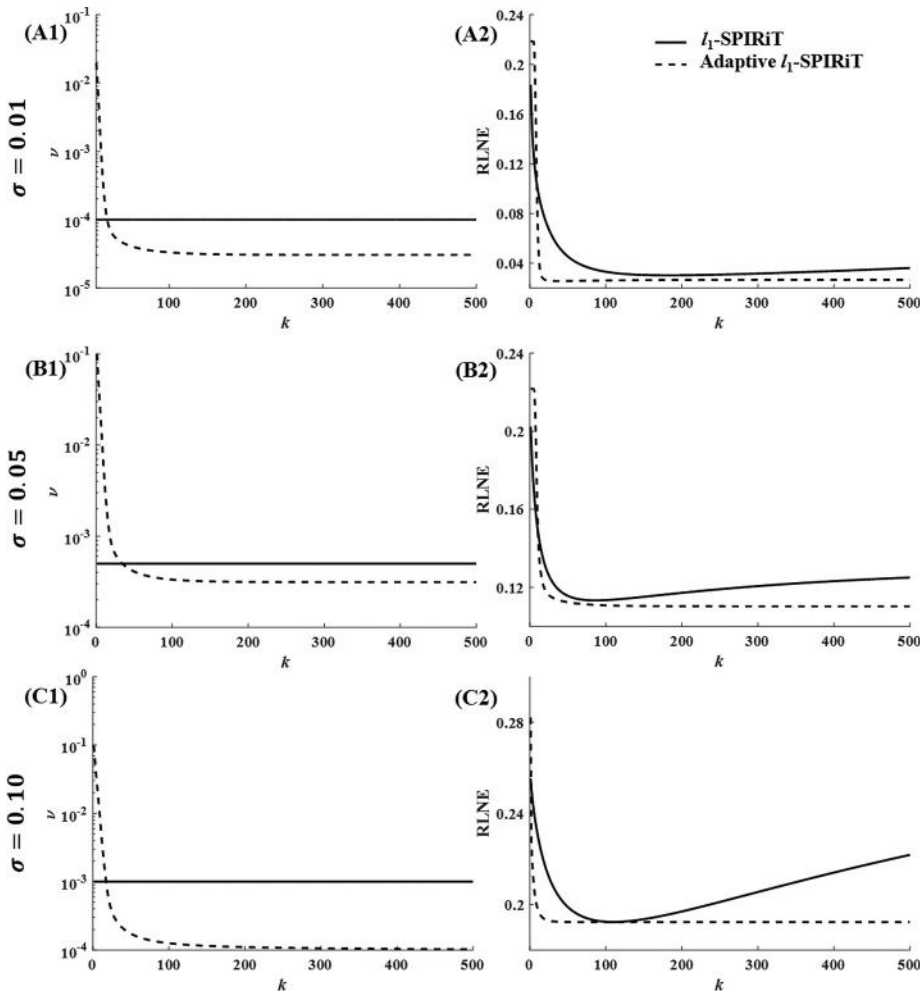
Figure 7.5 shows plots of thresholds and RLNEs as functions of the number of iterations for L1-SPIRiT and adaptive L1-SPIRiT, using synthetic data for three input noise levels with

**FIGURE 7.4**

Comparison of L1-SPIRiT reconstructions for dataset II using MASK 2. The top two panels show L1-SPIRiT reconstructed images and difference images with empirically chosen parameter value kept constant. The bottom four panels show adaptive L1-SPIRiT reconstructed images with the filter parameters chosen from the L-curve and empirically chosen as in standard L1-SPIRiT.

standard deviation $\sigma = 0.01, 0.05$, and 0.1 . The plots clearly indicate that the threshold at convergence increases with increase in input noise level. The semi-convergence behavior of the RLNE plots is more pronounced with increasing noise levels. The difference images obtained using standard L1-SPIRiT and adaptive L1-SPIRiT are shown in [Figure 7.6](#).

[Figure 7.7a](#) shows the $RLNE_{\min}$ (minimum RLNE) plotted as a function of filter parameter (λ) for both variants of L1-SPIRiT. In this case, minimum λ in the plot corresponds to

**FIGURE 7.5**

Plots of (A1,B1 and C1) thresholds and (A2,B2 and C2) RLNE as functions of iteration number for three input noise levels.

the empirically chosen parameter value. It is evident that the minimum RLNE increases linearly with increase in λ for standard L1-SPIRiT. In contrast, the minimum RLNE values are less sensitive to changes in λ for the adaptive L1-SPIRiT. The plot of RLNE at convergence versus initial thresholds ν^0 for adaptive L1-SPIRiT is shown in Figure 7.7b. In this case, the initial threshold, as given by the selection procedure described in Section 5.4.1 (VisuShrink), is 0.09. For initial thresholds higher than the VisuShrink threshold, the image quality remains unaltered until a certain higher threshold is used. But thresholds above this limiting value result in poor-quality reconstruction. Alternatively, on the lower side, the image quality deteriorates rapidly only when the initial thresholds are closer to zero compared to the VisuShrink threshold. Thus, the choice of the initial threshold in the proximity of the VisuShrink threshold guarantees the best possible performance. The illustrations reveal that the adaptation not only brings in empirical convergence of the algorithm but also reduces the sensitivity of the steady-state RLNE on the chosen parameters involved in the optimization problem.

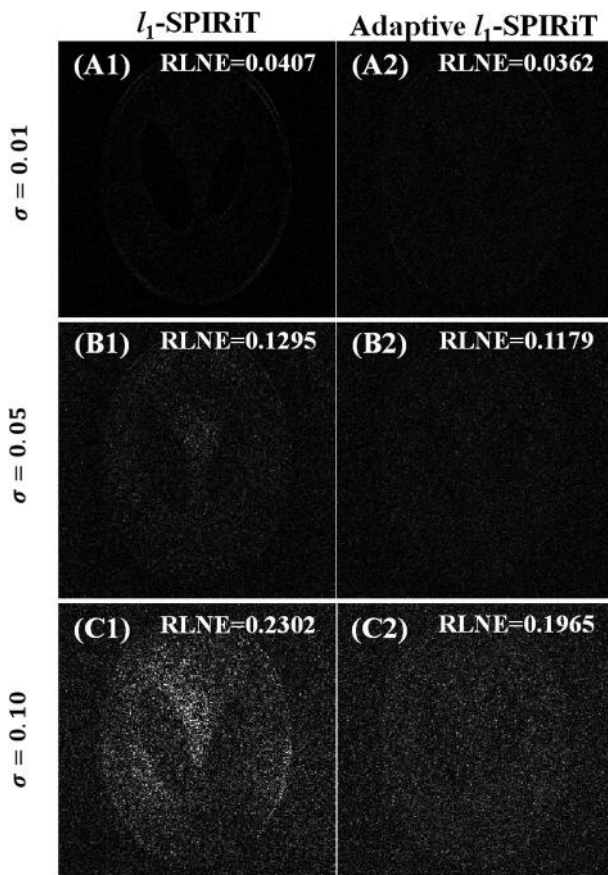


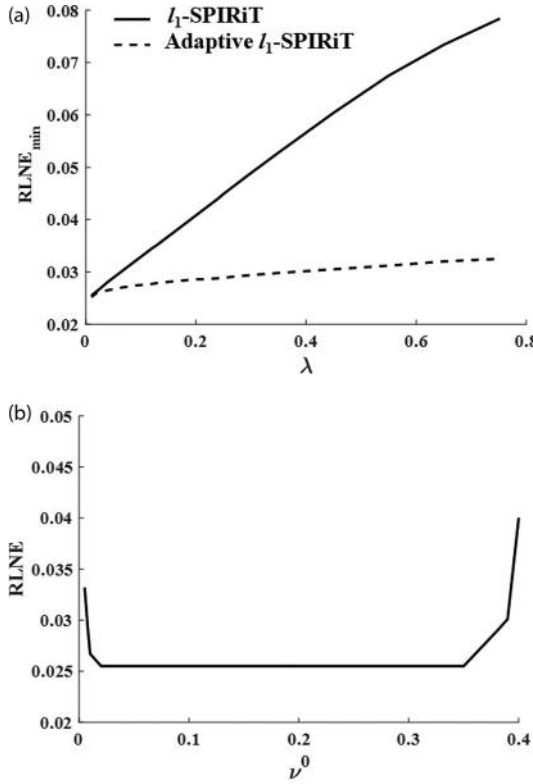
FIGURE 7.6

Difference images obtained using standard L1-SPIRiT and adaptive L1-SPIRiT for three input noise levels.

7.9 Sparsity and Low-Rank Enhanced SPIRiT (SLR-SPIRiT)

SLR-SPIRiT is an extension of L1-SPIRiT [10] with a locally low-rank constraint [38] that exploits the data redundancy in dynamic image series [39]. This joint parallel imaging and compressed-sensing (sparsity and low-rank) method has been potentially applied to 3D-acquisition schemes like dynamic contrast enhanced (DCE) MRI [40–42]. This scheme has been shown to yield higher spatial-temporal resolution for DCE MRI, in which a trade-off between spatial and temporal resolution is necessary for achieving better visual quality of anatomical structures.

In this approach, the dynamic images are reordered into a Casorati matrix in which each column of the matrix represents a single image at one time point, which is of low rank. Typically, under-sampled dynamic series can be reconstructed using a matrix completion approach that minimizes the rank of the Casorati matrix subject to a data consistency constraint. When partitioned into image blocks, the dynamic images can have even lower rank (i.e. locally low rank) [38].

**FIGURE 7.7**

(a) RLNE_{min} (minimum RLNE) plotted as a function of filter parameter (λ) for standard L1-SPIRiT and adaptive L1-SPIRiT, (b) RLNE at convergence versus ν^0 for adaptive L1-SPIRiT.

Consider a vector x_t representing images from all coils at a particular time point t , y_t a vector of acquired k -space samples from all coils at the same time point t , x a vector of images from all channels at all T time points, x_b a vector of images in image block b from all coils at all T time points, Z_t the SPIRiT kernel in the image domain at time point t , F a Fourier transform operator, D_t an under-sampling operator for time point t , Ψ a sparsifying transform operator applied to images from all coils and time points, and C an operator that reformats x_b into its corresponding Casorati matrix; the optimization problem for image reconstruction can be formulated as:

$$\min_x \text{Joint } L1(\Psi x) + \mu \left(\sum_b \|C x_b\|_* \right)$$

$$\text{subject to } D_t F x_t = y_t, Z_t x_t = x_t, t = 1, 2, \dots, T, \quad (7.43)$$

where $\|\cdot\|_*$ denotes the nuclear norm and μ is an empirically determined weighting constant. At each time point, the SPIRiT kernel in the image domain Z_t is computed using the ACS data in the central k -space region. A POCS-based iteration is adopted for obtaining the solution which includes the following operations: (1) to achieve low rank locally,

the singular values of the Casorati matrix in each image block is soft-thresholded; (2) joint sparsity constraint is enforced by soft thresholding the joint sparsity transformed data [6,10]; (3) image domain SPIRiT is applied; and (4) the data consistency is applied by replacing the reconstructed data at the acquired k -space locations using the acquired data. If the acquisition is 3D Cartesian, then an inverse Fourier transform in the readout direction is to be carried out on the acquired data. This implies that at each spatial location, the SLR-SPIRiT is to be performed separately in the readout direction. To ensure lower reconstruction times, coil compression [43] can be used before SLR-SPIRiT reconstruction. Furthermore, the inaccurate contrast enhancement dynamics can be overcome by computing the SPIRiT kernel separately in each phase. This method has been shown to yield higher image quality in highly under-sampled datasets along with a very high spatial-temporal resolution.

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Matrix Completion Methods

8.1 Introduction

The area of matrix completion is a recent field of study following the track of what has been explored in fields related to compressed sensing (CS). Matrix completion deals with the recovery of a data matrix that is incompletely filled, with the known samples corrupted with noise or perturbations. As in CS, matrix completion algorithms therefore involve reconstruction of the data matrix from a small subset of its noise-corrupted entries. The missing entries can be recovered under certain conditions when the data matrix has a low rank [1]. The conditions mainly stipulate that the number of available entries fall below a certain limit and that some of the rows and columns of the matrix are completely unknown. A large body of research is currently trying to focus on improving the accuracy of reconstruction of the unknown entries for a low-rank matrix with the rank a priori unknown.

8.2 Matrix Completion Problem

This section introduces readers to the basic definitions and notions followed in a matrix completion problem using a set of standard notations and terminology that will be used in later sections of the chapter. The data matrix is denoted as $\mathbf{M} \in \mathbb{R}^{m \times n}$, with only q of the entries mn are considered known. In the sequel, $\mathbf{M}$ is considered to be under-sampled, with an under-sampling ratio of $\frac{q}{mn}$. From the known samples, one would like to recover the missing samples to obtain the reconstructed matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$. Let Ω denote the set of observed entries or sampling set, that is, $(i, j) \in \Omega$ if the (i, j) th element of $\mathbf{M}$ is known. Thus, a matrix completion problem may be treated as a constrained minimization problem that states:

$$\begin{aligned} & \text{minimize } \text{rank}(\mathbf{X}) \\ & \text{s.t. } \mathbf{X}(i, j) = \mathbf{M}(i, j), \quad \forall (i, j) \in \Omega. \end{aligned} \tag{8.1}$$

The rank minimization problem is non-deterministic polynomial-time (NP)-hard, and the compute time is $\mathcal{O}(n^n)$. The connection between the problem in Equation (8.1) and the problem with the objective function in Equation (8.1) replaced with the nuclear norm is crucial. Under suitable conditions, Candès and Recht [1] have shown that the two problems can be treated as equivalent, and low-rank matrices can be completed from the set of sampled

entries even though the cardinality of the set is small. As the rank function is simply the number of non-zero singular values, it can be replaced by the nuclear norm:

$$\|\mathbf{X}\|_* = \sum_{i=1}^{\min(m,n)} \sigma_i, \quad (8.2)$$

where σ_i denotes the i th singular value of $\mathbf{X}$. Using the nuclear norm as a convex approximation of the rank, it is possible to reformulate the minimization in Equation (8.1) in terms of the nuclear norm. It is also convenient to express the constraint in Equation (8.1) using the sampling operator $P_\Omega : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$ defined by:

$$[P_\Omega(\mathbf{X})] = \begin{cases} \mathbf{X}_{ij}, & (i, j) \in \Omega \\ 0 & \text{otherwise.} \end{cases} \quad (8.3)$$

Using the sampling operator, the matrix completion problem may be expressed as:

$$\begin{aligned} & \text{minimize } \text{rank}(\mathbf{X}) \\ & \text{s.t. } P_\Omega(\mathbf{X}) = P_\Omega(\mathbf{M}). \end{aligned} \quad (8.4)$$

For an optimal solution, it is assumed that the locations $(i, j) \in \Omega$ are selected at random to avoid trivial situations in which either a row or a column is fully unsampled. In more specific terms, the matrix completion problem becomes trivial once the singular vectors of the observed data matrix become excessively sparse.

8.3 Conditions Required for Accurate Recovery

The existence of a unique low-rank matrix consistent with the measured samples requires the singular vectors of $\mathbf{M}$ to be spread across all coordinates. From the expression of the singular value decomposition (SVD) of $\mathbf{M}$ given by:

$$\mathbf{M} = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^*, \quad (8.5)$$

$\sigma_1 > \sigma_2 > \dots > \sigma_r \geq 0$ are the singular values and $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r \in \mathbb{R}^m$, $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r \in \mathbb{R}^n$ are the singular vectors. The notion of the spread of singular vectors across all coordinates requires satisfying the following conditions:

$$\|\mathbf{u}_i\|_\infty \leq \sqrt{\mu_B / m}, \quad \|\mathbf{v}_j\|_\infty \leq \sqrt{\mu_B / n}, \quad (8.6)$$

for some $\mu_B \geq 1$. However, the value of μ_B is considered to be small enough that it does not make the singular vectors spiky. With N_Ω elements of $\mathbf{M}$ uniformly sampled at randomly chosen locations $(i, j) \in \Omega$ and $N_m = \max(m, n)$, then nuclear norm minimization of $\mathbf{X}$ in Equation (8.4) recovers all entries of $\mathbf{M}$ with a high accuracy and probability of error $\leq 1 - N_m^{-3}$ if there exists a C such that [2]:

$$N_\Omega \geq C \mu_B^4 N_m \log^2 N_m. \quad (8.7)$$

8.3.1 Matrix Completion under Noisy Condition

When the entries of a data matrix are corrupted with noise, the observed matrix is given by:

$$\mathbf{M} = \mathbf{M}_{\text{true}} + \mathbf{Z}, \quad (8.8)$$

where $\mathbf{M}_{\text{true}} \in \mathbb{R}^{m \times n}$ is the true underlying data matrix to be estimated, and $\mathbf{Z} \in \mathbb{R}^{m \times n}$ is a noise matrix satisfying [3]:

$$P_{\Omega}(\mathbf{M}) = P_{\Omega}(\mathbf{M}_{\text{true}}) + P_{\Omega}(\mathbf{Z}) \quad (8.9)$$

In MRI, the elements form samples of k -space, so the entries of $\mathbf{Z}$ are determined from a Rician distribution. It is assumed that $\|P_{\Omega}(\mathbf{Z})\|_F \leq \delta$ for some $\delta > 0$. If either the real or imaginary component of $\{\mathbf{Z}_{ij}\}$ is a white noise sequence with standard deviation σ , then:

$$\delta^2 \leq (N_{\Omega} + \sqrt{8N_{\Omega}})\sigma^2. \quad (8.10)$$

Under noisy conditions, the matrix completion problem is formulated as follows:

$$\begin{aligned} & \text{minimize } \|\mathbf{X}\|_* \\ & \text{s.t. } |\mathbf{X}(i, j) - \mathbf{M}(i, j)| \leq \delta, \quad \forall (i, j) \in \Omega, \end{aligned} \quad (8.11)$$

where δ is proportional to the noise power. Noise suppression results from the relaxation in the constraint. Also for a given rank r , the lower singular values of $\mathbf{X}$ are such that $\sum_{k=r+1}^{\min(m,n)} \sigma_k$ are of the $\mathcal{O}(C_1\sigma^2)$ and $\sqrt{\sum_{k=r+1}^{\min(m,n)} \sigma_k^2}$ are of the $\mathcal{O}(C_2\sigma^2)$, with $C_1 > C_2$. An extension to the noisy formulation of matrix completion problems is the Dantzig selector [4]. Suppose one has observed linear measurements about $\mathbf{M}$, as follows:

$$\mathbf{b} = \mathbf{A}\mathbf{M} + \mathbf{z}, \quad (8.12)$$

where $\mathbf{z} = \text{vec}(\mathbf{Z})$, $\mathbf{z} \in \mathbb{R}^{mn}$ is a noise vector. In this case, the matrix completion corresponds to finding a matrix that minimizes the nuclear norm from the group of all matrices which are consistent with the observed data $\mathbf{b}$.

8.4 Algorithms for Matrix Completion

The minimization problem in Equation (8.11) is convex and can be expressed as semi-definite programming (SDP) [5]. The solvers for SDP use interior-point methods, which means that they cannot be used for large matrices similar to those encountered in pMRI image reconstruction. This is because of the need for solving large-scale linear equations for determining the Newton direction. Alternately, a low-rank approximation done by truncating the SVD and retaining the part of expansion involving singular values exceeding a threshold τ led to emergence of a general family of algorithms referred to as singular-value thresholding (SVT) methods. This provides a sequence

of iterates $\{\mathbf{X}^k\}$ converging to a unique low-rank approximation of the original data matrix $\mathbf{M}$. In these algorithms, the SVT operator $\mathcal{D}_\tau$ serves as a common tool that stands for application of a soft-thresholding operation at level τ to the singular values of the input matrix:

$$\mathcal{D}_\tau(\sigma_i) = \begin{cases} 0, & \text{if } |\sigma_i| < \tau \\ \text{sgn}(\sigma_i)(|\sigma_i| - \tau) & \text{if } |\sigma_i| \geq \tau. \end{cases} \quad (8.13)$$

Based on the definition in Equation (8.13), the matrix SVT operator is defined as $\mathcal{D}_\tau(\mathbf{X}) = \mathbf{U} \text{Diag}(\bar{\sigma}(\mathbf{X})) \mathbf{V}^T$, where $\bar{\sigma}(\mathbf{X}) = \mathcal{D}_\tau(\sigma_i)$ for $i = 1, 2, \dots, n$ with $m > n$. In SVT-based algorithms, it is required to compute this operator for each iteration. The SVT operator imposes a major computational cost in using the algorithm. For any fixed matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$ and for $\tau > 0$, $\|\mathcal{D}_\tau(\mathbf{X})\|_*$ is a piecewise linear and continuous decreasing function. In particular, $\|\mathcal{D}_0(\mathbf{X})\|_* = \|\mathbf{X}\|_*$ and $\|\mathcal{D}_\tau(\mathbf{X})\|_* = 0$ for $\tau \geq \max(\sigma_i(\mathbf{X}))$.

Algorithms for l_1 -norm minimization that use a soft-thresholding operator include iterative thresholding [6], proximal forward-backward splitting (PFBS) [7], split Bregman method [8], linearized Bregman method [9], Bregmanized operator splitting [10], accelerated proximal gradient [11], and alternating-direction method of multipliers [12]. Due to the similarity between the l_1 and nuclear norms, soft-thresholding algorithms can be easily extended for solving low-rank matrix recovery using nuclear norm minimization.

8.4.1 SVT Algorithm

SVT is a mathematical tool for computing the solution for nuclear norm minimization problems; it utilizes the fact that $\mathcal{D}_\tau$ is the proximal operator of the nuclear norm function. This implies that:

$$\mathcal{D}_\tau(\mathbf{Y}) = \min_{\mathbf{X} \in \mathbb{R}^{m \times n}} \tau \|\mathbf{X}\|_* + \frac{1}{2} \|\mathbf{Y} - \mathbf{X}\|_F^2. \quad (8.14)$$

In SVT algorithm, the minimization in Equation (8.4) is first approximated by:

$$\begin{aligned} & \min_{\mathbf{X}} \tau \|\mathbf{X}\|_* + \frac{1}{2} \|\mathbf{X}\|_F^2 \\ & \text{s.t. } P_\Omega(\mathbf{X}) = P_\Omega(\mathbf{M}), \end{aligned} \quad (8.15)$$

with a large parameter τ , and the solution is obtained by applying a gradient descent algorithm, or a gradient ascent algorithm to its dual problem. Starting with $\mathbf{Y}_0 = \mathbf{X}$, the SVT algorithm is implemented through a linearized Bregman iteration, as follows:

$$\begin{aligned} \mathbf{X}_k &= \mathcal{D}_\tau(\mathbf{Y}_{k-1}) \\ \mathbf{Y}_k &= \mathbf{Y}_{k-1} + \delta_k P_\Omega(\mathbf{M} - \mathbf{X}_k), \end{aligned} \quad (8.16)$$

where δ_k is the step size. An important property of the SVT algorithm is that the iteration matrix $\mathbf{Y}_k$ is sparse as its elements vanish outside Ω . Also, the computational cost of updating $\mathbf{Y}_k$ is $\mathcal{O}(N_\Omega)$.

8.4.2 FPCA Algorithm

This algorithm combines the fixed-point continuation (FPC) [13] (also known as forward-backward splitting [7]) with Bregman iteration [14]. Using an augmented Lagrangian formulation, Fixed Point Continuation with Approximate SVD (FPCA) updates $\mathbf{X}_k$ using a set of inner iterations [14]:

$$\begin{aligned}\mathbf{X}_i &= \mathcal{D}_\tau(\mathbf{Y}_{i-1}) \\ \mathbf{Y}_i &= \mathbf{X}_{i-1} + \delta_k P_\Omega(\mathbf{M} - \mathbf{X}_i + \mathbf{Z}_{k-1}),\end{aligned}\tag{8.17a}$$

with $\mathbf{Z}_k$ updated using the last iterate $\mathbf{X}_k = \mathbf{X}_i$:

$$\mathbf{Z}_k = \mathbf{Z}_{k-1} + \delta_k P_\Omega(\mathbf{M} - \mathbf{X}_k)\tag{8.17b}$$

8.4.3 Projected Landweber (PLW) Method

The idea of projected Landweber (LW) comes from the two alternative ways of solving the basis pursuit (BP) problem [15], $\min_{\mathbf{x} \in \mathbb{R}^n} \{\|\mathbf{x}\|_1 : \mathbf{A}\mathbf{x} = \mathbf{b}\}$, when $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$ are given. One method is to solve the Lagrangian version using $\min_{\mathbf{x} \in \mathbb{R}^n} \{\|\mathbf{x}\|_1 + \lambda \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2\}$, or using the least absolute shrinkage and selection operator (LASSO) approach [16] with the 1-norm constrained by a parameter τ : $\min_{\mathbf{x} \in \mathbb{R}^n} \{\|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2 : \|\mathbf{x}\|_1 \leq \tau\}$. Note that the objective function in the LASSO problem is a quadratic function which can easily be minimized using the LW iteration:

$$\mathbf{X}_{k+1} = \mathbf{X}_k + \beta P_\Omega(\mathbf{M} - \mathbf{X}_k),\tag{8.18}$$

with a fixed step size β . Also let $B_R := \{\|\mathbf{x}\|_1 \leq R\}$ be an l_1 -norm closed ball with radius R in finite dimensional space, and $\bar{B}_R := \{\mathbf{X} \in \mathbb{R}^{m \times n} : \|\mathbf{X}\|_* \leq R\}$ be a nuclear norm closed ball with radius R . Accordingly, the corresponding projected operators in the respective closed balls are denoted by P_R and the Frobenius norm projection $\bar{P}_R$ such that $\bar{P}_R(\mathbf{X}) = \mathbf{U} \cdot \text{Diag}(\sigma'(\mathbf{X})) \cdot \mathbf{V}^T$, with $\sigma'(\mathbf{X}) = P_R(\sigma(\mathbf{X}))$. In projected LW iteration, the LW update is projected into the ball $\bar{B}_R$ to satisfy the constrained condition that yields the following iterative form [17]:

$$\mathbf{X}_{k+1} = \bar{P}_R(\mathbf{X}_k + \beta P_\Omega(\mathbf{M} - \mathbf{X}_k))\tag{8.19}$$

The relationship between the matrix projected operator $\bar{P}_R$ and the matrix singular-value thresholding operator $\mathcal{D}_\tau$ is described using the property that if $\|\mathbf{X}\|_* > R$, then $\bar{P}_R(\mathbf{X}) = \mathcal{D}_\tau(\mathbf{X})$, where τ depending on $\mathbf{X}$ and R is chosen such that $\|\mathcal{D}_\tau(\mathbf{X})\|_* = R$. On the contrary if $\|\mathbf{X}\|_* \leq R$, $\bar{P}_R(\mathbf{X}) = \mathcal{D}_0(\mathbf{X}) = \mathbf{X}$. For a given ball radius R and rank r , the matrix projected operator $\bar{P}_R(\mathbf{X})$ is computed using the Algorithm 8.1 outlined in Zhang and Cheng [17], and included below for convenient reference.

Algorithm 8.1 Compute Matrix Projected Operator $\bar{P}_R(X)$

Step 1: Compute SVD:

$$X = U \cdot \text{Diag}(\sigma(X)) \cdot V^T, \text{ with } \sigma_1 > \sigma_2 > \dots > \sigma_r \geq 0, \text{ and } \sigma_{r+j} = 0 \text{ for } j = 1, \dots, n-r.$$

Step 2: If $\sum_{l=1}^r \sigma_l(X) > R$,

Step 2a: Compute $k \leq r$ such that:

$$\left\| \mathcal{D}_{\sigma_k(X)}(\sigma(X)) \right\|_1 = \sum_{l=1}^{(k-1)} (\sigma_l(X) - \sigma_k(X)) \leq R < \sum_{l=1}^k (\sigma_l(X) - \sigma_{(k+1)}(X))$$

Step 2b: Compute $v = k^{-1} \left(R - \left\| \mathcal{D}_{\sigma_k(X)}(\sigma(X)) \right\|_1 \right)$

$$\text{and } \mu = \sigma_k(X) + v,$$

elseif $\sum_{l=1}^r \sigma_l(X) \leq R$,

$$\text{set } \mu = 0.$$

Step 3: $\bar{P}_R(X) = U \cdot \text{Diag}(\mathcal{D}_\mu(\sigma(X))) \cdot V^T$

For a given radius R , the threshold parameter μ is adjusted by using the matrix $X_k + \beta P_\Omega(M - X_k)$. In this way, the PLW algorithm serves as a form of self-adaptive FPC approach because the threshold determined by the nuclear norm radius is self-adaptive and results in a comparatively decreased number of iterations. The fact that low-rank property is implicitly used to implement SVD drastically decreases the computation time in each iteration. One of the main disadvantages of PLW implementation is the need for fixing the nuclear norm radius. In practice, this is done by setting R to a large value by application of the continuation technique [13].

8.4.4 Alternating Minimization Schemes

While SVT-based methods try to solve an unconstrained nuclear norm minimization problem in Equation (8.14), the alternating direction using augmented Lagrangian, also known as alternating direction method of multipliers (ADMM), uses a constrained form $P_\Omega(X - M) = 0$ [18]. There are other forms of alternating minimization (AM)-based approaches that typically combine matrix factorization and the low rank matrix completion (LRMC). While algorithms discussed so far mainly addresses only LRMC problem, the combination addresses a problem of the following form:

$$\min_{X, Y} \left\{ \left\| P_\Omega(XY - M) \right\|_F^2 : X \in \mathbb{R}^{m \times q}, Y \in \mathbb{R}^{q \times n} \right\}. \quad (8.20)$$

To avoid confusion about notation, note that the matrix X now denotes a factor of the reduced rank matrix Z of unknown rank r and missing samples in Ω^c which used to be represented as X in the earlier sections. Also note that q is not necessarily set equal to r .

8.4.4.1 Non-linear Alternating Least Squares Method

In this approach, the unknown low-rank matrix $\mathbf{Z} \in \mathbb{R}^{m \times n}$ is expressed as a matrix product $\mathbf{Z} = \mathbf{X}\mathbf{Y}$ where $\mathbf{X} \in \mathbb{R}^{m \times r}$ and $\mathbf{Y} \in \mathbb{R}^{r \times n}$. With this representation, it is possible to formulate the low-rank matrix completion as follows:

$$\min_{\mathbf{X}, \mathbf{Y}, \mathbf{Z}} \frac{1}{2} \|\mathbf{X}\mathbf{Y} - \mathbf{Z}\|_F^2 \text{ s.t. } P_\Omega(\mathbf{Z}) = P_\Omega(\mathbf{M}). \quad (8.21)$$

Given the current iterates $\mathbf{X}, \mathbf{Y}$ and $\mathbf{Z}$, each variable is updated by minimizing Equation (8.21) with respect to that variable while fixing the other two. For example, by fixing the values of $\mathbf{Y}$ and $\mathbf{Z}$, one obtains the update $\mathbf{X}_+$:

$$\begin{aligned} \mathbf{X}_+ &\leftarrow \mathbf{Z}\mathbf{Y}^\dagger \equiv \mathbf{Z}\mathbf{Y}^T(\mathbf{Y}\mathbf{Y}^T)^{-1}, \\ \mathbf{Y}_+ &\leftarrow \mathbf{X}_+^\dagger \mathbf{Z} \equiv (\mathbf{X}_+^T \mathbf{X}_+)^{-1} (\mathbf{X}_+^T \mathbf{Z}), \\ \mathbf{Z}_+ &\leftarrow \mathbf{X}_+ \mathbf{Y}_+ + P_\Omega(\mathbf{M} - \mathbf{X}_+ \mathbf{Y}_+). \end{aligned} \quad (8.22)$$

Also from the first two equations in Equation (8.22), we have:

$$\mathbf{X}_+ \mathbf{Y}_+ = \left(\mathbf{X}_+ (\mathbf{X}_+^T \mathbf{X}_+)^{-1} \mathbf{X}_+^T \right) \mathbf{Z} \quad (8.23)$$

where the terms inside the bracket represent an orthogonal projection onto the range space of $\mathbf{X}$. This is the same as $\mathbf{Q}_X \mathbf{Q}_X^T$ with $\mathbf{Q}_X := \text{orth}(\mathbf{X})$ forming an orthonormal basis to the range space of $\mathbf{X}$. Numerically, $\mathbf{Q}_X$ is computed using the QR factorization of $\mathbf{X}$.

With introduction of a Lagrange multiplier $\Lambda \in \mathbb{R}^{m \times n}$ such that $P_{\Omega^c}(\Lambda) = \mathbf{0}$, the Lagrangian function for the optimization problem in Equation (8.24) takes the following form:

$$\mathcal{L}(\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \Lambda) = \frac{1}{2} \|\mathbf{X}\mathbf{Y} - \mathbf{Z}\|_F^2 + \langle \Lambda, P_\Omega(\mathbf{Z} - \mathbf{M}) \rangle, \quad (8.24)$$

where the inner product between two matrices $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{m \times n}$ is defined as $\langle \mathbf{A}, \mathbf{B} \rangle = \sum_{i,j} A_{i,j} B_{i,j}$. The first-order optimality conditions are then obtained as follows:

$$\partial \mathcal{L}_X = (\mathbf{X}\mathbf{Y} - \mathbf{Z})\mathbf{Y}^T = 0, \quad (8.25a)$$

$$\partial \mathcal{L}_Y = \mathbf{X}^T (\mathbf{X}\mathbf{Y} - \mathbf{Z}) = 0, \quad (8.25b)$$

$$\partial \mathcal{L}_\Lambda = P_\Omega(\mathbf{Z} - \mathbf{M}) = 0, \quad (8.25c)$$

and

$$\partial \mathcal{L}_Z = P_\Omega(\mathbf{X}\mathbf{Y} - \mathbf{Z}) - \Lambda = 0. \quad (8.25d)$$

From Equation (8.25d), it follows that:

$$P_\Omega(\mathbf{Z} - \mathbf{X}\mathbf{Y}) = \Lambda. \quad (8.26)$$

This indicates that the multiplier matrix Λ is the same as the residual $\mathbf{Z} - \mathbf{X}\mathbf{Y}$ in Ω , and therefore does not play a role in determining $\mathbf{X}, \mathbf{Y}$ and $\mathbf{Z}$. Furthermore, the solutions of

Equations (8.25a) to (8.25c) are exactly the non-linear alternating least squares (NALS) solutions of Equation (8.22). Interested readers may also refer to [19] for details of accelerating the NALS solutions by a successive over-relaxation (SOR) approach.

8.4.4.2 ADMM with Nonnegative Factors

The problem in Equation (8.21) is slightly modified for efficient use of the ADMM approach as follows:

$$\begin{aligned} \min_{\mathbf{U}, \mathbf{V}, \mathbf{X}, \mathbf{Y}, \mathbf{Z}} \quad & \|\mathbf{XY} - \mathbf{Z}\|_F^2, \\ \text{s.t.} \quad & \mathbf{X} = \mathbf{U}, \mathbf{Y} = \mathbf{V}, \\ & \mathbf{U} \geq 0, \mathbf{V} \geq 0 \\ & P_\Omega(\mathbf{Z} - \mathbf{M}) = 0, \end{aligned} \quad (8.27)$$

where $\mathbf{X}, \mathbf{U} \in \mathbb{R}^{m \times q}$ and $\mathbf{Y}, \mathbf{V} \in \mathbb{R}^{q \times n}$. The augmented Lagrangian of Equation (8.27) is:

$$\begin{aligned} \mathcal{L}(\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathbf{U}, \mathbf{V}, \mathbf{\Lambda}, \mathbf{\Pi}) = & \frac{1}{2} \|\mathbf{XY} - \mathbf{Z}\|_F^2 + \langle \mathbf{\Lambda}, \mathbf{X} - \mathbf{U} \rangle + \langle \mathbf{\Pi}, \mathbf{Y} - \mathbf{V} \rangle \\ & + \frac{\alpha}{2} \|\mathbf{X} - \mathbf{U}\|_F^2 + \frac{\beta}{2} \|\mathbf{Y} - \mathbf{V}\|_F^2, \end{aligned} \quad (8.28)$$

where $\mathbf{\Lambda} \in \mathbb{R}^{m \times q}$ and $\mathbf{\Pi} \in \mathbb{R}^{q \times n}$ are Lagrangian multipliers, and $\alpha, \beta > 0$ are penalty parameters. The solution using the alternating direction method is obtained by successively minimizing $\mathcal{L}$ with respect to each variable ($\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathbf{U}$ and $\mathbf{V}$) one at a time, while others are fixed using their most recent values, that is:

$$\mathbf{X}_{k+1} = \left(\mathbf{Z}_k \mathbf{Y}_k^T + \alpha \mathbf{U}_k - \mathbf{\Lambda}_k \right) \left(\mathbf{Y}_k \mathbf{Y}_k^T + \alpha \mathbf{I} \right)^{-1}, \quad (8.29a)$$

$$\mathbf{Y}_{k+1} = \left(\mathbf{X}_{k+1}^T \mathbf{X}_{k+1} + \beta \mathbf{I} \right)^{-1} \left(\mathbf{X}_{k+1}^T \mathbf{Z}_k + \beta \mathbf{V}_k - \mathbf{\Pi}_k \right), \quad (8.29b)$$

$$\mathbf{Z}_{k+1} \leftarrow \mathbf{X}_k \mathbf{Y}_k + P_\Omega(\mathbf{M} - \mathbf{X}_{k+1} \mathbf{Y}_{k+1}), \quad (8.29c)$$

$$\mathbf{U}_{k+1} = P_+(\mathbf{X}_{k+1} + \mathbf{\Lambda}_k / \alpha), \quad (8.29d)$$

$$\mathbf{V}_{k+1} = P_+(\mathbf{Y}_{k+1} + \mathbf{\Pi}_k / \beta), \quad (8.29e)$$

$$\mathbf{\Lambda}_{k+1} = \mathbf{\Lambda}_k + \gamma \alpha (\mathbf{X}_{k+1} - \mathbf{U}_{k+1}), \quad (8.29f)$$

$$\mathbf{\Pi}_{k+1} = \mathbf{\Pi}_k + \gamma \beta (\mathbf{Y}_{k+1} - \mathbf{V}_{k+1}). \quad (8.29g)$$

where $\gamma \in (0, 1.0)$ is a step size and $(P_+(\mathbf{X}))_{ij} = \max(x_{ij}, 0)$. The matrix inversions in Equations (8.29a) and (8.29b) are less expensive as $q < \min(m, n)$.

8.4.4.3 ADMM for Matrix Completion without Factorization

Since data matrices encountered in matrix completion problems related to pMRI acceleration are generally positive semi-definite, the ADMM-based matrix completion model

for positive semi-definite matrix completion (PSDMC) [20] can provide sufficient insights for understanding the implementation details of some pMRI acceleration techniques discussed later in this chapter. Recall Equation (8.11); when the vector of known elements $P_\Omega(\mathbf{M})$ is contaminated with noise, the PSDMC problem can be formulated as:

$$\begin{aligned} & \min_{\mathbf{X}} \text{tr}(\mathbf{X}), \\ & s.t. \|P_\Omega(\mathbf{X}) - P_\Omega(\mathbf{M})\|_F \leq \delta, \\ & \mathbf{X} \succeq \mathbf{0}. \end{aligned} \quad (8.30)$$

since $\|\mathbf{X}\|_* = \text{tr}(\mathbf{X})$ for a positive semi-definite matrix. The formulation in Equation (8.30) can be equivalently represented as a regularized least squares model:

$$\begin{aligned} & \min_{\mathbf{X}} \mu \text{tr}(\mathbf{X}) + \frac{1}{2} \|P_\Omega(\mathbf{X}) - P_\Omega(\mathbf{M})\|_F^2, \\ & s.t. \mathbf{X} \succeq \mathbf{0}. \end{aligned} \quad (8.31)$$

Once the values of μ and δ are set according to the noise level, Equation (8.30) is equivalent to Equation (8.31). Similar ADMM with factored variables discussed earlier, a new splitting variable $\mathbf{Y}$ is introduced to facilitate the ADMM formulation. With this modification, the augmented Lagrangian form of the optimization problem in Equation (8.31) can now be represented as follows:

$$\begin{aligned} \mathcal{L}(\mathbf{X}, \mathbf{Y}, \Pi) = & \mu \text{tr}(\mathbf{X}) + \frac{1}{2} \|P_\Omega(\mathbf{Y}) - P_\Omega(\mathbf{M})\|_F^2 \\ & + \langle \Pi, \mathbf{X} - \mathbf{Y} \rangle + \frac{\rho}{2} \|\mathbf{X} - \mathbf{Y}\|_F^2, \end{aligned} \quad (8.32)$$

where $\Pi \in \mathbb{R}^{m \times n}$ is a Lagrangian multiplier and $\rho > 0$ is a penalty parameter. The solution $\mathbf{X}$ is obtained by successively minimizing $\mathcal{L}(\cdot)$ with respect to $\mathbf{X}$ and $\mathbf{Y}$ in an alternating fashion, as follows:

$$\mathbf{X}_{k+1} = \arg \min_{\mathbf{X} \succeq \mathbf{0}} \mathcal{L}(\mathbf{X}, \mathbf{Y}_k, \Pi_k), \quad (8.33a)$$

$$\mathbf{Y}_{k+1} = \arg \min_{\mathbf{Y}} \mathcal{L}(\mathbf{X}_{k+1}, \mathbf{Y}, \Pi_k), \quad (8.33b)$$

$$\Pi_{k+1} = \Pi_k + \gamma \rho (\mathbf{X}_{k+1} - \mathbf{Y}_{k+1}), \quad (8.33c)$$

where γ is a step size. From Equation (8.32), the cost function for the sub-problem in Equation (8.33a) becomes:

$$\arg \min_{\mathbf{X} \succeq \mathbf{0}} \mu \text{tr}(\mathbf{X}) + \frac{\rho}{2} \|\mathbf{X} - \mathbf{G}_k\|_F^2, \quad (8.34)$$

where $\mathbf{G}_k = \mathbf{Y}_k - (1/\rho)\Pi_k$. Using Equation (8.14), the solution for Equation (8.34) is obtained as follows:

$$\mathbf{X}_{k+1} = \mathbf{U}_k \text{Diag}(\mathcal{D}_{\mu/\rho}(w_k)) \mathbf{V}_k^T, \quad (8.35)$$

where $\mathbf{G}_k = \mathbf{U}_k \text{Diag}(w_k) \mathbf{V}_k^T$ represents the SVD of $\mathbf{G}_k$, and $\mathcal{D}_{\mu/\rho}$ denotes the SVT operator with $\tau = \mu/\rho$ as the threshold. Similar to Equation (8.34), the cost function for the sub-problem in Equation (8.33b) becomes:

$$\arg \min_{\mathbf{Y}} \frac{1}{2} \|\mathbf{P}_{\Omega}(\mathbf{Y}) - \mathbf{P}_{\Omega}(\mathbf{M})\|_F^2 + \frac{\rho}{2} \|\mathbf{Y} - (\mathbf{X}_k + (1/\rho)\mathbf{\Pi}_k)\|_F^2 \quad (8.36)$$

To arrive at a closed-form solution, Equation (8.36) is recast into two separate sub-problems:

$$\arg \min_{\mathbf{Y}} \frac{1}{2} \|\mathbf{P}_{\Omega}(\mathbf{Y}) - \mathbf{P}_{\Omega}(\mathbf{M})\|_F^2 + \frac{\rho}{2} \|\mathbf{P}_{\Omega}(\mathbf{Y}) - \mathbf{P}_{\Omega}(\mathbf{X}_k + (1/\rho)\mathbf{\Pi}_k)\|_F^2, \quad (8.37a)$$

and

$$\arg \min_{\mathbf{Y}} \frac{\rho}{2} \|\mathbf{P}_{\Omega^c}(\mathbf{Y}) - \mathbf{P}_{\Omega^c}(\mathbf{X}_k + (1/\rho)\mathbf{\Pi}_k)\|_F^2. \quad (8.37b)$$

The solutions to the above two sub-problems are obtained as follows:

$$(\mathbf{Y}_{k+1})_{\Omega} = \frac{1}{(\rho+1)} \mathbf{P}_{\Omega}(\mathbf{M} + \rho \mathbf{E}_k), \quad (8.38a)$$

and

$$(\mathbf{Y}_{k+1})_{\Omega^c} = \mathbf{P}_{\Omega^c}(\mathbf{E}_k). \quad (8.38b)$$

The algorithmic form of matrix completion using ADMM is summarized in Algorithm 8.2.

Algorithm 8.2 Matrix Completion Using ADMM

Step 1: Input $\mathbf{P}_{\Omega}(\mathbf{M})$ and $\text{tol} > 0$.

Step 2: Set μ, γ and $\rho > 0$. Set $\mathbf{X}_0$ and $\mathbf{Y}_0$ as random matrices and $\mathbf{\Pi}_0$ as a zero matrix.

Step 3: *while* not converge *do*

Step 4: Update $\mathbf{X}_k$ and $\mathbf{Y}_k$ using Equations (8.35) and (8.38), and $\mathbf{\Pi}_k$ using Equation (8.33c).

8.5 Methods for pMRI Acceleration Using Matrix Completion

Matrix completion techniques have been widely applied to Magnetic Resonance (MR) image reconstruction, particularly to build calibration-less parallel imaging methods. One of the pioneering methods is simultaneous auto-calibrating and k -space estimation (SAKE) [21]. In SAKE, the unacquired k -space samples are estimated by imposing the structural maintenance constraints of the block Hankel structure of a data matrix constructed from the channel k -space data besides imposing data consistency as in conventional CS approaches. It was not clear whether SAKE could outperform the CS approach when it is applied to single-coil data as the origin of the low rankness in the Hankel structured matrix for a single-coil measurement was not extensively investigated. Based on Haldar's work on

low rankness of the Hankel structured matrix generated by single-coil k -space data, the low-rank modeling of local k -space neighborhoods algorithm (LORAKS) [22] and its parallel imaging version, P-LORAKS [23] were developed. Later, it was shown that SAKE and LORAKS are special cases of a family of algorithms based on the annihilating filter-based low-rank Hankel matrix approach (ALOHA) [24]. It is confirmed that sparsity in the image domain is directly linked to the existence of annihilating filters in the k -space. Extension of ALOHA to the general framework of CS is mainly founded on the relationship between sparsity in the spatial domain and the number of required MR k -space measurements [25].

In all aforementioned algorithms, the application of matrix completion methods to CS parallel MRI appears in the form of rank-reducing penalty functions used in conjunction with the cost functional used to encourage consistency of the ideal vector of Nyquist-sampled noiseless multi-channel k -space data $\mathbf{k} \in \mathbb{C}^S$ with the measured noisy under-sampled data $\mathbf{d} \in \mathbb{C}^L$, $L < S$, according to:

$$\mathbf{d} = \mathbf{F}\mathbf{k} + \mathbf{n}, \quad (8.39)$$

where $\mathbf{n} \in \mathbb{C}^L$ is the unknown noise vector and $\mathbf{F} \in \mathbb{C}^{L \times S}$ is a sub-sampling operator formed by keeping L rows from the $S \times S$ identity matrix. In this framework, a low-rank constrained CS parallel MRI reconstruction may be formulated as:

$$\hat{\mathbf{k}} = \min_{\mathbf{k} \in \mathbb{C}^S} \|\mathbf{F}\mathbf{k} - \mathbf{d}\|_2^2 + \lambda_X J_X(\mathcal{P}_X(\mathbf{k})), \quad (8.40)$$

where $J_X(\cdot)$ is a regularizing penalty functional to encourage the matrix X derived from k -space to have a rank-deficient structure, and $\mathcal{P}_X(\cdot)$ is a linear operator that maps a length- S vector of k -space samples into a matrix X with a pre-defined block Hankel or Toeplitz structure.

8.5.1 Simultaneous Auto-calibration and k -Space Estimation

One of the main motivations for development of SAKE is based on the difficulty of acquiring sufficient auto-calibration signals (ACSs) for accurate calibration in certain MR applications. One example is that of spectroscopic imaging for which matrix sizes in spatial dimensions are relatively small, and ACS acquisitions occupy a large proportion of the total imaging time. A second example points to the challenges imposed by the need for repeated acquisition of ACSs in dynamic MRI. Artifacts due to off-resonance often pose difficulties in non-Cartesian imaging such as spiral acquisition that require longer read-outs for acquiring sufficient ACSs. In view of these challenges, the SAKE algorithm aims at reconstruction of the full k -space from an under-sampled multi-channel dataset devoid of any ACSs. SAKE involves organizing the multi-channel data into a structured matrix with low rank and performs reconstruction using structured low-rank matrix completion methods. In its basic form, SAKE employs the projection-onto-sets type algorithm with Cadzow filtering [26].

One of the key steps is the construction of an enhanced data structure, referred to as a data matrix, with definite structure and low-rank property. The low rankness is mainly due to correlations within multi-channel k -space data. A region of size $N_x \times N_y$ from each channel k -space is initially chosen for the reconstruction process. Note that the computational cost of SAKE is mainly dependent on the size of this region. In the process of constructing the data matrix, a block of the channel k -space from the prescribed region is selected by first removing the last $(w-1)$ rows and $(w-1)$ columns; k -space values for each coil at locations within the block are chosen in vectorized form to fill one column of the data matrix.

In each step, the block is displaced one row at a time until the block contains elements from N_x th row of the region. From this point onwards, the block is displaced one column followed by repeated displacement of $(N_x - 1)$ rows. Again, the column displacements are performed only till the last series of row-wise displaced blocks contain elements from the N_y th column of the region. For each position of the block, one column of the data matrix is filled by arranging the channel k -space values corresponding to elements within the block as a single column vector. In this way, the matrix for each channel builds up as each column of the channel data matrix gets filled for every successive position of the sliding block. Each row of the matrix then represents k -space values at locations within contiguous positions of a sliding window of size $w \times w$ as it is slid through the entire region. Figure 8.1 illustrates the procedure for construction of the data matrix.

From $N_x \times N_y$ sized data and N_C number of coils, one can generate a data matrix with size $w^2 N_C \times (N_x - w + 1)(N_y - w + 1)$ by sliding a $w \times w \times N_C$ window across the entire k -space. The nature of the sliding-window operation results in a data matrix with a block-wise Hankel structure in which entries from identical k -space locations are cross-diagonally repeated. Besides the structural characteristics, the sliding window size has a direct implication on the upper bound of the data matrix rank. It is shown in [21] that the rank is bounded by $\text{rank}(A) \leq (w + s - 1)^2$, where A is a data matrix and s is the coil bandwidth measured in k -space pixels. With an increase in w , the normalized upper bound on rank $(w + s - 1)^2 / w^2$ approaches unity. Likewise, the normalized rank exceeds unity for lower window sizes. Several other factors that determine the rank include the spread of sensitivity spectrums, relative size of the object within the field of view (FOV), and the extent of blank signal regions within the object.

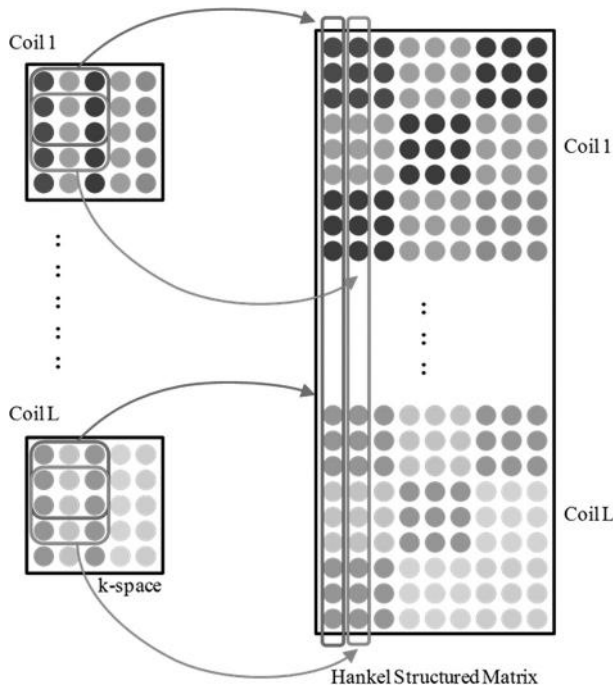


FIGURE 8.1

The low-rank matrix formation in SAKE. Elements inside the window are vectorized to form a single column of block Hankel structured matrix.

Thus, the sequence of operations in SAKE mainly aims at recovering the structured low-rank data matrix A when only a subset of its entries is known due to under-sampling in k -space. These can be summarized as a set of operators comprising a data matrix generation operator $H_w : \mathbb{C}^{N_x \times N_y \times N_C \times 1} \rightarrow \mathbb{C}^{w^2 N_C \times (N_x - w + 1)(N_y - w + 1)}$ and a reverse operator that provides a k -space dataset from a data matrix $H_w^\dagger : \mathbb{C}^{w^2 N_C \times (N_x - w + 1)(N_y - w + 1)} \rightarrow \mathbb{C}^{N_x \times N_y \times N_C \times 1}$. In the reverse operation, the data matrix is not expected to have a block Hankel structure. The role of $H_w^\dagger$ is to first enforce the block-wise Hankel structure by averaging the cross-diagonal entries originated from the same k -space points. Using the operator definitions, the parallel imaging reconstruction is formulated as a structured low-rank matrix completion problem [27,28]:

$$\begin{aligned} & \text{minimize } \text{rank}(A) \\ & \text{s.t. } \mathbf{x} = H_w^\dagger(A) \\ & \text{and } \|P_\Omega(\mathbf{x}) - \mathbf{y}\|_2^2 < \varepsilon, \end{aligned} \quad (8.41)$$

where $\mathbf{y}$ is the acquired k -space data, and ε denotes a bound on noise. If the rank of the data matrix is known beforehand, the formulation in Equation (8.41) can be recast as:

$$\begin{aligned} & \text{minimize } \|P_\Omega(\mathbf{x}) - \mathbf{y}\|_2^2 \\ & \text{s.t. } \mathbf{x} = H_w^\dagger(A) \\ & \text{and } \text{rank}(A) = k. \end{aligned} \quad (8.42)$$

Using the idea that MR images have sparse representation in the wavelet domain, the SAKE approach is shown to be more efficient when it is employed in a regularization model with an additional l_1 -norm penalty term consisting of the wavelet coefficients. Using the joint sparsity model that considers multi-channel images as jointly sparse [29], a regularized form of SAKE is obtained as:

$$\begin{aligned} & \text{minimize } \|P_\Omega(\mathbf{x}) - \mathbf{y}\|_2^2 + \lambda \sum_l \sqrt{\sum_c |\Psi\{\text{IFFT}(\mathbf{x})\}|^2} \\ & \text{s.t. } \mathbf{x} = H_w^\dagger(A) \\ & \text{and } \text{rank}(A) = k, \end{aligned} \quad (8.43)$$

where Ψ denotes the wavelet transform of an MR image with l and c indexing the spatial and coil dimensions. For implementing the SAKE algorithm formulated in Equation (8.42), the authors define the following projection operators that correspond to each of the constraints in Equation (8.42):

1. Low-rankness projection: Hard-threshold the singular values of data matrix formed from the current estimate of k -space. This is accomplished using SVD-based sub-space decomposition to break up the information into signal and noise sub-spaces [30], which are spanned by singular vectors corresponding to the dominant and non-dominant singular values, respectively.
2. Structural consistency projection: Project the data matrix onto the space of block-wise Hankel matrices by application of $H_w^\dagger$ to the data matrix.

3. Data consistency projection: Project the current estimate of k -space data onto a set in which the k -space values at acquired locations match the original values or, equivalently, the l_2 -norm of their difference in vectorized form is less than a bound on noise. For non-Cartesian sampling, this projection is represented by a linear operator $\mathcal{D}$ such that $\|\mathcal{D}\mathbf{x} - \mathbf{y}\|_2^2 < \varepsilon$. The main steps of the SAKE algorithm are summarized in Algorithm 8.3.

Algorithm 8.3 SAKE

Initialize: $\mathbf{x}_0$ is set as the k -space region to be reconstructed with unacquired locations assigned value of 0.

$n = 0$.

Do {

$n = n + 1$

$A_n = H_w(\mathbf{x}_{n-1})$ % construct data matrix.

$[U, \Sigma, V] = \text{SVD}(A_n)$ % Perform SVD of data matrix.

$A_n = U\Sigma_k V^H$ % Hard-threshold singular values (low-rankness projection).

$\mathbf{x}_n = H_w^\dagger(A_n)$ % Transform data matrix back to k -space data (structural consistency projection)

$\mathbf{x}_n = P_\Omega(\mathbf{x}_n)$ % data consistency projection

$err = \|\mathbf{x}_n - \mathbf{x}_{n-1}\|$

} while $err > Tol$.

For implementing SAKE with multi-channel under-sampled data, the data matrix is constructed from the central portion of k -space with sizes typically on the order of 80×80 . The centre cropping is done to reduce the SVD computation. Note, however, that SAKE requires a minimum number of samples around the centre of k -space. To comply with the conditions for accurate matrix recovery, it is necessary that the sampling pattern be of a random nature.

In practice, random under-sampling in k -space with elliptical, variable-density Poisson disk patterns was used [29]. The window sizes used typically range from 2×2 to 15×15 . In general, calibration and data matrices from real MR data have low rank with compact column space. With increasing window size, the data matrix tends to have decreasing window normalized rank. This is because the data matrices start possessing compact column spaces if the window size becomes larger than a certain value. Depending on the relative size of the object with respect to the FOV, the window-normalized rank converges to a value ≤ 1 . Smaller objects tend to result in lower rank of the data matrix.

As in SVT-based matrix completion algorithms, a major challenge in SAKE is the computation of SVD on the large data matrix in every single loop. The computational burden also increases in proportion to the data matrix rank, window size, and object size. Unlike the basic matrix completion methods, SAKE employs only a simple thresholding scheme (Cadzow's algorithm) for estimating the dominant singular values. One of the

main findings of SAKE is the dependence of bound for the rank of a data matrix on the bandwidth of the encoding function. Due to sensitivity encoding, any (vectorized) blocks of k -space data lie in a low-dimensional sub-space. The dimensionality of signal sub-space is also implicitly determined by object geometry. As there is no abstract approach to estimate the rank in under-sampled conditions, the authors of SAKE used estimates from other scans in order to predict the rank value for prospectively under-sampled datasets.

To circumvent the problem of computation time, SAKE is generally employed to reconstruct only a central portion of the under-sampled k -space. In this way, the SAKE reconstructed data can be further exploited as a form of synthetic calibration data to be used along with other auto-calibrating methods for full data reconstruction. Estimation of the low-dimensional sub-space is very similar to the iterative generalized autocalibrating partially parallel acquisitions algorithm [31] which involves calibration of a new GRAPPA kernel from the reconstructed data in every iteration.

8.5.2 Low-Rank Modeling of Local k -Space Neighborhoods

Compared to SAKE, which involves low-rank modeling with multiple channels, contrasts, or time points, LORAKS focuses on low-rank modeling of local k -space neighborhoods and corresponds to cases of low-dimensional single-channel, single-contrast, single-time-point data. Using linear dependencies within local neighborhoods of k -space data of images with limited spatial support and/or slowly varying phase, lower-dimensional spatial data is mapped onto higher-dimensional matrices. LORAKS applies the above constraints in a different way from other direct regularized methods in CS approaches.

Distinct from other, previously established calibrationless methods applied for low-rank reconstruction of single channel MRI [32,33], the algorithm leading to LORAKS is built primarily based on the idea of low-rank matrices constructed based on the linear dependencies in k -space induced by local image support and slowly varying phase. Support of the image $\rho(x, y)$ is defined as the set of points $\Omega_\rho = \{(x, y) \in \mathbb{R}^2 : |\rho(x, y)| \neq 0\}$. The size and shape of the imaging support determines how the FOV and Nyquist sampling rate are chosen. Analytic Fourier sampling relationship has been used to simulate k -space data sampled on a finite dimensional 256×256 Cartesian grid for different values of the k -space sampling rates $\Delta k = 1, 1/2, 1/4$ and $1/6$, as shown in Figure 8.2.

This is tantamount to saying that k -space data sampled at relative sampling rates $\Delta k \leq 1$ will have linear dependencies due to the existence of measurable regions within the FOV for which $\rho(x, y) = 0$ [34]. To show this, one must consider a non-zero function $f(x, y)$ over support $\Omega_f \subset \text{FOV}$ such that $f(x, y) = 0, \forall (x, y) \in \Omega_\rho$. With this representation, it is easily inferred that $\rho(x, y)f(x, y) = 0, \forall (x, y) \in \Omega_\rho \cup \Omega_f$. Using $\tilde{\rho}(n_x, n_y)$ and $\tilde{f}(n_x, n_y)$ to denote

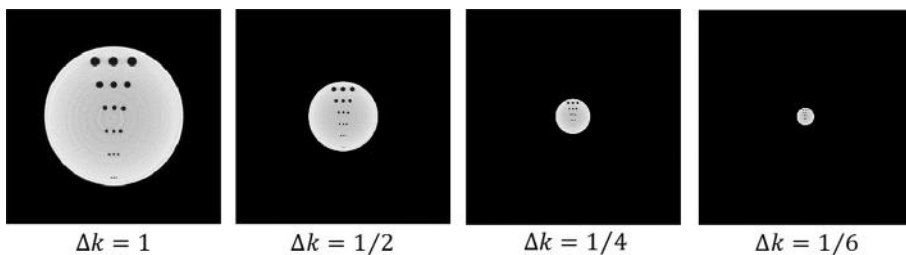


FIGURE 8.2

Illustration of how the relative k -space sampling rate determines the image support within the imaging FOV.

discrete Fourier transforms (DFTs) of the respective Nyquist sampled functions $\rho(x, y)$ and $f(x, y)$, and treating $f(x, y)$ to be a smooth low-pass function, it is possible to conceive of a region of finite support R in k -space with $\tilde{f}(n_x, n_y) \approx 0$ whenever $(n_x, n_y) \notin \Lambda_R$, where $\Lambda_R = \{(n_x, n_y) \in \mathbb{R}^2 : (n_x^2 + n_y^2) \leq R^2\}$ is the set of points within radius R about the origin. With N_R representing the number of elements in Λ_R and location (p_m, q_m) denoting the m th element of Λ_R , the discrete convolution relationship in the Fourier domain corresponding to the vanishing product of the spatial domain functions is given by:

$$\sum_{m=1}^{N_R} \tilde{\rho}(n_x - p_m, n_y - q_m) \tilde{f}(p_m, q_m) \approx 0, \quad (8.44)$$

$\forall (n_x, n_y)$ within the Nyquist sampled k -space limits. Note that the bandwidth of $f(x, y)$ can be simulated with different neighborhood systems Λ_R corresponding to different k -space radii, $R = 1, 2, 3$ and 5 , with respective N_R values of $5, 13, 29$ and 81 , as shown in Figure 8.3.

By selecting K distinct k -space points $\{(n_x^{(k)}, n_y^{(k)})\}_{k=1}^K$ with $K \geq N_R$, Equation (8.44) is used to construct a low-rank matrix $\mathbf{C} \in \mathbb{Z}^{K \times N_R}$ with the following elements:

$$[\mathbf{C}]_{km} = \tilde{\rho}[n_x^{(k)} - p_m, n_y^{(k)} - q_m], \quad (8.45)$$

for $k = 1, \dots, K$, and $m = 1, \dots, N_R$, such that:

$$\mathbf{C}\tilde{\mathbf{f}} \approx \mathbf{0}, \quad (8.46)$$

where $\tilde{\mathbf{f}} \in \mathbb{Z}^{N_R}$ denotes a vector of k -space values $\tilde{f}(n_x, n_y) \in \Lambda_R$. A graphical illustration for construction of $\mathbf{C}$ is provided in Figure 8.4. In practice, for a grid size of $N_x \times N_y$, K is chosen as $(\min(N_x, N_y) - 2R)^2$ corresponding to the maximum number of k -space points with their entire Λ_R neighborhood system contained fully within the sampling grid. Note

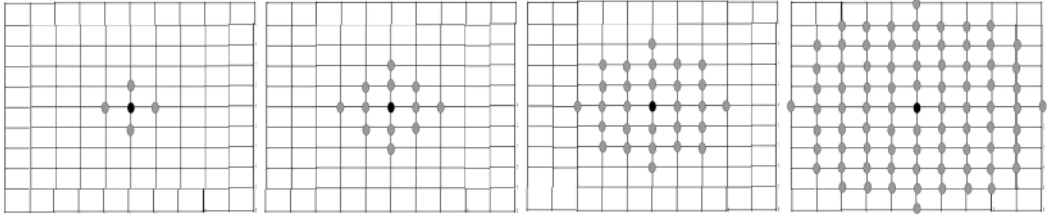


FIGURE 8.3

Neighborhood systems Λ_R for different k -space neighborhood radii R , with the centers of the neighborhood system marked in black. (Adapted from Haldar J, P., *IEEE Trans. Med. Imaging*, 33(3), 668–681, 2014.)

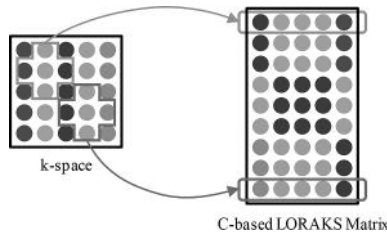


FIGURE 8.4

The C-based low-rank matrix formation in LORAKS. Elements inside the window are vectorized to form a single row of the C-matrix.

that the k th row of $\mathbf{C}$ contains elements from the Λ_R neighborhood of the k -space point $(n_x^{(k)}, n_y^{(k)})$. The rank of $\mathbf{C}$ depends on R and the size of Ω_ρ relative to FOV. For example, the N_R singular values of $\mathbf{C}$ decays faster for smaller values of Δk , indicating that $\mathbf{C}$ matrices constructed from images with smaller support have lower rank. Likewise, large R generally corresponds to larger dimension of the null space of $\mathbf{C}$ or, equivalently the larger proportion of near-zero singular values.

As in the SAKE algorithm, a simple form of matrix completion task is performed on $\mathbf{C}$ by computing a rank- l approximation $\hat{\mathbf{C}}_l$. This is obtained by setting $\sigma_k = 0$ for $k = l + 1, \dots, N_R$. Since k -space points repeat themselves in $\mathbf{C}$, and the low-rank approximation will in general result in perturbation of the repeated values, reconstruction of the new k -space data back from $\hat{\mathbf{C}}_l$ is performed by averaging all of the entries from $\hat{\mathbf{C}}_l$ where the k -space value of ρ at a specific location (n_x, n_y) repeated itself in $\mathbf{C}$. Accuracy of reconstruction improves when l is substantially lower than N_R and when Ω_ρ is small relative to FOV. The main steps of the LORAKS algorithm are summarized in Algorithm 8.4.

Algorithm 8.4 LORAKS

Initialize: Set the value of R, K and choose $\hat{\mathbf{k}}_0$ as the zero-filled k -space.

$i = 0$.

Do {

$i = i + 1$,

$\mathbf{Q}^{(i)} = \mathcal{P}_C(\mathbf{k}^{(i-1)})$

% Construct the $\mathbf{C}$ matrix for the given choice of R
% and K .

$[U, \Sigma, V] = \text{SVD}(\mathbf{Q}^{(i)})$

% Perform SVD of the $\mathbf{C}$ matrix.

$\mathbf{Q}^{(i)} = U \Sigma_k V^H$

% Hard-threshold singular values.

$\mathbf{k}^{(i)} = \mathcal{P}_C^*(\mathbf{Q}^{(i)})$

% Transform data matrix back to k -space data.

% Using adjoint operator $\mathcal{P}_C^*$.

$\mathbf{k}^{(i)} = P_\Omega(\mathbf{k}^{(i)})$

% data consistency projection

$err = \|\mathbf{k}^{(i)} - \mathbf{k}^{(i-1)}\|$

} while $err > Tol$.

8.5.3 Annihilating Filter–Based Low-Rank Hankel Matrix Approach

ALOHA is another structured matrix completion approach for parallel imaging reconstruction [24,25]. The SAKE and C-based LORAKS techniques were based on the idea that k -space measurement is low-ranked if the image has finite support or a slow-varying phase. If the image is not finite supported but can be sparsified using a transform domain, the generalized form of ALOHA is the most suitable form of matrix completion approach. In ALOHA, the transform domain sparsity of the image is linked to the existence of annihilating filters in the weighted k -space. This leads to the construction of a rank-deficient Hankel structured matrix using weighted k -space measurements where the rank depends on the sparsity of the image. Therefore, if the image is not finite supported but can be sparsified using a transform domain, the missing k -space data can be recovered using this low-rank Hankel structured matrix completion approach. Thus, the rank of the Hankel

structured matrix is decided from the sparsity level k in the image domain when, for example, a one-dimensional version of the image signal $f(\rho)$ can be represented as a k -Dirac signal within a support set $[0, \tau]$:

$$f(\rho) = \sum_{j=0}^{k-1} c_j \delta(\rho - \rho_j) \quad (8.47)$$

In accelerated MRI, only a sparse subset of the Fourier samples is acquired at $\{\omega_i\}_{i=1}^L$ from a k -space of dimension S and a grid size below the Nyquist limit $\Delta = 2\pi / \tau$ to avoid aliasing artifacts. Using the discrete spectral model, the measured k -space samples can be obtained using:

$$\hat{f}(m\Delta) = \sum_{j=0}^{k-1} c_j \exp(-i2\pi m\rho_j / \tau), \quad m \in \Omega, \quad (8.48)$$

where Ω denotes the index set of the sparse sampling positions. A limitation of the sampling model in Equation (8.48) is that it requires the image to be represented as a stream of Diracs with period τ , defined as a finite rate innovation (FRI) signal [35–37]. The linear dependence relation in ALOHA is established from the FRI sampling theory, which defines the existence of an annihilating filter $\hat{h}(n)$ that satisfies:

$$(\hat{h} * \hat{f})[n] = \sum_{l=0}^k \hat{h}[l] \hat{f}[n-l] = 0, \quad \forall n. \quad (8.49)$$

Using a matrix-vector notation, Equation (8.49) can be represented as:

$$\mathcal{H}(\hat{\mathbf{f}}) \hat{\mathbf{h}} = 0, \quad (8.50)$$

where the Hankel structure matrix is constructed as:

$$\mathcal{H}(\hat{\mathbf{f}}) = \begin{bmatrix} \hat{f}[0] & \hat{f}[1] & \dots & \hat{f}[\kappa-1] \\ \hat{f}[1] & \hat{f}[2] & \dots & \hat{f}[\kappa] \\ \vdots & \vdots & \ddots & \vdots \\ \hat{f}[S-\kappa] & \hat{f}[S-\kappa+1] & \dots & \hat{f}[S-1] \end{bmatrix} \in \mathbb{C}^{(S-\kappa+1) \times \kappa}, \quad (8.51)$$

and $\hat{\mathbf{h}} = [\hat{h}[0] \dots \hat{h}[\kappa-1]]^T$. However, the nature of MR images means that such a representation by Diracs is more appropriate in a transform domain, or when represented in terms of the generalized total variation (TV) framework. A generalized one-dimensional TV operator is defined as:

$$\mathcal{L} := a_K D^K + a_{K-1} D^{K-1} + \dots + a_1 D + a_0, \quad (8.52)$$

where a_K is a real number and D denotes the one-dimensional differential operator. Thus, treating the generalized TV signal to be k -sparse, we have:

$$\mathcal{L}f(\rho) = \sum_{j=0}^{k-1} a_j \delta(\rho - \rho_j), \rho_j \in [0, \tau]. \quad (8.53)$$

Fourier transformation of Equation (8.53) yields:

$$\mathcal{F}(\mathcal{L}f(\rho)) = \hat{l}(\omega)\hat{f}(\omega) = \sum_{j=0}^{k-1} a_j \exp(-i\omega\rho_j), \quad (8.54)$$

where $\hat{l}(\omega) = a_K(i\omega)^K + a_{K-1}(i\omega)^{K-1} + \dots + a_1(i\omega) + a_0$. From Equation (8.54), it is obvious that a Fourier domain filter $\hat{h}[n]$ exists that annihilates the discretized form of the weighted signal $\hat{l}(\omega)\hat{f}(\omega) \triangleq \hat{l}(2\pi m/\tau)\hat{f}(2\pi m/\tau)$. Thus, the unknown k -space samples are determined by solving the ALOHA problem:

$$\begin{aligned} & \min_{\mathbf{m} \in \mathbb{C}^S} \|\mathcal{H}(\mathbf{m})\|_* \\ \text{s.t. } & P_\Omega(\mathbf{m}) = P_\Omega(\hat{\mathbf{l}} \odot \hat{\mathbf{f}}), \end{aligned} \quad (8.55)$$

where $\odot$ denotes the Hadamard product, and $\hat{\mathbf{l}}$ and $\hat{\mathbf{f}}$ denote the vectors $\hat{\mathbf{l}} = [\hat{l}[0] \dots \hat{l}[S]]^T$ and $\hat{\mathbf{f}} = [\hat{f}[0] \dots \hat{f}[S]]^T$. In its most basic form, the underlying image is assumed to be piecewise constant, corresponding to which the operator $\mathcal{L}$ becomes a first-order derivative and $\hat{l}(\omega) = a_1 i\omega$.

Unlike the TV model, MR images sparsely represented using dyadic wavelet transform can be recovered more efficiently using the generalized ALOHA with a pyramidal structure. If an image is sparse in the wavelet domain, the algorithm down-samples the k -space data from fine to coarse scale, and low-rank matrix completion is performed at these scales. At every scale, the following steps are performed.

Step 1: Extract the k -space data corresponding to the scale s such that $\hat{f}^s[m] = \hat{f}(\omega)|_{\omega=\frac{2\pi m}{\tau}}$, where $m = 0, \dots, S/2^s - 1$, and S denotes the space dimension.

Step 2: The k -space is weighted using an s -scale wavelet spectrum $\hat{l}^s[m] = \frac{1}{\sqrt{2^s}} \hat{\psi}^*(2^s \omega)|_{\omega=\frac{2\pi m}{\tau}}$, $m = 0, \dots, S/2^s - 1$, where $\hat{\psi}^*(\cdot)$ represent the complex conjugate of the Fourier spectrum of wavelet. While constructing the k -space weighting vector $\hat{\mathbf{l}}^s$, the sampling interval for $\hat{\psi}^*(\omega)$ increases by a factor of 2 for each successive scale s .

Step 3: Construct the low-rank Hankel matrix, and perform matrix completion using an s -scale generalized ALOHA formulation:

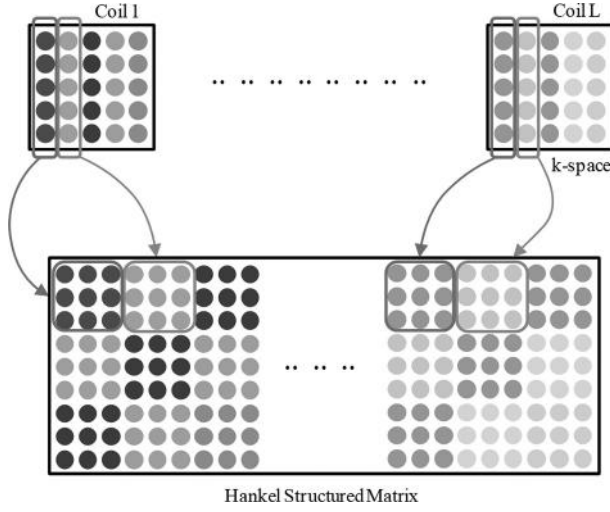
$$\begin{aligned} & \min_{\mathbf{m} \in \mathbb{C}^{\left(\frac{S}{2^s}\right)}} \|\mathcal{H}(\mathbf{m})\|_* \\ \text{s.t. } & P_\Omega(\mathbf{m}) = P_\Omega(\hat{\mathbf{l}}^s \odot \hat{\mathbf{f}}^s), \end{aligned} \quad (8.56)$$

A schematic diagram demonstrating the formation of a low-rank matrix using ALOHA is shown in [Figure 8.5](#).

Step 4: Perform inverse weighting by dividing the interpolated k -space data using the same s -scale wavelet spectrum used in Step 2.

Step 5: Unacquired samples of the k -space are replaced with the interpolated data obtained in Step 4.

The extension of ALOHA to a multi-channel Hankel matrix for pMRI introduces an additional annihilation relationship which is a generalization of that used in SAKE and

**FIGURE 8.5**

The low-rank matrix formation in ALOHA. Each column of the coil corresponds to a block in the low-rank Hankel structured matrix.

C-based P-LORAKS. The multi-channel version of g ALOHA is derived based on the pMRI model for an i th coil image $g_i(\rho)$ given by:

$$g_i(\rho) = C_i(\rho)f(\rho), \text{ for } i = 1, \dots, N_c, \quad (8.57)$$

where $C_i(\rho)$ denotes the i th coil sensitivity map, and $f(\rho)$ is the unknown MR image to be reconstructed. Assuming smooth variation of coil sensitivity, we have:

$$\mathcal{L}g_i(\rho) = C_i(\rho)\mathcal{L}f(\rho). \quad (8.58)$$

Also, any pair of channels (i, j) , $i \neq j$ that satisfies the condition:

$$C_i(\rho)\mathcal{L}g_j(\rho) = C_j(\rho)\mathcal{L}g_i(\rho), \quad (8.59)$$

can be equivalently represented in Fourier domain as:

$$C_i(\omega)^* \left(\hat{l}(\omega) \hat{g}_j(\omega) \right) - C_j(\omega)^* \left(\hat{l}(\omega) \hat{g}_i(\omega) \right) = 0, i \neq j, \forall \omega. \quad (8.60)$$

The individual coil sensitivity spectra take the form of an array of annihilating filters when the weighted channel k -spaces in the matrix-vector formulation of Equation (8.60) is expressed as an augmented matrix $\mathcal{Y}$ defined as:

$$\mathcal{Y} = \left[\mathcal{H} \left(\hat{l} \odot \hat{\mathbf{g}}_1 \dots \hat{l} \odot \hat{\mathbf{g}}_{N_c} \right) \right] \in \mathbb{C}^{(S-\kappa+1) \times \kappa N_c}. \quad (8.61)$$

Using Equation (8.61), the multi-channel formulation of g ALOHA is given by:

$$\begin{aligned} \min_{\mathbf{m} \in \mathbb{C}^S} & \left\| \mathcal{H}(\mathbf{m}_1 \cdots \mathbf{m}_{N_c}) \right\|_* \\ \text{s.t. } & P_\Omega(\mathbf{m}_i) = P_\Omega(\hat{\mathbf{1}} \odot \hat{\mathbf{g}}_i), i = 1, \dots, N_c. \end{aligned} \quad (8.62)$$

In pMRI, if L denotes the number of sampling locations, then k -space data are sampled simultaneously from N_c coils, and the total number of k -space samples will be LN_c . If Equation (8.60) holds for every pair $(i, j), i \neq j, \forall j = 1, \dots, N_c$ and $\forall i = 1, \dots, N_c$, the total degrees of freedom for the k -sparse FRI signal model using $\mathcal{Y}$ is $N_c(2k - N_c + 1)$. As the number of observations required should exceed the number of degrees of freedom, we have:

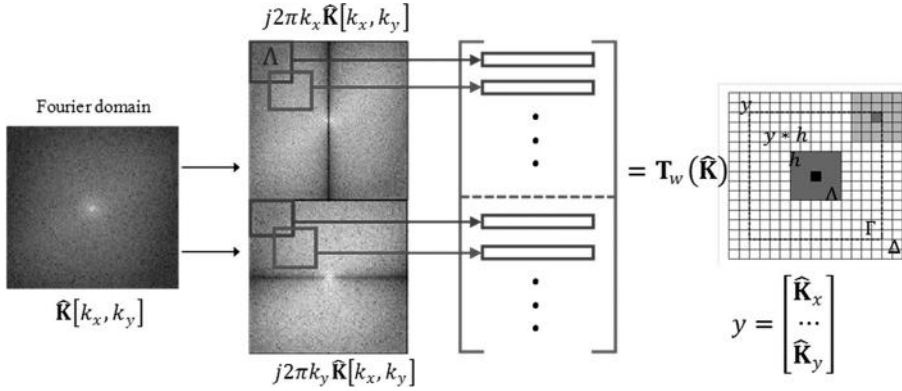
$$LN_c \geq N_c(2k - N_c + 1) \Leftrightarrow k \leq \frac{L + N_c - 1}{2}. \quad (8.63)$$

8.6 Non-convex Approaches for Structured Matrix Completion Solution for CS-MRI

With the matrix lifting approaches discussed so far, the aim is to recover an array $\hat{\mathbf{k}} \in \mathbb{R}^{N_x N_y}$ of the Fourier coefficients of an image from missing or corrupted k -space samples $\mathbf{k}_0$. With two-dimensional representation, the problem is that of recovering the k -space $\mathbf{K} \in \mathbb{R}^{N_x \times N_y}$ from a given set of measurements that constitute the non-zero elements of the zero-filled k -space $\mathbf{K}_0$ using the sampling relationship $P_\Omega(\hat{\mathbf{K}}) = \mathbf{K}_0$. By representing the gradients of an image $\rho(x, y)$ using an FRI signal model with sparsity level r , it is possible to construct a block Hankel structured matrix $\mathbf{H}$ by block-wise sampling the weighted k -spaces $\hat{\mathbf{K}}_x = j2\pi k_x \hat{\mathbf{K}}(k_x, k_y)$ and $\hat{\mathbf{K}}_y = j2\pi k_y \hat{\mathbf{K}}(k_x, k_y)$. This is implemented by application of the data matrix lifting operator H_w with kernel size w , as detailed in [Section 8.5.1](#). With each block vectorized as a row vector, the lifted matrix $\mathbf{T}_w \in \mathbb{C}^{(N_x-w+1)(N_y-w+1) \times w^2}$ can be represented in block Toeplitz form in which the blocks from $\hat{\mathbf{K}}_y$ are cascaded below those from $\hat{\mathbf{K}}_x$. Using a bandlimited function whose Fourier transform can be represented using a finite array $\mathbf{h}(k) \in \mathbb{C}^{w^2 \times 1}, k = 1, 2, \dots, w^2$ and zero valued inside the image support, $\mathbf{T}_w$ satisfies the convolution annihilation relationship:

$$\mathbf{T}_w \mathbf{h} = \mathcal{T}_w(\hat{\mathbf{K}}) \mathbf{h} = \mathcal{T}_w \begin{pmatrix} \hat{\mathbf{K}}_x \\ \dots \\ \hat{\mathbf{K}}_y \end{pmatrix} \mathbf{h} = 0, \quad (8.64)$$

where $\mathcal{T}_w$ denotes the operator that arranges the row-wise vectorized blocks into a Toeplitz form, and is alternately represented as operator $\mathbf{T}_w$ operating on $\hat{\mathbf{K}}$, resulting in the Toeplitz matrix $\mathbf{T}_w$. The left panel in [Figure 8.6](#) shows the lifting operation resulting in Toeplitz form. The right panel shows the support Δ for the weighted k -space; the support of filter $\mathbf{h}$ designated by Λ ; and the region Γ bounded by the dashed line, indicating the valid set of indices for which the convolution operation is well defined. As any filter derived by a linear phase shift of $\mathbf{h}$ lies in the null space of $\mathbf{T}_w$, it is true that $\mathbf{T}_w$ has a low rank. Thus, the solution of the image recovery problem using matrix lifting can be generalized as the solution that minimizes a cost function consisting of data fidelity

**FIGURE 8.6**

Low-rank property of a structured matrix $T_w(\hat{\mathbf{K}})$. The matrix $T_w(\hat{\mathbf{K}})$ is constructed by weighting the k -space two arrays of simulated Fourier derivative coefficients $\hat{\mathbf{K}}_x$ and $\hat{\mathbf{K}}_y$. (Adapted from Ongie, G., and Jacob, M., *IEEE Trans. Comp. Imaging*, 3(4), 535–550, 2017.)

with respect to the known points in k -space, and a regularization term expressing the degree to which the lifted matrix may be assumed to be of low rank. As discussed earlier, the trace (nuclear) norm penalty is similar to the l_1 -norm regularization of the singular values and provides a low-rank solution. However, the l_1 -norm over-penalizes large entries of vectors and results in a biased solution. Since the nuclear norm penalty shrinks all the singular values by the same amount, large singular values can be over-penalized. This indicates that the trace norm may make the solution deviate from the original solution as the l_1 -norm does.

Although non-convex, the Schatten- p quasi-norm for $0 < p < 1$ provides a closer approximation to the rank function. Using the Schatten- p norm of the convolutional structured lifted matrix as a regularizer, the recovered k -space $\mathbf{K}$ minimizes the objective function [38]:

$$\frac{1}{2} \|P_{\Omega}(\mathbf{K}) - \mathbf{K}_0\|_2^2 + \lambda \left\| T_w(\hat{\mathbf{K}}) \right\|_p^p, 0 < p \leq 1, \quad (8.65)$$

where λ is a tunable regularization parameter, and $\|\cdot\|_p$ denotes the Schatten- p norm. As the Schatten- p quasi-norm minimization problems are non-convex, non-smooth and non-Lipschitz, alternative formulations of the Schatten- p quasi-norm that uses iteratively reweighted least-squares (IRLS) algorithms [39–41] have been employed to solve Equation (8.65).

8.6.1 Solution Using IRLS Algorithm

The IRLS approach relies on the relationship between the Schatten- p quasi-norm and a weighted form of the Frobenius norm. This implies that, for a matrix $\mathbf{X}$ with full rank, the Schatten- p quasi-norm can be expressed as:

$$\|\mathbf{X}\|_p^p = \left\| \mathbf{X} \mathbf{H}^{\frac{1}{2}} \right\|_F^2, \quad (8.66)$$

where $\mathbf{H} = (\mathbf{X}^* \mathbf{X})^{\left(\frac{p-1}{2}\right)}$. The representation of Schatten- p quasi-norm as a weighted Frobenius norm leads to an iterative algorithm that alternately updates a weight matrix $\mathbf{H}$ and solves a least-squares problem for a fixed $\mathbf{H}$. Application of this method to MR image recovery from sub-sampled data gives rise to the following iterative scheme:

$$\mathbf{H}^{(n)} = \left[\left(\mathbf{T}_w^{(n-1)} \right)^* \left(\mathbf{T}_w^{(n-1)} \right) + \varepsilon_n \right]^{\left(\frac{p-1}{2}\right)}, \quad (8.67a)$$

$$\mathbf{K}^{(n)} = \min_{(\mathbf{K})} \frac{1}{2} \left\| P_{\Omega}(\mathbf{K}) - \mathbf{K}_0 \right\|_2^2 + \lambda p \left\| \mathbf{T}_w \left(\mathbf{H}^{(n)} \right)^{1/2} \right\|_F^2, \quad (8.67b)$$

where ε_n is an iteration-dependent, smoothing parameter whose value is reduced monotonically in each iteration. Although the IRLS- p algorithm has several advantages, its application to MR-CS using low-rank structured matrix completion is highly computation-intensive. This is because the weight matrix update involves computing an inverse power of the Gram matrix $\left(\mathbf{T}_w^{(n-1)} \right)^* \left(\mathbf{T}_w^{(n-1)} \right) \in \mathbb{C}^{w^2 \times w^2}$, which is expensive when the inner dimension $(N_x - w + 1)(N_y - w + 1) \gg w^2$. Instead, a more stable and efficient method to compute inverse is by using an SVD of the $(N_x - w + 1)(N_y - w + 1) \times w^2$ matrix $\mathbf{T}_w$. Due to the low rank of $\mathbf{T}_w$, it is enough to compute the top r singular values and singular vectors either using deterministic methods, such as Lanczos bidiagonalization [42], or randomized methods [43]. Compared to the direct SVD that requires $\mathcal{O}((N_x - w + 1)(N_y - w + 1)w^4)$ flops, computation using the first r components requires only $\mathcal{O}((N_x - w + 1)(N_y - w + 1)w^2 r)$ flops. Note that, even with rank reduction, the computational cost is still linear in the number of rows of $\mathbf{T}_w$, which can be excessively large. In addition, the IRLS- p approach also requires solving the weighted least-squares problem in Equation (8.67b) in each iteration.

8.6.2 Solution Using Extension of Soft Thresholding

The generalization of the soft-thresholding (or shrinkage) operator for any arbitrary p value was proposed by Chartrand [44–46] together with its modification using the generalization of the Huber function [47]. When applied to a vector $\mathbf{z} \in \mathbb{C}^N$, the modified soft-thresholding operator for minimization of the Schatten- p quasi-norm is defined element-wise as:

$$\mathcal{S}_{\alpha}^p(\mathbf{z}) = \left\{ \mathcal{S}_{\alpha}^p(z_n) \right\}_{n=1}^N = \text{sgn}(z_n) \left(|z_n| - \alpha |z_n|^{p-1} \right)_+ \quad (8.68)$$

Following the above definition, the generalized singular value thresholding operator is given by:

$$\mathcal{D}_{\tau}^p(\mathbf{Y}) = \mathbf{U} \mathcal{S}_{\tau}^p(\text{Diag}(\boldsymbol{\Sigma})) \mathbf{V}^T, \quad (8.69)$$

where $\mathbf{U}\Sigma\mathbf{V}^T$ represents the SVD of $\mathbf{Y}$. A simple solution of Equation (8.65) consists of the following update equation:

$$\mathbf{K}^{(n)} = \mathbf{T}_w^\dagger \left(\mathcal{D}_{\rho\tau}^p \left(\mathbf{T}_w \left(\mathbf{K}^{(n-1)} - \rho \left(P_\Omega \left(\mathbf{K}^{(n-1)} \right) - \mathbf{K}_0 \right) \right) \right) \right), \quad (8.70)$$

where $\mathbf{T}_w^\dagger$ denotes the inverse of the lifting operation that maps elements in the matrix $\mathbf{T}_w = \mathbf{T}_w \left(\mathbf{K}^{(n-1)} - \rho \left(P_\Omega \left(\mathbf{K}^{(n-1)} \right) - \mathbf{K}_0 \right) \right)$ after SVT onto the k -space.

8.7 Applications to Dynamic Imaging

The sliding window method is one of the earliest approaches introduced for reconstruction of dynamic MRI data from partial (k, t) -space samples. The most basic version of sliding window is the zeroth-order hold technique [48], which estimates the unacquired k -space samples at a given time frame, with the most recent data at the consecutive time frame from the same k -space location. A more sophisticated technique was put forward by Madore et al. [49]; this technique uses a lattice under-sampling scheme, called unaliasing, by Fourier-encoding the overlaps using the temporal dimension (UNFOLD). The lattice under-sampling scheme introduces aliasing artifacts that can be easily removed by filtering in the temporal Fourier domain.

In the broad-use linear acquisition speed-up technique (k - t BLAST) by Tsao et al. [50], a variable-density sampling scheme that consist of acquired random k -space lines (called an under-sampled dataset) along with samples acquired from the central part of the k -space (called a training dataset) is used to reconstruct the missing values in k - t space. These datasets are Fourier-transformed along the temporal dimension to obtain a low-resolution estimate that contains aliasing artifacts. The training dataset is utilized in k - t BLAST to guide the reconstruction and remove artifacts in the under-sampled dataset. Later, Lustig et al. [51] proposed k - t SPARSE, which uses a sparsifying transform adapted for dynamic MR imaging reconstruction using l_1 -norm penalty.

To take advantage of the computational efficiency using the l_2 -norm penalty, Jung et al. proposed k - t FOCUSS [52–54], whose solution relies on the focal under-determined system solver (FOCUSS) [55]. In this method, an estimation of the low-resolution version of the dynamic MRI signal in the temporal Fourier domain known as the $(y$ - f)-space signal is first made and then a FOCUSS reconstruction is performed to recover the missing samples. Together with the evolution of penalties based on low rank and sparsity, methods involving a combination of both penalties have also gained much attention in the dynamic imaging reconstruction problems. This requires that the underlying matrix $\mathbf{X}$ not possess low rank and sparsity at the same time. In a dynamic imaging sequence consisting of N_t images, each having dimension $N_x \times N_y$, such a matrix can be conceived as one whose columns consist of the vectorized images of the dynamic sequence with dimension $N_x N_y \times 1$. Using the N_t vectorized images in the dynamic imaging sequence, the Casorati matrix is defined as:

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{N_t}] \in \mathbb{C}^{N_x N_y \times N_t}. \quad (8.71)$$

Since this matrix can be represented using the dominant singular values due to the high correlation between images acquired at consecutive time points, it can be approximated as one having low rank. In the partial separability-sparse (PS-sparse) algorithm [56], the Casorati matrix can be expressed using a matrix factorization $\mathbf{X} = \mathbf{UV}$, where $\mathbf{U}$ represents a basis for a spatial sub-space of $\mathbf{X}$, and $\mathbf{V}$ represents a basis for a temporal sub-space of $\mathbf{X}$. Another reconstruction method for dynamic MRI using a low-rank plus sparse approach was proposed by Otazo et al. [57]. Compared to the PS-sparse approach, which considers $\mathbf{X}$ as low rank and sparse, this method considers a decomposition of $\mathbf{X}$ into a linear combination of a low-rank $\mathbf{L}$ and sparse $\mathbf{S}$ components.

The reconstruction model for dynamic imaging based on the low-rank plus sparse ($L + S$) matrix decomposition model is also known as robust principal component analysis (RPCA). In this analysis, the low-rank component models the background, and the sparse component models the dynamic variations [58]. Under specific conditions, the possibility of recovery of low-rank plus sparse components using only a fraction of observations has been well studied [58–64]. For dynamic images, the method using RPCA involves reconstruction of images from partially acquired (k, t) -space [65–67].

8.7.1 RPCA

RPCA, also known as low-rank plus sparse decomposition, decomposes a given matrix $\mathbf{X} \in \mathbb{C}^{N_x \times N_y}$ into its low-rank $\mathbf{L}$ and sparse $\mathbf{S}$ components such that $\mathbf{X} = \mathbf{L} + \mathbf{S}$. Mathematically, this is equivalent to solving:

$$\begin{aligned} \min_{\mathbf{L}, \mathbf{S}} \text{rank}(\mathbf{L}) + \lambda \|\mathbf{S}\|_0 \\ \text{s.t. } \mathbf{L} + \mathbf{S} = \mathbf{X} \end{aligned} \quad (8.72)$$

As the formulation in Equation (8.72) is intractable, as explained earlier in Section 8.2, a more tractable form employs the convex relation for each term, as in the following:

$$\begin{aligned} \min_{\mathbf{L}, \mathbf{S}} \|\mathbf{L}\|_* + \lambda \|\mathbf{S}\|_1 \\ \text{s.t. } \mathbf{L} + \mathbf{S} = \mathbf{X} \end{aligned} \quad (8.73)$$

Note that the solution to Equation (8.73) may not be unique if the matrix $\mathbf{X}$ is both low-rank and sparse at the same time.

8.7.2 Solution Using ADMM

With $\mathbf{Z}$ as the Lagrange multiplier for the linear constraint $\mathbf{L} + \mathbf{S} = \mathbf{X}$, and δ as the penalty parameter, the augmented Lagrange function of Equation (8.73) is expressed as:

$$\begin{aligned} \mathcal{L}_\delta^A(\mathbf{L}, \mathbf{S}, \mathbf{Z}) = \|\mathbf{L}\|_* + \lambda \|\mathbf{S}\|_1 \\ + \langle \mathbf{Z}, \mathbf{L} + \mathbf{S} - \mathbf{X} \rangle + \frac{\delta}{2} \|\mathbf{L} + \mathbf{S} - \mathbf{X}\|_F^2. \end{aligned} \quad (8.74)$$

Using ADMM, the solution updates are obtained as:

$$\mathbf{L}_{k+1} = \arg \min_{\mathbf{L}} \mathcal{L}_d^A(\mathbf{L}, \mathbf{S}_k, \mathbf{Z}_k), \quad (8.75a)$$

$$\mathbf{S}_{k+1} = \arg \min_{\mathbf{S}} \mathcal{L}_s^A(\mathbf{L}_{k+1}, \mathbf{S}, \mathbf{Z}_k), \quad (8.75b)$$

$$\mathbf{Z}_{k+1} = \mathbf{Z}_k + \delta (\mathbf{L}_{k+1} + \mathbf{S}_{k+1} - \mathbf{X}). \quad (8.75c)$$

Usually, the linear and quadratic terms in the augmented Lagrange functions in Equations (8.75a) to (8.75c) are combined to produce the scaled form given by:

$$\mathbf{L}_{k+1} = \arg \min_{\mathbf{L}} \frac{1}{\delta} \|\mathbf{L}\|_* + \frac{1}{2} \|\mathbf{L} - [\mathbf{X} - \mathbf{S}_k - \mathbf{U}_k]\|_F^2, \quad (8.76a)$$

$$\mathbf{S}_{k+1} = \arg \min_{\mathbf{S}} \frac{\lambda}{\delta} \|\mathbf{S}\|_1 + \frac{1}{2} \|\mathbf{S} - [\mathbf{X} - \mathbf{L}_{k+1} - \mathbf{U}_k]\|_F^2, \quad (8.76b)$$

$$\mathbf{U}_{k+1} = \mathbf{U}_k + \mathbf{L}_{k+1} + \mathbf{S}_{k+1} - \mathbf{X}, \quad (8.76c)$$

where $\mathbf{U}_k = \frac{1}{\delta} \mathbf{Z}_k$ is the scaled dual variable. As discussed in Candès et al. [58] and Yuan and Yang [59], the value of δ is fixed to $N_x N_y / 4 \|\mathbf{X}\|_1$, or dynamically updated as suggested by Lin et al. [68]. Since Equations (8.76a) and (8.76b) are the proximal operators of $\|\cdot\|_*$ and $\|\cdot\|_1$, respectively, evaluated at $\mathbf{X} - \mathbf{S}_k - \mathbf{U}_k$ and $\mathbf{X} - \mathbf{L}_{k+1} - \mathbf{U}_k$, the implementation of Equations (8.76a) to (8.76c) can be achieved by SVT and soft-thresholding operators, as described in Algorithm 8.5.

Algorithm 8.5 RPCA

Input: $\mathbf{X}$, $\lambda > 0$.

Initialize: $k = 0$, $\mathbf{S}_0 = \mathbf{U}_0 = 0$, $\delta = \frac{1}{4} \frac{N_x N_y}{\|\mathbf{X}\|_1}$.

Step 1: $\mathbf{L}_{k+1} \leftarrow \mathcal{D}_{(1/\delta)}(\mathbf{X} - \mathbf{S}_k - \mathbf{U}_k)$

Step 2: $\mathbf{S}_{k+1} \leftarrow \mathcal{S}_{(1/\delta)}(\mathbf{X} - \mathbf{S}_k - \mathbf{U}_k)$

Step 3: $\mathbf{U}_{k+1} \leftarrow \mathbf{U}_k + \mathbf{L}_{k+1} + \mathbf{S}_{k+1} - \mathbf{X}$

Step 4: Repeat Steps 1 to 3 until convergence.

Output: $\hat{\mathbf{L}} = \mathbf{L}_k$, $\hat{\mathbf{S}} = \mathbf{S}_k$.

An appropriate choice of the input parameter λ provides the desired separation between the low-rank and sparse components. Based on the dimensions of $\mathbf{X} \in \mathbb{C}^{M \times N}$, Candès et al. [58] have shown the theoretical upper limit for λ to be:

$$\lambda = \max \left(\frac{1}{\sqrt{M}}, \frac{1}{\sqrt{N}} \right). \quad (8.77)$$

Depending on the application, scaling this to a lower value can help adjust the separation between the two components. With RPCA applied to the breath-hold cardiac MRI sequence, the low-rank part is found to appear as a static component (background), while the sparse component captures the dynamic part due to heart beats.

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MATLAB Codes

L.1 Summary of MATLAB Codes

L.1.1 Determination of Cross-over

```
function [Nerr, pert_err, crossover_dist, sig_k, S, Min_sig] = Perturb_
kspace(kspace,nACS, s, b)
% Inputs :
% kspace : input k-space data ( Npe x Nfe x C ) where Npe is the number
% of phase encode lines, Nfe is the number of frequency-encode points, C
% is the number of coils.
% nACS : number of autocalibrating lines
% s : under-sampling factor
% b : number of neighbors for computing the GRAPPA kernel
% Outputs :
% Nerr : residual error
% pert_err : perturbation error
% crossover_dist : Cross-over distance
% sig_k : singular value corresponding to the truncation parameter
% S : minimum singular value of the initially unperturbed calibration matrix
% Min_sig : vector containing minimum singular value at each perturbation
step
```

L.1.2 Retrospective Under-Sampling of Fully Sampled *k*-Space Data

```
function [subkpace, ACS, acqACS, acq] = micsplkspacesubsample (kspace,
nACS, s)
% Inputs :
% kspace : input k-space data ( Npe x Nfe x C ) where Npe is the number
% of phase encode lines, Nfe is the number of frequency-encode points, C
% is the number of coils.
% nACS : number of autocalibrating lines
% s : under-sampling factor
% Outputs :
% subkpace : retrospectively under-sampled data ( Npe x Nfe x C ), where Npe is
% the number of phase encode lines, Nfe is the number of frequency-encode
% points, C is the number of coils, with zero values in the unacquired points
% ACS : autocalibrating signal data (nACSx Nfe x C), fully sampled region
% acqACS : phase encoding indices of lines that follow the acquired
% points in the ACS
% acq : phase encode indices of all the acquired lines in the k-space
```

L.1.3 Calibration Matrix Formation

```
function [PI, PE_index] = getPI (subkspace, s, b, indexing, acqACS)
% Inputs :
% subkspace : retrospectively under-sampled data ( Npe x Nfe x C ), where Npe is
% the number of phase encode lines, Nfe is the number of frequency-encode
% points, C is the number of coils, with zero values in the unacquired points
% s : under-sampling factor
% b : number of neighbors for computing the GRAPPA kernel
% indexing : Calibration matrix indexing is 1 for linear indexing or 2
% for circular indexing
% acqACS : phase encoding indices of lines that follow the acquired
% points in the ACS
% Outputs :
% PI : calibration matrix of size (length(acqACS)-1)*Nfe) x (b* C)
% PE_index : phase-encode indices of the 'b' acquired neighbors for each
k-space point
```

L.1.4 Picard's Value Calculation

```
function [PIC] = picardsval (subkspace, acqACS, C, s)
% Inputs :
% subkspace : retrospectively under-sampled data ( Npe x Nfe x C ), where Npe is
% the number of phase encode lines, Nfe is the number of frequency-encode
% points, C is the number of coils, with zero values in the unacquired points
% acqACS : phase encoding indices of lines that follow the acquired
% points in the ACS
% C : Number of Coils
% s : under-sampling factor
% Outputs :
% PIC : Picard's values stack of size (length(acqACS)-1)*Nfe) x r x C, where
% acqACS is the indices of lines that follow the acquired lines in the
% ACS, Nfe is the number of frequency encode points, r=s-1, where 's' is
% the under-sampling factor and C is the number of coils
```

L.1.5 Coil Coefficient Calculation Using Truncated Singular Value Decomposition (SVD)

```
function [Z, Err, PI, KrC, PE_index]=tsvdestm (subkspace, s, b, indexing,
acqACS, k)
% Inputs :
% subkspace : retrospectively under-sampled data ( Npe x Nfe x C ), where Npe is
% the number of phase encode lines, Nfe is the number of frequency-encode
% points, C is the number of coils, with zero values in the unacquired points
% s : under-sampling factor
% b : Number of neighbors for computing the GRAPPA kernel
% indexing : Calibration matrix indexing is 1 for linear indexing or 2
% for circular indexing
% acqACS : phase encoding indices of lines that follow the acquired
% points in the ACS
% k: truncation parameter
% Outputs :
% Z : GRAPPA coil coefficients of size ((b*C)×1×s-1), where b is the
% number of neighbors for computing the GRAPPA kernel, C is the number of
% coils and s is the under-sampling factor
```

```
% Err : Calibration error of size (length(acqACS-1)*Nfe) × r × C, where
% acqACS is the indices of lines that follow the acquired lines in the
% ACS, Nfe is the number of frequency encode points, r=s-1, where 's' is
% the under-sampling factor and C is the number of coils
% PI : calibration matrix of size (length(acqACS-1)*Nfe) × (b* C)
% PE_index : phase-encode indices of the 'b' acquired neighbors for each
% k-space point
% KrC : the observations of size (length(acqACS-1)*Nfe) × r × C, where
% acqACS is the indices of lines that follow the acquired lines in the
% ACS, Nfe is the number of frequency encode points, r=s-1, where 's' is
% the under-sampling factor and C is the number of coils
```

L.1.6 Frequency-Dependent Regularized Reconstruction

```
function [Kstk, IndMap]=Recon (subspace, Err_bound, Zbank, zgcv, s, b,
PE_index, acq)
% Inputs :
% subspace : retrospectively under-sampled data ( Npe × Nfe × C ), where Npe is
% the number of phase encode lines, Nfe is the number of frequency-encode
% points, C is the number of coils, with zero values in the unacquired points
% Err_bound : error bound measure of size ( Npe × Nfe × C ) obtained by
% calculating the differences in the reconstructed k-spaces at the cross-over
% and the initial unperturbed data
% Zbank : Bank of GRAPPA coil coefficients of size ((b*C)×1×s-1×N), where b is the
% number of neighbors for computing the GRAPPA kernel, C is the number of
% coils and s is the under-sampling factor, formed from each  $\lambda_i$  in the range
% [ $\lambda_{gcv}$ ,  $\lambda_L$ ]
% zgcv : GRAPPA coil coefficients of size ((b*C)×1×s-1), where b is the number
% of neighbors for computing the GRAPPA kernel, C is the number of coils and
% s is the under-sampling factor formed from regularization parameter
% calculated using GCV
% s : under-sampling factor
% b : No. of neighbors for computing the GRAPPA kernel
% PE_index : phase-encode indices of the 'b' acquired neighbors for each
% k-space point
% acq : phase encode indices of all the acquired lines in the k-space
% Outputs :
% Kstk : reconstructed k-space obtained from the frequency dependent
% regularization data ( Npe × Nfe × C ), where Npe is the number of phase
% encode lines, Nfe is the number of frequency-encode points, C is the
% number of coils
% IndMap : frequency dependent regularization parameter map ( Nuacq × Nfe × C ),
% where Nuacq is the number of phase encode lines which are not acquired,
% Nfe is the number of frequency-encode points, C is the number of coils
```

L.1.7 Filter Coefficient Bank Formation

```
function [Zbank] = Get_filters (subspace, s, b,PI)
% Inputs :
% subspace : retrospectively under-sampled data ( Npe × Nfe × C ), where Npe is
% the number of phase encode lines, Nfe is the number of frequency-encode
% points, C is the number of coils, with zero values in the unacquired points
% s : under-sampling factor
% b : Number of neighbors for computing the GRAPPA kernel
```

```
% PI: calibration matrix
% Outputs :
% Zbank : Bank of GRAPPA coil coefficients of size ((b*C)×1×s-1×N), where b is
% the number of neighbors for computing the GRAPPA kernel, C is the number of
% coils and s is the under-sampling factor, formed from each  $\lambda_i$  in the range
% [ $\lambda_{\text{gcv}}$ ,  $\lambda_{\text{L}}$ ]
```

L.1.8 Single-Channel Wavelet-Based Threshold Adaptation

```
function [Irec, E, Tim, Th, delt] = alph_delt_UpdteVecNom (I, lambda,
alpha, Nit, mask,XFM,FT)
% Inputs :
% I : Input fully sampled data
% lambda : regularization parameter
% alpha : step size, typically set as 1
% Nit : number of iterations
% mask : Sampling mask
% XFM : Wavelet operator
% FT : under-sampling Fourier operator
% Outputs :
% Irec: reconstructed image
% E : relative  $l_2$ -norm error (RLNE)
% Tim : computation time in seconds
% Th : iteration dependent regularization parameter or threshold
% delt : discrepancy level
```

L.1.9 Soft Thresholding

```
function y = soft(x,T)
% Inputs
% x : input to be thresholded
% T : threshold
% Output
% y: thresholded output
```

L.1.10 Relative l_2 -Norm Error (RLNE) Computation

```
function [res]=RLNE(I1,I2)
% Inputs
% I1 : reference image
% I2 : reconstructed image
% Output
% res: RLNE value
```

L.1.11 Parallel Magnetic Resonance Imaging (pMRI) Reconstruction with Update Calculation Using Error Information from Combined Image (Method I)

```
function [Irecon, E, Th, Tim] = Fista_
ImSoSMul (Istk, T, alpha, Nit, mask, XFM, FT)
% Inputs
% Istk : multi-channel image
% T : regularization parameter or threshold
```

```

% alpha : step size
% Nit : number of iterations
% mask : under-sampling mask
% XFM : Wavelet transform operator
% FT: Fourier transform operator
% Outputs
% Irecon : reconstructed SoS image
% E : Reconstruction error computed using RLNE
% Th : Threshold
% Tim : reconstruction time

```

L.1.12 Parallel Magnetic Resonance Imaging (pMRI) Reconstruction with Update Calculation Using Sum-of-Squares (SoS) of Channel-Wise Errors (Method II)

```

function [Irecon, E, Tim, Th] = Fista_SosMul (Kz,T, Nit,mask, XFM, FT)
% Inputs
% Kz : multi-channel k-space data
% T : initial regularization parameter
% Nit : number of iterations
% mask : under-sampling mask
% XFM : Wavelet transform operator
% FT : Fourier transform operator
% Outputs
% Irecon : reconstructed image from all channels
% E : reconstruction error computed using RLNE
% Tim : computation time
% Th : Threshold values in all iterations

```

L.1.13 Parallel Magnetic Resonance Imaging (pMRI) Reconstruction with Update Calculation Using Covariance Matrix (Method III)

```

function [Irecon, E, Tim, Th, delt, ll_res, ll_pert] = Fista_
CovMul(Kz,T,Nit,mask,XFM, FT)
% Inputs
% Kz : multi-channel k-space data
% T : initial regularization parameter
% Nit : number of iterations
% mask : under-sampling mask
% Outputs
% Irecon : reconstructed image from all channels
% Th : Threshold values in all iterations

```

L.1.14 Basic Codes for Adaptive Total Variation Iterative Shrinkage (ATVIS)

L.1.14.1 Derivative

```

function [res] = derivative(image)
% Input
% image : input image or matrix
%Output
% res : derivative of the input image

```


L.1.14.2 Divergence

```
function [res] = divergence(dI)
%Input
% dI : derivative of an image
%Output
% res : output image
```

L.1.14.3 Initial Threshold Computation for TVIS

```
function [T]=find_initThresh(I)
% Input
% I : image
% Output
% T : threshold
```

L.1.14.4 Initial Threshold Computation for Adaptive Total Variation Iterative Shrinkage (ATVIS)

```
function [T]=thresh_test(I)
% Input
% I : input image of size N x N
% Output
% T : Threshold
```

L.1.14.5 Gradient to Image Operator (Anisotropic)

```
function [I]=Grad2Im(dI)
% Input
% dI : derivative of an image
% Output
% I : image
```

L.1.15 Comparison of Single-Channel Total Variation Iterative Shrinkage (TVIS) and Adaptive Total Variation Iterative Shrinkage (ATVIS (Anisotropic))

```
function [Output]=Demo_Single (Iref,Nit, mask,tol)
% Inputs
% Iref : Reference Image of size N x N
% Nit : number of iterations
% mask : Sampling Mask
% tol : tolerance value typically 1e-4

% Outputs
% Irec : TVIS reconstructed Image
% E : RLNE for TVIS
% Tim : Reconstruction time for TVIS
% IrecA : ATVIS reconstructed Image
% EA : RLNE for ATVIS
% TimA : Reconstruction time for ATVIS
% ThA : threshold for ATVIS
```

L.1.15.1 Single-Channel Total Variation Iterative Shrinkage (TVIS) Reconstruction

```
function [Irec, E, Tim, Th] = TV_ISTAreDdemo(I,T,Nit,mask, tol,FT)
% Inputs
% I      : Reference Image
% T      : regularization parameter/threshold
% Nit    : number of iterations
% mask   : Sampling Mask
% tol    : tolerance value, typically 1e-4
% FT     : Fourier transform operator
% Outputs
% Irec   : reconstructed image
% E      : RLNE
% Tim    : CPU time for reconstruction
% Th     : Adaptive Threshold
```

L.1.15.2 Single-Channel Adaptive Total Variation Iterative Shrinkage (ATVIS) Reconstruction

```
function [Irec, E, Tim, Th] = TV_alph_updte_rep_
Demo(I, T, Nit, mask, tol,FT)
% Inputs
% I      : Reference Image
% T      : regularization parameter/ threshold
% Nit    : number of iterations
% mask   : Sampling Mask
% tol    : tolerance value typically 1e-4
% FT     : Fourier transform operator
% Outputs
% Irec   : reconstructed image
% E      : RLNE
% Tim    : CPU time for reconstruction
% Th     : Adaptive Threshold
```

L.1.16 Comparison of Multi-channel Total Variation Iterative Shrinkage (TVIS) and Adaptive Total Variation Iterative Shrinkage (ATVIS)

```
function[Output]=Demo_Parallel(Iref, Nit, mask,FT,tol)
% Inputs
% Iref   : Reference Image of size N x N
% Nit    : number of iterations
% mask   : Sampling Mask
% FT     : Fourier transform operator
% tol    : tolerance value, typically 1e-4
% Outputs
% Irec   : TVIS reconstructed Image
% E      : RLNE for TVIS
% Tim    : Reconstruction time for TVIS
% IrecA  : ATVIS reconstructed Image
% EA     : RLNE for ATVIS
% TimA   : Reconstruction time for ATVIS
% ThA    : threshold for ATVIS
```

L.1.16.1 Multi-channel Total Variation Iterative Shrinkage (TVIS) Reconstruction

```
function [Irec, E, Tim] = TV_ISTArep_multi_Demo(I, T, Nit, mask, tol, FT)
% Inputs
% I      : Reference Image
% T      : regularization parameter/ threshold
% Nit    : number of iterations
% mask   : Sampling Mask
% tol    : tolerance value, typically 1e-4
% FT     : Fourier transform operator
% Outputs
% Irec   : reconstructed image
% E      : RLNE
% Tim    : CPU time for reconstruction
```

L.1.16.2 Multi-channel Adaptive Total Variation Iterative Shrinkage (ATVIS) Reconstruction

```
function [Irec, E, Tim, Th] = TV_alph_updte_rep_Multi_
Demo (I, T, Nit, mask, tol, FT)
% Inputs
% I      : Reference Image
% T      : regularization parameter/ threshold
% Nit    : number of iterations
% mask   : Sampling Mask
% tol    : tolerance value, typically 1e-4
% FT     : Fourier transform operator
% Outputs
% Irec   : reconstructed image
% E      : RLNE
% Tim    : CPU time for reconstruction
% Th     : Adaptive Threshold
```

L.1.17 Comparison of Single-Channel Total Variation Iterative Shrinkage (TVIS) and Adaptive Total Variation Iterative Shrinkage (ATVIS) (Isotropic)

```
function[Output]=Demo_iso( Iref, Nit, mask, FT, tol, L )
% Inputs
% Iref   : Reference Image of size N × N
% Nit    : number of iterations
% mask   : Sampling Mask
% FT     : Fourier transform operator
% tol    : tolerance value, typically 1e-4
% L      : typically 3 or 5 is chosen
% Outputs
% Irec   : TVIS reconstructed Image
% E      : RLNE for TVIS
% Tim    : Reconstruction time for TVIS
% IrecA  : ATVIS reconstructed Image
% EA     : RLNE for ATVIS
% TimA   : Reconstruction time for ATVIS
% ThA    : threshold for ATVIS
```

L.1.17.1 Single-Channel Total Variation Iterative Shrinkage (TVIS)

```
function [Irec, E, Tim] = TVIS(I, T, Nit, mask, tol, L, FT)
% Inputs
% I      : Reference Image
% T      : regularization parameter/ threshold
% Nit    : number of iterations
% mask   : Sampling Mask
% tol    : tolerance value typically 1e-4
% L      : typically 3 or 5 is chosen
% FT     : Fourier transform operator
% Outputs
% Irec   : reconstructed image
% E      : RLNE
% Tim    : CPU time for reconstruction
% Th     : Adaptive Threshold
```

L.1.17.2 Single-Channel Adaptive Total Variation Iterative Shrinkage (ATVIS)

```
function [Irec, E, Tim, Th] = ATVIS(I, T, Nit, mask, tol, L, FT)
% Inputs
% I      : Reference Image
% T      : initial regularization parameter/ threshold
% Nit    : number of iterations
% mask   : Sampling Mask
% tol    : tolerance value typically 1e-4
% L      : typically chosen as 3 or 5
% FT     : Fourier transform operator
% Outputs
% Irec   : reconstructed image
% E      : RLNE
% Tim    : CPU time for reconstruction
% Th     : Adaptive Threshold
```

L.1.17.3 Gradient to Image Operator

```
function [I] = Grad2Im_iso(dI, L)
% Inputs
% dI    : gradient image
% L     : typically chosen as 3 or 5
% Output
% I     : image
```

L.1.17.4 Multi-directional Gradient Operator

```
function [v] = multi_dir_grad(u, L)
% Inputs
% u     : input image
% L     : typically chosen as 3 or 5
% Output
% v     : gradient image
```

L.1.17.5 Multi-directional Divergence Operator

```
function[u]=multi_dir_divergence(v,L)
% Inputs
% v : gradient image of size NxMx2L
% L : typically chosen as 3 or 5
% Output
% u :image of size NxM
```

L.1.18 Code for Adaptive L1-SPIRiT

```
function [I_recon, E, Th]=main_thadapt_spirit_
JS(DATA, kSize, nIter, mask, T)
% Inputs
% DATA      : multi-channel k-space data
% kSize      : SPIRiT kernel size
% nIter      : number of iteration; phantom requires twice as much as the brain.
% mask       : mask can be uniform or random
% lambda     : Tikhonov regularization in the calibration
% T          : Soft-thresholding parameter
% Outputs
% I_recon    : Reconstructed Image (SoS combined)
% E          : Reconstruction Error
```

L.1.18.1 Generate Under-Sampled Data Using Random Mask

```
function[DATA, CalibSize]=get_SampledData(DATA,mask)
% Inputs
% DATA      : multi-channel k-space data
% mask       : under-sampling mask
% Outputs
% DATA      : under-sampled k-space data
% CalibSize  : size of calibration data
```

L.1.18.2 Computation of SPIRiT Filter in the Image Domain

```
function[Z, z]=Imfilt_Calib(DATA,CalibSize,kSize,lambda)
%Inputs
% DATA      : Under-sampled multi-channel data
% CalibSize  : size of calibration data
% kSize      : SPIRiT kernel size
% lambda     : Tikhonov regularization in the calibration
% Outputs
% Z          : calibration filter in the image domain
% z          : calibration filter in the k-space
```

L.1.18.3 Code for Adaptive SPIRiT Reconstruction

```
function [K, E, Th] = TVthadapt_JS_A (Kz, Z, nIter, T, im, tol)
% Inputs
% Kz       : Under-sampled k-space data.
% Z        : SPIRiT image domain filter
% nIter    : Maximum number of iterations
```

```
% T      : thresholding parameter
% im     : combined reference image
% tol    : tolerance value, typically 1e-4
% Outputs
% K      : Full k-space matrix
```

L.2 Detailed MATLAB Codes

L.2.1 Determination of Cross-over

```
function [Nerr, pert_err, crossover_dist, sig_k, S, Min_sig] = Perturb_kspace
% (kspace,nACS, s,b) retrospectively under-sample fully acquired Data
[subkspace, ACS, acqACS, acq] = micsplkspaceSubsample( kspace, nACS, s);
% calibration matrix (PI) formation
PI = getPI (subkspace, s, b, indexing, acqACS);
% SVD Calculation
[U S V] = svd(PI);
S = diag(S);
% Picard' s value calculation
PIC = picardsval ( subkspace, acqACS, C, s);
% Enter the truncation parameter as indicated by Picard's plot
k = input('enter k');
% reference for perturbation ||E||<sig_k
sig_k = S(k);
% variance within range 0 to one tenth of minimum singular value
el = linspace(0,.1*min(S),50);
% Coil coefficient estimation using truncated SVD
[Z, Err, PIn, KrC, PE_index] = tsvdestm (subkspace, s, b, indexing, acqACS, k);
CHI = PI'*PI;
flag=0;
% Perturbing k-space by incremental noise addition
for i =1:length(el)-1,
subkspace_n = subkspace + (el(i+1) *randn(size(subkspace)) + 1i* el(i+1)
*randn(size(subkspace)));
% calculation of Coil coefficients using tsvd,
% truncation parameter
[Z, Err, PIn, KrCn, PE_index] = tsvdestm (subkspace_n, s, b, indexing, acqACS, k);
CHIn = PIn'*PIn;
if flag==0,
    vec = get_ps(CHI,CHIn);
    flag = 1;
end
% Calculating the amount of perturbation
En = PI - PIn;
% updating the singular values using singular value update
sig = sig+(vec*norm(En));
% minimum singular value after each perturbation saved to an array
Min_sig(i) = min(diag(sn));
% residual error in one channel, say channel no.2
Nerr(i) = norm((Err(:,1,2)));
% Calculation of perturbation error, (error in the observation)
First_term = norm((KrC(:,1,2)-KrCn(:,1,2))); % norm of observation error
```

```

    % Calculation of error in the matrix
    [uE sE vE] = svd(En);
    Second_term = max(diag(sE))*norm(Z(:,1,2));
% Total perturbation error
pert_err(i) = First_term + Second_term;
% Terminate perturbation either when the norm of perturbation is greater than
% kth singular value or when the perturbation error exceeds the residual error
if (pert_err(i) > sig_k) || Nerr(i)<pert_err(i)),
    disp( 'Cross Over reached' );
    break;
end
end
% Cross-over distance Calculation
crossover_dist = (Min_sig(end)-min(S))/min(S);

```

L.2.2 Retrospective Under-Sampling of Fully Sampled k -Space Data

```

function [subkspace, ACS, acqACS, acq] = micsplkspacesubsample (kspace,
nACS, s)
[Npe,Nfe,C]=size(kspace);
% The index for ACS lines in the PE direction
ACS = transpose([floor(Npe/2 - nACS/2 + 1):floor(Npe / 2 + nACS/2)]);
% The index for subsampled PE lines in the row direction
acq = 1:s:Npe;
if ismember(ACS(1) - 1, acq)
    ACS = union(ACS(1) - 1, ACS);
end
if ismember(ACS(end) + 1, acq)
    ACS = union(ACS, ACS(end) + 1);
end
% The index of neighbouring points within ACS lines.
acqACS = intersect(acq, ACS);
% The index of subsampled PE and ACS lines
acq = union(acq, ACS);
% Subsampling k-space data along row-direction. ACS lines are included in
% the subsampled data. The size of PE lines is incremented by one in order
% to reconstruct data in the final block
subkspace = zeros(Npe, Nfe, C);
for index = 1: C
    subkspace(acq, :, index) = kspace(acq, :, index);
    mask(acq, :, index)=1;
end
return;

```

L.2.3 Calibration Matrix Formation

```

function [PI, PE_index] = getPI (subkspace, s, b, indexing, acqACS)
[Npe,Nfe,C] = size(subkspace);
% Acquired neighbor indices for each k-space row
PE_index = twoacqneighbourindex (s, Npe); %size of PE_index will be Npe x 2
lim=( (b/2)-1 ) /2;
subkspacepad =padarray (subkspace, [lim,lim]);
subkspacepad = subkspacepad(lim+1:end-lim,:,:);
[P, Q, C] =size(subkspacepad);

```



```

PI=[];
for i=lim+1:Q-lim,
    for j=acqACS(1:end-1),
        temp=subspacepad (PE_index(j,:) ,i-lim:i+lim,:);
        [x,y,z]=size (temp);
        PI = [PI; reshape(temp, 1, x*y*z)];
    end
end

```

L.2.4 Picard's Value Calculation

```

function [PIC] = picardsval (subspace, acqACS, C, s)
indexing=1;
[PI, PE_index] = getPI (subspace, s, b, indexing, acqACS);
[U,S,V]=svd(PI);
for c=1:C,
    for r = 1:s - 1,
        krindex=r + (acqACS(1):s:acqACS(length(acqACS) - 1));
        Kr = subspace(krindex,:, c);
        %Kr is a matrix with rows corresponding to the k-space
        %values of unacquired lines within ACS
        [x y z] = size(Kr);
        Kr = reshape(Kr, 1, x*y*z);
        PIC(r,:,c) = picard(U,S,transpose(Kr)); % picard is a sub function from Hansen's
        %Regularization toolbox
    end
end

```

L.2.5 Coil Coefficient Calculation Using Truncated Singular-Value Decomposition (SVD)

```

function [Z, Err, PI, KrC, PE_index]=tsvdestm (subspace, s, b, indexing,
acqACS, k)
[Npe,Nfe,C]=size(subspace);
% Acquired neighbour indices for each k-space row
PE_index = acqneighbours (s, b, Npe); %size of PE_index will be Npe x b
lim = ( (b/2)-1 ) /2;
subspacepad = padarray (subspace ,[lim,lim]);
subspacepad = subspacepad (lim+1:end-lim,:,:)
[P, Q, C]=size (subspacepad);
Z=[]; % Z is a coil coefficient stack having C slices. Each slice is of
size (b*C) X (s-1)
[PI, PE_index] = getPI (subspace, s, b, indexing, acqACS);
[U S V]=svd(PI);
sig=diag(S);
for c = 1:C
    % disp(['coil coefficient ' num2str(c)]);
    for r = 1:s - 1,
        krindex=(r + [acqACS(1):s:acqACS(length(acqACS) - 1)]);
        Kr = subspace(krindex,:, c);
        %Kr is a matrix with rows corresponding to the k-space
        %values of unacquired lines within ACS
        [x y z] = size(Kr);
        Kr = reshape(Kr, 1, x*y*z);
    end
end

```

```

    [z] = tsvd(U,sig,V,transpose(Kr),len_sig); % tsvd is a sub function
    from Hansen's
    %Regularization toolbox
    Z(:,r,c) = z;
    KrC(:,r,c)=Kr;
    KrhatC(:,r,c)=PI*z;
    end
end
Err=KrC-KrhatC;

```

L.2.6 Frequency-Dependent Regularization Reconstruction

```

function [Kstk, IndMap]=Recon (subkspace, Err_
bound, Zbank, zgcv, s, b, PE_index, acq)
[Npe,Nfe,C]=size(subkspace);
Npe=Npe-1;
lim=( (b/2)-1 ) /2;
subkspacepad = padarray (subkspace, [lim,lim]);
subkspacepad = subkspacepad(lim+1:end-lim,:,:)
[P,Q,C] = size(subkspacepad);
Errstk = [];
Indexmap = zeros(Npe, Nfe, C);
for r=1:s-1,
    for c=1:C,
        zg=zgcv(:,r,c);
        for p=1:size(Z,4),
            delz(:,r,c,p)=Zbank(:,r,c,p)-zg;
        end
    end
end
for r = 1:s - 1,
    for i = lim+1: Q-lim,
        for j = 1:s:Npe-s+1,
            if ismember(j + r, acq) < 1
                Kqprime = subkspacepad(PE_index(r + j, :), i-lim:i+lim, :);
                [x y z] = size(Kqprime );
                Kqprime = reshape(Kqprime, 1, x*y*z);
                for c = 1:C,
                    err_cv=Err_bound(j+r,i-lim,c);
                    for p=1:size(Z,4),
                        z=delz(:,r,c,p);
                        krprime=Kqprime * z;
                        err_vec(p)=krprime;
                    end
                ind=find(abs(err_vec)<=sc*abs(err_cv));
                if isempty(ind),
                    Ind=1;
                else
                    Ind=ind(end);
                end
                z1=Z(:,r,c,Ind);
                kprime=Kqprime * z1;
                subkspace(j+r,i-lim,c)=kprime;
                Indexmap(j + r, i-lim, c)=Ind;
            end
        end
    end
end

```

```

        end
    end
end
end
end
n1=1:Npe;
uacq=setdiff(n1,acq);
IndMap=Indexmap(uacq, :, :);
Kstk(1:Npe, :, :)=subspace(1:Npe, :, :);

```

L.2.7 Filter Coefficient Bank Formation

```

function [Zbank] = Get_filters (subspace, s, b)
[U S V]=svd(PI);
for c = 1:C,
    for r = 1:s - 1,
        Kr=KrC(:,r,c);
        reg_corner = l_curve(U,diag(S),Kr); % l_curve is a sub function from
% Hansen's Regularization toolbox
        reg_min = gcv(U,diag(S),Kr,'Tikh'); % gcv is a sub function from
% Hansen's %Regularization toolbox
        Reg_gcv(:,r,c)=reg_min;
        Reg_L(:,r,c)=reg_corner;
        reg_range=linspace(reg1,reg2,25);
        for p=1:length(reg_range),
            z = tikhonov(U,diag(S),V,KrC(:,r,c),reg_range(p)); % tikhonov is a sub
% function from %Hansen's Regularization toolbox
        Zbank(:,r,c,p)=z;
    end
end
end
end

```

L.2.8 Single-Channel Wavelet-Based Threshold Adaptation

```

[Irec, E, Tim, Th, delt,l1_wres,l1_wpert,xnorm,mu_res,mu_pert,mval,rval,
ft,st,Em_con] = alph_delt_UpdteVecNom(I, lambda, alpha, Nit, mask)
Kz = fft2c(I).*mask; %Under-sampled k-space
Iz = ifft2c(Kz);
E = []; %Reconstruction error in k-space
Kref = fft2c(I);
x = zeros(size(I)); %Initializing Wavelet coefficients
T = lambda; %Threshold in wavelet domain
xz = XFM*Iz;
h=waitbar(0,'Iterations..');
mu_pert = []; %Mean of Wavelet modulus perturbation error
mu_res = []; %Mean of Wavelet modulus residual error
Xstk = []; %Stack of Denoised Wavelet coeffs x(k)
delt(1) = 0;
time_start=cputime;
for k = 1:Nit

    waitbar(k/(Nit),h,sprintf('Iterations... %d/%d',k,Nit))
    Eres = Kz - FT*(XFM'*x); %Data consistency error where x
denotes x(k-1)

```

```

wres          = XFM*(FT'*(Eres));      %Wavelet domain consistency error
xhat          = x + wres;              % Landweber update step
x             = soft(xhat/alpha, T);   % soft thresholding
wpert         = xhat - x;              % wavelet perturbation error
l1_wpert(k)   = norm(wpert(:),1)/256; % scaled  $l_1$ - norm of the sparse
approximation error
l1_wres(k)     = norm(wres(:),1)/256; % scaled  $l_1$ - norm of the consistency error
[hres, bin_res] = hist(abs(wres(:)),1001); % Histogram of modulus of
consistency error
hres          = hres/sum(hres); % Normalization

mu_res(k)     = sum(bin_res.*hres); % Mean (Expectation) of
consistency error

hpert         = histc(abs(wpert(:)),bin_res); % Histogram of modulus
of sparse approximation % error
hpert         = hpert/sum(hpert); % Normalization
mu_pert(k)    = sum(bin_res.*hpert'); % Mean (Expectation) of consistency
error
Im            = XFM'*x;
if(k>1),
    delt(k) = abs(l1_wres(k)-l1_wpert(k-1)); %discrepancy level
    T       = mu_res(k)/(delt(k)+(mu_pert(k-1)/Th(k-1)));
    %Threshold update
    Em_con(k) = norm(Im-Im_old)/norm(Im_old);
end
Th(k)        = T;
Im_old       = Im;
E(k)         = RLNE (ifft2c(Kref), (XFM'*x));
Tim(k)       = cputime-time_start;
end
Irec = XFM'*x;
delete(h)

```

L.2.9 Soft Thresholding

```

function y = soft(x,T)
y = max(abs(x) - T, 0);
y = y./(y+T) .* x;

```

L.2.10 Relative l_2 -Norm Error (RLNE) Computation

```

function [res]=RLNE(I1,I2)
L2_error=I1-I2;
res=norm(L2_error(:),2)/norm(I1(:),2);

```

L.2.11 Parallel Magnetic Resonance Imaging (pMRI) Reconstruction with Update Calculation Using Error Information from Combined Image (Method I)

```

function [Irecon, E, Th, Tim] = Fista_ImSoSMul(Istk, T, alpha, Nit, mask,
XFM, FT)
[P Q R] = size(Istk); % size of the multi-channel image
Iorg    = zeros(P,Q); % initialization

```

```

Icom      = zeros(P,Q);    % initialization
E         = [ ];          % initialization
for i=1:R,
    Iorg   = Iorg + abs(Istk(:,:,i)) .^ 2;
    Kz(:,:,i) = fft2c(Istk(:,:,i)).*mask;
    Iz(:,:,i) = ifft2c(Kz(:,:,i));
    Icom    = Icom + abs(Iz(:,:,i)) .^ 2;
end
Kcom = fft2c(Icom);
xc    = zeros(P,Q);
xoc   = zeros(P,Q);
x      = zeros(size(Kz)); % Initializing Wavelet coefficients
xo     = zeros(size(Kz));
h      = waitbar(0,'Iterations..');
t       = ones(1,R);
tc     = 1;
time_start = cputime;

for k = 1:Nit

    waitbar (k/(Nit), h, sprintf('Iterations... %d/%d', k, Nit))
    Th(k) = T;
    Eres   = Kcom - FT*(XFM'*x); % Data consistency error
    wres    = XFM*(FT*(Eres)); % Data consistency error in Wavelet
domain
    xhat    = xc + wres; % Landweber update step
    xc      = soft(xhat/alpha, T); % Soft Thresholding
    wpert   = xhat - xc; % Sparse approximation error in
wavelet domain
    tn      = (1+sqrt(1+4*(t^2)))/2; % FISTA step
    xn      = x + ((t-1)/tn)*(x-xoc); % FISTA step
%-----Expectations of errors-----
    [hres, bin_res] = hist(abs(wres(:)),1001); %Histogram of consistency
error in wavelet domain
    hres      = hres/sum(hres); % normalization
mu_res      = sum(bin_res.*hres); % expectation of
consistency error
st_res      = sqrt(sum(hres.*((bin_res-mu_res).^2))); % standard
deviation of consistency error
    if (k>1)
        hpert      = histc(abs(wpert_p(:)),bin_res); %Histogram of sparse
approximation error in wavelet domain
        hpert      = hpert/sum(hpert); % normalization
mu_pert     = sum(bin_res.*hpert'); % expectation of sparse
approximation error
st_pert     = sqrt(sum(hpert'.*((bin_res-mu_pert).^2))); %standard
deviation of sparse approximation error
    llres(k) = norm(wres,1); %  $l_1$ -norm of
consistency error
    llpert(k) = norm(wpert,1); %  $l_1$ -norm of sparse
approximation error
    delt      = abs(llres(k)-llpert(k-1)); % discrepancy level
T            = sqrt(st_res^2/(delt+(st_pert^2/(T^2)))); % Threshold
update step

```

```

end
Irecon = zeros(P,Q);
for c = 1:R,
    [x(:,:,c), xo(:,:,c), t(k+1,c)] = Recon(Kz(:,:,c), FT, XFM, x(:,:,c),
xo(:,:,c), T, t(k,c));
    Ic = XFM'*x(:,:,c);
    Irecon = Irecon + abs(Ic) .^ 2;
end
wpert_p = wpert;
E(k) = RLNE(Iorg, Irecon);
Tim(k) = cputime-time_start;
end
delete(h)
%-----Calling function-----
function [xn, x, t] = Recon (Kz, FT, XFM, x, xo, T, t)

Eres = Kz - FT*(XFM'*x); % Data consistency error where x
denotes x(k-1)
wres = XFM*(FT'*(Eres)); % Wavelet residual error
xhat = x + wres; % xhat(k) = x(k-1) + wres(k)
x = soft(xhat, T); %x(k) % soft thresholding
tn = (1+sqrt(1+4*(t^2)))/2; % FISTA steps
xn = x + ((t-1)/tn)*(x-xo);

```

L.2.12 Parallel Magnetic Resonance Imaging (pMRI) Reconstruction with Update Calculation Using Sum-of-Squares (SoS) of Channel-Wise Errors (Method II)

```

function [Irecon, E, Tim, Th] = Fista_SosMul(Kz, T, Nit, mask, XFM, FT)
Istk = ifft2c(Kz);
[P Q R] = size(Istk);
Iorg = zeros(P,Q);
E = []; % Reconstruction error in k-space (RLNE initialization)
for i = 1:R,
    Iorg = Iorg + abs(Istk(:,:,i)) .^ 2;
    Kz(:,:,i) = fft2c(Istk(:,:,i)).*mask;
    Iz(:,:,i) = ifft2c(Kz(:,:,i));
end
x = zeros(size(Kz)); %Initializing Wavelet coefficients
xo = zeros(size(Kz));
h = waitbar(0, 'Iterations..');
t = ones(1,R);
xmat = [];
time_start = cputime;
for k = 1:Nit
    waitbar(k/(Nit), h, sprintf('Finding Histogram.. %d/%d', k, Nit))
    Th(k) = T;
    for c = 1:R,
        [x(:,:,c), xo(:,:,c), wres(:,:,c), wpert(:,:,c), t(k+1,c)] = Err_Calc
K z(:,:,c), FT, XFM, x(:,:,c), xo(:,:,c), alpha, T, t(k,c));
    end
    if (k>1),
        [T, ll_res(k), ll_pert(k), delt(k)] = Update_Thresh(wres, wpert_pre, T);
    end
end

```

```

end
wpert_pre = wpert;
%-----Obtaining combined image-----
Irecon=zeros(P,Q);
for r=1:R,
    Ic = XFM'*x(:,:,r);
    Irecon = Irecon + abs(Ic) .^ 2;
end
E(k)          = RLNE(Iorg,Irecon);
Tim(k)        = cputime-time_start;
end
delete(h)

%----- calling function -----
function [ xn, x, wres, wpert,tn]= Err_Calc(Kz,FT,XFM,x,xo,alpha,T,t)
Eres      = Kz - FT*(XFM'*x);    %Data consistency error where x denotes x(k-1)
wres      = XFM*(FT'*(Eres));    %Wavelet residual error
xhat      = x + wres;            %xhat(k) = x(k-1) + wres(k)
x         = soft(xhat/alpha, T); %x(k) =soft(xhat(k),T) --> Denoised image
wpert     = xhat - x;            %wavelet perturbation error
tn        = (1+sqrt(1+4*(t^2)))/2;
xn        = x + ((t-1)/tn)*(x-xo);
%-----calling function-----
function[T,llres,llpert,delt] = Update_Thresh(wres,wpert,T)
wres_sq   = wres.^2;
wpert_sq  = wpert.^2;
wres      = sum(wres_sq,3);
wpert     = sum(wpert_sq,3);
[hres, bin_res] = hist(abs(wres(:)),1001); %Hist of Wavelet modulus
residual error
hres      = hres/sum(hres);
mu_res    = sum(bin_res.*hres);
st_res    = sqrt(sum(hres.*((bin_res-mu_res).^2)));
hpert     = histc(abs(wpert(:)),bin_res); %Hist of Wavelet modulus
perturbation error
hpert     = hpert/sum(hpert);
mu_pert   = sum(bin_res.*hpert');
st_pert   = sqrt(sum(hpert'.*((bin_res-mu_pert).^2)));
llres     = norm(wres,1);
llpert    = norm(wpert,1);
delt      = abs(llres-llpert);
T         = sqrt(st_res^2/(delt+(st_pert^2/(T^2))));

```

L.2.13 Parallel Magnetic Resonance Imaging (pMRI) Reconstruction with Update Calculation Using Covariance Matrix (Method III)

```

function [Irecon, E, Tim, Th, delt, ll_res, ll_pert] = Fista_
CovMul(Kz,T,Nit,mask,XFM, FT)
Istk      = ifft2c(Kz);
[P Q R]   = size(Istk);
Iorg      = zeros(P,Q);
E         = []; % Reconstruction error in k-space (RLNE) initialization
for i=1:R,
    Iorg   = Iorg + abs(Istk(:,:,i)) .^ 2;

```



```

Kz(:,:,i) = fft2c(Istk(:,:,i)).*mask;
Iz(:,:,i) = ifft2c(Kz(:,:,i));
end

x = zeros(size(Kz)); %Initializing Wavelet coefficients
xo = zeros(size(Kz));
h=waitbar(0,'Iterations..');
t=ones(1,R); % initialize FISTA constant
xmat=[];
time_start = cputime; % CPU time
for k = 1:Nit
    waitbar(k/(Nit),h,sprintf('Finding Histogram.. %d/%d',k,Nit))
    Th(k) = T;
    for c=1:R,
        [x(:,:,c), xo(:,:,c), wres(:,:,c), wpert(:,:,c),t(k+1,c)]= Err_
Calc (Kz(:,:,c), FT, XFM, x(:,:,c), xo(:,:,c), T, t(k,c));
    end
    if(k>1),
        [T,ll_res(k),ll_pert(k),delt(k)] = Update_Thresh(wres,wpert_
pre,T);
    end
    wpert_pre = wpert;
    Irecon=zeros(P,Q);
    for r=1:R,
        Ic = XFM'*x(:,:,r);
        Irecon = Irecon + abs(Ic) .^ 2;
    end
    E(k) = RLNE(Iorg,Irecon);
    Tim(k) = cputime-time_start;
end
delete(h)
%----- calling function -----
function [ xn, x, wres, wpert,tn]= Err_Calc(Kz,FT,XFM,x,xo,T,t)

Eres = Kz - FT*(XFM'*x); %Data consistency error
wres = XFM*(FT'*(Eres)); %consistency error in wavelet domain
xhat = x + wres; % Landweber update step
x = soft(xhat, T); % soft thresholding
wpert = xhat - x; % sparse approximation error in wavelet domain
tn = (1+sqrt(1+4*(t^2)))/2; % FISTA update steps
xn = x + ((t-1)/tn)*(x-xo);
%-----calling function-----
function[T1,llres,llpert,delt] = Update_Thresh(wres,wpert,T)
llres = norm(sum(wres,3),1); %  $l_1$ -norm of consistency error
llpert = norm(sum(wpert,3),1); %  $l_1$ -norm of sparse approximation error
delt = abs(llres-llpert); % discrepancy level
Cres = get_cov(wres); % covariance computation of consistency error
Cpert = get_cov(wpert); % covariance computation of sparse approximation error
temp1 = sum(sqrt(Cres(:))); % Threshold update steps
temp2 = 1/(T^2)*sum(sqrt(Cpert(:)));
T1 = sqrt(temp1/(delt+temp2));
%-----calling function-----
function[C]=get_cov(Estk)

```

```

R = size(Estk,3);
C = zeros(R,R);
for i=1:R,
    Er = Estk(:,:,i);
    Era = abs(Er.^2);
    for j=1:R,
        Ec = Estk(:,:,j);
        Eca = abs(Ec.^2);
        Mat = Era.*Eca;
        [h, bin] = hist(Mat(:),1001);
        hn = h/sum(h);
        me = sum(hn.*bin);
        var_e = sum(hn.*((bin-me).^2));
        C(i,j) = var_e;
    end
end

```

L.2.14 Basic Codes for Adaptive Total Variation Iterative Shrinkage (ATVIS)

L.2.14.1 Derivative

```

function [res] = derivative(image)
Dx = image([2:end,end],:) - image;
Dy = image(:, [2:end,end]) - image;
res = cat(3,Dx,Dy);

```

L.2.14.2 Divergence

```

function [res] = divergence(dI)
x = dI(:,:,1);
y = dI(:,:,2);
res = adjDx(x) + adjDy(y);
%---- calling function----
function res = adjDy(x)
res = x(:, [1,1:end-1]) - x;
res(:,1) = -x(:,1);
res(:,end) = x(:,end-1);
%---- calling function----
function res = adjDx(x)
res = x([1,1:end-1],:) - x;
res(1,:) = -x(1,:);
res(end,:) = x(end-1,:);

```

L.2.14.3 Initial Threshold Computation for Total Variation Iterative Shrinkage (TVIS)

```

function [T]=find_initThresh(I)
dI = derivative(I);
dN = sqrt(dI(:,:,1).^2+dI(:,:,2).^2);
[h,bin]= hist(abs(dN(:)),1001);
h = h/sum(h);
varnce = sum(bin.^2.*h);
bet = sqrt(.5*varnce);
sig_est=1.4826* mad(dI(:,1))/sqrt(2);
T = sig_est^2/(2*bet);

```

L.2.14.4 Initial Threshold Computation for Adaptive Total Variation Iterative Shrinkage (ATVIS)

```
function [T]=thresh_test(I)
N   = size(I,1);
a_m = -.395; % Initialization of constants
b_m = .552;
a_b = -1.512;
b_b = -.247;
dI  = derivative(I); % derivative of image
mu  = exp(a_m+b_m*log(log(N)));
be  = exp(a_b+b_b*log(log(N)));
p_M = 2*N*(N-1);
p    = 1-2/(sqrt(log(p_M)));
sig_est = 1.4826* mad(dI(:),1)/sqrt(2);
inv_gumb = mu - be*log(-log(p));
lam = sig_est*inv_gumb;
```

L.2.14.5 Gradient to Image Operator (Anisotropic)

```
function [I]=Grad2Im(dI)
[N,M,c] = size(dI);
Wi = Int_filtr(N,M);
I = idct2(dct2(-divergence(dI)).*Wi);

%----- calling function -----
function [Wi]=Int_filtr(N,M)
k=0:1:N-1;
l=0:1:M-1;
[K,L]=meshgrid(k,l);
K=K';
L=L';
W=2*cos(pi*K/N)+ 2*cos(pi*L/M)-4;
ind = find(W==0);
Wi = 1./W;
Wi(ind)=0;
```

L.2.15 Code for Comparison of Single-Channel Total Variation Iterative Shrinkage (TVIS) and Adaptive Total Variation Iterative Shrinkage (ATVIS)

```
function [Output]=Demo_Single (Iref,Nit, mask,tol)
% This implementation needs codes (ifft2c, fft2c, sos) available for download at
% http://people.eecs.berkeley.edu/~mlustig/Software.html
% To get image of the under-sampled k-space
I = ifft2c(fft2c(Iref).*mask);
% Compute Threshold for TVIS
[T] = find_initThresh(I);
% Compute Initial Threshold for ATVIS
[TA]= thresh_test(I);
% TVIS reconstruction
[Irec, E, Tim] = TV_ISTAreparep_Demo(Iref,T,Nit,mask,tol);

% ATVIS reconstruction
[IrecA, EA, TimA, ThA] = TV_alph_updte_rep_Demo(Iref, TA, Nit, mask, tol);
```

```

Output.Irec    = Irec;
Output.E       = E;
Output.Tim     = Tim;
Output.IrecA   = IrecA;
Output.EA      = EA;
Output.TimA    = TimA;
Output.ThA     = ThA;

```

L.2.15.1 Single-Channel Total Variation Iterative Shrinkage (TVIS) Reconstruction

```

function [Irec, E, Tim, Th] = TV_ISTAreplem_Demo(I,T,Nit,mask, tol,FT)
% Initialization
Kz = fft2c(I).*mask; % Under-sampled k-space
E = []; % Reconstruction error in k-space
U = zeros(size(I)); % Initializing zero image
U_p = U;
t = 1;
Un = U;
h=waitbar(0,'Iterations..');
time_start=cputime;
for k = 1:Nit
    waitbar(k/(Nit),h,sprintf('Iterations.. %d/%d',k,Nit))
    Eres = Kz - FT*(Un); % Data consistency error
    res = FT'(Eres); % Data consistency error in image domain
    g = Un+res;
    me_g = mean(g(:));
    dIhat = derivative(g);
    dI = soft(dIhat,lambda);
    U = Grad2Im(dI)+me_g;
    E(k) = RLNE(I,U);
    % FISTA steps
    tn = (1+sqrt(1+4*(t^2)))/2;
    Un = U + ((t-1)/tn)*(U-U_p);
    t = tn;
    U_p = U;
    if(k>1),
        Em_con = norm(U-Im_old)/norm(Im_old); % Computing tolerance
        if Em_con <tol,
            break;
        end
    end
    Im_old = U;
    Tim(k) = cputime-time_start;
end
Irec = U;
delete(h)

```

L.2.15.2 Single-Channel Adaptive Total Variation Iterative Shrinkage (ATVIS) Reconstruction

```

function [Irec, E, Tim, Th] = TV_alph_updte_rep_
Demo(I, T, Nit, mask, tol,FT)
% Initialization
Kz = fft2c(I).*mask; % Under-sampled k-space
E = []; % Initialize Reconstruction error in k-space

```

```

U = zeros(size(I)); % Initializing zero image
U_p = U;
t = 1;
Un = U;
h = waitbar(0,'Iterations..');
time_start = cputime;
for k = 1:Nit
    waitbar(k/(Nit),h,sprintf('Finding Histogram.. %d/%d',k,Nit))
    Eres = Kz - FT*(Un); % Data consistency error
    res = FT'(Eres); % consistency error in image domain
    g = Un+res;
    me_g = mean(g(:));
    dIhat = derivative(g);
    dI = soft(dIhat,T);
    U = Grad2Im(dI)+me_g;
    E(k) = RLNE(I,U);
    % FISTA steps
    tn = (1+sqrt(1+4*(t^2)))/2;
    Un = U + ((t-1)/tn)*(U-U_p);
    t = tn;
    U_p = U;
    % Computing errors in difference domain
    dres = derivative(res);
    dpert = dIhat-dI;
    ll_dres(k) = norm(dres(:),1);
    ll_dpert(k) = norm(dpert(:),1);
    %-----Histogram Computation-----
    [hres, bin_res] = hist(abs(dres(:)),1001); %Histogram of consistency error
    hres = hres/sum(hres);
    mu_res(k) = sum(bin_res.*hres);
    hpert = histc(abs(dpert(:)),bin_res);
    %Histogram of sparse approximation error
    hpert = hpert/sum(hpert);
    mu_pert(k) = sum(bin_res.*hpert');
    ind = find(dpert>=T);
    num_e(k) = length(ind);
    X(:, :, :, k) = dI;
    if(k>1),
        delt(k) = abs( ll_dres(k)-ll_dpert(k-1));
    end
    %-----Threshold Update-----
    if(k>1),
        T = mu_res(k)/(delt(k)+(mu_pert(k-1)/Th(k-1))); %Adapt
        Em_con = norm(U-Im_old)/norm(Im_old); % Computing tolerance
        if Em_con < tol,
            break;
        end
    end
    Th(k) = T;
    Im_old = U;
    Tim(k) = cputime-time_start;
end
Irec = U;
delete(h)

```

L.2.16 Comparison of Multi-channel Total Variation Iterative Shrinkage (TVIS) and Adaptive Total Variation Iterative Shrinkage (ATVIS) (Anisotropic)

```
function [Output] = Demo_Parallel(Iref, Nit, mask, FT)
% This implementation needs codes (ifft2c, fft2c, sos) available for download at
% http://people.eecs.berkeley.edu/~mlustig/Software.html
% To get image of the Under-sampled k-space
Mask = repmat (mask, [1,1,size(Iref,3)]);
I = ifft2c(fft2c(Iref).*Mask);
for c=1:size(Iref,3),
    T(c) = find_initThresh(I(:,:,c)); % Compute Threshold for TVIS
    TA(c) = thresh_
test(I(:,:,c)); % Compute Initial Threshold for ATVIS
end
% TVIS reconstruction
[Irec, E, Tim] = TV_ISTAreple_multi_Demo(Iref,T,Nit,mask,tol);
% ATVIS reconstruction
[IrecA, EA, TimA, ThA] = TV_alph_updte_rep_Multi_
Demo(Iref, TA, Nit, mask, tol);
Output.Irec = Irec;
Output.E = E;
Output.Tim = Tim;
Output.IrecA = IrecA;
Output.EA = EA;
Output.TimA = TimA;
Output.ThA = ThA;
```

L.2.16.1 Multi-channel Total Variation Iterative Shrinkage (TVIS) Reconstruction

```
function [Irec, E, Tim,Th] = TV_ISTAreple_multi_
Demo(I, T, Nit, mask, tol, FT)
% Initialization
C = size(I,3);
Kref = fft2c(I);
for c=1:C,
    Kz(:,:,c) = fft2c(I(:,:,c)).*mask; %Under-sampled k-space
end
E = []; %Reconstruction error in k-space
U = zeros(size(I)); %Initializing zero image
U_p = U;
t = 1;
Un = U;
h=waitbar(0,'Iterations..');
time_start=cputime;
for k = 1:Nit
    waitbar(k/(Nit),h,sprintf('Iterations... %d/%d',k,Nit))
    for c=1:C,
        Eres = Kz(:,:,c) - FT*(Un(:,:,c)); %Data consistency error
    where x denotes x(k-1)
    res = FT'(Eres); % residual error
    g = Un(:,:,c)+res;
    me_g = mean(g(:));
    dIhat = derivative(g);
    dI = soft(dIhat, lambda(c));
```

```

U(:,:,c) = Grad2Im(dI)+me_g;
end
E(k)      = RLNE(sos(I),sos(U));
% FISTA steps
tn        = (1+sqrt(1+4*(t^2)))/2;
np(k)     = norm(sum(U-U_p,3));
Un        = U + ((t-1)/tn)*(U-U_p);
t         = tn;
tp(k)     = ((t-1)/tn);
U_p       = U;
if (k>1),
    Em_con = norm(U(:,:,1)-Im_old(:,:,1))/norm(Im_old(:,:,1));
% Computing tolerance
    if Em_con < tol,
        break;
    end
end
Im_old    = (U);
Tim(k)    = cputime-time_start;
end
Irec = sos(U);
delete(h)

```

L.2.16.2 Multi-channel Adaptive Total Variation Iterative Shrinkage (ATVIS) Reconstruction

```

function [Irec, E, Tim, Th] = TV_alph_updte_rep_Multi_
Demo (I, T, Nit, mask, tol, FT)
% Initialization
C = size(I,3);
Kref = fft2c(I);
for c=1:C,
    Kz(:,:,c) = fft2c(I(:,:,c)).*mask; %Under-sampled k-space
end
E = []; % Reconstruction error in k-space
U = zeros(size(I)); %Initializing zero image
h=waitbar(0,'Iterations..');
time_start=cputime;
for k = 1:Nit
    waitbar(k/(Nit),h,sprintf('Iterations.. %d/%d',k,Nit))
    for c=1:C,
        Eres      = Kz(:,:,c) - FT*(U(:,:,c)); % Data consistency error
        res       = FT'*(Eres); % Image domain residual error
        g         = U(:,:,c)+res;
        me_g      = mean(g(:));
        dIhat     = derivative(g);
        dI        = soft(dIhat,T(c));
        U(:,:,c)  = Grad2Im(dI)+me_g;
        % Residual error in difference domain
        res       = derivative(res);
        dpert     = dIhat-dI;
        ll_dres   = norm((dres(:)),1);
        ll_dpert(k,c) = norm((dpert(:)),1);
    end
end

```



```

hres, bin_res] = hist(abs(dres(:)),1001); %Histogram of consistency error
hres          = hres/sum(hres);
mu_res(k,c)   = sum(bin_res.*hres);

hpert = histc(abs(dpert(:)),bin_res); %Histogram of sparse
approximation error
hpert          = hpert/sum(hpert);
mu_pert(k,c)   = sum(bin_res.*hpert');
if(k>1),
    delt = abs( l1_dres-l1_dpert(k-1,c));
end
%-----Threshold Update-----
if(k>1),
    T(c)          = mu_res(k,c) / (delt/256 + (mu_pert(k-1,c)/Th(k-1,c))); %Adapt
    Em_con(k) = norm(U(:, :, 1)-Im_old(:, :, 1))/norm(Im_old(:, :, 1));
% Computing tolerance
    if Em_con < tol,
        break;
    end
end
Th(k,c) =T(c);
end
E(k)          = RLNE(sos(ifft2c(Kref)), sos(U));
Im_old = U;
Tim(k)        = cputime-time_start;
end
Irec = sos(U);
delete(h)

```

L.2.17 Comparison of Single-Channel Total Variation Iterative Shrinkage (TVIS) and Adaptive Total Variation Iterative Shrinkage (ATVIS) (Isotropic)

```

function[Output]=Demo_iso( Iref, Nit, mask, FT, tol, L )
% This implementation needs codes available for download at
% http://people.eecs.berkeley.edu/~mlustig/Software.html
% To get image of the Under-sampled k-space
I = ifft2c(fft2c(Iref).*mask);
% Compute Threshold for TVIS
[T] = find_initThresh(I);
% Compute Initial Threshold for ATVIS
[TA] = thresh_test(I);
% TVIS reconstruction
[Irec, E, Tim] = TVIS(Iref, T, Nit, mask,L);
% ATVIS reconstruction
[IrecA, EA, TimA, ThA] = ATVIS(Iref, TA, Nit, mask,L);
Output.Irec = Irec;
Output.E = E;
Output.Tim = Tim;
Output.IrecA = IrecA;
Output.EA = EA;
Output.TimA = TimA;
Output.ThA = ThA;

```

L.2.17.1 Single-Channel Total Variation Iterative Shrinkage (TVIS)

```

function [Irec, E, Tim] = TVIS(I, T, Nit, mask, tol, L, FT)
% Initialization
Kz = fft2c(I).*mask; % Under-sampled k-space
E = []; % Initialize Reconstruction error in k-space
U = zeros(size(I)); % Initializing zero image
[N,M]= size(Kz);
U_p = U;
t = 1;
Un = U;
h = waitbar(0,'Iterations..');
time_start = cputime;
for k = 1:Nit
    waitbar(k/(Nit),h,sprintf('Finding Histogram.. %d/%d',k,Nit))
    Eres = Kz - FT*(U); % Data consistency error
    res = FT'*(Eres); % consistency error in image domain
    g = U+res;
    me_g = mean(g(:));
    dIhat = multi_dir_grad(g,L);
    dI = soft(dIhat,T);
    U = Grad2Im_iso(dI,L)+me_g;
    E(k) = RLNE(I,U);
    Tim(k) = cputime-time_start;
end
Irec = U;
delete(h)

```

L.2.17.2 Single-Channel Adaptive Total Variation Iterative Shrinkage (ATVIS)

```

function [Irec, E, Tim, Th] = ATVIS(I, T, Nit, mask,tol,L, FT)
% Initialization
Kz = fft2c(I).*mask; % Under-sampled k-space
E = []; % Initialize Reconstruction error in k-space
U = zeros(size(I)); % Initializing zero image
[N,M]= size(Kz);
U_p = U;
t = 1;
Un = U;
h = waitbar(0,'Iterations..');
time_start = cputime;
for k = 1:Nit
    waitbar(k/(Nit),h,sprintf('Finding Histogram.. %d/%d',k,Nit))
    Eres = Kz - FT*(U); % Data consistency error
    res = FT'*(Eres); % consistency error in image domain
    g = U+res;
    me_g = mean(g(:));
    dIhat = multi_dir_grad(g,L);
    dI = soft(dIhat,T);
    U = Grad2Im_iso(dI,L)+me_g;
    E(k) = RLNE(I,U);
    % Computing errors in difference domain

```

```

dres          = multi_dir_grad(res,L);
dpert         = dIhat-dI;
l1_dres(k)    = norm(dres(:),1)/L^2;
l1_dpert(k)   = norm(dpert(:),1)/L^2;
%-----Histogram Computation-----
[hres, bin_res] = hist(abs(dres(:)),1001); %Histogram of consistency error
hres          = hres/sum(hres);
mu_res(k)     = sum(bin_res.*hres);
hpert = histc(abs(dpert(:)),bin_res); %Histogram of sparse approximation error
hpert         = hpert/sum(hpert);
mu_pert(k)    = sum(bin_res.*hpert');
if(k>1),
    delt(k) = abs( l1_dres(k)-l1_dpert(k-1));
end
%-----Threshold Update-----
if(k>1),
    T = mu_res(k)/(.05*delt(k)/256+(mu_pert(k-1)/Th(k-1))); %Adapt
    Em_con = norm(U-Im_old)/norm(Im_old); % Computing relative
difference for tolerance check
    if Em_con < tol,
        break;
    end
end
Th(k) = T;
Im_old = U;
Tim(k) = cputime-time_start;
end
Irec = U;
delete(h)

```

L.2.17.3 Gradient to Image Operator

```

function[I]=Grad2Im_iso(dI,L)
[N,M,c] = size(dI);
Wi = Int_filttr(N,M);
I = (1/L)*ifft2(fft2(multi_dir_divergence(dI,L)).*Wi);
% the calling function Int_filttr is same as that in A.2.14.5.

```

L.2.17.4 Multi-directional Gradient Operator

```

function[v]=multi_dir_grad(u,L)
theta = (pi/2/L)*(0:L-1);
alpha = cos(theta);
beta = sin(theta);
[N,M] = size(u);
ux = u-u([N,1:N-1],:);
uy = u-u(:, [M,1:M-1]);
v = zeros(N,M,2*L);
for k=1:L,
    v(:, :, 2*k-1) = alpha(k)*ux+beta(k)*uy;
    v(:, :, 2*k) = alpha(k)*uy-beta(k)*ux;
end

```

L.2.17.5 Multi-directional Divergence Operator

```
function [u]=multi_dir_divergence(v,L)
theta = (pi/2/L)*(0:L-1);
alpha = cos(theta);
beta = sin(theta);
[N,M,K] = size(v);
[ux,uy] = deal(zeros(N,M));

for k=1:L,
    ux = ux+(alpha(k)*v(:,:,2*k-1)-beta(k)*v(:,:,2*k));
    uy = uy+(beta(k)*v(:,:,2*k-1)+alpha(k)*v(:,:,2*k));
end
u = (ux([2:N,1],:)-ux)+(uy(:,[2:M,1])-uy);
```

L.2.18 Adaptive L1-SPIRiT

```
function [I_recon, E, Th]=main_thadapt_spirit_
JS(DATA, kSize, nIter, mask, T)
% This implementation needs codes (ifft2c, sos, getCalibSize, crop, dat2AtA,
% zpad, softThresh, fft2c) available for download at http://people.eecs.berkeley.edu/~mlustig/Software.html
im = ifft2c(DATA);
%Under-Sampling
[DATA, CalibSize, scale_fctr]=get_SampledData(DATA,mask);
%Perform Calibration
[Z,z]=Imfilt_Calib(DATA,CalibSize,kSize);
%Reconstruction
[Kstk, E, Em_con, mu_res1, mu_pert1, madx1, Th] = TVthadapt_
JS_A(DATA, Z, nIter, T, im);
Istk = ifft2c(Kstk);
I_recon=sos(Istk);
```

L.2.18.1 Generate Under-Sampled Data Using Random Mask

```
function [DATA, CalibSize]=get_SampledData(DATA,mask)
C=size(DATA,3);
%Under-Sampling
DATA = DATA.*repmat(mask,[1,1,C]);
% estimating the largest region in the center of k-space
% to be used for calibration
[CalibSize, dcomp] = getCalibSize(mask);
```

L.2.18.2 Computation of SPIRiT Filter in the Image Domain

```
function [Z, z]=Imfilt_Calib(DATA,CalibSize,kSize,lambda)
% Perform Calibration
[Nfe, Npe,C] = size(DATA);
% to get the calibration data
kCalib = crop(DATA,[CalibSize,C]);
kernel = zeros([kSize,C,C]);
% Formation of Calibration Matrix
[AtA] = dat2AtA(kCalib, kSize);
%Calculation of coil coefficients
```

```

for n=1:C
    z(:,:,:,n) = calibrate(AtA, kSize, C, n, lambda);
end
for n=1:size(z,4)
    Z(:,:,:,n) = (ifft2c(zpad(z(end:-1:1,end:-
1:1,:,:),n)*sqrt(Nfe*Npe), Nfe, Npe, size(z,3)));
end

```

L.2.18.3 Adaptive SPIRiT Reconstruction

```

function [K, E, Th] = TVthadapt_JS_A (Kz, Z, nIter, T, im, tol)
acq_ind = find(abs(Kz)>0);
K        = Kz;
Eres1    = zeros(size(Kz));
Eres2    = zeros(size(Kz));
C        = size(Kz,3);
h =waitbar(0,'Iterations..');
for n=1:nIter
    waitbar(n/(nIter),h,sprintf('Iterations... %d/%d',n,nIter))
    %----- Apply filter for reconstructing initially----
    Ko    = recon(K,Z);
    Eres1(acq_ind) = Kz(acq_ind)-Ko(acq_ind);
    Pert1    = K-Ko;
    K        = Ko+Eres1;
    Uhat     = ifft2c(K);
    %----- apply TV thresholding-----
    [dUhat,me_g] = get_grad(Uhat);
    dU(:,:,1,:) = softThresh (dUhat(:,:,1,:),T); % Lustig's toolbox
    dU(:,:,2,:) = softThresh (dUhat(:,:,2,:),T); % Lustig's toolbox
    U          = grad_To_Im(dU, me_g);
    K          = fft2c(U);
    Eres2(acq_ind) = Kz(acq_ind)-K(acq_ind);
    K          = K + Eres2; %data consistency after soft thresholding
    Istk       = ifft2c(K);
    I_recon    = sos(Istk);
    E(n)       = RLNE(sos(im),I_recon);
    %-----check  $l_1$ -norms of the respective errors-----
    for c=1:C,
        % consistency Error and sparse approximation errors
        dres1(:,:,:,c) = derivative(ifft2c(Eres1(:,:,:,c)));
        dpert1(:,:,:,c) = derivative(ifft2c(Pert1(:,:,:,c)));
        pertch = derivative(ifft2c(Pert1(:,:,:,c)));
    end
    dresx    = dres1(:,:,1,:);
    dresy    = dres1(:,:,2,:);
    Dres (:,:,1) = (sum(abs(squeeze(dresx)),3));
    Dres (:,:,2) = (sum(abs(squeeze(dresy)),3));

    dpertx    = dpert1(:,:,1,:);
    dperly    = dpert1(:,:,2,:);
    Dpert (:,:,1) = (sum(abs(squeeze(dpertx)),3));
    Dpert (:,:,2) = (sum(abs(squeeze(dperly)),3));

    l1_dres(n) = norm(Dres(:),1);

```

```

    ll_dpert(n) = norm(Dpert(:),1);
    %-----Histograms and Means -----
    [hres1, bin_
res1] = hist(abs(Dres(:)),1001); % histogram of consistency error
    hres1 = hres1/sum(hres1);
    mu_res1(n) = sum(bin_res1.*hres1);
    hpert1 = histc(abs(Dpert(:)),bin_res1); % histogram of
filter approximation error
    hpert1 = hpert1/sum(hpert1);
    mu_pert1(n) = sum(bin_res1.*hpert1');
    % -----Threshold Adaptation-----
    if(n>1),
        delt(n) = abs(ll_dres(n)-ll_dpert(n));
        T = mu_res1(n)/(delt(n)/(size(Kz,2))+(mu_pert1(n)/
Th(n-1))); %threshold adaptation
    end
    Th(n) =T;
    if(n>1),
        Em_con(n) = norm(I_recon-Im_old)/norm(Im_old);
        if(Em_con<=tol)
            break;
        end
    end
    Im_old = I_recon;
end
delete(h)
%-----calling function-----
function [K]= recon(K,Z)
res = zeros(size(K));
C = size(K,3);
I = ifft2c(K);
for n=1:C
    Ztmp = Z(:,:,n);
    res(:,:,n) = sum(Ztmp.*I,3);
end
K = fft2c(res);
%-----calling function-----
function[dUhat,me_g]=get_grad(Uhat)
[M,N,C]=size(Uhat);
for c=1:C,
    g = Uhat(:,:,c);
    me_g(c) = mean(g(:));
    dUhat(:,:,c) = derivative(Uhat(:,:,c));
end
%-----calling function-----
function[U]=grad_To_Im(dU, me)
C = size(dU,4);
for c=1:C,
    der_im = squeeze(dU(:,:,c));
    U(:,:,c) = Grad2Im(der_im)+me(c);
end

```

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